

Mathematically, the economic factor model defines the return to stock i , r_i , as

$$r_i = \alpha_i + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i \quad (7.1)$$

where $f_1, \dots, f_K$ are factor premiums (which do not vary across stocks and so do not have the subscript i), and $\beta_{i1}, \dots, \beta_{iK}$ are factor exposures (which do vary across stocks and have the subscript i). The term α_i is the constant. The term $\beta_{i1}f_1 + \dots + \beta_{iK}f_K$ represents the nondiversifiable risk of the stock, and ϵ_i , the error, reflects the diversifiable risk of the stock.³ In terms of the basic equation, there is no difference between the fundamental factor model and the economic factor model.⁴

As with the fundamental factor model, we again define K -dimensional column vector $\mathbf{f}$ and $\boldsymbol{\beta}_i$ as follows:

$$\mathbf{f} = (f_1, \dots, f_K)' \quad (7.2)$$

$$\boldsymbol{\beta}_i = (\beta_{i1}, \dots, \beta_{iK})' \quad (7.3)$$

Using this notation,

$$r_i = \alpha_i + \boldsymbol{\beta}_i' \mathbf{f} + \epsilon_i \quad (7.4)$$

As we saw in [Chapter 6](#), the average stock return is the product of the factor exposure and the factor premium plus the constant term,

$$E(r_i) = \alpha_i + \boldsymbol{\beta}_i' E(\mathbf{f}) \quad (7.5)$$

whereas the total risk is the sum of the nondiversifiable risk and the diversifiable risk, that is,

$$V(r_i) = \boldsymbol{\beta}_i' V(\mathbf{f}) \boldsymbol{\beta}_i + V(\epsilon_i) \quad (7.6)$$

The model also can express the stock return at a particular time interval. We use lowercase letter t to indicate one time interval and the capital letter

T to indicate the number of time intervals. Thus the return to stock i at time t , r_{it} , is written as

$$r_{it} = \alpha_i + \beta_i' \mathbf{f}_t + \epsilon_{it} \quad t = 1, \dots, T \quad (7.7)$$

where α_i is a K -dimensional column vector of the factor exposure of stock i , $\mathbf{f}_t$ is a K -dimensional column vector of the factor premium at time t , and ϵ_{it} is the diversifiable component of the return to stock i at time t .⁵ Note that the factor premium has time subscript t because it changes over time. That is, we interpret the factor premium as a random variable that has different values at different times. On the other hand, the factor exposure does not have time subscript t because it is not assumed to change over time. We interpret the factor exposure as an unknown parameter rather than a random variable.

In academia, the economic factor model has long been the only factor model deemed valid, with the fundamental factor model only recently gaining some credence. In fact, the academic literature fails to distinguish between economic and fundamental factor models, using instead the term *multifactor pricing model* to refer generally to the framework of the economic factor model. We choose to distinguish between the two types of models using the terms *economic factor model* and *fundamental factor model* because many practitioners use the former for modeling stock returns against economic indicators and the latter for modeling stock returns against stock fundamentals. [Table 7.1](#) summarizes the process of creating an economic factor model.

TABLE 7.1

Creating an Economic Factor Model

Preliminary work:

1. Choose the factors for the model.
2. Determine the treatment of the risk-free rate.
3. Define the investment universe.
4. Decide on the time interval and time period of the data.

Main steps for modeling total stock return (for nonbenchmark managers):

1. Stock returns are estimated with the following equation: $r_i = \alpha_i + \beta_{i1}f_1 + \dots + \beta_{iK}f_K + \epsilon_i$.
2. Collect stock return and factor premium data for the time period at each time interval. factor premium data are readily available for economic/behavioral/market factors. To calculate factor premiums for fundamental/technical/analyst factors, use the zero-investment portfolio method. To calculate factor premiums for statistical factors, use principal-component analysis.
3. If there are sufficient factor premium data, estimate the factor exposure from a time-series regression of the stock return on the factor premium.
4. If there are insufficient factor premium data to perform a time-series regression, use characteristic matching to identify similar firms that can be used as proxies for the firms lacking data. Collect stock return and factor premium data for the proxies. Using the proxies' data, estimate the factor exposure from a time-series regression of the stock return on the factor premium.
5. Calculate each stock's average return as the product of the factor exposure and the vector containing the stock's factor premiums.
6. Decompose the risk of the stock return into its diversifiable and nondiversifiable components.
7. Calculate the correlation between the returns of the stocks in the investment universe.

Main steps for modeling residual stock return (for benchmark managers):

1. Stock returns are estimated with the following equation: $r_i = \tilde{\alpha}_i + \tilde{\beta}_i r_B + \tilde{\epsilon}_i$.
 2. Collect stock return and benchmark return data for the given time period at the given time intervals.
 3. After estimating $\tilde{\alpha}$ and $\tilde{\beta}$, calculate the residual return as $\tilde{r}_i \equiv \tilde{\alpha}_i + \tilde{\epsilon}_i$.
 4. Estimate residual return with the following equation: $\tilde{r}_i = \alpha_i + \beta_{i1}f_1 + \dots + \beta_{iK}f_K + \epsilon_i$.
 5. Follow steps 2 through 7 of the main steps for modeling total stock return, substituting the residual return for the total stock return.
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7.2 PRELIMINARY WORK

As with the fundamental factor model, constructing the economic factor model begins with good planning. The general decisions are the same—the choice of factors, the treatment of the risk-free rate, and the makeup of the

sample (in terms of time interval, time period, and stock universe). Since the criteria for deciding how to treat the risk-free rate, for determining the investment universe, and for defining the time interval and period are the same for either type of model, we will just discuss the factor choice here.⁶

The strength of the economic factor model is that it can include practically all kinds of factors. In terms of how the model treats them, there are three categories of factors:

1. *Economic/behavioral/market factors*: Gross domestic product (GDP), inflation, unemployment, interest rates, and other macroeconomic variables; consumer sentiment index, business confidence index, investor sentiment index, or other survey-based indexes; returns on broad market indexes such as the Standard & Poor's (S&P) 500 or returns on other market group/industry indexes
2. *Fundamental/technical/analyst factors*: Log of market capitalization, book-to-price ratio, earnings-to-price ratio, debt-to-equity ratio, and other firm characteristics available through financial statements; momentum, trading volume, and other information reflected in trading data; analyst rating changes, earnings revisions, or other information provided by analysts
3. *Statistical factors*: Factors obtained from principal-component analysis applied to historical returns

For economic/behavioral/market factors, computation is minimal. In most cases, relevant information is publicly available or provided by data vendors. On the other hand, for fundamental/technical/analyst factors, computation can be somewhat demanding, and even though they may be obtained from an external source, the vendor's methodology should be examined carefully. Some studies suggest, though, that thoroughly vetted fundamental/technical/analyst factors predict stock returns more effectively than other factor groups do.⁷ Statistical factors require the highest level of computation.

7.3 BENCHMARK AND α

Many quantitative portfolio managers manage their portfolios against a benchmark. As we mentioned in [Chapter 6](#), it is useful but not necessary to account for the benchmark at the outset of creating a model. Some portfolio managers prefer to account for the benchmark later on—by controlling for the tracking error at the portfolio construction stage, for instance.⁸ As a measure of portfolio performance, however, it can be useful to create a model that reflects, or adjusts for, the effects of the benchmark. One way to do this is to remove the benchmark-related return from the stock return before building the model.⁹ We first regress stock returns on the benchmark returns and calculate the residual, the part of stock returns that is not correlated with the benchmark return. Then we construct an economic factor model for the residual stock return rather than the total stock return. The resulting model shows the relationship between the factor premium, the factor exposure, and the portion of the stock return that is above and beyond the benchmark return.

Alternatively, we can include the benchmark in the economic factor model explicitly. If the benchmark is the broad-market index, the capital asset pricing model (CAPM) supports its inclusion in the model as a predictor of stock returns. Even if the benchmark is something other than the broad-market index, including it in the model will clarify the relationship between the portfolio and the benchmark.

Let r_B be the benchmark return. If we add the benchmark as the K th factor in the economic factor model, then the economic factor model becomes

$$r_i = \alpha_i + \beta_{i,1}f_1 + \cdots + \beta_{i,K-1}f_{K-1} + \beta_{i,K}r_B + \epsilon_i \quad (7.8)$$

Given this formulation, we can interpret the right-hand side of the equation up to the term $\beta_{i,K-1}f_{K-1}$ as the expected return of stock i that is not related to the benchmark. Collectively, these terms represent the benchmark alpha, or α^B , that is,

$$\alpha^B = \alpha_i + \beta_{i,1}f_1 + \cdots + \beta_{i,K-1}f_{K-1} \quad (7.9)$$

The benchmark α shows the contribution of the portfolio manager (and the model he or she creates) to the stock return and is one measure of the

portfolio manager's performance.¹⁰

7.4 THE FACTOR PREMIUM

Recall that to create the fundamental factor model, the portfolio manager collects a data set of many observations of stock return and factor exposure. The factor exposure changes over time (and the factor premium—a single, statistically estimated proportion—remains constant). By contrast, to create the economic factor model, the portfolio manager gathers pairs of stock returns and factor premiums, and the factor premiums change over time. If there are K factors, one needs to find the factor premiums of the K factors at each time interval t , $\mathbf{f}_t = (f_{1t}, \dots, f_{Kt})$.

In the economic factor model, the factor premium is the *known* value (as opposed to the factor exposure, which is a regression *estimate*). This does not mean that one can always observe a factor premium directly, though. For economic/behavioral/market factors, the computation is rather trivial. For fundamental/technical/analyst factors, though, the computation is somewhat more demanding, and for statistical factors, it poses quite a bit of a challenge.

7.4.1 Factor Premium for Economic/Behavioral/Market Factors

Obtaining factor premiums for economic/behavioral/market factors does not involve any computation. All we need to do is simply “copy and paste” the values.¹¹

Table 7.2 presents the premiums for three factors: unemployment, consumer sentiment, and overall market return. To find the factor premium for unemployment, we simply take the unemployment rate published by the Bureau of Labor Statistics (BLS). For example, the unemployment rate in December 2019 was 3.6%, so after allowing for a one-month reporting gap, the factor premium for January 2020 is 3.6%. For consumer sentiment, we take the consumer sentiment index compiled by the University of Michigan, which shows the level of consumer sentiment for each month. The factor premium is simply the growth rate of this index from one month to the next month. For example, the consumer sentiment index for December 2019 was

99.3, whereas the index for January 2020 was 99.8. Thus, our consumer sentiment factor premium is 0.50% $\left[= \left(\frac{99.8}{99.3} - 1 \right) \cdot 100 \right]$. The market factor premium is the S&P 500 total return in excess of the one-month U.S. Treasury bill return.¹² For example, in January 2020, the S&P 500 total return was -0.04% , whereas the one-month U.S. Treasury bill return was 0.13% . Thus, the factor premium for January 2020 was -0.17% ($= -0.04 - 0.13$).

Table 7.2

Premium of Unemployment, Consumer Sentiment, and Market Factors

	Unemployment Rate	Consumer Sentiment Growth	Excess Market Return
Jan 2020	3.60	0.50	-0.17
Feb 2020	3.50	1.20	-8.35
Mar 2020	3.50	-11.78	-12.47
Apr 2020	4.40	-19.42	12.82
Apr 2020	14.80	0.70	4.75
Jun 2020	13.30	8.02	1.98
Jul 2020	11.10	-7.17	5.63
Aug 2020	10.20	2.21	7.18
Sep 2020	8.40	8.50	-3.81
Oct 2020	7.80	1.74	-2.67
Nov 2020	6.90	-5.99	10.94
Dec 2020	6.70	4.94	3.83

Note: All numbers are in percent. The specific formulas for each factor can be found in Appendix 16A.

After we have all the factor premiums for a certain time interval, we may store them in a set of vectors $\{\mathbf{f}_1, \dots, \mathbf{f}_T\}$, where $\mathbf{f}_t$ is the factor premium for time t . In this example, we can write the factor premium for January 2020 as

$$\mathbf{f}_{\text{Jan20}} = (3.6, 0.50, -0.17) \quad (7.10)$$

The first element of $\mathbf{f}_t$ always reflects the existence of the constant term in the return equation.

7.4.2 Factor Premium for Fundamental/Technical/Analyst Factors

A little more computation is necessary to find the factor premiums for fundamental/technical/analyst factors. The computation involves constructing zero-investment portfolios and calculating their returns. A zero-investment portfolio simultaneously takes a long position in a portfolio of stocks with high factor exposures and a short position in a portfolio of stocks with low factor exposures. Suppose that we want to find the factor premium on value, and suppose that we use the book-to-price (B/P) ratio as a proxy for value. We need to identify a portfolio of stocks with high exposure to the value factor (i.e., high B/P ratio) and a portfolio of stocks with low exposure to the value factor (i.e., low B/P ratio). To be systematic about it, we can start by ranking all the stocks at time t in order of their B/P ratio, with the highest ratio stock ranked first. We can create a high-value portfolio by equally weighting the stocks in the top 33% of the list and a low-value portfolio by equally weighting the stocks in the bottom 33%. The zero-investment portfolio return equals the difference between the return on the high-value portfolio and the return on the low-value portfolio.

The frequency of the zero-investment portfolio construction needs to be carefully selected if the sorting variable is not frequently updated. In the case of the book-to-price ratio, book values are only reported every quarter, while price is updated daily. While it is possible to calculate the book-to-price ratio every day, the daily fluctuation in this ratio would only reflect daily changes in the stock price rather than daily changes in the book value. Therefore, the ratio is likely to have a big jump when the book values are updated. Thus, creating a zero-investment portfolio too frequently may not reflect information an investor needs and, consequently, may not be a wise choice.

It is important to note that a single piece of data can represent more than one type of factor. The same B/P ratio value, for instance, can either be considered high exposure to a value factor or low exposure to a growth factor. There are many instances of this sort of symmetry. We could use market capitalization figures to measure a small-size factor, in which case a large market capitalization would mean low exposure to the small-size factor. Conversely, we could interpret the same large market cap figure as a high exposure to a large-size factor. If we are considering including a

momentum factor in the model, a low exposure to the momentum factor means the same thing as a high exposure to the contrarian factor. The same numerical figure can stand in for either factor. In fact, we could use either factor to represent the continuity or change in direction of a stock's return from one year to the next. The sign and name of the factor do not matter as long as we assign consistent meanings (low exposure or high exposure, as the case may be) to high and low values.

In general, these are the steps to calculate the factor premium for fundamental/technical/analyst factors:

1. Rank all the stocks at time 1 in terms of the factor.
2. Create high-exposure and low-exposure portfolios by equally weighting the stocks in the top 33% of the list and in the bottom 33% of the list. (A critical value other than 33% may be justifiable.¹³)
3. Calculate the zero-investment portfolio return as the difference between the returns on the high-exposure and low-exposure portfolios. The return on the zero-investment portfolio is the factor premium for time 1.

Repeat these steps for time 2. If lack of data prevents the construction of new portfolios for time 2, calculate the returns as of time 2 on the same portfolios used for time 1. Repeat the procedure for each factor at each time interval. In other words, the total number of iterations of the procedure will equal the number of factors multiplied by the number of time intervals.

[Table 7.3](#) presents the premium for the size and value factors.¹⁴

TABLE 7.3

Premium of Market Capitalization and Book-to-Price Factors

	Log of Market Capitalization (LOGSIZE)	Book-to-Price Ratio (B/P)
Jan 2020	3.67	−8.35
Feb 2020	−1.00	−3.34
Mar 2020	6.71	−10.83
Apr 2020	−6.55	2.99
May 2020	1.22	−7.85
Jun 2020	−5.01	−1.67
Jul 2020	2.66	−6.96
Aug 2020	−1.74	1.98
Sep 2020	1.79	−3.81
Oct 2020	−0.83	4.74

Note: The figures are in percent. To create a zero-investment portfolio, we include the stocks with low values (i.e., small market capitalization or low B/P ratio) in the short portfolio and stocks with high values (large market capitalization or high B/P ratio) in the long portfolio. The zero-investment portfolio return is the long portfolio returns minus the short portfolio returns. Thus, a positive number for B/P or value means that value stocks did well in that month and a positive number for market capitalization (also known as SIZE or LOGSIZE) indicates that large companies did better than small companies in that month. In this sense, our LOGSIZE portfolio is different than, for example, the Fama–French SMB, which is the small-cap portfolio minus the large-cap portfolio. In addition, our measure of market capitalization is actually the natural log of the market capitalization of a company.

7.4.3 Factor Premium for Statistical Factors

Obtaining factor premiums for statistical factors involves a rather intensive computation. The computational procedure is known as *principal-component analysis* and is available on standard computer software packages.

The starting point for principal-component analysis is estimating the variance-covariance matrix of stock returns. We will express N stock returns at time t as an N -dimensional column vector; that is, $\mathbf{r}_t = (r_{1t}, \dots, r_{Nt})'$. We have a total of T such vectors $\{\mathbf{r}_1, \dots, \mathbf{r}_T\}$. The variance-covariance matrix of returns Σ is estimated as

$$\hat{\Sigma} = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t \mathbf{r}_t' - \bar{\mathbf{r}} \bar{\mathbf{r}}' \quad (7.11)$$

where $\bar{\mathbf{r}}$ is the average return vector, that is,

$$\bar{\mathbf{r}} = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t \quad (7.12)$$

Once we have the variance-covariance matrix, we “diagonalize” it by finding an orthogonal matrix $\mathbf{Q}$ (that is, $\mathbf{Q}^{-1} = \mathbf{Q}'$) such that

$$\mathbf{Q}' \hat{\Sigma} \mathbf{Q} = \mathbf{D} \quad (7.13)$$

where $\mathbf{D}$ is a diagonal matrix whose diagonal elements are eigenvalues (i.e., characteristic values) of $\hat{\Sigma}$. It turns out that each column of $\mathbf{Q}$ is an orthonormal (i.e., of unit length) eigenvector corresponding to eigenvalues of $\hat{\Sigma}$.

To be more specific, let $\lambda_1, \dots, \lambda_N$ be the eigenvalues of $\hat{\Sigma}$ such that $\lambda_1 \geq \dots \geq \lambda_N \geq 0$. (Since $\hat{\Sigma}$ is a positive definite matrix, all the eigenvalues are positive.) Then matrix $\mathbf{D}$ is constructed as

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_N \end{pmatrix} \quad (7.14)$$

Let $\mathbf{q}_1, \dots, \mathbf{q}_N$ be the orthonormal eigenvectors corresponding to $\lambda_1, \dots, \lambda_N$. Then matrix $\mathbf{Q}$ is

$$\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_N) \quad (7.15)$$

If we want to find K factors, then we obtain K factor premiums by weighting individual stock returns using the first K columns of $\mathbf{Q}$. That is, factor premiums $f_1, \dots, f_K$ are defined as

$$\begin{aligned}
f_{1,t} &= \mathbf{q}'_1 \mathbf{r}_t \\
&\vdots \\
f_{K,t} &= \mathbf{q}'_K \mathbf{r}_t
\end{aligned} \tag{7.16}$$

These K factors together have the highest in-sample explanatory power for N stock returns among any set of K explanatory variables constructed from linear combinations of N stock returns.¹⁵

7.5 FACTOR EXPOSURE

7.5.1 The Standard Approach

In the economic factor model, factor exposures typically are determined from the time-series regression of stock returns on factor premiums. Since the regression coefficients (the factor exposures) measure the sensitivity of the dependent variable (the stock return) to the change in the independent variables (the factor premiums), factor exposures sometimes are called *factor sensitivities*. They are also sometimes referred to as *factor loadings*.

Given the returns of stock i and the factor premium for T periods of time, $\{r_{i1}, \dots, r_{iT}\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_T\}$, we can estimate the following equation¹⁶:

$$r_{it} = \alpha_i + \boldsymbol{\beta}'_i \mathbf{f}_t + \epsilon_{it} \quad t = 1, \dots, T \tag{7.17}$$

where “coefficient” $\boldsymbol{\beta}_i$ is the factor exposure that we wish to discover, and ϵ_{it} is the error term reflecting the diversifiable risk of stock returns. The ordinary least squares (OLS) estimator of $\boldsymbol{\beta}_i$ is given by

$$\hat{\boldsymbol{\beta}}_i = \left[\sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(\mathbf{f}_t - \bar{\mathbf{f}})' \right]^{-1} \left[\sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(r_{it} - \bar{r}_i) \right]^{-1} \tag{7.18}$$

where

$$\bar{\mathbf{f}} = \frac{1}{T} \sum_{t=1}^T \mathbf{f}_t \quad (7.19)$$

and

$$\bar{r}_i = \frac{1}{T} \sum_{t=1}^T r_{it} \quad (7.20)$$

By repeating the regression for each of the N stocks, we will obtain all the factor exposures we need: $\{\hat{\beta}_1, \dots, \hat{\beta}_N\}$. The standard error of the factor exposure is the square root of (the diagonal elements of) the following variance:

$$\hat{V}(\hat{\beta}_i) = \hat{\sigma}_i^2 \left[\sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(\mathbf{f}_t - \bar{\mathbf{f}})' \right]^{-1} \quad (7.21)$$

where $\hat{\sigma}_i^2$ is the estimated variance of ϵ_{it} :

$$\hat{\sigma}_i^2 = \frac{1}{T} \sum_{t=1}^T (r_{it} - \hat{\alpha}_i - \hat{\beta}_i' \mathbf{f}_t)^2 \quad (7.22)$$

Table 7.4 presents the factor exposure estimates and standard errors for selected stocks for a five-factor model. These factor exposures were estimated using monthly data from January 2016 to December 2020. The unemployment factor exposures are negative for Microsoft, Johnson & Johnson, and Amazon but positive for Apple and Walmart. This reflects Apple's and Walmart's positions as business-cycle-defensive stocks: stocks whose values tend to go up when the economy slows down.¹⁷ Although one would expect all five stocks in Table 7.4 to have a positive exposure to the log of market capitalization factor, they do not. That is, Microsoft, Walmart, and Johnson & Johnson have a positive exposure, meaning these stock returns tend to be positive when the portfolio of large-cap companies does better than a portfolio of small-cap companies. However, Amazon and

Apple have a slightly negative reading. The value or book-to-price exposures indicate that Walmart is a value stock (with a positive exposure to the value or book-to-price factor), whereas the other companies are growth stocks (with negative exposures to the book-to-price factor).

TABLE 7.4

Factor Exposure of Selected Stocks

Ticker	Unemployment Rate (UR)	Consumer Sentiment Growth (CSG)	Excess Market Returns (MKT)	Log of Market Capitalization (LOGSIZE)	Book-to-Market Ratio (B/P)
AAPL	0.215 (0.411)	0.226 (0.191)	1.418 (0.218)	-0.066 (0.354)	-0.647 (0.310)
MSFT	-0.297 (0.233)	0.127 (0.108)	1.008 (0.124)	0.203 (0.201)	-0.320 (0.176)
WMT	0.231 (0.280)	-0.302 (0.130)	0.473 (0.148)	0.660 (0.241)	0.244 (0.211)
JNJ	-0.309 (0.233)	-0.141 (0.108)	0.768 (0.123)	0.191 (0.200)	-0.025 (0.175)
AMZN	-0.356 (0.379)	-0.181 (0.176)	1.345 (0.201)	-0.088 (0.326)	-0.841 (0.285)

Note: For the period from January 2016 to December 2020. Standard errors inside the parentheses. Company names for ticker symbols are Apple (AAPL), Microsoft (MSFT), Walmart (WMT), Johnson & Johnson (JNJ), and Amazon (AMZN).

It is worth noting that while the factor exposure estimates in [Eq. \(7.4\)](#) can be explained with common sense, this does not necessarily imply that they are identical to the factor exposures we used for the fundamental factor model in [Chapter 6](#). For example, the ranking of stocks according to estimated size exposure does not exactly correspond to the ranking of stocks according to actual market capitalization, which was what we used as the size factor exposure in [Chapter 6](#). As a result, here Walmart ranks first in terms of size exposure, whereas it ranks fourth after Apple, Microsoft, Amazon, and Johnson & Johnson in terms of actual market capitalization. Similarly, the ranking of stocks by the estimated B/P ratio or

value exposure does not correspond exactly to the ranking of stocks by the observed B/P ratio of each stock.

7.5.2 When the Standard Approach Fails

To run time-series regressions, portfolio managers need to have enough data on stock returns and factor premiums to thoroughly cover a reasonable time period at regular time intervals. Recent initial public offerings (IPOs) and the stocks of recently merged or divested companies lack sufficient data for meaningful regressions. For a stock with insufficient data, we may infer the factor exposures by weighting groups of similar stocks and using the weighted factor exposures as proxies for the original stock's exposures.

When two firms merge, the natural thing to do is to find the weighted average of the factor exposures of the two premerger firms.¹⁸ The appropriate weights would be the market capitalizations of the premerger firms. Suppose that firm A and firm B merged recently. From stock returns of firm A , r_{At} , we can find the factor exposures of firm A , $\hat{\beta}_A$, by regressing factor premiums on stock returns, that is,

$$\hat{\beta}_A = \left[\sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(\mathbf{f}_t - \bar{\mathbf{f}})' \right]^{-1} \left[\sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(r_{At} - \bar{r}_A)' \right] \quad (7.23)$$

We can find the factor exposures of firm B , $\hat{\beta}_B$, in a similar way. Then the factor exposures of the merged firm $\hat{\beta}_{AB}$ can be calculated by

$$\hat{\beta}_{AB} = \frac{s_A}{s_A + s_B} \hat{\beta}_A + \frac{s_B}{s_A + s_B} \hat{\beta}_B \quad (7.24)$$

where s_A is the premerger market capitalization of firm A , and s_B is the premerger market capitalization of firm B .

For a recent IPO, the only available information about the firm is the observable firm characteristics (i.e., what is reported in financial statements). We can use this information to find similar firms and take the average factor exposures of those similar firms. Suppose that we want to find similar firms using L characteristics. Then we calculate the Z-score for

each of the L characteristics of each firm for which we already have factor exposures (see [Chapter 5](#) for Z-score). Let a young firm i 's Z-score be $\mathbf{z}_i = (z_{i1}, \dots, z_{iL})$. To identify similar firms, we can choose a small critical level e and find all firms j such that $(\mathbf{z}_i - \mathbf{z}_j)'(\mathbf{z}_i - \mathbf{z}_j) < e$. We choose a critical level e that will give us more than one similar firm. Once we identify the similar firms, we can take the equal-weighted average of the factor exposures of the similar firms as the factor exposure of firm i . That is, if firms $1, \dots, M$ are similar to young firm C , then the factor exposure of the young firm C , $\hat{\beta}_C$, is given by

$$\hat{\beta}_C = \frac{1}{M} (\hat{\beta}_1 + \dots + \hat{\beta}_M) \quad (7.25)$$

where $\hat{\beta}_1, \dots, \hat{\beta}_M$ are the factor exposures of firms $1, \dots, M$.

This process of identifying similar firms is called *characteristic matching*, and it applies to searching for other characteristics, such as expected stock return, as well.¹⁹ An alternative to characteristic matching is using the industry-average figure. For example, the average factor exposure of the entire pharmaceutical industry can stand in for the factor exposure of one new pharmaceutical company's stock. This approach is used quite often to guess the market "beta" of IPO stocks. Financial economic research suggests, however, that characteristic matching works better than using the industry average.

7.6 DECOMPOSITION OF RISK

7.6.1 The Standard Approach

The total risk of the stock return is the sum of nondiversifiable risk and diversifiable risk. The nondiversifiable risk depends on the factor exposure and the variance of the factor premium, whereas the diversifiable risk equals the variance of the error term, that is,

$$V(r_i) = \beta_i' V(f) \beta_i + V(\epsilon_i) \quad (7.26)$$

We already have the estimate for β_i . Given the factor premium data $\{\mathbf{f}_1, \dots, \mathbf{f}_T\}$, finding the estimate for $V(\mathbf{f}_t)$ is straightforward:

$$\hat{V}(\mathbf{f}) = \frac{1}{T} \sum_{t=1}^T (\mathbf{f}_t - \bar{\mathbf{f}})(\mathbf{f}_t - \bar{\mathbf{f}})' \quad (7.27)$$

The estimate for $V(\epsilon_i)$ is obtained naturally from estimation of the factor exposure if the factor exposure was estimated by the standard approach in Section 7.5.1. However, if the factor exposure was estimated by the approach in Section 7.5.2, we need to use an alternative approach. Given the estimate of the factor exposure and constant $\hat{\beta}_i$ and $\hat{\alpha}_i$, the estimate for $V(\epsilon_i)$ is as follows:

$$\hat{V}(\epsilon_i) = \frac{1}{T} \sum_{t=1}^T (r_{it} - \hat{\alpha}_i - \hat{\beta}_i' \mathbf{f}_t)^2 \quad (7.28)$$

Table 7.5 shows the decomposition of risk for selected stocks estimated from January 2016 to December 2020. We show the risk as measured by the variance and the standard deviation. Note that the relative size of the diversifiable risk varies quite a bit across stocks.

TABLE 7.5

Risk Decomposition for Selected Stocks

Ticker	Total Risk		Nondiversifiable Risk		Diversifiable Risk	
	V	SD	V	SD	V	SD
AAPL	911.73	30.19	456.93	21.38	454.80	21.33
MSFT	334.74	18.30	188.32	13.72	146.42	12.10
WMT	302.20	17.38	91.42	9.56	210.78	14.52
JNJ	278.29	16.68	133.00	11.53	145.29	12.05
AMZN	823.41	28.70	437.56	20.92	385.85	19.64

Note: Based on the period from January 2016 to December 2020. V refers to the variance, and SD refers to the standard deviation. Both variance and standard deviation are annualized and expressed as percentages. Company names for ticker symbols are reported in Table 7.4.

When we get to the stage of finding the optimal portfolio, we also need to know the correlation among stock returns. The correlation between two stocks' returns has two parts: the correlation between the nondiversifiable components and the correlation between the diversifiable components. The covariance between two stock returns r_i and r_j is

$$C(r_i, r_j) = \beta_i' V(f) \beta_j + C(\epsilon_i, \epsilon_j) \quad (7.29)$$

The estimate for $C(\epsilon_i, \epsilon_j)$ can be found from estimation of the factor exposure, that is,

$$\hat{C}(\epsilon_i, \epsilon_j) = \frac{1}{T} \sum_{t=1}^T (r_{it} - \hat{\alpha}_i - \hat{\beta}_i' f_t) (r_{jt} - \hat{\alpha}_j - \hat{\beta}_j' f_t) \quad (7.30)$$

In practice, however, unless T is quite large, it is conventional to assume that $C(\epsilon_i, \epsilon_j)$ is zero. If there are N stocks, then $N(N-1)/2$ covariances should be estimated. Unless T is large, not all these covariances can be estimated precisely. If the model is good, it does not create too much distortion to assume that the covariance is zero. Diversifiable risks, the ϵ terms, can be diversified out of the portfolio and ignored, so their covariance ought to be negligible as well.

7.6.2 When the Standard Approach Fails

If a stock is new and lacks data, then $V(\epsilon_i)$ cannot be estimated in a conventional way. However, even if a stock is new, it usually has at least a short trading history. Thus it may be possible to estimate the variance of the error term at a higher frequency (shorter time interval) and recover the variance of the frequency that we need.

For the purposes of illustration, let us assume that the time interval of the analysis is a month. Suppose that for young stock A the available data cover only a few days. In this situation, we can recover the variance of the monthly error term from the variance of the daily error term. We use letter s to indicate the daily time interval. The data for stock A are available for S days, $s = 1, \dots, S$. Let $\tilde{r}_{As}$ be the return to stock A on day s . Given the factor

exposure and constant estimate $\hat{\beta}_A$ and $\hat{\alpha}_A$, we can estimate the variance of the daily error term $\tilde{\epsilon}_{As}$, that is,

$$\hat{V}(\tilde{\epsilon}_A) = \frac{1}{S} \sum_{s=1}^S (\tilde{r}_{As} - \hat{\alpha}_A - \hat{\beta}_A' \tilde{\mathbf{f}}_s)^2 \quad (7.31)$$

where $\tilde{\mathbf{f}}_s$ is the factor premium for day s .²⁰

To recover the variance of the monthly error term, first we identify the stocks that are similar to stock A based on the characteristics, as explained in the preceding section. Suppose that stocks $1, \dots, M$ are such similar stocks. Then we estimate the variance of the daily error term for those similar stocks, $\hat{V}(\tilde{\epsilon}_1), \dots, \hat{V}(\tilde{\epsilon}_M)$ as well as the variance of the monthly error term for those similar stocks, $\hat{V}(\epsilon_1), \dots, \hat{V}(\epsilon_M)$. Assuming that the ratio of the monthly variance to the daily variance is similar for similar stocks, we can estimate the monthly variance of stock A as

$$\hat{V}(\epsilon_A) \equiv \hat{V}(\tilde{\epsilon}_A) \frac{1}{M} \left[\frac{\hat{V}(\epsilon_1)}{\hat{V}(\tilde{\epsilon}_1)} + \dots + \frac{\hat{V}(\epsilon_M)}{\hat{V}(\tilde{\epsilon}_M)} \right] \quad (7.32)$$

This procedure will allow the estimator to have less extreme values because we are scaling the monthly variance from similar stocks.

7.7 CONCLUSION

In this chapter we turned our attention from one major tool of QEPM—the fundamental factor model—to the other—the economic factor model. The economic factor model, like its counterpart, relates a stock's return to its factor exposures and the factor premiums that the marketplace assigns them, but it emphasizes the fact that stock returns react to changes in the overall economic environment. The economic factor model is one of the most theoretically sound models of stock returns and, at the same time, one of the most flexible in terms of the variety of factors it can handle.

The conceptual distinctions between the economic and fundamental factor models can confuse even the most experienced practitioners. We

pointed out that the major difference between the two models is the fact that in the economic factor model the factor exposure of each stock must be estimated from the data, whereas in the fundamental factor model statistical estimation is instead necessary for the factor premium. For an economic factor model, for instance, we must estimate how much a company's stock reacts to a shift in unemployment or real GDP growth. Another difference between the models is that the economic factor model encapsulates a straightforward, intuitively appealing concept: stocks react to external risks in the marketplace. Even when data are scarce, such as with recent IPOs or recently merged companies, this model can help us to gauge how stocks will respond to the economic environment.

The two factor models achieve similar ends, though. Like the fundamental factor model, the economic factor model can be a model of either overall stock returns or of benchmark α , providing a performance metric for benchmarked portfolios. It can be used to predict the expected return and expected risk of any stock in the portfolio manager's investment universe, as well as decompose the risk of any stock into diversifiable and nondiversifiable components. Both factor models serve as essential tools as we move toward [Chapter 9](#) and strive for the final frontier of portfolio construction, the optimal portfolio. Before we can construct a portfolio, though, we need to gather inputs that are necessary in order to make predictions about future stock returns.

QUESTIONS

- 7.1. Explain why the capital asset pricing model (CAPM) is a special case of the economic factor model.
- 7.2. Explain why the economic factor model is typically estimated from time-series data.
- 7.3. Explain why the factor premium is treated as observable in the economic factor model but not in the fundamental factor model.
- 7.4. Discuss the strengths and weaknesses of the economic factor model compared with the fundamental factor model.
- 7.5. If the benchmark is a broad market index, can benchmark α be interpreted as CAPM α ?
- 7.6. Consider a simple economic factor model:

$$r_i = \alpha_i + \beta_i f + \epsilon_i \quad \epsilon_i \sim N(0, \sigma_i^2)$$

Suppose that the estimation produced $\hat{\alpha}_i = 2$ and $\hat{\beta}_i = 0.5$. The portfolio manager has a benchmark whose return is denoted as r_B .

- (a) Calculate the benchmark α when the benchmark return and the factor premium are identical.
 - (b) Calculate the benchmark α when the benchmark return and the factor premium are perfectly correlated.
 - (c) Express the benchmark α as a function of the following three quantities: the correlation between r_B and f , the variance of r_B , and the variance of f .
 - (d) Specify the exact condition under which multifactor α is identical to benchmark α .
- 7.7.** Describe three ways to calculate the factor premium.
- 7.8.** Based on the capital asset pricing model (CAPM), a broad market index is often included as a factor in the economic factor model. Would it be a good idea to use the Dow Jones Industrial Average for this purpose?
- 7.9.** Consider an economic factor model with the following three factors: the size factor (i.e., based on the market capitalization), the return on the S&P 500, and the return on the S&P 600. What problem would you encounter in estimating such a model?
- 7.10.** In the financial economic literature, the factor premium is assumed to have a mean of zero. One way to satisfy this assumption is to write the economic factor model as

$$r_i = \alpha_i + \beta_{i1}[f_1 - E(f_1)] + \dots + \beta_{iK}[f_K - E(f_K)] + \epsilon_i$$

Would the estimation of this equation result in different estimates of $\beta_{i1}, \dots, \beta_{iK}$? How about the estimates of α_i ?

- 7.11.** Suppose that there is an exact linear relationship between the expected stock return and firm size; that is,

$$E(r_i) = 5 + 0.01x_i$$

where x_i is the size of firm i . Assume that x_i is uniformly distributed between 100 and 900. (That is, every value between 100 and 900 has equal probability of being realized.)

- (a) To calculate the size premium, we may sort all the firms by size and create two size-sorted portfolios. That is, we assign the top 50% of the firms to the large-size portfolio and the remaining 50% of the firms to the small-size portfolio. Each portfolio is equal-weighted. The factor premium is defined as the difference between the two portfolios' returns. What would be the expected value of the factor premium?
- (b) Instead of making two portfolios, we may create three size-sorted portfolios, each containing 33.3% of firms in the sample. Each portfolio is equal-weighted. The factor premium can be defined as the difference between the first portfolio return and the last portfolio return. What would be the expected value of the factor premium?
- (c) Does it matter how many portfolios we create to calculate the factor premium? Do we always get the same expected factor premium?

7.12. Principal-component analysis is used quite often in psychology. For example, a psychologist could have given a test with 100 multiple-choice questions to 1,000 subjects. The answers to each question are treated as a single series, and principal-component analysis can identify a small number of factors that together explain most of the variations in the answers. Then the psychologist is able to interpret each factor by examining the components of each factor (e.g., "This factor represents aversion to the color red," "This factor represents childhood trauma," etc.). In applying principal-component analysis to stock returns, however, portfolio managers cannot easily interpret what each factor really means. Explain why portfolio managers have more difficulty in interpreting factors than psychologists do.

7.13. Discuss whether principal-component analysis is free from the danger of data mining.

7.14. When firm A and firm B merge, we may use the weighted average of the premerger factor exposures of the two firms as the factor exposure for the new merged firm.

- (a) Show that the expected return implied by the factor exposure of the merged firm is the weighted average of the premerger expected returns of the two original firms.
 - (b) If there is synergy from the merger, would the procedure just described be reasonable?
- 7.15.** After estimating an economic factor model for stock A , we obtained the following decomposition of the risk:
- Nondiversifiable risk 90
Diversifiable risk 60
- The risk is measured in variance.
- (a) What is the R^2 of the regression?
 - (b) Explain the relationship between R^2 and the decomposition of risk.

¹ The material we present in this chapter is based on Chen et al. (1986), Fama and French (1993), Lehmann and Modest (1988), and Connor and Korajczyk (1988), who based their arguments on Ross (1976). Chen et al. developed a model for economic factors, whereas Fama and French developed a model for fundamental factors. Lehmann and Modest (1988) and Connor and Korajczyk (1988) developed models for statistical factors. All these academic studies are based on the arbitrage pricing theory of Ross (1976).

² The economic factor model can process fundamental factors, but it still frames them this way. Since the basic equations of both the economic and fundamental factor models are the same, though, they yield similar results when they use the same factors. Please see [Chapter 3](#) for further explanation of the equivalence of the models.

³ Bear in mind that the only risk that matters to a diversified investor is the nondiversifiable risk.

⁴ If you compare [Eq. \(7.1\)](#) to [Eq. \(6.2\)](#) very carefully, you will find one difference: the constant term α does not have subscript i in [Eq. \(6.2\)](#). Subscript i was not necessary in [Eq. \(6.2\)](#) as we presented the simplest version of the fundamental factor model. A more sophisticated version of the fundamental factor model is discussed in [Chapter 16](#), where subscript i is necessary.

⁵ Typically, when we estimate these factor models, we use information at time t to predict returns at time $t + 1$. For convenience, we sometimes write in this book the equation with time subscript t everywhere, as we do here. The factor exposure and the returns are at time t . The equation should still be interpreted as using information at time t to predict returns at time $t + 1$. In other words, the factor exposure at time t is really the factor exposure value as of the end of the previous period, $t - 1$, which we may refer to, sometimes, as the beginning-of-month exposure for time t .

⁶ Refer to [Chapter 6](#) for a discussion of the risk-free rate and investment universe decisions.

⁷ Chen et al. (1986), Lehmann and Modest (1988), and Connor and Korajczyk (1988), who used nonfundamental factors, report rather limited out-of-sample predictive power of their models.

On the other hand, Fama and French (1993) and Daniel and Titman (1997) and others who used fundamental factors generally report higher predictive power. See also [Chapter 16](#).

- ⁸ See [Chapter 9](#) for a discussion of how to control for the tracking error.
- ⁹ See Section 6.3 of [Chapter 6](#).
- ¹⁰ The reader should consult [Chapter 2](#), where we discussed the various types of α 's that are used in the portfolio management world.
- ¹¹ People sometimes use the variation from the mean as the factor premium to reflect the idea that what matters is the surprise of the variable, not the level of the macro variable. This surprise approach, however, does not improve the model unless the deviation from the mean really reflects the surprise. It is not easy to measure the magnitude of the surprise.
- ¹² We used the one-month Treasury bill return downloaded from the website of Professor Kenneth R. French of Tuck School of Business:
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.
- ¹³ If there are clear breaks elsewhere in the list, then divide the list accordingly. In the absence of other clear breaks, use 33% as the default cutoff value.
- ¹⁴ For size, we use the log of market capitalization and for value, we use the B/P ratio.
- ¹⁵ For further details, see page 399 of Srivastava (2002).
- ¹⁶ We are assuming that the number of time intervals is constant across stocks. This does not have to be the case in general. If the number of time intervals is different for different stocks, we would indicate the number of time intervals T with the subscript i .
- ¹⁷ One might not think of Apple as a defensive stock, but it could be that the iPhone has become a staple for individuals and thus has altered the way Apple's stock price behaves.
- ¹⁸ This is one method to approach the problem, especially when running models on a large database of firms. In some cases, the analysis is much more complex, such as when the acquiring firm does so by issuing much more debt.
- ¹⁹ See Chincarini and Kim (2001) for the application.
- ²⁰ When factor premiums are not available on a daily frequency, we have to make certain approximations, such as calculating a daily premium by dividing the monthly number by the number of trading days in the month.

CHAPTER 8

Forecasting Factor Premiums and Exposures

All things appear and disappear because of the concurrence of causes and conditions. Nothing ever exists entirely alone; everything is in relation to everything else.

—Buddha

8.1 INTRODUCTION

In the two preceding chapters we established the framework for factor models, our main quantitative equity portfolio management (QEPM) tools for assessing stock returns and risk. As we have seen, one idea underlies all factor models: The average stock return equals the product of factor premiums and factor exposures. We have shown how to build both the fundamental factor model and the economic factor model on this idea using past data and estimates based on the data. While the model is built from past data, though, it runs on future values because managers want to predict future stock returns. The predicted future average stock return equals the product of future factor premiums and future factor exposures. Put more formally, the return on stock i at time $T + 1$, $r_{i,T+1}$, is determined by

$$r_{i,T+1} = \alpha_i + \beta'_{i,T+1} \mathbf{f}_{T+1} + \epsilon_{i,T+1} \quad (8.1)$$

where α_i is the constant term, $\beta_{i,T+1}$ is the K -dimensional vector of the exposure for stock i at time $T+1$, $\mathbf{f}_{T+1}$ is the K -dimensional vector of the factor premium at time $T+1$, and $\epsilon_{i,T+1}$ is the deviation of stock i 's return from its average.

To generalize the stock return equation here to all types of factor models, we have added a time subscript to both the factor exposure and factor premium terms. Depending on the type of factor model, however, the time subscript can be dropped from one of the variables but kept for the other. In the fundamental factor model that we presented in [Chapter 6](#), for example, only the factor exposure changes over time, and, given stock returns and factor exposures up to time T , we estimated a fixed factor premium through time T .¹ In [Chapter 7](#), on the other hand, we saw that in the economic factor model, only the factor premium changes over time, and, given stock returns and factor premiums up to time T , we estimated a fixed factor exposure through T .

Now we must look beyond the given data to time $T+1$ and determine the future factor premium ($\mathbf{f}_{T+1}$) and the future factor exposure ($\beta_{i,T+1}$), together with the constant term (α_i) from the information available at time T . In order to do this, we sometimes need to forecast. The term *forecast* takes on a very specific meaning in this chapter. Assuming that a variable follows a normal distribution, forecasting the variable means specifying the variable's distribution (a.k.a. *predictive distribution*) by finding the distribution mean and variance. Once we have found—whether by forecasting or otherwise—the future values of the factor premium and factor exposure, we can forecast future stock returns. Predicting future stock returns, after all, is our goal; if we can predict returns, we can build a high-return portfolio.

8.2 WHEN IS FORECASTING NECESSARY?

To predict future stock returns, we need future values of the factor premium and factor exposure. In some cases we need to forecast these values; in other cases we can skip the forecasting step. Whether we need to forecast depends largely on the type of factor model we are using.

In fact, if we are using the simple version of the fundamental factor model from [Chapter 6](#), we do not need to forecast anything. We do not need

to forecast the explanatory variable, the factor exposure. The fundamental factor model is dynamic in the sense that the explanatory variable is realized before the dependent variable. The explanatory variable (factor exposure β_{it}) is measured at the beginning of time t , whereas the dependent variable (stock return r_{it}) is measured during time t . Thus, at the end of time T , we actually have the value of the factor exposure for time $T + 1$, so we do not need to forecast $\beta_{i,T+1}$ to predict the dependent variable for time $T + 1$, namely, $r_{i,T+1}$.

We do not need to forecast the factor premium either. For any type of factor model, it is conventional to assume that the model estimates will remain constant for a while. Thus, in the fundamental factor model, we can assume that the factor premium $\mathbf{f}$ estimated from time periods 1, ..., T will be valid for time $T + 1$. This assumption follows logically from the way we estimated the factor premium from time 1 to time T ; since we assumed that the parameters stay constant from time 1 to time T , we can assume that these parameters stay constant for one more period, time $T + 1$.

Forecasting becomes a necessary step when we are using the economic factor model from [Chapter 7](#). The economic factor model is not dynamic, so we need to forecast the value of the explanatory variable, the factor premium. In the economic factor model, the explanatory variable (factor premium $\mathbf{f}_t$) is known only at the end of time t , as is the dependent variable (stock return r_{it}). Thus, at the end of time T , we do not have the value of the factor premium for time $T + 1$, and we cannot predict the return for time $T + 1$, namely, $r_{i,T+1}$, without forecasting $\mathbf{f}_{T+1}$.

However, since we can always assume that the estimates in a model remain fairly constant, we can avoid forecasting the factor exposure. The factor exposure $\hat{\beta}_i$ estimated from time periods 1, ..., T still should be valid for time $T + 1$.

The principal point is that, regardless of the factor model selected, there is no need to forecast the factor exposure. However, depending on the factor model selected, one may need to forecast the factor premium.

8.3 COMBINING EXTERNAL FORECASTS

When reliable external forecasts are available, the portfolio manager should use them rather than attempt to do forecasts in-house. Factor premium forecasts can be obtained from economists who publish their forecasts, from various economic forecasting agencies, and from research analysts at various firms. This section discusses the way to combine multiple external forecasts.

Suppose that we obtained the predicted value of the factor premium $\mathbf{f}_{T+1}$ from J different forecasters. We denote these predicted values by $\mathbf{f}_{T+1}^{(1)}, \dots, \mathbf{f}_{T+1}^{(J)}$. Assuming that each forecaster is equally reliable, the best prediction of $\mathbf{f}_{T+1}$ is the simple average of the J predictions.² This average prediction is our expected value of $\mathbf{f}_{T+1}$; thus we may write

$$\hat{E}(\mathbf{f}_{T+1}) = \frac{1}{J} \sum_{j=1}^J \mathbf{f}_{T+1}^{(j)} \quad (8.2)$$

Not only should a portfolio manager care about the expected value of the factor premium from the various forecasts, but he or she also should care about how reliable it is and how much confidence he or she can place in it. If the J forecasters more or less agree, we can be more confident about a prediction than we would be if their predictions were widely divergent. One way to measure the confidence of the combined forecast, therefore, is to measure the inverse of the variance of the J predictions. The variance of $\mathbf{f}_{T+1}$ is given by

$$\hat{V}(\mathbf{f}_{T+1}) = \frac{1}{J} \sum_{j=1}^J \mathbf{f}_{T+1}^{(j)} \mathbf{f}_{T+1}^{(j)'} - \hat{E}(\mathbf{f}_{T+1}) \hat{E}(\mathbf{f}_{T+1}') \quad (8.3)$$

and the inverse of this variance can be used as a measure of our confidence. Thus, if the variance of the predictions is large, we have little confidence, and if it is small, we have more confidence. If we assume that $\mathbf{f}_{T+1}$ has a normal distribution, the mean and the variance of $\mathbf{f}_{T+1}$ given above completely specify the distribution of $\mathbf{f}_{T+1}$, and our task of forecasting the factor premium is done.

8.4 MODEL-BASED FORECAST

Forecasting is not usually a portfolio manager's forte. It is a challenging task even for highly quantitatively oriented portfolio managers, and, in any case, it is not the best use of their time. However, there may be an in-house model that seems to forecast a particular factor very well. In this case, the portfolio could benefit from time spent on forecasting. Let us consider a simple example of a model that could be used to forecast the factor premium for an economic factor model that includes a market factor:

$$r_{t+1}^{\text{S\&P500}} = a + bLI_t + \epsilon_{t+1} \quad (8.4)$$

where $r_{t+1}^{\text{S\&P500}}$ is the next month's return on the Standard and Poor's (S&P) 500, and LI_t is the value of a leading economic indicator. The leading economic indicator can be the actual leading indicator published by the Bureau of Economic Analysis or one component of it, such as average workweek, initial unemployment claims, new orders for consumer goods, plant and equipment orders, sensitive material prices, changes in consumer confidence, changes in gross domestic product (GDP), and so on. Testing whether or not the model can serve as the basis for a forecast is simple. One first estimates the historical relationship that the model describes. If it turns out the estimate for b is significant, then the model can be used to forecast the market factor in the economic factor model. If the economic factor model contains factors besides the market factor, similar regressions can be run to forecast the values of the other factors. Given a leading economic indicator for month T , the return and estimated variance determined from the preceding model can be used to forecast the factor premium for time $T + 1$.

8.5 ECONOMETRIC FORECAST

A portfolio manager might not find external factor premium forecasts satisfactory either because he or she does not believe the forecasters are reliable or because he or she simply cannot find outside forecasts for the particular factors in his or her model. At the same time, there may be no satisfactory in-house forecasting model. In this situation, the manager might

consider using some basic econometric or statistical forecasting techniques.³ We have chosen to focus on the most basic technique, called *vector autoregression* (VAR).⁴ Although the VAR procedure is rather mechanical, it has been shown to work well at predicting out of sample in many contexts.

In VAR, we model the factor premiums as a vector autoregressive process. That is, the current factor premiums are a linear function of past factor premiums, that is,

$$\mathbf{f}_t = \boldsymbol{\gamma}_0 + \boldsymbol{\gamma}_1 \mathbf{f}_{t-1} + \cdots + \boldsymbol{\gamma}_L \mathbf{f}_{t-L} + \boldsymbol{\omega}_t \quad t = L + 1, \dots, T \quad (8.5)$$

Each of $\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_L$ is a $K \times K$ matrix of coefficients, where K is the number of factors (the size of vector $\mathbf{f}_t$). L is the number of past factor premiums relevant for the current factor premium. This model is sometimes denoted as VAR(L) because there are L lags included on the right-hand side.

Given this modeling strategy, the only thing to decide is the number of lags L —in other words, how many past values should enter the equation.⁵ Including many past values reduces the potential for misspecification.⁶ At the same time, including many past values substantially increases the number of parameters to be estimated. The number of lags therefore should be decided based on the amount of available data. Note that given K factors and L lags, the number of parameters to be estimated is $(KL + 1) \times K$ because $\boldsymbol{\gamma}_0$ has K elements and all other $\boldsymbol{\gamma}$'s have K^2 elements. For three factors, the number of parameters to be estimated would be 12, 21, and 30 for $L = 1, 2, 3$. For four factors, the number of parameters to be estimated would be 20, 36, and 52. Note also that the number of observations available for estimation declines as the lag size grows. In practical situations, it is not very common to include more than one lagged value. For example, if there are 60 observations, estimating even 12 or 20 parameters is already challenging because the precision of the estimates will be quite low.

By running regressions, we obtain the estimates $\hat{\boldsymbol{\gamma}}_0, \dots, \hat{\boldsymbol{\gamma}}_L$. We also obtain the variance-covariance matrix estimates of $\boldsymbol{\omega}$. We will call this estimate $\hat{\boldsymbol{\Sigma}}_{\boldsymbol{\omega}}$. Then the predicted factor premium (i.e., the expected factor premium) for $T + 1$ is

$$\hat{E}(\mathbf{f}_{T+1}) = \hat{\gamma}_0 + \hat{\gamma}_1 \mathbf{f}_T + \cdots + \hat{\gamma}_L \mathbf{f}_{T-L+1} \quad (8.6)$$

The estimated variance of the factor premiums for $T + 1$ is

$$\hat{V}(\mathbf{f}_{T+1}) = \hat{V}(\boldsymbol{\omega}_{T+1}) = \hat{\Sigma}_{\omega} \quad (8.7)$$

This completes our task of forecasting factor premiums.

[Table 8.1](#) presents the estimates of VAR(1) for five factor premiums: the premiums of the unemployment rate, consumer sentiment growth, excess market return, the log of market capitalization, and book-to-price ratio. To estimate this model, we used five years of monthly data from January 2016 to December 2020. A VAR(1) means that we chose a lag length of 1; thus L is 1, and T is 59 ($= 60 - 1$). Each column in the table represents an equation for one factor. Since there are five factors, there are five equations. The first row corresponds to γ_0 , and each row from the second to the sixth row corresponds to each column in γ_1 .

TABLE 8.1

Estimates of the VAR Factor Forecasting Equation

Explanatory Variables	Dependent Variables				
	Unemployment (UR)	Consumer Sentiment Growth (CSG)	Excess Market Return (MKT)	Log of Market Capitalization (LOGSIZE)	Book-to-Price (B/P)
Constant	0.854 (0.339)	-1.543 (1.224)	-1.223 (1.297)	1.778 (0.968)	-1.100 (1.089)
Unemployment rate ($t-1$)	0.814 (0.066)	0.190 (0.239)	0.588 (0.253)	-0.424 (0.189)	0.164 (0.212)
Consumer sentiment growth ($t-1$)	-0.169 (0.031)	0.105 (0.112)	-0.339 (0.119)	0.147 (0.089)	-0.011 (0.100)
Excess market return ($t-1$)	0.041 (0.035)	0.567 (0.127)	-0.261 (0.134)	0.064 (0.100)	-0.129 (0.113)
Log of market capitalization ($t-1$)	-0.076 (0.058)	0.091 (0.207)	0.045 (0.220)	-0.219 (0.164)	0.374 (0.185)
B/P ($t-1$)	-0.004 (0.050)	0.226 (0.181)	0.254 (0.192)	-0.103 (0.143)	0.214 (0.161)

Note: Estimated for the period from January 2016 to December 2020. Standard errors are listed inside parentheses.

The premium for the unemployment factor is the U.S. unemployment rate of the previous month in percentage terms. The premium for the consumer sentiment growth factor is the percentage change in the University of Michigan Consumer Sentiment Index from the previous month. The premium for the excess market factor is the S&P 500 total return in excess of the one-month Treasury bill return. The premiums for the log of market capitalization factor and the book-to-price factor are calculated by constructing zero-investment portfolios of those factors (see [Chapter 7](#) for more details).

Given the estimates in [Table 8.1](#), we can easily calculate the expected value of the factor premium for January 2020, which is time $T+1$. Using [Eq. \(8.6\)](#), we obtained the expected value reported in [Table 8.2](#). [Table 8.3](#) reports the estimate of the variance-covariance matrix Σ_ω . Since Σ_ω is a symmetric matrix, we report only the lower triangular part of the matrix. For example, the unemployment rate variance is 1.072, which is roughly

equal to $(1.035)^2$. Thus the standard deviation of the unemployment factor is 1.035% per month.

TABLE 8.2

Predicted Value of Factor Premium from VAR

Unemployment Rate	Consumer Sentiment Growth	Excess Market Return	Log of Market Capitalization	Book-to-Price
5.865	1.964	−0.301	0.664	−1.871

Note: Prediction for January 2021 based on the data from January 2016 to December 2020.

TABLE 8.3

Variance–Covariance of Factor Premium Forecasts from VAR

	Unemployment Rate	Consumer Sentiment Growth	Excess Market Return	Log of Market Capitalization	Book-to-Price
Unemployment rate	1.072				
Consumer sentiment growth	−0.707	13.934			
Excess market return	0.115	−2.907	15.642		
Log of market capitalization	0.248	0.006	−3.846	8.712	
Book-to-price	−0.509	2.711	3.769	−6.042	11.043

Note: Prediction for January 2021 based on the data from January 2016 to December 2020.

8.6 PARAMETER UNCERTAINTY

The variance of the factor premiums calculated in the preceding section reflects the uncertainty of the future factor premiums. This uncertainty, like the uncertainty of stock returns themselves, is a component of investment risk and must be recognized as such. Even by recognizing the variance-related risk, though, we still have not fully acknowledged the risk inherent in forecasting until we consider parameter uncertainty. The variance computed above shows the part of the variation in the future factor premiums that cannot be explained by the model. This would be all that mattered if the model were exact and the parameters we estimated

$(\hat{\gamma}_0, \dots, \hat{\gamma}_L)$ were exact, but the parameters we estimated are not exact. All we can say is that the true value of $\gamma_0, \dots, \gamma_L$ is likely to be near what we have estimated with the VAR, and our confidence in the estimates is included in the standard error of the estimation. If the standard error of the estimation is large, there is more uncertainty regarding the future factor premiums, and, as a result, the investment risk is higher. To fully account for the parameter uncertainty in the computation of investment risk, we need to include the standard error of $\gamma_0, \dots, \gamma_L$ in the calculation of the variance of the future factor premiums. We will examine two approaches to do this. One is an exact method, and the other is an approximate method.

In the exact method, we recognize that the future factor premium has two sources of variation, one coming from the estimation of $\gamma_0, \dots, \gamma_L$ and the other coming from the variation allowed in the model. The first component goes in a $K(L+1) \times K(L+1)$ -dimensional variance-covariance matrix $V(\hat{\gamma}_0, \dots, \hat{\gamma}_L)$, and the second component goes in a $K \times K$ -dimensional variance-covariance matrix $\hat{V}(\omega) = \hat{\Sigma}_\omega$. Once we arrive at the VAR estimator, we can compute both these components in a straightforward way. The correct variance of the future factor premium is

$$\hat{V}(\mathbf{f}_{T+1}) = (1+d)\hat{\Sigma}_\omega \quad (8.8)$$

where d is a constant computed from $\mathbf{f}_T, \dots, \mathbf{f}_{T-L}$ (see Appendix 8A).⁷

Table 8.4 is a recalculation of the variance-covariance matrix of $\mathbf{f}_{T+1}$ reported in Table 8.3. Note that each diagonal element of Table 8.4 is greater than the corresponding figure in Table 8.3. This shows that as we account for parameter uncertainty, our ignorance (measured by the variance) increases.

TABLE 8.4

Variance–Covariance of Factor Premium Forecasts, Parameter Uncertainty Considered

	Unemployment Rate	Consumer Sentiment Growth	Excess Market Return	Log of Market Capitalization	Book-to- Price
Unemployment rate	1.152				
Consumer sentiment growth	−0.760	14.973			
Excess market return	0.123	−3.124	16.808		
Log of market capitalization	0.267	0.006	−4.132	9.362	
Book-to-price	−0.547	2.913	4.050	−6.493	11.866

Note: Prediction for January 2021 based on the data from January 2016 to December 2020.

If the complexity of the setup increases and the number of parameters increases, this approach may no longer be feasible. The alternative method is known as *bootstrapping*. Given the sample of T observations, we can create T pseudosamples of $T - 1$ observations by dropping one observation at a time. From each pseudosample, we estimate $\gamma_0, \dots, \gamma_L$ and Σ_ω and obtain T sets of estimates. From each set of estimates, we calculate the predicted value of the future factor premiums and obtain T sets of future factor premiums. These T sets of future factor premiums can be treated as if each were obtained from an independent forecaster (as we discussed at the beginning of this section). That is, the variance of the T sets of future factor premiums includes parameter uncertainty, as well as the variance of the model error.⁸

8.7 FORECASTING THE STOCK RETURN

Up to this point we have discussed ways to forecast the distribution of the factor premium for time $T + 1$, namely, $\mathbf{f}_{T+1}$. As we mentioned earlier, the factor exposure for time $T + 1$, $\beta_{i,T+1}$, is either observable (in the case of the fundamental factor model) or is assumed to be the same as the factor exposure that was estimated from the data through time T (in the case of the economic factor model). Given our forecasts for $\mathbf{f}_{T+1}$ and $\beta_{i,T+1}$, we can now proceed to the forecasting of the stock return. Assuming a normal distribution as usual, the distribution of the stock return is specified by estimating the mean and variance. From the relationship

$$r_{i,T+1} = \alpha_i + \beta'_{i,T+1} \mathbf{f}_{T+1} + \epsilon_{i,T+1} \quad (8.9)$$

the mean and variance of the stock return are

$$E(r_{i,T+1}) = \alpha_i + \beta'_{i,T+1} E(\mathbf{f}_{T+1}) \quad (8.10)$$

$$V(r_{i,T+1}) = \beta'_{i,T+1} V(\mathbf{f}_{T+1}) \beta_{i,T+1} + V(\epsilon_{i,T+1}) \quad (8.11)$$

We have the estimates for $E(\mathbf{f}_{T+1})$ and $V(\mathbf{f}_{T+1})$ from the previous sections, and we know the value (or the estimate) of α_i and $\beta_{i,T+1}$. $V(\epsilon_{i,T+1})$ was estimated in previous chapters. Thus we simply may substitute the estimates for each component and obtain the estimates for the mean and variance of the stock return, that is,

$$\hat{E}(r_{i,T+1}) = \alpha_i + \beta'_{i,T+1} \hat{E}(\mathbf{f}_{T+1}) \quad (8.12)$$

$$\hat{V}(r_{i,T+1}) = \beta'_{i,T+1} \hat{V}(\mathbf{f}_{T+1}) \beta_{i,T+1} + \hat{V}(\epsilon_{i,T+1}) \quad (8.13)$$

If we estimate $\alpha_i + \beta_{i,T+1}$, this expression does not account for the uncertainty regarding the true value of α_i and $\beta_{i,T+1}$. We have to add the estimation error generated in the estimation of α_i and $\beta_{i,T+1}$ to the preceding formula. Note that the estimation error associated with $\beta_{i,T+1}$ is captured by the variance–covariance matrix $\hat{V}(\hat{\beta}_{i,T+1})$ as presented in the preceding chapter. Similarly, the estimation error associated with α_i is represented by the variance $\hat{V}(\hat{\alpha}_i)$. The only remaining question is how these terms enter into the preceding equation.⁹

Table 8.5 reports the mean and standard deviation estimates of selected stock returns for January 2021, based on the economic factor model of Chapter 7. The last column in the table shows the standard deviation estimate considering parameter uncertainty. Consideration of parameter uncertainty increases the standard deviation estimate, hence the increase between the third and the last columns.

TABLE 8.5

Predictive Return Distribution for Selected Stocks

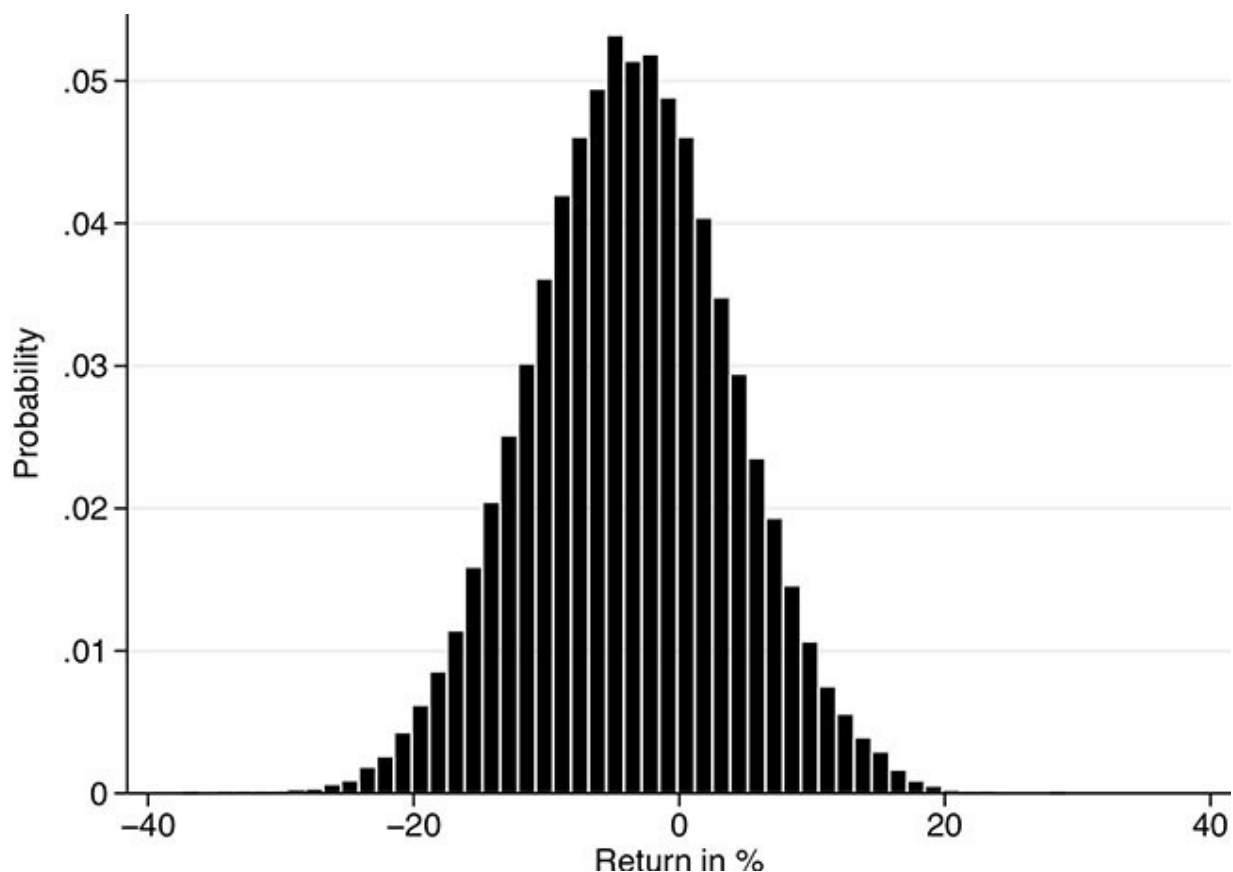
Ticker	Mean	SD	SD*
AAPL	2.58	8.43	8.65
MSFT	1.71	5.26	5.38
WMT	0.83	5.10	5.27
JNJ	-0.48	4.81	4.93
AMZN	1.31	8.21	8.40

Note: Prediction for January 2021 based on the data from January 2016 to December 2020. SD is the standard deviation, and SD* is the standard deviation considering the parameter uncertainty. All numbers are expressed in percentage terms. Company names for ticker symbols are Apple (AAPL), Microsoft (MSFT), Walmart (WMT), Johnson & Johnson (JNJ), and Amazon (AMZN).

Once we have estimates of the mean and variance of the stock returns, we can plot the whole distribution of the stock return. [Figure 8.1](#) shows the predictive distribution of the return to ExxonMobil's stock. The figure is based on the variance estimate, not accounting for parameter uncertainty. The height of each bar corresponds to the probability of a particular stock return. For example, there is about an 4.5% chance that the stock return in January 2021 will be around 0% and approximately a 1% chance that the stock return will be around 10%.

FIGURE 8.1

Predictive return for Walmart stock (January 2021). (Based on the data from January 2016 to December 2020.)



8.8 CONCLUSION

We have arrived at the point at which we can set up and run a factor model to generate predictions of stock returns and risk. In [Chapters 6 and 7](#) we had formulated fundamental and economic factor models from past data on stock returns, factor premiums, and factor exposures. In this chapter we showed how the models relate future stock returns to future values of the factor exposure and factor premium. In the case of the simple fundamental factor model, we did not have to worry about forecasting the factor exposure or factor premium because the model uses the existing values of both variables. In the case of the economic factor model, we also could use the existing factor exposure estimate, but we needed a forecast of the future factor premiums. Although we recommend that portfolio managers use outside forecasts whenever possible, we realize that in some cases they want to do forecasts in-house. For this reason, we went through the steps of different forecasting methods and also discussed the need to consider the parameter uncertainty of forecasts. Finally, and most importantly, we

showed how to forecast the future returns and risks of stocks regardless of the type of factor model used. Forecasting a stock's return and risk is always the last step of modeling. The goal of this process, after all, is to predict risk-adjusted returns in order to decide which stocks to include in the portfolio. There may be no crystal ball to gaze into for a picture of future stock movements, but a good model may be the closest thing to it. Once we know what a stock might yield and know that we want to include it in the portfolio, we can then decide exactly how much of the portfolio to dedicate to it. Portfolio weighting means building the portfolio, and we start building in the next chapter.

QUESTIONS

- 8.1. Whether to forecast a factor exposure depends on whether the exposure is likely to change in the near future. Discuss, for each of the following factors, whether the exposure is likely to change in 1 month: size, price-to-earnings (P/E) ratio, dividend yield, and return on equity (ROE).
- 8.2 Explain why forecasting factor exposures is not necessary in a fundamental factor model if the model is in dynamic form, that is,

$$r_{i,t+1} = \alpha + \beta_i f_t + \epsilon_{i,t+1}$$

- 8.3. Consider an economic factor model:

$$r_{it} = \alpha_i + \beta_{i1}f_{1t} + \dots + \beta_{iK}f_{Kt} + \epsilon_{it}$$

Suppose that we estimated this model using time-series data $(r_{i1}, f_{11}, \dots, f_{K1}), \dots, (r_{iT}, f_{1T}, \dots, f_{KT})$ and obtained the estimates $\hat{\alpha}_i, \hat{\beta}_{i1}, \dots, \hat{\beta}_{iK}$. Using this model, we would like to predict the future stock return $r_{i,T+1}$, especially its expected value.

- (a) Instead of forecasting $f_{1,T+1}, \dots, f_{K,T+1}$ as discussed in the chapter, if we use the last value of the factor premium $f_{1,T}, \dots, f_{K,T}$, what would be the expected value of $r_{i,T+1}$?

- (b) Instead of forecasting $f_{1,T+1}, \dots, f_{K,T+1}$ as discussed in the chapter, if we use the sample average of the factor premium $1/T \sum_{t=1}^T f_{1t}, \dots, 1/T \sum_{t=1}^T f_{Kt}$, what would be the predicted value of $r_{i,T+1}$?
- (c) Explain why forecasting factor premiums is necessary in the economic factor model.

8.4. Suppose that a portfolio manager obtained GDP forecasts $x_1, \dots, x_{10}$ from 10 independent forecasters.

- (a) What would be the expected GDP of the portfolio manager?
- (b) If it turns out that the tenth forecaster simply averaged the forecasts of the other nine forecasters, what would be the expected GDP of the portfolio manager?
- (c) Suppose that the tenth forecaster made her own forecast x_{10}^* , but instead of reporting her own forecast, she used the formula:

$$x_{10} = \frac{1}{18} (x_1 + \dots + x_9) + \frac{1}{2} x_{10}^*$$

and reported x_{10} . What would be the expected GDP of the portfolio manager?

- (d) Discuss why it is important to obtain independent forecasts.

8.5. What are the strengths and the weaknesses of model-based forecasts compared with econometric forecasts?

8.6. Suppose that the unemployment rate is forecast with the econometric forecasting technique.

- (a) Is it reasonable to assume a normal distribution for the error?
- (b) Explain how one may obtain a more normal looking distribution by transforming the variable.

8.7. After estimating an AR(1) model for a factor premium, we obtained the following estimates:

$$f_{t+1} = 0.2 + 0.7f_t + \eta_t \quad \eta_t \sim N(0,1)$$

where η_t is an error, and $N(\cdot)$ indicates a normal distribution.

- (a) If $f_T = 1.1$, what would be the predicted distribution of f_{T+1} ?
- (b) If $f_T = 1.1$, what would be the predicted distribution of f_{T+2} ?

(c) Find the general formula for the distribution of f_{T+s} .

8.8. Consider an AR(1) model of a factor premium:

$$f_{t+1} = \gamma + \delta f_t + \eta_{t+1} \quad \eta_{t+1} \sim N(0, \sigma^2)$$

Suppose that we estimate this model using T observations $f_1, \dots, f_T$.

(a) Write down the estimates $\hat{\gamma}$ and $\hat{\delta}$ as a function of data $f_1, \dots, f_T$.

(b) Show that $\hat{\gamma}$ and $\hat{\delta}$ are independent of η_{T+1} , the future error.

(c) Show that the predicted variance of f_{T+1} is the sum of the error variance and the parameter uncertainty. Explain why it is important here to have the answer to part (b).

8.9. Describe the strengths and weaknesses of the vector autoregression (VAR) technique.

8.10. In a VAR model, the number of parameters grows very fast as the number of variables K or the number of lags L increases.

(a) If there are five factors and we want to estimate a VAR(2) for these factors, how many parameters do we need to estimate?

(b) If we have 50 observations, what would be the degrees of freedom?

(c) What is the practical implication of having negative degrees of freedom?

8.11. What is parameter uncertainty? Why is it important in QEPM? Explain how the parameter uncertainty can be accounted for in QEPM.

8.12. Consider an economic factor model of the stock return:

$$r_{it} = \alpha_i + \beta_i f_t + \epsilon_{it} \quad \epsilon_{it} \sim N(0, \sigma^2)$$

We estimate this model using T observations $(r_{i1}, f_1), \dots, (r_{iT}, f_T)$.

Using the same data, we also estimate an AR(1) model of the factor premium:

$$f_{t+1} = \gamma + \delta f_t + \eta_{t+1} \quad \eta_{t+1} \sim N(0, \sigma_\eta^2)$$

(a) Express $\hat{\delta}$ in terms of data and errors.

- (b) Express $\hat{\beta}_i$ in terms of data and error.
- (c) Are $\hat{\delta}$ and $\hat{\beta}_i$ independent of each other conditional on the data? What would happen if they were not independent?
- 8.13.** Consider two random variables A and B whose joint distribution is multivariate standard normal. That is, each variable has a standard normal distribution and is independent of the other variable.
- (a) Is the distribution of the product AB still normal?
- (b) Explain why the predictive distribution of the stock return may not be normal when considering parameter uncertainty if the factor premium has to be forecasted.
- 8.14.** Consider an economic factor model:

$$r_{it} = \alpha_i + \beta_i f_t + \epsilon_{it} \quad \epsilon_{it} \sim N(0, \sigma^2)$$

where the factor premium is an AR(1) process:

$$f_{t+1} = \gamma + \delta f_t + \eta_{t+1} \quad \eta_{t+1} \sim N(0, \sigma_n^2)$$

- (a) Show that the stock return is serially correlated as long as $\delta \neq 0$.
- (b) Specify the condition under which we may observe momentum profit (i.e., positive serial correlation in the stock return) and contrarian profit (i.e., negative serial correlation in the stock return).
- 8.15.** The bootstrap standard error is often calculated when calculating the exact standard error is too complicated. Let x_t be the data observed in time t . The sample is a collection of observations made at different times. Suppose that our sample is made of observations from $t = 1$ to $t = T$, that is, $(x_1, \dots, x_T)$. Given the sample of size T , we may create T subsets of the sample, where each subset has $T - 1$ elements. If we index the subsets by j , then subset j is a set of all observations except x_j . We may carry out the estimation using each subset and call the resulting estimate $\theta^{(j)}$. The bootstrap standard error is defined as the sample standard deviation of $\theta^{(1)}, \dots, \theta^{(T)}$.
- (a) Explain the relationship between the bootstrap standard error and the regular standard error.

- (b) Under what situation would one prefer the bootstrap standard error to the regular standard error?
- (c) What modifications are necessary in order to calculate the bootstrap standard error in a VAR model?

¹ In [Chapter 6](#), we presented two types of fundamental factors models. One, which we call the simple one, takes the factor exposures of stocks over time along with the stock returns over time and estimates a factor premium over the historical sample period. In this case, there is only one factor premium estimated from the data and one estimate of the factor premium at time $T + 1$, which is simply the historical estimate up to time T . However, in the last part of [Chapter 6](#), and also in [Chapter 16](#), we estimate the fundamental factor model as a series of cross-sectional regressions. That is, for each period or month in our historical data, we use stock returns and factor exposures of stocks to estimate a different factor premium in each period or month. We thus have a series of factor premiums for each time period. In this case, if we choose, we can use a VAR to forecast the period $T + 1$ factor premiums based on the time series of factor premiums through time T . Thus, in the more “complicated” fundamental factor model, both factor premiums and factor exposures change through time. Finally, as a reminder, when we say a fixed factor premium is estimated, we are referring to $\{\mathbf{f}_1, \dots, \mathbf{f}_T\}$, and when we say a fixed factor exposure is estimated, we are referring to $\{\beta_{i,1}, \dots, \beta_{i,T}\}$.

² For combining forecasts from forecasters of differing abilities, we refer the reader to [Chapter 9](#) of Granger and Newbold (1986).

³ There are many sophisticated econometric techniques to forecast factor premiums. The reader should be warned, though, that more sophisticated models do not necessarily produce better forecasts. Sophisticated models come with an extra set of assumptions, which may or may not apply well to a particular factor model. Unless there is a strong reason for an econometric model, we suggest sticking with a simpler forecasting technique.

⁴ Most standard statistical packages can perform VAR estimations and forecasts.

⁵ Several selection criteria can be used to determine the number of lags. The two most common are the Akaike information criterion (AIC) and the Schwarz Bayesian information criterion (SIC/BIC/SBIC). Please consult an econometrics textbook on how to use these tools.

⁶ Omitting relevant variables creates a problem, whereas including irrelevant variables does not. See Appendix C at www.ludwigbc.com under QEPM Exclusive Content for more details on this matter.

⁷ Appendix 8A can be found at www.ludwigbc.com under QEPM Exclusive Content.

⁸ When an observation is dropped in the creation of the pseudosample, the order of the remaining $T - 1$ observations should not change. The time distance among observations should not change either. This reduces the effective sample size by more than one. For example, if $\mathbf{f}_2$ is dropped, we have to ignore not only $\mathbf{f}_2 = \gamma_0 + \gamma_1 \mathbf{f}_1 + \omega_2$ but also $\mathbf{f}_3 = \gamma_0 + \gamma_1 \mathbf{f}_2 + \omega_3$. Thus the effective sample size is reduced by 2. See Appendix 8A at www.ludwigbc.com under QEPM Exclusive Content for a general discussion of bootstrapping.

⁹ We provide the exact formula in Appendix 8A, which can be found at www.ludwigbc.com under QEPM Exclusive Content.

CHAPTER 9

Portfolio Weights

A little caution outflanks a large cavalry.

—Otto von Bismarck

9.1 INTRODUCTION

In [Chapters 5](#) through [8](#) we discussed how to create models for stock selection.¹ Models serve as essential tools for identifying “good” and “bad” stocks. However, to construct a portfolio, the manager ultimately needs to do more than separate the wheat from the chaff. A good manager does not just throw a bunch of stocks into the same basket. Rather, he or she assigns them relative weights so that they make sense as a single portfolio.

There are many ways to generate stock weights. Some methods are very rudimentary, such as choosing the best stocks in one’s universe and weighting them equally. Another simple method is to weight the stocks by market capitalization, as many indices do. A slightly more complicated twist on market-capitalization weighting is to weight by the square root of the market capitalization. The most complex weighting methods draw on ideas from modern portfolio theory and attempt to weight stocks in a way that maximizes the portfolio’s expected return and minimizes its overall risk.

For the majority of portfolio managers who manage their portfolios against a benchmark, the weighting possibilities multiply. For instance, a manager could weight the stocks such that the weighted-average factor

exposures of the stocks equal the benchmark's weighted-average factor exposures. Extending the concepts of modern portfolio theory to the weighting decision, the manager could try to maximize the excess expected return while placing a limit on the allowable tracking error of the portfolio versus the benchmark.

In order to weight stocks in a way that minimizes tracking error or risk, the manager needs to know each stock's expected return, each stock's risk or variance, and the covariances among stocks.² A manager also may wish to put constraints on the portfolio creation process. He or she may, for instance, prohibit short sales (i.e., not allow negative stock weights) or specify the range of permissible weights for individual stocks (e.g., to uphold mutual fund diversification rules). The manager then puts all these parameters and constraints into a *quadratic optimizer* and solves for the optimal weights of the portfolio.

This chapter discusses the mathematical and statistical concepts of portfolio weighting. For portfolio managers who use commercial software, the discussion will be helpful for understanding the mechanics of combining their preferred stocks with the vendor's risk models to determine the portfolio weights. For those who want to build their own in-house risk models and portfolio optimizers, this chapter will walk you through the important methods.

We begin by discussing the two ways to create a portfolio that is not managed against a benchmark. Ad hoc methods (Section 9.2) use various rules of thumb to weight stocks. The mean-variance optimization method (Section 9.3) minimizes the portfolio's total risk given its expected return.

We then consider the four potential approaches to creating a portfolio that is managed against a specific *benchmark*. The portfolio manager again could use an ad hoc weighting method that only roughly tracks the benchmark. Alternatively, he or she could use a method of sampling known as *stratification*, which involves choosing representative samples of the benchmark. Or he or she could use *factor exposure targeting*, in which the stocks are weighted so that the average factor exposures of the portfolio and the benchmark match. Finally, the manager could follow the most common weighting method of professional money managers—*tracking-error minimization*. The stock weights are chosen so as to achieve the highest possible expected return or α of the portfolio while keeping the tracking error below a specified threshold. Of the four weighting techniques, only

the ones that minimize risk have theoretical rigor. They also require, though, a larger skill set, more time, and more effort than do the other less quantitatively precise techniques.

9.2 AD HOC METHODS

Once the portfolio manager decides which stocks to include in his or her portfolio, there are two very easy ways to determine the portfolio weights: equal weighting and value weighting. *Equal weighting* is simply assigning the same weight to every stock. If there are 10 stocks, then each stock will have a weight of 0.1 ($= 1/10$). If there are 50 stocks, then each stock will have a weight of 0.02 ($= 1/50$). In general, if there are N stocks, each stock will have the weight of $1/N$. Equal weighting is certainly quick and simple, but it has one major deficiency: It reflects neither the risks nor the expected returns of stocks. Equal weighting makes sense only when the portfolio manager has very poor information about the expected return and the risk of stocks selected.

Value weighting is assigning weights proportional to the stocks' market capitalizations. For example, if the market capitalization of stock A is twice as great as the market capitalization of stock B , then the weight of stock A also should be twice the weight of stock B . In general, if there are N stocks with market capitalizations $x_1, \dots, x_N$, then the weight of stock i , w_i , is

$$w_i = \frac{x_i}{\sum_{j=1}^N x_j} \quad (9.1)$$

Just like equal weighting, value weighting does not reflect the expected returns or the risks of the selected stocks, so it is also not the best method when there is good information about stock returns and risks. In the absence of such information, value weighting may be an improvement on equal weighting. The performance of the value-weighted portfolio is at least guaranteed to match the market average performance because market capitalizations are the weights, so to speak, that the market assigns to stocks.

There are a number of variations on value weighting.³ Some portfolio managers have gone so far as to weight stocks by the square root of the

market capitalization. Using the square root reduces the weighting bias toward stocks with very large market capitalizations. In this scheme, stock i would get a weight of

$$w_i = \frac{\sqrt{x_i}}{\sum_{j=1}^N \sqrt{x_j}} \quad (9.2)$$

One could get even more creative and do the cubed root, and so on and so forth.

Instead of equal weighting or value weighting, portfolio managers also can use *price weighting*. In the price-weighting scheme, the manager buys the same number of shares in each stock so that the weights are proportional to the prices of stocks. If the share price of a stock is high, this stock will have relatively more weight in the portfolio. This is the way the Dow Jones Industrial Average and the Nikkei 225 are calculated.

We will discuss a few more ad hoc weighting schemes when we discuss the weighting of benchmarked portfolios.

9.3 STANDARD MEAN-VARIANCE OPTIMIZATION

Given the mean and the variance of future stock returns, we can use the quadratic programming technique to find the portfolio that has the *minimum risk*. That is, we can find the portfolio that has the lowest ex-ante risk among portfolios with identical expected returns. This is known as *mean-variance optimization* or simply, *MVO*.

The underlying idea of this method is the comparison of all the portfolios that potentially could be built given a list of stocks. We theoretically compute each portfolio's ex-ante risk from the variances and covariances of the returns of all stocks and each portfolio's expected return from the mean returns of individual stocks. Then we theoretically compare the ex-ante risks and the expected returns of all the portfolios and choose the one with the lowest risk for a given level of expected return. (Or, conversely, we can choose the portfolio with the highest expected return for a given level of risk.⁴)

Of course, actually calculating the expected return and the risk of every potential portfolio would take forever because there are infinite potential portfolios. Thus we use quadratic programming to find the minimum-risk portfolio without having to explicitly calculate every portfolio's risk and return.

One common objection of MVO is that the selected portfolio often assigns very large weights to certain outlier stocks, including stocks with very low variances or very high means. This in itself should not be a concern. If the stock indeed has a low variance for a given return, then it is smart to overweight it in the portfolio. However, in portfolio construction, we must estimate the means and variances of stocks, so we cannot be sure of their true values. Some outliers could be mismeasurements and throw off our results ("Garbage in, garbage out," as the saying goes). The portfolio manager therefore must do his or her best to account for estimation error.⁵ If this is not practical, the manager may include additional portfolio constraints in the quadratic optimization problem that put limitations on the maximum and minimum stock weights. The most common additional portfolio constraints are short-sale constraints, which do not allow for a negative weight on any stock; diversification constraints, which restrict any stock from having more than a certain threshold weighting; and sector constraints, which do not allow the composite weight of any group of stocks in a certain sector of the economy to be beyond a certain level.

The portfolio manager must be careful not to impose so many constraints that some of them begin to contradict each other. As a rather obvious example, not all stocks in a portfolio can weigh less than 5% if the portfolio is limited to 10 stocks; the portfolio will not be fully invested. Typically, portfolio managers impose only the most obvious or needed constraints and let the optimizer do the rest.

9.3.1 No Constraints

Let us first look at how to use MVO without imposing additional constraints. We assume that the portfolio manager already has built a model of expected stock returns and risk either by using one of the models discussed in previous chapters or by using commercial software risk models combined with his or her excess-return models.

The first step is to include all the relevant information about individual stock returns in a vector $\boldsymbol{\mu}$ and a matrix $\boldsymbol{\Sigma}$. $\boldsymbol{\mu}$ is an N -dimensional column vector of expected returns of individual stocks, where N is the number of stocks in the investment universe. $\boldsymbol{\Sigma}$ is an $N \times N$ matrix of variances and covariances of individual stock returns. That is,

$$\boldsymbol{\mu} = \begin{bmatrix} E(r_1) \\ \vdots \\ E(r_N) \end{bmatrix} \quad (9.3)$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} V(r_1) & C(r_1, r_2) & \cdots & C(r_1, r_N) \\ C(r_2, r_1) & V(r_2) & \cdots & C(r_2, r_N) \\ \vdots & & & \\ C(r_N, r_1) & C(r_N, r_2) & \cdots & V(r_N) \end{bmatrix} \quad (9.4)$$

where $E(r_i)$ is the expected return of stock i , $V(r_i)$ is the variance of stock i 's return, and $C(r_i, r_j)$ is the covariance between stock i 's return and stock j 's return.

A portfolio is specified by a weight vector $\mathbf{w}$. $\mathbf{w}$ is an N -dimensional column vector of stock weights:

$$\mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_N \end{bmatrix} \quad (9.5)$$

where w_i is the weight of stock i in the portfolio. For $\mathbf{w}$ to be a valid weight, the sum of all the elements in $\mathbf{w}$ should be 1. We may define $\mathbf{1}$ as an N -dimensional vector of 1 such that

$$\iota = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad (9.6)$$

The sum of elements in $\mathbf{w}$ is $\sum_{i=1}^N w_i$ or simply $\mathbf{w}'\iota$, which should be 1. For the portfolio specified by weights $\mathbf{w}$, the expected return of the portfolio is $\sum_{i=1}^N w_i E(r_i)$ or simply $\mathbf{w}'\boldsymbol{\mu}$, whereas the risk of the portfolio (i.e., the variance of the portfolio return) is $\sum_{i=1}^N w_i^2 V(r_i) + 2\sum_{i=1}^N \sum_{j=i+1}^N w_i w_j C(r_i, r_j)$ or, in matrix notation, $\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$. Thus the portfolio that has the minimum risk with the expected return of μ_p is the solution to the following minimization problem:

$$\min_{\mathbf{w}} \mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} \quad s.t. \quad \mathbf{w}'\boldsymbol{\mu} = \mu_p \quad \text{and} \quad \mathbf{w}'\iota = 1 \quad (9.7)$$

Note that the objective function ($\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w}$) is a quadratic function of $\mathbf{w}$ (i.e., the weight terms are squared), whereas the constraints ($\mathbf{w}'\boldsymbol{\mu} = \mu_p$) are linear in $\mathbf{w}$. Mathematicians call this problem a *quadratic optimization problem* and have developed a way of dealing with this called *quadratic programming*. In fact, mathematicians would prefer to rewrite the constraints in the following way:

$$\mathbf{A}\mathbf{w} = \mathbf{b} \quad (9.8)$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \mu_1 & \mu_2 & \cdots & \mu_N \end{bmatrix} \quad (9.9)$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ \mu_p \end{bmatrix} \quad (9.10)$$

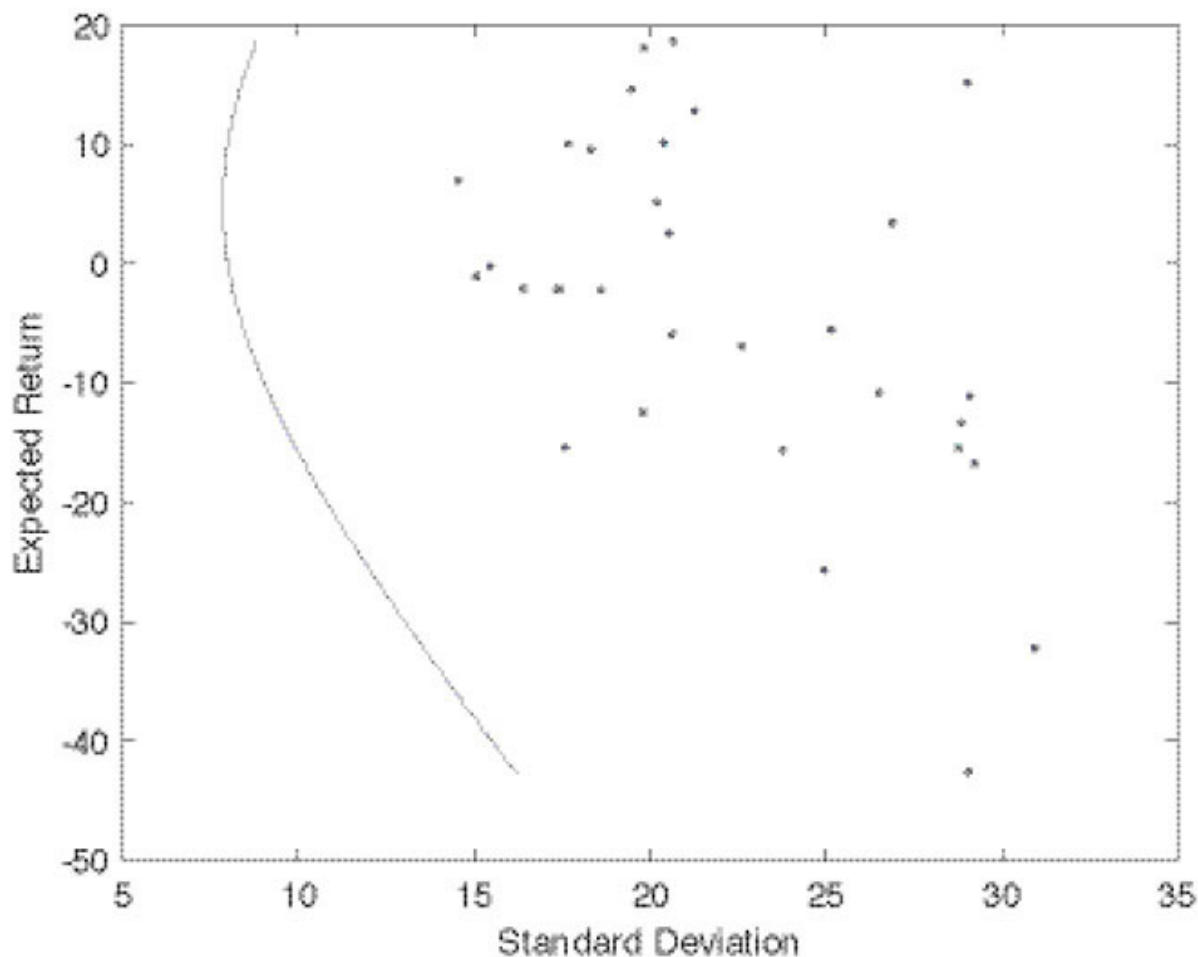
We show in [Appendix 9A](#) that this particular form of the quadratic minimization problem with equality constraints has a closed-form solution. The solution is

$$\mathbf{w} = \Sigma^{-1} \mathbf{A}' (\mathbf{A} \Sigma^{-1} \mathbf{A}')^{-1} \mathbf{b} \quad (9.11)$$

[Figure 9.1](#) provides a visual example of the results from a mean-variance optimization. In this particular optimization, we focused on the 32 stocks that comprise the consumer staples sector of the Standard & Poor's (S&P) 500. We created a five-factor model of security returns using the unemployment rate, consumer sentiment growth, excess market returns, log of market capitalization, and book-to-price factor. We estimated the economic factor model for the period from January 2016 to December 2020,⁶ and we produced forecasts for the expected returns and variances of all stocks for January 2021.

FIGURE 9.1

Efficient frontier for consumer staples sector (January 2021). Based on data from January 2016 to December 2020. Dots indicate the expected returns and the standard deviations of individual stocks. Expected return and standard deviation are annualized and in percent.



After that, we used our mean-variance optimization to create the minimal-risk portfolios for a variety of expected portfolio returns. We first found the stocks with the lowest and highest mean returns. We then began the optimization procedure to minimize risk for a given level of expected return using the expected return of the lowest expected return stock. Storing the optimal portfolio weights that would create this portfolio, we incremented the mean return and performed the optimization again, finding the optimal weights for that given mean return. We continued doing this until we reached the expected return of the security with the highest mean return. At this point we stopped running the optimization and plotted all the points in the diagram and connected them. The curve that results from these plotted points is typically called the *efficient frontier* in modern portfolio theory, although, technically, the efficient frontier is only the part of the curve whose gradient is positive (the upper half of the curve). Also, since

we forecasted the expected returns and variances, we should call this curve the *predictive efficient frontier*.

Typically, we decide how many intervals of expected return and variance we would like to plot. Smaller intervals necessitate more computation but also result in a smoother plot of the efficient frontier. The interval is usually chosen by taking the highest expected return $\max(\mu)$, subtracting the minimum expected return $\min(\mu)$, and dividing by the number of intervals one desires (that is, $[\max(\mu) - \min(\mu)] / \text{number of intervals} = \text{increment}$). This value becomes the incremental value to the expected portfolio return in the mean-variance optimization problem. Thus one starts with $\min(\mu)$, computes the optimal portfolio weights, and then recomputes the optimization problem using the portfolio mean of $\min(\mu)$ plus increment. And the process continues until one reaches the maximum expected return.

We also plotted the expected returns and variances of all the individual stocks in the consumer staples sector. (These are represented by dots.) This plot of the efficient frontier completes the process and maps out our portfolio choices. We should decide on the expected-return–expected-risk profile we wish to have and then pick the weights of the stocks corresponding to that profile on the efficient frontier.

Table 9.1 shows the characteristics and composition of selected portfolios along the efficient frontier that we created. We chose to show five portfolios with varying annual expected returns from -5.35% to 18.65% . Every stock in the original universe is included in the portfolios. Stock weights vary from about minus 9.88% to plus 17.64% . Since we did not impose the short-sale restriction, a number of stocks have negative weights. In this table, we show the three largest-weight stocks and the three smallest-weight stocks.

TABLE 9.1

Selected Efficient Portfolios from the Consumer Staples Sector

Portfolios	A	B	C	D	E
Expected return	−5.35	0.65	6.65	12.65	18.65
SD	8.43	7.95	7.86	8.16	8.82
Market beta	0.38	0.37	0.36	0.35	0.35
Number of stocks	32	32	32	32	32
Max. weight	13.56	13.27	12.99	15.31	17.64
Min. weight	−6.24	−5.75	−5.27	−6.96	−9.88
Selected constituents (weight)	HRL	HRL	HRL	PG	PG
	(13.56)	(13.27)	(12.99)	(15.31)	(17.64)
	PEP	PG	PG	HRL	HRL
	(9.48)	(10.66)	(12.99)	(12.70)	(12.42)
	K	PEP	CLX	CLX	CLX
	(9.28)	(9.26)	(9.90)	(11.03)	(12.15)
	EL	BF.B	KHC	KHC	GIS
	(−4.20)	(−2.12)	(−3.05)	(−4.52)	(−5.68)
	COST	COST	TAP	MNST	KHC
	(−4.67)	(−2.71)	(−4.04)	(−4.78)	(−6.00)
	MNST	MNST	MNST	TAP	TAP
	(−6.24)	(−5.75)	(−5.27)	(−6.96)	(−9.88)

Note: The investment universe consists of 32 consumer staples stocks in the S&P 500. The expected returns and risk are estimated from the economic factor model of Chapter 7. The computation is based on data from January 2016 to December 2020. Expected return and standard deviation (SD) are annualized and in percent. Weights are in percent. Ticker symbols are for the following companies: Hormel Foods Corp (HRL), Pepsico Inc. (PEP), Kellogg Co. (K), Estee Lauder Companies Inc. (EL), Costco Wholesale Corp (COST), Monster Beverage Corp (MNST), Procter & Gamble Co. (PG), Brown Forman Corp (BF.B), Clorox Co. (CLX), Kraft Heinz Co. (KHC), Molson Coors Beverage Co. (TAP), and General Mills Inc. (GIS). In the table we show the three largest-weight stocks and the three smallest-weight stocks for each portfolio.

Let's look at portfolio C on the efficient frontier. We expect that portfolio C will generate an annualized expected return of 6.65%. The annualized standard deviation is expected to be 7.86%. This optimal portfolio contains 32 stocks. Hormel Foods (HRL) has the highest weighting (12.99%) in this portfolio, and Monster Beverage Corp (MNST) has the lowest weighting (−5.27%). A portfolio manager might like to create a portfolio with a 6.65% annual expected return and a standard deviation of 7.86%, but he or she may not be able to implement the results of this mean-variance optimization. For instance, if he or she is not allowed to short stocks, a 5.27% short of MNST is out of the question. This illustrates why the manager might wish to perform a mean-variance

optimization with additional constraints. We discuss some of the most common additional constraints in the next subsections.

9.3.2 Short-Sale and Diversification Constraints

Portfolio managers may face constraints on their investment portfolios for various reasons, such as legal restrictions or prospectus mandates. The main constraint on a long-only portfolio manager is the short-sale restriction that prevents shorting securities. Mathematically, we can represent the short-sale restriction as the condition that

$$\mathbf{w} \geq 0 \tag{9.12}$$

which indicates that each stock has at least a weight of zero.

This restriction simply can be added to the minimization problem as an additional constraint. This type of constraint, however, is an *inequality constraint*. We show in [Appendix 9A](#) that when dealing with inequality constraints, the quadratic optimization problem does not have a simple analytical solution and instead requires numerical methods to solve for the optimal portfolio weighting.

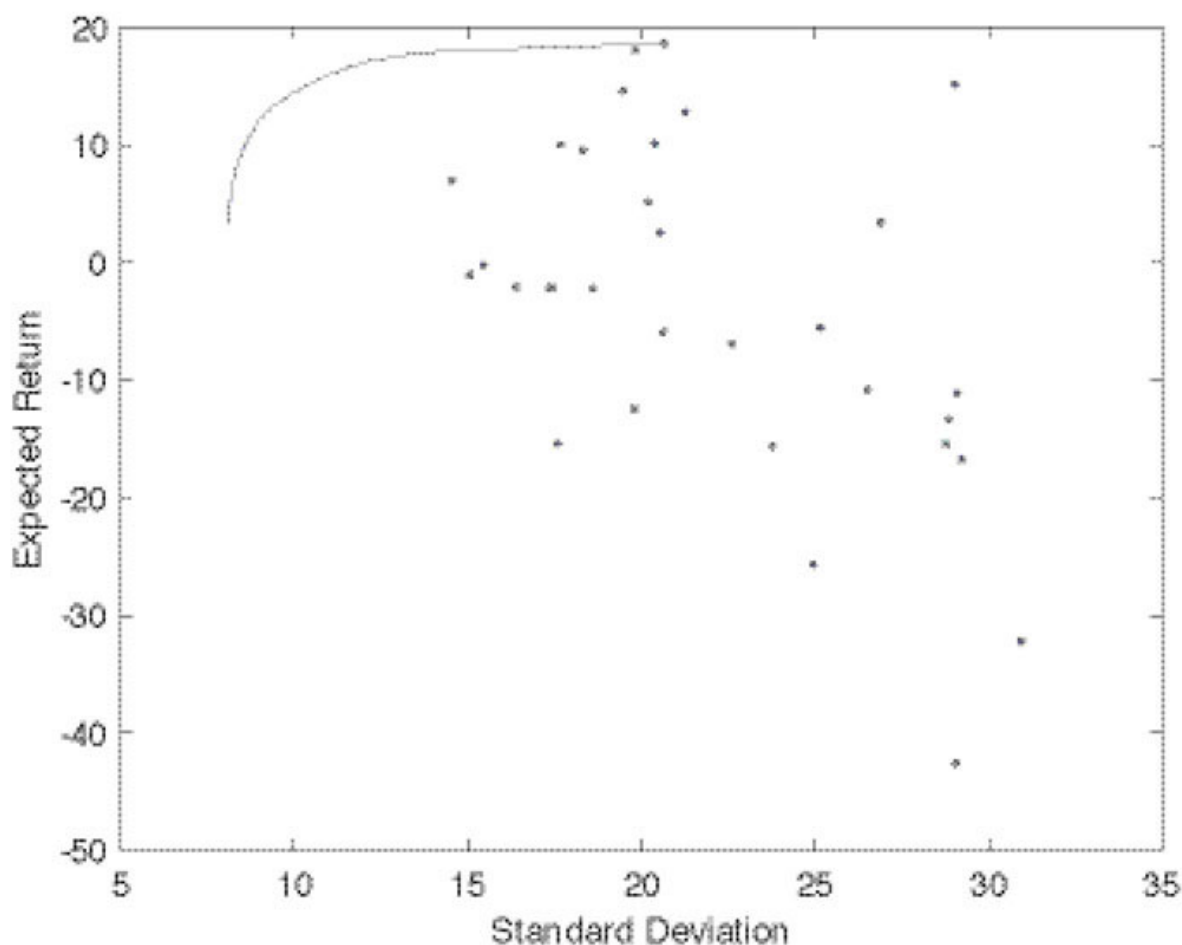
Techniques designed to solve this type of quadratic minimization with inequality constraints are known as *quadratic programming*. Most portfolio managers simply need to formulate the constraints, enter them into the commercial software or quadratic optimizer, and let the software/optimizer do the rest. For readers who are more interested in the mathematics of quadratic programming, we discuss it in [Appendix 9A](#).

Using the same data as before and a quadratic optimization programming tool, we recalculated the set of efficient portfolios with the additional short-sale constraints using the same approach. [Figure 9.2](#) plots the efficient frontier. As before, we focused on the same consumer staples sector of the S&P 500 and used the same time period for estimation, January 2016 to December 2020, and the same model for security returns. We then produced forecasts for the expected returns and variances of all stocks for January 2021.

FIGURE 9.2

Efficient frontier for consumer staples sector with short-sale constraints (January 2021). Based on data from January 2016 to December 2020. Dots indicate the expected returns and the standard

deviations of individual stocks. Expected return and standard deviation are annualized and in percent.



Compared with [Figure 9.1](#), the efficient frontier has shifted to the right owing to the additional short-sale constraint. This makes sense. By adding an additional constraint to the optimization, the lowest-risk portfolio we can create will have a higher risk than the lowest-risk portfolio we were able to create without the short-sale constraint.

[Table 9.2](#) shows the characteristics and composition of selected portfolios along the efficient frontier. We chose to show three portfolios with target monthly mean returns identical to the portfolios listed in [Table 9.1](#). We did not include two portfolios with expected returns of -5.35% and 0.65% . The variances of these portfolios are higher than the variance of portfolio C, so, in the strictest sense, they are no longer efficient portfolios. The minimum-risk portfolio includes fewer stocks to achieve the same expected return. Its maximum stock weights are also higher than in the

optimization without constraints. In this table, we show the weights for the five stocks with the largest weights in the optimal portfolio. Portfolio E is an extreme case in which the entire portfolio consists of just one stock, CHD.

TABLE 9.2

Selected Efficient Portfolios from the Consumer Staples Sector with Short-Sale Constraints

Portfolios	C	D	E
Expected return	6.65	12.65	18.65
SD	8.23	9.19	20.65
Market beta	0.41	0.43	0.45
Number of stocks	18	9	1
Max. weight	13.81	16.17	100.00
Selected constituents (weight)	HRL (13.81) PG (13.61) CLX (11.71) MKC (7.99) LW (7.40)	CLX (16.17) MKC (15.23) CHD (13.07) PG (12.91) LW (11.61)	CHD (100.00)

Note: The investment universe consists of 32 consumer staples stocks in the S&P 500. The expected returns and risk are estimated from the economic factor model of Chapter 7. The computation is based on data from January 2016 to December 2020. Expected return and standard deviation (SD) are annualized and in percent. Weights are in percent. Ticker symbols are for the following companies: Hormel Foods Corp (HRL), Procter & Gamble Co. (PG), Clorox Co. (CLX), McCormick & Co. (MKC), Lamb Weston Holdings Inc. (LW), and Church & Dwight Inc. (CHD). In this table, we show the weights for the five stocks with the largest weights in each portfolio.

Let's look again at portfolio C on the efficient frontier. We expect that portfolio C will generate an annualized expected return of 6.65%. This is comparable to portfolio C in the unconstrained case reported in [Table 9.1](#). The annualized standard deviation is expected to be 8.23%. Even though this optimal portfolio's expected return is similar to that of portfolio C in [Table 9.1](#), the risk is clearly higher. This optimal portfolio contains 18 stocks compared with the 32 in the unconstrained portfolio, with the

highest-weighted stock representing 13.81% of the portfolio. The constraint has removed the extremely negative weighting of MNST, as it should. At first glance, one might be unsatisfied with the results of the mean-variance optimization because this portfolio is not as optimal as the portfolio in the unconstrained case. However, the new portfolio will satisfy the needs of a long-only portfolio manager, demonstrating why it is useful to perform a mean-variance optimization with short-sale constraints.

In addition to short-sale constraints, portfolio managers also may wish to have *diversification constraints*. These are constraints that a mutual fund accepts in compliance with the diversification requirements of the Investment Company Act of 1940.⁷ Of course, even if a portfolio manager is not regulated by the Securities and Exchange Commission (SEC), he or she may wish to follow maximum-weight constraints because too much exposure to very few stocks also increases the portfolio's diversifiable risk. Restrictions of this kind generally can be expressed as

$$\underline{\mathbf{w}} \leq \mathbf{w} \leq \overline{\mathbf{w}} \quad (9.13)$$

where $\underline{\mathbf{w}}$ and $\overline{\mathbf{w}}$ are N -dimensional vectors of maximum and minimum allowed weights. This constraint can be added easily to the optimization problem and satisfies both the short-sale constraint and the diversification constraint simultaneously.

9.3.3 Sector or Industry Constraints

Many portfolio managers—especially those who manage against a benchmark—wish to constrain the sector weightings of the portfolio. Here is a simple modification to the framework to constrain sector weightings:

$$\underline{w}_j \leq w_j \leq \overline{w}_j \quad (9.14)$$

where w_j represents the weight of sector j in the portfolio.⁸

9.3.4 Trading-Volume Constraint

One of the typical constraints that portfolio managers add to the optimization is the trading-volume constraint. This is particularly relevant

when the value of the portfolio is large and the portfolio manager's transactions have a large price impact.⁹ To avoid creating negative price impact, the portfolio manager may restrict the holding of each stock to keep it below some threshold amount, typically a fraction of the average trading volume of each stock.

Suppose that the value of the portfolio is \$500 million and that the portfolio manager wants to keep the holding of one stock below 10% of the average daily trading volume of the stock (ADV). If w_i is the portfolio weight of stock i and x_i is the average trading volume of stock i measured in millions of dollars, then the constraint is $500w_i \leq 0.1x_i$ or $w_i \leq (0.1/500)x_i$.

In general, the trading volume constraint can be expressed as

$$\mathbf{w} \leq c\mathbf{x} \quad (9.15)$$

where $\mathbf{x}$ is a vector of average daily trading volume in dollar terms, and c is a constant indicating the threshold.

9.3.5 Risk-Adjusted Return

So far we have formulated the mean-variance optimization problem as a risk minimization. Some portfolio managers may prefer an alternative formulation with expected return maximization. The expected return maximization can be written as

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} \quad s.t. \quad \mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} = \sigma_p^2 \quad (9.16)$$

and other constraints. The expected return maximization may be more useful if the portfolio manager has a specific target risk level σ_p^2 , whereas the risk minimization may be more useful if the portfolio manager has a target expected return.¹⁰

When the portfolio manager has neither a target risk nor a target expected return, the mean-variance optimization can be expressed in terms of *risk-adjusted* expected return. We can adjust the expected return for the risk by subtracting some multiple of the risk, that is, $\mu_p - A\sigma_p^2$. The multiplier A is called the *risk-aversion parameter* because a high value of A indicates that the portfolio manager considers the risk very costly. If the

value of A is 2, it means that the portfolio manager equates a 1% increase in the variance with a 2% decrease in the expected return. Once the value of the risk-aversion parameter is decided, the mean-variance problem becomes

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} - A\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} \quad (9.17)$$

This formulation turns out to be quite useful in certain applications.¹¹

9.4 BENCHMARK

Most portfolio managers manage the portfolio versus a benchmark. Managers who stick very closely to the benchmark sometimes are referred to as *index managers*, but managers who manage very loosely with respect to the benchmark are described more accurately as *active managers* or *enhanced index managers*. The goal of active managers is to produce a portfolio of stocks that is broadly similar to the underlying benchmark but that outperforms the benchmark by some amount. The manager must walk a fine line in order to outperform the benchmark without diverging dramatically from it. Fortunately, when the benchmark itself is not efficient, these two goals do not conflict.¹²

There is a whole set of tools to deal with tracking a benchmark while producing α (outperformance) over the benchmark. We will describe a number of these in the following sections, including ad hoc methods, stratification, factor exposure targeting, and tracking error minimization. The last method is the most popular because, in theory, it provides the tightest control on risk versus the benchmark while still allowing the portfolio manager to select his or her favorite stocks.

9.5 AD HOC METHODS AGAIN

One rather simplistic approach to weighting a benchmarked portfolio so that it follows the benchmark closely is to select the largest holdings of the benchmark for the portfolio. If the portfolio is going to include 50 stocks, this would amount to choosing the 50 stocks with the largest market capitalization in the benchmark and then possibly computing the weights of the 50 stocks according to their relative market capitalizations. A manager

could further tilt the weights slightly toward his or her preferred stocks. There are a number of ad hoc methods for tilting the weights toward preferred stocks. Let's assume that the portfolio manager has ranked the stocks by the aggregate Z-score methodology described in [Chapter 5](#).¹³ The portfolio manager should renormalize the aggregate Z-scores so that the sum of the Z-scores of the subset of 50 stocks (or whatever number of stocks he or she chooses) is equal to 0. The steps to alter the relative market capitalization weighting are as follows: First, the portfolio manager should decide what is the maximum percentage deviation in weight he or she is willing to allow from the market-cap weighting for the maximum absolute Z-score value. Call this η . The second step is to find the maximum absolute Z-score of all the stocks in the universe of 50 stocks. Thus, take the absolute value of all the individual Z-scores and find the maximum; call this $z^{\max}$. The third step is to compute the Z-score multiplier $m = \eta/z^{\max}$. The fourth step is to compute the new weights w_i^* of the portfolio such that ($w_i^* = w_i + mz_i$), where w_i is the relative market capitalization weight within the benchmark.¹⁴ The adjusted portfolio is complete.

As a brief example of this method for creating an altered market-capitalization portfolio to track the benchmark with the top N stocks by market capitalization, we selected the top 30 S&P 500 stocks and modified the relative market capitalization weights using the Z-scores methodology used in [Table 5.2](#). The modified weights as well as the relative weights and the Z-scores are shown in [Table 9.3](#). By altering the portfolio in this way, the portfolio manager can continue to track the benchmark relatively closely while still altering the stock weights to take advantage of his or her α model.

TABLE 9.3

Ad Hoc Modified Market-Cap Weights to Reflect Z-Score Model

Ticker	Relative Market-Cap Weight (w_i)	Z-Score	Z-Score- Adjusted Weight (w_i^*)	Change in Weight ($w_i^* - w_i$)
AAPL	0.1498	-1.6817	0.1473	-0.0025
MSFT	0.1117	-1.9532	0.1088	-0.0029
AMZN	0.1085	-1.4515	0.1064	-0.0021
GOOGL	0.0734	-1.3965	0.0713	-0.0021
TSLA	0.0444	13.5251	0.0644	0.0200
FB	0.0436	-0.8509	0.0424	-0.0013
BRK.B	0.0361	0.3480	0.0366	0.0005
JNJ	0.0275	-1.0351	0.0260	-0.0015
WMT	0.0271	-0.6762	0.0261	-0.0010
JPM	0.0257	-0.5420	0.0249	-0.0008
V	0.0246	-1.0393	0.0231	-0.0015
MA	0.0234	-1.3853	0.0214	-0.0020
PG	0.0229	-0.8318	0.0217	-0.0012
UNH	0.0221	-0.4488	0.0214	-0.0007
DIS	0.0218	-0.5556	0.0210	-0.0008
NVDA	0.0215	1.1247	0.0231	0.0017
HD	0.0190	-1.0188	0.0175	-0.0015
PYPL	0.0182	1.1093	0.0199	0.0016
BAC	0.0174	0.6922	0.0184	0.0010
VZ	0.0161	-0.3645	0.0156	-0.0005
ADBE	0.0159	-0.0586	0.0158	-0.0001
CMCSA	0.0159	0.1797	0.0162	0.0003
NFLX	0.0159	0.1099	0.0160	0.0002
KO	0.0157	-1.0718	0.0141	-0.0016
MRK	0.0137	-0.7408	0.0126	-0.0011
PEP	0.0136	-0.9175	0.0123	-0.0014
T	0.0136	0.8621	0.0149	0.0013
PFE	0.0136	-0.2055	0.0133	-0.0003
INTC	0.0136	0.1305	0.0138	0.0002
CRM	0.0135	0.1437	0.0137	0.0002

Note: In this particular example, the maximum absolute Z-score is 13.53 (TSLA), $\eta = 0.02$, and $m = 0.001479$. w_i is the relative market capitalization weights, Z-scores were computed as in Table 5.2, and w^* is based upon the ad hoc method. Finally, the last column represents the change in weights from w_i to w^* . All the data are as of the end of December 2020. The ticker symbols are for the following companies: Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), Alphabet (GOOGL), Tesla (TSLA), Facebook (FB), Berkshire Hathaway (BRK.B), Johnson & Johnson (JNJ), Walmart (WMT), J.P. Morgan Chase & Co. (JPM), Visa (V), Mastercard (MA), UnitedHealth Group (UNH), Disney (DIS), Procter & Gamble (PG), American Express (AXP), Home Depot (HD), PayPal (PYPL), Bank of America (BAC), Verizon (VZ), Adobe (ADBE), Comcast (CMCSA), Netflix (NFLX), Coca-Cola (KO), Merck (MRK), PepsiCo (PEP), Target (T), Pfizer (PFE), Intel (INTC), and Salesforce (CRM).

Mastercard (MA), Procter & Gamble (PG), United Health Group (UNH), Disney (DIS), Nvidia (NVDA), Home Depot (HD), PayPal (PYPL), Bank of America (BAC), Verizon Communications (VZ), Adobe (ADBE), Comcast (CMCSA), Netflix (NFLX), Coca-Cola (KO), Merck & Co. (MRK), PepsiCo (PEP), AT&T (T), Pfizer (PFE), Intel Corp (INTC), and Salesforce (CRM).

Although this is certainly one type of ad hoc method to compute portfolio weights while still tracking the benchmark, it will not be the best solution. For one, there was no attempt to choose the portfolio based on minimum tracking error versus the index. Thus there will be no control of other risk factors or of asset-specific risk in the portfolio. Almost surely it will not be the optimal portfolio. Moreover, the relatively small number of stocks in the portfolio (30 in our example) represent only a fraction of the total market capitalization of the benchmark (less than 50% in our example). Though simple, this is an inefficient way to choose the stocks.

Other ad hoc methods are possible, but a professional portfolio manager will be dissatisfied with them despite their simplicity. Ultimately, a manager wants to quantify his or her risk versus the benchmark, something that simply is not possible with these other, rather amateurish mechanisms.

9.6 STRATIFICATION

Stratification or stratified sampling is another simple way to build portfolios while maintaining a very rudimentary risk control mechanism. Stratified sampling was devised primarily for empirical statisticians who wanted to understand the characteristics of a population but could not afford to gather observations on every member of the population. One method to create a representative sample of the population would be to randomly select members of the universe. While this method converges to the true mean and standard deviation of the population, we can improve on it by first stratifying the sample and then choosing a certain number of observations within each stratum.¹⁵

We assume that the portfolio manager already has predicted the excess returns of all the stocks in his or her universe. His or her goal is to choose the high- α stocks while controlling risk versus the benchmark. The stratification method involves minimizing the exposures of the portfolio along many dimensions of risk. The first stage of stratification is to stratify the universe of stocks by dividing it into J nonoverlapping groups. We might want to stratify, for instance, by dividing the stocks into industry

buckets so that each stratum represents a different industry. Call the number of stocks in stratum j , N_j . The total stocks in the universe can be represented by N . Thus $\sum_{j=1}^J N_j = N$.

The next step in stratified sampling with one subgroup classification is to select representative stocks from each stratum. At this point the portfolio manager will have to decide how many stocks he or she wants in the portfolio. Let's call this number N_P . The idea of stratification is to select a proportion of stocks from each stratum that is representative of the universe of stocks. Thus, from each stratum, the portfolio manager should choose $N_j N_P / N$ stocks. This value may need to be rounded to the nearest integer. In traditional stratified sampling, an index portfolio manager would select the stocks at random, which is a valid procedure for just replicating the characteristics of the index. Beyond pure mimicry, though, a manager ought to maintain some risk controls while picking the high- α stocks. Thus, before making selections of stocks from each stratum, he or she should rank the stocks in each stratum according to aggregate Z-score, expected return, excess expected return, or α and then choose the stocks with the highest values of the chosen criterion. For example, if the stock universe is divided into sectors and four stocks must be chosen from the transportation sector, those four stocks could be the ones with the highest relative ranking according to future risk-adjusted returns.

Of course, a portfolio manager might wish to create more than one subgroup classification to control risk. This can be accomplished by dividing the universe of N stocks into a group of J categories based on one classification and then dividing each group into subgroups, and so on and so forth. For example, to divide the universe of stocks into two subgroup classifications, we could divide them by industry and then by size.

Figure 9.3 depicts the stratification of the universe of stocks into two subgroup classifications based on 9 industry groups and 3 size groups (large, medium, and small). With the stocks separated into 27 separate groups, or buckets, the manager ranks them according to highest Z-score or some other excess-return criterion. He or she selects stocks from each subgroup based on the formula $N_j N_P / N$ rounded to the nearest integer.

FIGURE 9.3

The stratified-sample approach.

	Industry								
	1	2	3	4	5	6	7	8	9
Large	•	•	•	•	•	•	•	•	•
Medium	•	•	•	•	•	•	•	•	•
Small	•	•	•	•	•	•	•	•	•

Stratification is an easy way to choose one's preferred stocks while also controlling the risk via broad diversification. A portfolio manager might have an α model that, left to its own devices, tends to select stocks in one particular industry. Stratifying across industries or sectors ensures broad diversification and risk control versus the benchmark. The biggest drawback to stratification is that it is a fairly rudimentary method for controlling risk. Most professional portfolio managers would be averse to using it because it does not precisely quantify ex ante how much risk they are taking. Managers ought to be able to quantify their risks using all the available information. Clearly, stratification lacks a precise, quantitative control mechanism.

9.7 FACTOR EXPOSURE TARGETING

Another way to bring the portfolio in line with the benchmark is to set the benchmark's factor exposures as target factor exposures for the portfolio. Or one may set the portfolio's overall beta with respect to the benchmark equal to (or very close to) 1.¹⁶

The benchmark beta of any portfolio is simply the weighted average of the benchmark betas of the individual stocks. Let β be an N -dimensional column vector of the benchmark beta of individual stocks, that is,

$$\beta = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_N \end{bmatrix} \quad (9.18)$$

where β_i is the beta of stock i with respect to the benchmark. The beta of the portfolio equals $\mathbf{w}'\beta$. We can add the following constraint to the optimization problem:

$$\mathbf{w}'\beta = 1 \quad (9.19)$$

Or we can specify some range for the portfolio β instead:

$$0.75 \leq \mathbf{w}'\beta \leq 1.25 \quad (9.20)$$

Let us return to the consumer staples example. In the example we created the efficient frontier given a short-sale constraint. Now let us add a constraint that requires the β of the portfolio with respect to the benchmark to be no less than 0.85. The benchmark for the portfolio manager in this case is the value-weighted (i.e., market-cap-weighted) return of all stocks in the consumer staples sector. Other than our new constraint, the computation here is based on the same model and the same data as in Tables 9.1 and 9.2.

Table 9.4 shows the selected minimum-risk portfolios created from 32 stocks in the consumer staples sector with the restriction on the benchmark β . Portfolio A (with a mean return of -5.35%) and portfolio B (with a mean return of 0.65%) are not reported because there is no minimum-variance portfolio with either of those mean returns owing to the additional constraint.

TABLE 9.4

Selected Efficient Portfolios from the Consumer Staples Sector with Short-Sale and Benchmark β Constraints

Portfolios	C	D	E
Expected return	6.65	12.65	18.65
SD	8.25	9.22	20.65
Benchmark beta	0.85	0.85	0.86
Number of stocks	20	10	1
Max. weight	13.70	15.14	100.00
Selected constituents (weight)	PG (13.70) HRL (11.82) CLX (11.03) MKC (8.32) PEP (7.44)	MKC (15.14) CLX (14.71) PG (14.04) CHD (12.46) LW (11.25)	CHD (100.00)

Note: The investment universe consists of 32 consumer staples stocks in the S&P 500. The benchmark is the market-cap-weighted index of the same 32 stocks. The expected returns and risk are estimated from the economic factor model of Chapter 7. The benchmark β is restricted to be no less than 0.85. The computation is based on data from January 2016 to December 2020. Expected return and standard deviation (SD) are annualized and in percent. Weights are in percent. Ticker symbols are for the following companies: Procter & Gamble Co. (PG), Hormel Foods Corp (HRL), Clorox Co. (CLX), McCormick & Co. (MKC), Pepsico Inc. (PEP), Church & Dwight Inc. (CHD), and Lamb Weston Holdings Inc. (LW). In this table, we show the weights for the five stocks with the largest weights in each portfolio.

We imposed an inequality constraint on the portfolio requiring that the market β be no less than 0.85. In the case of portfolios A and B, since the betas of the portfolios turned out to be exactly 0.85, which hit our minimum requirement, the constraint was binding. Since the constraint was binding, the variances of these portfolios are greater than the variances of the portfolios constructed in Tables 9.1 and 9.2. The compositions of these new portfolios also differ significantly from the compositions of the previous portfolios.

While we began with the simple case of targeting the portfolio's benchmark β to a specified range, we may prefer to stipulate a range for each of the portfolio's other factor exposures as well. This is sometimes referred to as *factor tilting* because we are tilting the portfolio toward greater exposure to certain factors and less exposure to others, according to

our view on the factors and upcoming market conditions. If a portfolio manager believes that the market will rally, for instance, he or she may wish to have a higher market β than that of the benchmark while keeping all other factor exposures equal to those of the benchmark. He or she would tilt the portfolio toward the market factor.

The factor exposure of a portfolio is the weighted average of the factor exposures of individual stocks. Let $\mathbf{B}$ be an $N \times K$ matrix of factor exposures of individual stocks. That is,

$$\mathbf{B} = \begin{bmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1K} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2K} \\ \vdots & & & \\ \beta_{N1} & \beta_{N2} & \cdots & \beta_{NK} \end{bmatrix} \quad (9.21)$$

where K is the number of relevant factors, and β_{ik} is the exposure of stock i to factor k . Then the factor exposure of a portfolio with weight $\mathbf{w}$ is simply $\mathbf{B}'\mathbf{w}$. Thus we may add the following general factor exposure constraint:

$$\underline{\beta} \leq \mathbf{B}'\mathbf{w} \leq \bar{\beta} \quad (9.22)$$

The portfolio manager can exercise his or her particular management style through this sort of constraint. Assigning a minimum exposure (say, 0.9) to the growth factor orients the portfolio toward growth investments. If the first factor is the growth factor, then the manager would set the first element of $\underline{\beta}$ to 0.9.

9.8 TRACKING-ERROR MINIMIZATION

9.8.1 Direct Computation

Most professional portfolio managers with a benchmark use the minimization-of-tracking-error approach to weighting stocks and building a portfolio. There are two ways to formulate the optimization problem using this approach. One way is to minimize the tracking error for a given expected excess return over the benchmark, and the other is to maximize

the expected excess return over the benchmark without exceeding a maximum tracking-error constraint. The former approach is explained in this subsection, and the latter approach is explained in Section 9.8.4. We prefer the latter formulation because it is more realistic. In the practical application part of this book ([Chapters 16 and 17](#)), we use the latter approach.

Typically, the portfolio manager would like to use all his or her tracking-error constraints so long as he or she is adding to the expected excess return over the benchmark. Tracking-error constraints vary from portfolio to portfolio but can range from as little as 0.5% per annum to 10% per annum. The investment committee of the investment management firm often comes up with the tracking-error constraint, but sometimes portfolio managers choose to be even more conservative than required by the committee's constraint. Fortunately, the quadratic optimization framework we have already discussed is applicable to tracking-error minimization. We need only make slight modifications to the optimization problem.

Tracking error (TE) is defined by most portfolio managers as the standard deviation of portfolio returns minus benchmark returns.¹⁷ Thus

$$TE = S(r_P - r_B) = \sqrt{V(r_P - r_B)} \quad (9.23)$$

where $S(\cdot)$ is the standard deviation function.

The first method for minimizing the tracking error of the portfolio is to minimize it given some expected level of excess returns over the benchmark. Now consider the components of the variance term¹⁸:

$$V(r_P - r_B) = V(r_P) - 2C(r_P, r_B) + V(r_B) \quad (9.24)$$

The last term, the variance of the benchmark, is beyond our control. Thus, to find a portfolio that minimizes the tracking error of the portfolio, we minimize $V(r_P) - 2C(r_P, r_B)$.

Given the portfolio weight $\mathbf{w}$, the variance of the portfolio is given as $\mathbf{w}'\Sigma\mathbf{w}$. The covariance between the portfolio and the benchmark can be computed from the covariances between individual stock returns and the benchmark return. Let γ be an N -dimensional vector of the covariances between individual stock returns and the benchmark return:

$$\gamma = \begin{bmatrix} C(r_1, r_B) \\ \vdots \\ C(r_N, r_B) \end{bmatrix} \quad (9.25)$$

Then the covariance between the portfolio return and the benchmark return equals $\mathbf{w}'\gamma$.

To find the portfolio that minimizes tracking error, we need to solve the following quadratic minimization problem:

$$\min_{\mathbf{w}} \mathbf{w}'\Sigma\mathbf{w} - 2\mathbf{w}'\gamma \quad s.t. \quad \mathbf{w}'\mu = \mu_P \quad (9.26)$$

The same quadratic programming routine that was used in the preceding section can solve this problem as well. Typically, the chosen portfolio mean μ_P will be some excess return over the benchmark. Practically, we should think of $\mu_P = \mu_B + \delta$. That is, we should take the expected return of the benchmark and add a small amount to it according to our desire and then run the optimization to find the portfolio weights. Just as in the case of a portfolio with no benchmark, we will see shortly that we can again add various additional constraints, such as the short-sale, diversification, and style constraints.

9.8.2 Tracking by Factor Exposure

There is an alternative but equivalent representation of the tracking-error minimization problem. Recall from [Chapters 6](#) and [7](#) that the variance of an individual stock i can be estimated as

$$V(r_i) = \beta_i' V(\mathbf{f})\beta_i + V(\epsilon_i) \quad (9.27)$$

Assuming that the covariance between the residuals of the stocks is 0, we can write the variance-covariance matrix of all stock returns as

$$\begin{aligned}
\Sigma &= \begin{bmatrix} \beta_{1,1} & \dots & \beta_{1,K} \\ \vdots & \ddots & \vdots \\ \beta_{N,1} & \dots & \beta_{N,K} \end{bmatrix} \begin{bmatrix} V(f_1) & \dots & C(f_1, f_K) \\ \vdots & \ddots & \vdots \\ C(f_K, f_1) & \dots & V(f_K) \end{bmatrix} \begin{bmatrix} \beta_{1,1} & \dots & \beta_{N,1} \\ \vdots & \ddots & \vdots \\ \beta_{1,K} & \dots & \beta_{N,K} \end{bmatrix} \\
&+ \begin{bmatrix} V(\epsilon_1) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & V(\epsilon_N) \end{bmatrix} \\
&= \mathbf{B}V(\mathbf{f})\mathbf{B}' + V(\boldsymbol{\epsilon})
\end{aligned} \tag{9.28}$$

where $\mathbf{B}$ is an $N \times K$ matrix of factor exposures, $V(\mathbf{f})$ is a $K \times K$ matrix of factor premium variances and covariances, and $V(\boldsymbol{\epsilon})$ is an $N \times N$ diagonal matrix of error variances. Given this, the squared tracking error is

$$\begin{aligned}
TE^2 &= (\mathbf{w}^P - \mathbf{w}^B)' \mathbf{B}V(\mathbf{f})\mathbf{B}'(\mathbf{w}^P - \mathbf{w}^B) + \\
&\quad (\mathbf{w}^P - \mathbf{w}^B)' V(\boldsymbol{\epsilon})(\mathbf{w}^P - \mathbf{w}^B)
\end{aligned} \tag{9.29}$$

where $\mathbf{w}^P$ is the weight of the portfolio, and $\mathbf{w}^B$ is the weight of the benchmark. Once other relevant constraints are added, this tracking-error minimization problem can be solved using the quadratic optimizer.¹⁹

Table 9.5 shows the minimum-tracking-error portfolio created from 32 stocks in the consumer staples sector. The table is based on the same model and data as in Tables 9.1, 9.2, and 9.4. The benchmark is the value-weighted portfolio of 32 stocks in the consumer staples sector, which is identified as portfolio A in the table. As we try to obtain higher expected returns, the tracking error increases. To obtain a 5.29% (6.65% – 1.36%) excess return over the benchmark (portfolio C), the portfolio manager can expect a tracking error of 1.79%. To obtain a 17.29% excess return over the benchmark (portfolio E), the portfolio manager must accept an expected tracking error close to 20%.

TABLE 9.5

Selected Portfolios from the Consumer Staples Sector by Minimizing Tracking Error with Short-Sale Constraints

Portfolios	A	B	C	D	E
Expected return	1.36	0.65	6.65	12.65	18.65
SD	10.14	10.17	10.07	10.24	20.65
Tracking error	0.00	0.21	1.79	5.03	19.62
Number of stocks	32	32	26	11	1
Max. weight	17.08	16.80	19.52	21.59	100.00
Selected constituents (weight)	WMT (17.08)	WMT (16.80)	WMT (19.52)	WMT (21.59)	CHD (100.00)
	PG (14.45)	PG (14.18)	PG (16.55)	PG (13.31)	
	KO (9.87)	KO (9.83)	COST (9.27)	MKC (12.37)	
	PEP (8.59)	PEP (8.61)	KO (9.13)	CHD (12.18)	
	COST (6.99)	COST (6.76)	PEP (7.25)	EL (11.19)	

Note: The investment universe consists of 32 consumer staples stocks in the S&P 500. Portfolio A is the benchmark portfolio. The benchmark is the market-cap-weighted index of the same 32 stocks. The expected returns and risk are estimated from the economic factor model of Chapter 7. The computation is based on data from January 2016 to December 2020. Expected return and standard deviation (SD) are annualized and in percent. Weights are in percent. Ticker symbols are for the following companies: Walmart (WMT), Procter & Gamble Co. (PG), Coca-Cola (KO), Pepsico (PEP), Costco (COST), McCormick & Company (MKC), Estee Lauder Companies (EL), and Church & Dwight Co. (CHD). In this table we show the weights for the five stocks with the largest weights in each portfolio.

Minimizing tracking error will force the factor exposures of the portfolio to be quite close to those of the benchmark, so the first term in [Eq. \(9.29\)](#) will be less significant than the second term. In fact, when the portfolio manager has specific desired values for portfolio factor exposures, the first term can be completely dropped. In this case the optimization problem is specified simply as minimization of the variance of error terms subject to the additional constraint on the portfolio factor exposure. This is a type of *factor tilting*.

Suppose that the portfolio manager wants to tilt the portfolio toward small growth stocks but in all other respects wants the portfolio's exposures to remain identical to those of the benchmark. He or she can achieve this by setting the portfolio's exposures to the size factor and the growth factor higher than the benchmark's exposures to those factors while keeping its other factor exposures equal to those of the benchmark. For example, if

there are five factors—market return, log of market capitalization, GDP growth, unemployment, and consumer sentiment growth—then he or she can add the following constraints:

$$\mathbf{B}'(\mathbf{w}^P - \mathbf{w}^B) = \mathbf{d} \quad (9.30)$$

where

$$\mathbf{d} = \begin{pmatrix} 0 \\ 0.1 \\ 0.1 \\ 0 \\ 0 \end{pmatrix} \quad (9.31)$$

The zeros in vector $\mathbf{d}$ will make sure that the portfolio's exposures to the market return, unemployment, and sentiment factors are identical to the benchmark's exposures to those factors. The value of 0.1 in vector $\mathbf{d}$ will make sure that the portfolio's exposures to the size and growth factors will be higher than the benchmark's exposure to those factors by 0.1. With factor tilting, the optimization problem becomes

$$\min_{\mathbf{w}^P} (\mathbf{w}^P - \mathbf{w}^B)' V(\epsilon) (\mathbf{w}^P - \mathbf{w}^B) \quad (9.32)$$

subject to [Eq. \(9.30\)](#) and any other constraints.

9.8.3 Ghost Benchmark Tracking

There may be cases in which the portfolio manager does not know the weights of the underlying benchmark's securities or he or she cannot estimate all the benchmark's factor exposures. In this case, the minimization of tracking error involves minimizing the tracking error with respect to the stream of historical returns on the benchmark.

The tracking-error formula must undergo some modification because we do not know the benchmark weights. After some manipulation of the

equation, the squared tracking error becomes

$$\begin{aligned}
 TE^2 = & \begin{pmatrix} \mathbf{w}^P \\ -1 \end{pmatrix}' \begin{pmatrix} \mathbf{B} \\ \beta_B \end{pmatrix} V(\mathbf{f}) \begin{pmatrix} \mathbf{B} \\ \beta_B \end{pmatrix}' \begin{pmatrix} \mathbf{w}^P \\ -1 \end{pmatrix} \\
 & + \begin{pmatrix} \mathbf{w}^P \\ -1 \end{pmatrix}' \begin{pmatrix} V(\epsilon) & 0 \\ 0 & V(\epsilon_B) \end{pmatrix} \begin{pmatrix} \mathbf{w}^P \\ -1 \end{pmatrix}
 \end{aligned} \tag{9.33}$$

In this equation, β_B is the benchmark's factor exposure, and ϵ_B is the benchmark's error term. Once the tracking error is defined, the tracking-error minimization can proceed as described earlier.

9.8.4 Risk-Adjusted Tracking Error

In practical situations, the portfolio manager may have some maximum tracking-error constraint, say, 3% per annum. He or she can keep increasing the expected excess return until he or she reaches the tracking-error constraint. This problem is essentially identical to what we discussed so far. The objective function will be the expected return of the portfolio rather than the tracking error, and the constraint will include the tracking error rather than the expected return. Mathematically speaking, given the target tracking error of σ_x ,

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} \quad s.t. \quad V(r_P - r_B) = \sigma_x^2 \tag{9.34}$$

and other constraints.

If there is neither a target tracking error nor a target mean, then the problem might be expressed in terms of the tracking-error-adjusted expected return. As we did for the risk-adjusted return, we can adjust the expected return for the tracking risk by subtracting some multiple of the squared tracking error $\mu_P - A\sigma_x^2$. The multiplier A is called the *tracking-error aversion parameter* because a high value of A indicates that the portfolio manager considers the tracking error very costly. Then the problem becomes

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} - AV(r_P - r_B) \quad (9.35)$$

subject to other additional constraints.

Note that there is an important relationship between the two formulations. The set of maximum-return portfolios that we obtain as we vary the target tracking error σ_x is identical to the set of optimal portfolios that we obtain as we vary the tracking-error aversion parameter A . That is, we can always choose the target tracking error σ_x or the tracking-error aversion parameter A so that the two formulations are equivalent. This property can be quite useful in the optimization process. A commercial software package may not support maximizing the expected return subject to a quadratic constraint.²⁰ In such a case we can maximize the tracking-error-adjusted expected return [i.e., Eq. (9.35)] for a certain value of A . After the maximization, we can check whether the tracking-error constraint is satisfied. If not, we change the value of A and continue this process until we satisfy the tracking-error constraint. The value of A that makes the optimal portfolio satisfy the tracking-error constraint can be found in a small number of iterations using the property that a higher value of A reduces the tracking error of the optimal portfolio. This is the approach we follow in Part V of this book.

9.9 CONCLUSION

In this chapter we showed how to turn the factor model's predictions of stock return and risk into a portfolio through various methods of stock weighting. As we saw, there are a number of very simple ad hoc ways to assign weights in a portfolio that are not measured against a benchmark, but they do not allow the portfolio manager to maintain good control of the risk of the portfolio. For nonbenchmark portfolios, we instead recommended mean-variance optimization (MVO), which maximizes the portfolio's return and minimizes its risk while also allowing the manager to impose certain constraints on the portfolio's characteristics. More often than not, though, managers try to track a benchmark with their portfolios, so we turned our attention to weighting methods that take into account the relationship between the portfolio and the benchmark. Again, we found that ad hoc methods failed to provide anything more than quick-and-easy but

incomplete fixes to the weighting problem. The stratification method was an improvement over ad hoc methods because it offered a very rough risk control mechanism by allowing us to divide stocks into groups and select samples from each group. The factor exposure method also controlled for risk, and it let us “tilt” the portfolio’s beta toward the factors that we thought were most important. We saw, however, that the best method of weighting stocks in a benchmarked portfolio is tracking-error minimization because it minimizes the portfolio’s tracking error, or risk, without compromising returns (or, conversely, it maximizes the portfolio’s return without exceeding a certain tracking error). This method of weighting achieves the highest possible return for the least possible relative risk, works even when the portfolio manager does not know the exact makeup of the benchmark, and allows him or her to place constraints on the portfolio. We have completed the portfolio construction process by showing how to design a portfolio that balances the weights of high-quality stocks in the optimal way. Yet even a well-designed portfolio is always a work in progress. As conditions change, the portfolio needs to be rebalanced, and the transactions costs of rebalancing must be considered. The portfolio also should be refined to operate as best as it can within the tax environment. Many managers fail to counteract the draining effect of taxes, yet there are strategies for fortifying a portfolio against the tax drain. We discuss these sorts of improvements in the next two chapters.

¹ The reader may wish to review [Chapters 5, 6, 7, and 8](#) before proceeding with this chapter.

² The choice of how to determine these values is up to the manager. Some managers use models to forecast the expected stock returns (or α) but rely on commercial software to calculate the implicit risks of the portfolio. Other managers build their own risk models.

³ Many portfolios and indices are also *float-weighted*, which has the same sort of characteristics as value weighting, except stocks are weighted by their floating shares, not their shares outstanding.

⁴ Mathematicians and software programs, however, prefer the formulation in which the objective function is quadratic and the constraint is linear (rather than the other way around). See [Appendix 9A](#) and [Appendix 9B](#) for the mathematical details of typical quadratic optimization problems.

⁵ [Chapter 8](#) discussed methods to measure estimation error. The reader also may be interested in a book that deals with estimation error, such as Michaud (1993). This book is particularly critical of MVO. On page 3, it reads, “MV optimizers function as a chaotic investment decision system. Small changes in input assumptions often imply large changes in the optimized

portfolio. The procedure overuses statistically estimated information and magnifies the impact of estimation errors. The result is that the optimized portfolio is ‘error maximized.’”

- ⁶ We produce forecasts of the factor premiums using a VAR model as described in [Chapter 8](#). We then use the estimated factor exposures along with the forecasts to calculate expected returns and variances.
- ⁷ The Investment Company Act of 1940 specified various rules on investment companies, including Section 12d and Section 5b. Section 12d specifies certain investment constraints for diversified and nondiversified mutual funds. One rule prohibits mutual funds from owning more than 5% of other investment companies, which are firms that derive more than 15% of their revenue from securities-related activity and a maximum of 10% of the investment company’s total AUM from such activity. This might be a constraint for any portfolio manager holding any type of banking stocks or investment banking stocks. Section 5b specifies rules for funds that advertise themselves as *diversified*. These funds may not hold more than 5% of their assets in any particular company or hold more than 10% of the voting stock of any company for 75% of the portfolio’s assets. Due to this rule, we like to call the restriction on the maximum weight of a stock in the portfolio the *diversification rule*.
- ⁸ Most software packages, such as MATLAB, have a simple way to add upper- and lower-bound constraints to their quadratic optimizer.
- ⁹ In [Appendix 10C](#), we discuss more direct ways to deal with market impact.
- ¹⁰ In fact, there is one more practical reason why this formulation may be preferred. In the minimization of the risk with a target expected return, there is a danger that the target expected return may not be feasible given the parameters. Thus the portfolio manager may end up getting a portfolio he or she did not wish to obtain, or, even worse, the optimization may fail altogether.
- ¹¹ This formulation has a theoretical appeal as well. The objective function can be interpreted as an investor’s utility function, and the optimization can be interpreted as an investor’s utility maximization. This is how theory says investors make portfolio decisions. On the other hand, maximizing the information ratio or some ratio of expected return to risk does not have such theoretical appeal.
- ¹² In practice, the benchmark is rarely efficient. A simple way to see whether the benchmark is efficient is to make a plot similar to the one in [Figure 9.1](#). On the plot, add a dot corresponding to the benchmark. If the dot is close to the efficient frontier, then the benchmark is nearly efficient. If not, then the benchmark is not efficient. This efficiency, sometimes called *mean-variance efficiency*, may not be the only measure of efficiency. Some prefer to define efficiency with respect to the benchmark. In such a case, of course, the benchmark is always efficient. Minimizing tracking errors when the benchmark is inefficient may lead to inefficient portfolios in a variety of cases. For more on this, see Roll (1992).
- ¹³ It is not important for this ad hoc weighting mechanism that the QEPM use the aggregate Z-score methodology. In fact, the manager can use any relative ranking method for the stocks. However, so that the ad hoc conversion mechanism we describe will function properly, the manager should normalize the ranking method so that the sum of all the ranks equals 0.
- ¹⁴ One can see that the new weights will add up to 1 as well so long as $\sigma \sum_{i=1}^N z_i = 0$. This is why it is critical to normalize the aggregate Z-scores or other ranking method among the subset of benchmark stocks.
- ¹⁵ There really are two types of stratified sampling: proportional and disproportional. The one we describe here is proportional random sampling, in which the same proportion is chosen for the

sample as is represented in the population. Disproportional random sampling may overweight certain strata with respect to their size in the population for a variety of reasons that will not be discussed here.

- ¹⁶ This, of course, does not control the asset-specific risk, and thus the tracking error is not bounded. That is, controlling directly for asset-specific risk with similar stocks and similar stock weights will imply a rather tight control of factor risk, but controlling for factor risk will not necessarily imply a tight control on asset-specific risk.
- ¹⁷ For further details on tracking error, including how to measure it ex post, see [Chapter 15](#). If the standard deviation of the difference between the portfolio return and the benchmark return is zero, but the expected values of the two returns are different, do we have a perfect tracking portfolio? Of course, the answer is no. The tracking error, as defined above, however, would misleadingly equal zero because it ignores the expected difference in returns. A better way of measuring the tracking error is to use the root mean squared error (RMSE). RMSE is defined as

$$\text{RMSE}(r_P, r_B) = \sqrt{V(r_P - r_B) + [E(r_P - r_B)]^2}$$

RMSE accounts for the expected value of the difference as well as the variation in the difference. The RMSE equals the tracking error as previously defined only if there is no difference between the expected return of the portfolio and the benchmark. Fortunately, however, whether we use RMSE or the standard deviation to measure tracking error, the optimal portfolio selection problem is not affected (see the questions at the end of this chapter). For this reason, and also to follow convention, we use the standard deviation as the measure of the tracking error.

- ¹⁸ We obtain this by the standard statistical result that $V(a - b) = V(a) + V(b) - 2C(a, b)$.
- ¹⁹ Since minimizing squared tracking error is equivalent to minimizing tracking error, we will refer to them interchangeably when discussing the optimization problem.
- ²⁰ IBM's CPLEX package does provide this functionality. It is discussed in more depth throughout this book, especially in [Appendix 9B](#).

Quadratic Programming

For most of the portfolio optimization problems that a portfolio manager will encounter, a general quadratic optimization routine will enable him or her to construct the portfolio given his or her objectives.¹ Most commercial softwares packages employ some form of the concepts we will describe below. For portfolio managers who build their own risk models and optimizers, the standard packages have versions of the quadratic programming techniques we will discuss below.²

The general quadratic programming problem can be expressed as

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} \quad \text{s.t.} \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (9A.1)$$

where $\mathbf{x}$ is the vector of unknowns in the problem, $\mathbf{Q}$ is a symmetric positive semidefinite matrix supplying the coefficients on the quadratic terms of the optimization problem, $\mathbf{c}$ is a vector of coefficients related to the linear objective function, $\mathbf{A}$ is a matrix of coefficients for the equality and inequality constraints, and $\mathbf{b}$ is a vector of constraint values.³

The general quadratic optimization problem works for both quadratic and linear optimization problems. For linear optimization problems, one can make $\mathbf{Q} = \mathbf{0}$, and then the problem becomes a linear programming problem. For quadratic optimizations, one uses the appropriate $\mathbf{Q}$. In layperson's terms, the preceding quadratic programming problem says the following: Minimize the function $\frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c}$ by choosing the values of $\mathbf{x}$ that make

it minimized. In addition, make sure that the chosen values of $\mathbf{x}$ satisfy some other *constraints*, some of which might be equality constraints and some of which might be inequality constraints (i.e., *s.t.* $\mathbf{Ax} \leq \mathbf{b}$, where *s.t.* stands for “subject to”). The objective function and the constraints can be almost anything. In the general optimization problem, these are just mathematical concepts. It is only when we apply these tools to actual real-world problems that they make more sense. We will do this in the next section of this appendix.

Let’s consider two special cases of the general quadratic optimization program. In one case, we specify only equality constraints. In the other case, we will allow for inequality as well as equality constraints. The reason we separate these into two categories is that with equality constraints, we actually can solve for the optimal weights with a *closed-form solution*.⁴

9A.1 QUADRATIC PROGRAMMING WITH EQUALITY CONSTRAINTS

With only equality constraints, the quadratic optimization can be solved in closed form. The problem becomes

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} \quad \text{s.t.} \quad \mathbf{Ax} = \mathbf{b} \quad (9A.2)$$

If matrix $\mathbf{A}$ is of full rank and matrix $\mathbf{Q}$ is positive definite, then a unique solution exists for $\mathbf{x}$.⁵ By a unique solution, we mean that there is one set of values for $\mathbf{x}$ that creates the minimum value of our objective function. Using the Lagrange method, we can find the first-order optimality conditions to solve the minimization problem.⁶ The Lagrangian is

$$\mathcal{L} = \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} - \lambda' (\mathbf{b} - \mathbf{Ax}) \quad (9A.3)$$

By taking the partial derivatives with respect to $\mathbf{x}$ and λ , we obtain the Lagrange necessary (or first-order) conditions for a solution:

$$\mathbf{Qx} + \mathbf{A}'\lambda + \mathbf{c} = \mathbf{0} \quad (9A.4)$$

$$\mathbf{Ax} - \mathbf{b} = 0 \quad (9A.5)$$

The solution for the optimal $\mathbf{x}$ can be found by algebraically manipulating these equations. From the first equation, we have

$$\mathbf{x} = -\mathbf{Q}^{-1}\mathbf{A}'\boldsymbol{\lambda} - \mathbf{Q}^{-1}\mathbf{c} \quad (9A.6)$$

Substitution of this into the second equation will give us

$$\boldsymbol{\lambda} = -(\mathbf{AQ}^{-1}\mathbf{A}')^{-1} (\mathbf{AQ}^{-1}\mathbf{c} + \mathbf{b}) \quad (9A.7)$$

As a final step, we substitute this value of $\boldsymbol{\lambda}$ back into the original expression for $\mathbf{x}$ so that we have a closed-form solution for $\mathbf{x}$. The value of $\mathbf{x}$ that minimizes the earlier general minimization problem with equality constraints is

$$\begin{aligned} \mathbf{x} = & -\mathbf{Q}^{-1}[\mathbf{I} - \mathbf{A}'(\mathbf{AQ}^{-1}\mathbf{A}')^{-1} \mathbf{AQ}^{-1}]\mathbf{c} \\ & + \mathbf{Q}^{-1}\mathbf{A}'(\mathbf{AQ}^{-1}\mathbf{A}')^{-1}\mathbf{b} \end{aligned} \quad (9A.8)$$

where $\mathbf{I}$ is the identity matrix.⁷

9A.1.1 A Numerical Example

Typically, in a portfolio risk-minimization problem, a portfolio manager would like to choose the weights of his or her stocks so as to minimize the variance of the portfolio for a given expected return level. The portfolio manager typically will have an equality constraint that the weights of the portfolio sum to 1. We can translate this easily into the quadratic optimization problem. The risk of a portfolio of stocks with weights $\mathbf{w}$ is given by

$$\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} \quad (9A.9)$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of the stock returns, and $\mathbf{w}$ is the vector of stock weights. We also would like to express the mean return for any given selection of weights. The mean return of the portfolio will be $\mu_p = \mathbf{w}'\boldsymbol{\mu}$, where $\mathbf{w}$ represents the stock weights and $\boldsymbol{\mu}$ is a vector of mean

returns for each of the stocks. We also would like to constrain the weights of the portfolio to sum to 1. This can be represented as $\mathbf{w}'\mathbf{1} = 1$, where $\mathbf{1}$ is just a vector of 1s.

We can see how easily this fits into our quadratic optimization framework. Simply replace $\mathbf{Q}$ with Σ , replace $\mathbf{x}$ with $\mathbf{w}$, and let $\mathbf{c} = \mathbf{0}$. We must be careful how we modify the matrix $\mathbf{A}$ and the vector $\mathbf{b}$. In particular, we want

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \mu_1 & \mu_2 & \cdots & \mu_N \end{bmatrix} \quad (9A.10)$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ \mu_p \end{bmatrix} \quad (9A.11)$$

Now we can substitute these values into our standard quadratic optimization problem with equality constraints, and the solution is readily available from [Eq. \(9A.8\)](#). Since $\mathbf{c} = \mathbf{0}$, the solution will be

$$\mathbf{w} = \Sigma^{-1}\mathbf{A}'(\mathbf{A}\Sigma^{-1}\mathbf{A}')^{-1}\mathbf{b} \quad (9A.12)$$

Let's create a simple six stock portfolio to illustrate the application in more detail. Suppose we have six stocks with annualized mean returns of $\mu_1 = 14.4$, $\mu_2 = 10.19$, $\mu_3 = 9.87$, $\mu_4 = 7.52$, $\mu_5 = 20.05$, and $\mu_6 = 2.66$.⁸ We also assume that we have an estimate of the annualized variance–covariance matrix of these stock returns. In our example, the matrix is:

$$\begin{aligned}
\Sigma &= \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} & \sigma_{15} & \sigma_{16} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} & \sigma_{25} & \sigma_{26} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} & \sigma_{35} & \sigma_{36} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} & \sigma_{45} & \sigma_{46} \\ \sigma_{51} & \sigma_{52} & \sigma_{53} & \sigma_{54} & \sigma_{55} & \sigma_{56} \\ \sigma_{61} & \sigma_{62} & \sigma_{63} & \sigma_{64} & \sigma_{65} & \sigma_{66} \end{bmatrix} \\
&= \begin{bmatrix} 452.33 & 249.33 & 189.23 & 70.75 & 481.14 & 106.50 \\ . & 1094.09 & 356.85 & 93.51 & 1216.91 & 135.05 \\ . & . & 617.57 & 161.82 & 1304.29 & 110.74 \\ . & . & . & 372.35 & 462.57 & 107.52 \\ . & . & . & . & 5658.42 & 425.35 \\ . & . & . & . & . & 244.31 \end{bmatrix}
\end{aligned} \tag{9A.13}$$

where σ_{ii} is the variance of stock i , and σ_{ij} is the covariance of stock i with stock j . We can see that this matrix is symmetric. The variances and covariances are written in percentage terms. Thus, for stock 1, the annualized variance is 452.33, which is equivalent to a variance of 452% per year (or standard deviation of 21.26% per year). Finally, we must construct the matrix **A** and vector **b**; they are given by

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 14.40 & 10.19 & 9.87 & 7.52 & 20.05 & 2.66 \end{bmatrix} \tag{9A.14}$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 8 \end{bmatrix} \tag{9A.15}$$

where we have explicitly chosen the value of μ_P that we want to reflect an 8% annualized return. Thus, we can now use our formula to find the optimal weights of the six stocks that will achieve the lowest risk for obtaining an expected mean return of 8% per annum. The solution is

$$\begin{aligned} \mathbf{w} &= \Sigma^{-1} \mathbf{A}' (\mathbf{A} \Sigma^{-1} \mathbf{A}')^{-1} \mathbf{b} \\ &= \begin{bmatrix} 0.305 \\ 0.057 \\ 0.204 \\ 0.274 \\ -0.085 \\ 0.245 \end{bmatrix} \end{aligned} \quad (9A.16)$$

Thus, the optimal solution for a portfolio with capital of \$1 is to buy \$0.305 of stock 1, buy \$0.057 of stock 2, buy \$0.204 of stock 3, buy \$0.274 of stock 4, short sell $-\$0.085$ of stock 5, and buy \$0.245 of stock 6.⁹ The short position in stock 5 might not be feasible for many portfolio managers for a variety of reasons. Thus, the portfolio manager might like to add inequality constraints—for instance, that the weight of security 2 should be greater than 0.10, or 10%. In addition, the portfolio manager may wish to have the weights of all the securities be greater than 0. In our preceding optimization, we did not apply these restrictions. We will continue with our present example in the following application adding these restrictions.

9A.2 QUADRATIC PROGRAMMING WITH INEQUALITY CONSTRAINTS

The quadratic optimization problem with inequality constraints cannot be solved with a closed-form solution. The solution is obtained through numerical solution methods. Numerical solutions have become much more reliable and feasible owing to the increased power of computers and some advances in linear and quadratic programming. The most typical method

that is used is the *active-set method* or *projection method*. Other methods are used as well, including the *interior-point method*.¹⁰ Since most portfolio managers and quantitative researchers will have access to robust optimizers, we will not go into detail on the mathematics of each numerical method.¹¹

9A.2.1 A Numerical Example

We will continue the previous numerical example, adding some inequality constraints. We will use the active-set method to solve the problem. The three inequality constraints that we will add are that the weights of each individual stock cannot be less than 0, except for stock 2, which will not be allowed to have a weight of less than 0.10. Thus $\mathbf{w} \geq 0$ and $w_2 \geq 0.10$. These inequality constraints can be added easily to our matrix $\mathbf{A}$.¹² In fact, the new optimization problem with the first two rows of $\mathbf{A}$ representing equality constraints and the last seven rows representing inequality constraints is¹³

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}' \Sigma \mathbf{w} \quad s.t. \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (9A.24)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 14.40 & 10.19 & 9.87 & 7.52 & 20.05 & 2.66 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad (9A.25)$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 8 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.90 \end{bmatrix} \quad (9A.26)$$

Notice that to express the constraint that $w_2 \geq 0.10$, we stated that $w_1 + w_3 + w_4 + w_5 + w_6 \leq 0.90$. Sometimes, depending on the types of programs we use, the most obvious constraints need a bit of reengineering. The solution using a standard optimization routine is¹⁴

$$\mathbf{w} = \begin{bmatrix} 0.256 \\ 0.100 \\ 0.026 \\ 0.287 \\ 0.000 \\ 0.331 \end{bmatrix} \quad (9A.27)$$

¹ For those who wish to have more insight into the mathematics of optimization, we recommend two books, Luenberger (1989) and Gill et al. (1981).

² For instance, RATS (Regression Analysis of Time Series), produced by Estima Corporation, has a procedure known as LQPROG that performs both linear and quadratic optimizations quite easily. MATLAB, produced by MathWorks Inc., has an optimization toolbox that includes the function QUADPROG, which can handle standard quadratic optimization problems. For a more complete suite of optimization tools that can also be used with other software programs, we recommend IBM's CPLEX software. For quadratic optimizations, this software has functions CPLEXQP, CPLEXMIQP, CPLEXQCP, and CPLEXMIQCP that are useful for

advanced optimizations. The open-source programming language R has a function SOLVE.QP that can handle standard quadratic optimizations.

- ³ The matrix $\mathbf{Q}$ in most portfolio problems actually will be positive definite. The symmetry of the matrix is very important. This refers to a matrix for which entry $a_{i,k} = a_{k,i}$ for all i and k . In layperson's terms, the matrix on either side of the diagonal terms looks like a mirror reflection of the other side.
- ⁴ In mathematics, the term *closed-form solution* refers to an expression for a given function or quantity that can be expressed in terms of known and well-understood quantities. These solutions are preferable because, given a set of variables, you can plug them into an equation and know the optimal solution immediately.
- ⁵ A symmetric matrix is called *positive definite* if $\mathbf{x}'\mathbf{Q}\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$. A matrix has *full rank* if it has rank equal to the number of rows or columns, whichever is less. If a nonzero $\mathbf{x}$ for which $\mathbf{A}\mathbf{x} = \mathbf{0}$ exists, then $\mathbf{A}$ is not of full rank. It also means that the columns of $\mathbf{A}$ are linearly dependent. That is, at least one row of $\mathbf{A}$ can be expressed as a simple linear transformation of another. This will lead to nonunique solutions. The reader should consult a book on linear algebra for more details, for example, Anton (1991) or Greene (2002).
- ⁶ The Lagrange method is a method of solving these types of constrained optimization problems. It is a clever method of reducing a constrained optimization problem to an unconstrained optimization problem by introducing *Lagrange multipliers*. It is named after Joseph Louis Lagrange, an Italian-born French mathematician and physicist who was a professor at the University of Turin in 1755. He later succeeded Euler as director of mathematics at the Berlin Academy of Science. He made many contributions to mathematics, including probability theory, number theory, the theory of equations, and the foundations of group theory.
- ⁷ The identity matrix is a diagonal matrix in which every diagonal term is equal to 1, and all other elements in the matrix are equal to 0.
- ⁸ In practice, these can be expected future returns, historical mean returns, or returns from one's factor models. The mean returns are expressed in percentage terms; thus, 14.4 represents an annual return of 14.4%.
- ⁹ One can compute the transpose of the matrices (i.e., $\mathbf{A}'$) and the inverse of the matrices (i.e., Σ^{-1}) using a software program or manually.
- ¹⁰ The state-of-the-art quadratic programming algorithms with inequality constraints are two kinds of approaches called the *active-set method* and the *interior-point method*. Both kinds of approaches solve a series of subproblems where there are only equality constraints. The two approaches differ only in how they arrange the order of the subproblems to be solved. In the active-set method, we proceed along the boundary of the feasible set defined by the constraints. In the interior-point method, we proceed within the feasible set.
- ¹¹ For a book on the details of the different types of numerical optimization procedures, see Nocedal and Wright (1999). Some software optimizers use a combination of techniques, such as RATS' LQPROG, which uses the active-set method with the conjugate gradient method. MATLAB's QUADPROG uses the interior-point algorithm. IBM's CPLEX suite uses a variety of techniques. See also Sharpe (1987) for an interesting algorithm for portfolio optimization improvement.
- ¹² Many software applications that have optimizers allow the user to specify upper- and lower-bound weights rather than having to include them in a constraint matrix.

- ¹³ Many quadratic programming optimizers allow the user to specify the equality $\mathbf{A}_{eq}$ and inequality matrices $\mathbf{A}_{ineq}$ as two different matrices rather than one ($\mathbf{A}$).
- ¹⁴ Using MATLAB, this optimization took 0.0068 second to reach a solution.

Advanced Techniques for Quadratic Optimization

Most portfolio optimization problems are covered by the basic techniques discussed in this chapter. However, there are many situations in which the portfolio optimization requires an expansion of the basic optimization setup. For example, a portfolio manager may wish to add transactions costs (discussed in the appendix to [Chapter 10](#)) or create a market-neutral portfolio with leverage constraints (discussed in the appendix to [Chapter 13](#)). There are other situations in which a portfolio manager wishes to have a limitation on the number of stocks in the portfolio, or in which the number of stocks in the portfolio is between some minimum and maximum, or, alternatively, in which the weights of any security are either zero or between a minimum and maximum weight. These and many other conceivable preferences in the optimization will require an expanded optimization framework. In other situations the portfolio optimization problem might require quadratic constraints, which are not part of the typical optimization framework. In this appendix we will discuss the basic building blocks for expanding the optimization framework to deal with these advanced optimization situations.

9B.1 PHANTOM WEIGHTS

In the portfolio optimization problems discussed in [Chapter 9](#), we typically had only N unknowns, which were the portfolio weights. In some nonstandard portfolio problems, it is useful to create what we term “phantom weights.” The idea of phantom weights is to create additional weights, in addition to the actual weights of the portfolio, for which the optimizer will also find optimal values. These can be used for a variety of purposes. Perhaps the simplest example is to create a set of buy and sell weights. Suppose the optimization problem has N stocks. In addition to the portfolio weights w_1 to w_N , we create an additional $2N$ weights $w_1^+, \dots, w_N^+$, and $w_1^-, \dots, w_N^-$. The new optimization is now more complicated because we have $3N$ weights to choose. These phantom weights offer many benefits in a portfolio optimization. Oftentimes, the phantom weights have a specific relationship to the underlying weights, such as $w_i = w_i^+ - w_i^-$.

9B.2 BINARY WEIGHTS

In addition to creating phantom weights for the portfolio optimization, it may also be useful to create binary variables as optimization weights. These binary weights are very similar to the concept of phantom weights. They are additional weights for the optimizer to select, but the weights are constrained to be binary (i.e., have a value of 0 or 1).

One very practical use of the binary variables is to force the phantom weights to be orthogonal. That is, if the actual weights and phantom weights have the following relationship, $w_i = w_i^+ - w_i^-$, we could have a situation where $w_i > 0$ and both $w_i^+ > 0$ and $w_i^- > 0$. This is a wasteful solution because we have a long position and a short position on the same stock. We can use binary variables to prevent this occurring.

Suppose one creates the following binary variables, v_i^+ and v_i^- for each one of the N stocks. One can then add a constraint of the following form:

$$v_i^+ \kappa_l \leq w_i^+ \leq v_i^+ \kappa_h \quad (9B.1)$$

$$v_i^- \gamma_l \leq w_i^- \leq v_i^- \gamma_h \quad (9B.2)$$

If we set $\kappa_l = \gamma_l = 0$ and $\kappa_h = \gamma_h = 1$ so that we allow the weights to fluctuate between 0 and 1, and add the constraint that $v_i^+ + v_i^- \leq 1$, we will force the phantom weights to be orthogonal. That is, for every stock i , if $w_i^+ > 0$, then $w_i^- = 0$ and vice versa.

Of course, this useful portfolio setup is not without costs. The additional phantom and binary weights and associated constraints make the optimization problem more difficult and complicated to solve.

9B.3 QUADRATIC CONSTRAINTS

Conventional quadratic optimizers typically allow for linear equality and inequality constraints. In some instances, it will be useful to specify quadratic constraints, in addition to a quadratic objective function.¹ The portfolio optimization structure is very similar to the standard one; however, the user typically can also specify constraints of the following form: $\mathbf{l}'\mathbf{x} + \mathbf{x}'\mathbf{Q}\mathbf{x} \leq r$.

9B.4 PRACTICAL EXAMPLES

9B.4.1 Market Neutrality with Leverage Constraints

If we add the following constraints that $w_i = w_i^+ - w_i^-$, $\sum_{i=1}^N w_i^+$, $\sum_{i=1}^N w_i^-$, and that $\mathbf{w}^+ \geq 0$ and $\mathbf{w}^- \geq 0$, the resulting optimization will create a market-neutral portfolio that is dollar neutral (i.e., the sum of weights of the long stocks equals the sum of weights of the shorted stocks) and such that the leverage of the market-neutral portfolio is limited to 2 (i.e., the portfolio is 100% long and 100% short of the assets under management). If the market-neutral manager wanted more excessive leverage, this could easily be changed by the constraints on the sum of the phantom long and short weights (i.e., $\sum_{i=1}^N w_i^+ = l_l$, $\sum_{i=1}^N w_i^- = l_s$, where l_l is the long exposure and l_s is the short exposure). For example, a 130–30 long–short portfolio could be constructed by making $l_l = 1.3$ and $l_s = 0.3$. More on the exact portfolio setup will be discussed in [Appendix 13A](#).

9B.4.2 Transactions Costs

Fixed Costs

If we want to rebalance a portfolio and we have to select new target weights while acknowledging our current portfolio weights, the use of phantom weights and binary variables will enable us to find an exact solution to the portfolio optimization problem while considering transactions costs. In particular, we can add the following constraints $w_i^a = w_i^+ - w_i^- + w_i^b$, where w_i^b represents the current weights of the portfolio, and w_i^a represents the after-rebalancing weights of the portfolio. We can set the same relationship on the binary variables such that $\kappa_l = \gamma_l = 0$ and $\kappa_h = \gamma_h = 1$ so that we allow the weights to fluctuate between 0 and 1 and add the constraint that $v_i^+ + v_i^- \leq 1$, which will force the phantom weights to be orthogonal. Stocks whose weight is reduced vis-à-vis the prior portfolio will have negative net weights, but their phantom weights (w_i^-) will be positive—thus the transactions cost vector can be multiplied by their value. Similarly, the stocks for which the portfolio increases the weight will be positive (w_i^+), and these can also be multiplied by the transactions cost vector. The optimized final weights of the portfolio will be w_i . Thus, in the case of transactions costs, the phantom weights simply serve as a mechanism to denote the change to the current weights, storing the positive changes in the positive phantom weights and the negative changes in the negative phantom weights. Since both positive and negative phantom weights are positive, our transactions cost vector can remain positive, and the transactions cost rebalance problem is resolved. More on the exact portfolio setup will be discussed in [Appendix 10A](#).

Market Impact Costs

These techniques along with quadratic optimization with quadratic constraints can be used to construct portfolios while accounting for market impact costs from a variety of market impact cost models.² More on the exact details will be discussed in [Appendix 10C](#).

9B.4.3 Elimination of Small-Weight Stocks

Sometimes a portfolio manager will wish to reduce the number of securities. That is, the portfolio manager might have a desire to construct an optimized portfolio that forces the weights of individual stocks to be above

a minimum or below a maximum weight. The traditional portfolio optimization allows this very easily. One simply adds an inequality constraint that the optimal weights are above and below a given weight, which can differ by stock or be the same for all stocks. The problem with the traditional method is that there might be many circumstances with no solution to the optimization because it forces all stocks to be within a given range—which may be far from optimal and will not resolve the initial problem of eliminating many stocks with trivial weights. However, the use of binary variables can create the optimization that forces the optimizer to only find weights between the minimum and maximum or forces the weight of that particular stock to zero. This will result in more successful optimizations and will be more in line with the portfolio manager's thought process. For simplicity, we will focus only on the situation where the portfolio manager would like all stock weights to be between a lower bound (κ_l) and an upper bound (κ_h). This constraint can be implemented on just the long portion of the portfolio or the long and short portions of the portfolio. Here we consider only the long portion of the portfolio. Using the inequality relationship of the binary variables and phantom weights, $v_i^+ \kappa_l \leq w_i^+ \leq v_i^+ \kappa_h$ and $v_i^- \gamma_l \leq w_i^- \leq v_i^- \gamma_h$, once we specify our values for κ_l , γ_l , κ_h , and γ_h , we have effectively accomplished our goal. If the portfolio is just a long portfolio, phantom weights are not needed and the one set of binary variables should be constructed with respect to the actual weights, w_i . Since the binary variables are either 0 or 1, the weights of the portfolio will be chosen to be within the minimum and maximum weights or to equal 0.

A Numerical Example

We will continue the numerical example in [Appendix 9A](#). Thus we will seek a portfolio with an average annualized return of 8%, with no short sales allowed, and the weights of the portfolio must sum to 1. We will then add the constraint that a stock can only have a weight greater than 0.03 (i.e., 3%) or less than 0.30 (i.e., 30%) or else it must have a value of 0.

The quadratic programming problem can be expressed as³

$$\min_x \frac{1}{2} \mathbf{x}' \Sigma \mathbf{x} \quad \text{s.t.} \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (9B.3)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 14.40 & 10.19 & 9.87 & 7.52 & 20.05 & 2.66 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -0.3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -0.3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -0.3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -0.3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -0.3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -0.3 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0.03 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0.03 \end{bmatrix} \quad (9B.4)$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 8 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (9B.5)$$

The first two rows of this $\mathbf{A}$ matrix are the equality constraints that the sum of weights equals 1 and that the target mean is 8%. For the binary weights, a 0 is placed in the matrix since binary weights are not relevant for these constraints. The rest of the rows represent inequality constraints to

restrict the weights of every stock between 0.03 and 0.30, or a weight of 0. That is, we chose the values of κ_l and κ_h such that $v_i^+ 0.03 \leq w_i \leq v_i^+ 0.30$. Since v_i^+ is a binary choice variable, if this variable equals 1, then the weight of stock i will be forced to lie in the range of 0.03 and 0.30; however, if it's more optimal to make its weight 0, then $v_i^+ = 0$ and w_i will also be equal to 0.

With many constraints and more stocks, these matrices can become very large. As a result, we will sometimes represent these matrices as a series of matrices. For example, it is convenient to write matrix $\mathbf{A}$ in Eq. (9B.4) as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\iota}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \boldsymbol{\mu}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \mathbf{I}_{6 \times 6} & -0.3 \cdot \mathbf{I}_{6 \times 6} \\ -1 \cdot \mathbf{I}_{6 \times 6} & 0.03 \cdot \mathbf{I}_{6 \times 6} \end{bmatrix} \quad (9B.6)$$

$$\mathbf{b} = \begin{bmatrix} 1 \\ 8 \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \end{bmatrix} \quad (9B.7)$$

where $\boldsymbol{\mu}_{1 \times 6} = [14.40 \ 10.19 \ 9.87 \ 7.52 \ 20.05 \ 2.66]$, $\boldsymbol{\iota}_{1 \times 6}$ is a row of six ones (that is, the first six entries of matrix $\mathbf{A}$'s row 1 are 1s), $\mathbf{0}_{1 \times 6}$ is a row of 6 entries of 0, $\mathbf{I}_{6 \times 6}$ is the identity matrix (that is, a 6×6 matrix with 1s on the diagonal and 0s everywhere else), and so on and so forth.

The solution using an optimization routine with continuous and binary variables is⁴

$$\mathbf{w} = \begin{bmatrix} 0.217 \\ 0.045 \\ 0.137 \\ 0.300 \\ 0.000 \\ 0.300 \end{bmatrix} \quad (9B.8)$$

9B.4.4 Restricting the Number of Stocks

Sometimes portfolio managers will wish to restrict the number of stocks in their portfolio. There could be a variety of reasons for this preference. For example, if it's a regularly rebalanced quantitative model, it might be beneficial from a logistical point of view to manage the portfolio with fewer stocks. Whatever the reason, the portfolio optimization can be done such that the portfolio managers can limit the number of stocks within a range of n_l and n_h , where n_l is the smallest number of stocks they want in their portfolio and n_h is the largest number of stocks they want in their portfolio. In this particular example, phantom weights are not needed, unless the managers are doing this for a long-short portfolio. However, if the portfolio managers are simply restricting the number of stocks, they need only use the binary variables. For a long-only portfolio, the managers should create one set of N binary variables, v_i^+ . For a long-short portfolio, the managers should create two sets of binary variables, v_i^+ and v_i^- , allowing them to specify the range of stocks in both the long and short portfolios, respectively. For both situations, the portfolio managers should add two inequality constraints. The first is that $\sum_{i=1}^N v_i^+ \leq \eta_h$ (i.e., that the number of stocks in the long portfolio is less than η_h) and $\eta_l \leq \sum_{i=1}^N v_i^+$ (i.e., that the number of stocks in the long portfolio is greater than η_l). In reality, most optimization frameworks will require a transformation of the second inequality into $-\sum_{i=1}^N v_i^+ \leq -\eta_l$. Of course, with a long-short portfolio, the portfolio manager would add similar constraints on the short portfolio with the corresponding binary variables.

A Numerical Example

We will continue the previous numerical example. Thus we will seek a portfolio with an average annualized return of 8%, with no short sales allowed, and the weights of the portfolio summing to 1. We will allow the weights to have any value between 0 and 1; however, we will not allow the portfolio to have more than three stocks (even though there are six stocks available to purchase).

The quadratic programming problem can be expressed as⁵

$$\min_x \frac{1}{2} \mathbf{x}' \Sigma \mathbf{x} \quad s.t. \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (9B.9)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 14.40 & 10.19 & 9.87 & 7.52 & 20.05 & 2.66 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix} \quad (9B.10)$$

$$\mathbf{b} = \begin{bmatrix} \mathbf{b}_{eq} \\ \mathbf{b}_{ineq} \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 3 \\ 0 \end{bmatrix} \quad (9B.11)$$

The first two rows of this $\mathbf{A}$ matrix are the equality constraints that the sum of weights equals 1 and that the target mean is 8%. For the binary weights, a 0 is placed in the matrix since binary weights are not relevant for these constraints. The rest of the rows represent inequality constraints to restrict the weight of every stock to between 0 and 1. The last two rows represent inequality constraints on the binary variables—in particular, that the sum of the binary variables be between 0 and 3. This is equivalent to constraining the number of stocks to be less than or equal to three.

The solution using an optimization routine with continuous and binary variables is⁶

$$\mathbf{w} = \begin{bmatrix} 0.326 \\ 0.000 \\ 0.000 \\ 0.312 \\ 0.000 \\ 0.362 \end{bmatrix} \quad (9B.12)$$

QUESTIONS

- 9.1. The solution to the mean-variance optimization problem with no constraints can be expressed as

$$\mathbf{w}^* = \frac{Y\Sigma^{-1}\boldsymbol{\iota} - X\Sigma^{-1}\boldsymbol{\mu}}{YZ - X^2} + \frac{Z\Sigma^{-1}\boldsymbol{\mu} - X\Sigma^{-1}\boldsymbol{\iota}}{YZ - X^2}\mu_p$$

where

$$X = \boldsymbol{\iota}' \Sigma^{-1} \boldsymbol{\mu}$$

$$Y = \boldsymbol{\mu}' \Sigma^{-1} \boldsymbol{\mu}$$

$$Z = \boldsymbol{\iota}' \Sigma^{-1} \boldsymbol{\iota}$$

Show that this can be derived from the general solution to the quadratic optimization solution with equality constraints, that is, $\mathbf{x} = -\mathbf{Q}^{-1} [\mathbf{I} - \mathbf{A}' (\mathbf{A}\mathbf{Q}^{-1}\mathbf{A}')^{-1} \mathbf{A}\mathbf{Q}^{-1}] \mathbf{c} + \mathbf{Q}^{-1} \mathbf{A}' (\mathbf{A}\mathbf{Q}^{-1}\mathbf{A}')^{-1} \mathbf{b}$.

- 9.2. Explain the circumstances under which shorts sales are not feasible.
- 9.3. Explain why true diversification may be achieved even when the conventional diversification constraints are violated.
- 9.4. Some computer software packages require every constraint to the minimization to be in inequality form. Rewrite the following constraints in the inequality form: $\mathbf{w}'\boldsymbol{\mu} = \mu_p$, $\mathbf{w}'\boldsymbol{\iota} = 1$
- 9.5. Consider the following variance-minimization problem:

$$\min_{\mathbf{w}} \mathbf{w}' \Sigma \mathbf{w} \quad s.t. \quad \mathbf{w}' \boldsymbol{\mu} = \mu_p, \mathbf{w}' \boldsymbol{\iota} = 1$$

- (a) Show that this problem is equivalent to minimization of the standard deviation.
- (b) In general, we may consider the minimization of a function of the variance; that is,

$$\min_{\mathbf{w}} f(\mathbf{w}'\Sigma\mathbf{w}) \quad s.t. \quad \mathbf{w}'\boldsymbol{\mu} = \mu_p, \mathbf{w}'\boldsymbol{\iota} = 1$$

What is the property of function $f(\cdot)$ that makes the preceding minimization equivalent to the variance minimization?

9.6. Consider the variance minimization

$$\min_{\mathbf{w}} \mathbf{w}'\Sigma\mathbf{w} \quad s.t. \quad \mathbf{w}'\boldsymbol{\mu} = \mu_p, \mathbf{w}'\boldsymbol{\iota} = 1$$

and the expected return maximization

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} \quad s.t. \quad \mathbf{w}'\Sigma\mathbf{w} = \sigma_p^2, \mathbf{w}'\boldsymbol{\iota} = 1$$

- (a) If the two problems produce the same solution $\mathbf{w}$, what are the portfolio variance and portfolio expected return?
- (b) Can we say that the set of solutions to the variance minimization is identical to the set of solutions to the expected-return maximization as we vary the values of μ_p and σ_p^2 ?

9.7. Consider the following minimization problem:

$$\min_{\mathbf{w}} \mathbf{w}'\Sigma\mathbf{w} \quad s.t. \quad \mathbf{w}'\boldsymbol{\mu} = \mu_p, \mathbf{w}'\boldsymbol{\iota} = 1$$

The Lagrangian function is defined as

$$\mathcal{L} = \mathbf{w}'\Sigma\mathbf{w} + \gamma(\mu_p - \mathbf{w}'\boldsymbol{\mu}) + \gamma'(1 - \mathbf{w}'\boldsymbol{\iota})$$

The first-order condition of this minimization problem is identical to the first-order condition for minimizing the Lagrangian with respect to $\mathbf{w}$, γ , and γ' . What are the roles of γ and γ' in the first-order condition?

9.8. The portfolio optimization may be expressed as

$$\min_{\mathbf{w}} \mathbf{w}'\Sigma\mathbf{w} + \lambda\mathbf{w}'\boldsymbol{\mu} \quad s.t. \quad \mathbf{w}'\boldsymbol{\iota} = 1$$

- (a) Show that given μ_P , there exists γ such that the solution $\mathbf{w}^*$ to this problem satisfies $\mathbf{w}^{*'}\boldsymbol{\mu} = \mu_P$.
 - (b) Explain why γ may be interpreted as a measure of risk tolerance of the portfolio manager.
- 9.9.** The quality of stratified sampling critically depends on the number of strata and the number of observations drawn from each stratum. Suppose that the portfolio manager wants to select 100 stocks out of a universe of 10,000 stocks. One alternative is to make 5 strata and select 20 stocks from each stratum. The second alternative is to make 10 strata and select 10 stocks from each stratum. Under which conditions would the portfolio manager prefer the first alternative to the second?
- 9.10.** What is factor tilting? Explain when factor tilting is desirable and how to accomplish it.
- 9.11.** Show that minimizing tracking error is equivalent to minimizing the mean squared error defined as

$$\text{MSE}(r_P - r_B) = V(r_P - r_B) + [E(r_P - r_B)]^2$$

- 9.12.** Tracking error can be defined alternatively by looking at the downside deviation only. That is, instead of calculating the standard deviation of $r_P - r_B$, we could calculate the standard deviation of $\min(r_P - r_B, 0)$. Would this alternative definition change the optimal portfolio?
- 9.13.** It is easy to combine the standard risk minimization with tracking-error minimization. Write down the minimization problem of the portfolio manager who wants to minimize the sum of the portfolio risk and the tracking error.
- 9.14.** Suppose that we constructed a portfolio through factor exposure targeting. Then the return of this portfolio can be expressed as follows:

$$r_P = \alpha_P + \beta_B' \mathbf{f} + \epsilon_P$$

where β_B is the factor exposure of the benchmark.

- (a) Express the tracking error of this portfolio in terms of $\mathbf{w}_P$, $\mathbf{w}_B$, and Σ_ϵ (the variance-covariance matrix of the stock-return errors).
 - (b) Is the size of the tracking error bounded?
- 9.15.** Describe the situation in which the ghost benchmark tracking technique may be used.
- 9.16.** This question concerns phantom weights and binary weights.
- (a) What are phantom weights?
 - (b) What are binary weights?
 - (c) Why are these weights important in advanced portfolio optimizations? In particular, what problems can they help solve?

¹ For example, IBM's CPLEX optimization system has a function called CPLEXQCP for standard optimization with quadratic constraints and CPLEXMIQCP for mixed-integer quadratic optimizations with quadratic constraints.

² See Chincarini (2017).

³ The $\mathbf{Q}$ matrix containing the variance–covariance matrix must also be expanded to account for the additional $\mathbf{x}$ variables by placing zeros throughout the larger matrix. That is, in this case with the N actual weights and N binary variables, $\mathbf{Q} = \begin{bmatrix} 2\Sigma_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix}$. Also, $\mathbf{x} = [\mathbf{w} \ \mathbf{v}^*]'$.

⁴ Using the CPLEX function CPLEXMIQP, available through the CPLEX library in MATLAB, this optimization took 0.0141 second to reach a solution. This technique is better than placing limits on the minimum and maximum value of stocks, which is an easier optimization to set up. The reason is that rather than eliminating small, unnecessary stocks (as in this case, stock 5), the optimizer will be forced to keep them at the minimum weight, which will lead to a higher risk for a given return. For the portfolio constructed here, the standard deviation was 13.51% per year, while the other type of optimization had a standard deviation of 14.77% with the following weights: $w_1 = 0.149$, $w_2 = 0.058$, $w_3 = 0.163$, $w_4 = 0.3$, $w_5 = 0.03$, and $w_6 = 0.3$.

⁵ The $\mathbf{Q}$ matrix containing the variance–covariance matrix must also be expanded to account for the additional $\mathbf{x}$ variables by placing zeros throughout the larger matrix. That is, in this case with the N actual weights and N binary variables, $\mathbf{Q} = \begin{bmatrix} 2\Sigma_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix}$. Also, $\mathbf{x} = [\mathbf{w} \ \mathbf{v}^*]'$.

⁶ Using MATLAB and CPLEX's CPLEXMIQP function, this optimization took 0.0178 seconds to reach a solution. There are weights for only three stocks, as desired by the portfolio manager.

CHAPTER 10

Rebalancing and Transactions Costs

Changes aren't permanent, but change is.

—Neil Peart

10.1 INTRODUCTION

In April 2020, Invesco reported a rebalancing error with its Invesco Equally-Weighted S&P 500 Fund. Invesco simply forgot to rebalance the fund on April 24, 2020, which cost Invesco a total of \$105 million over 5 days.¹ Even when funds rebalance correctly, transactions costs can severely deteriorate their returns. In 2019, the small-cap growth managers who traded most often had annual trading costs of more than 2.5%, while those who traded more cautiously had trading costs below 1%. This difference in costs and ultimately net returns could be the difference between investment success and failure. While the mistake in rebalancing by Invesco and the story of small-cap growth funds magnifies the rebalancing problem considerably, the lesson applies to all funds: managers cannot afford to ignore the intricacies of rebalancing and transactions costs.

Almost all portfolios need to be adjusted during their lifetimes, so incurring periodic transactions costs is inevitable—it is simply a question of how often and how much. The portfolio manager's decision regarding an adjustment to the portfolio is called the *rebalancing decision*. Three types of events potentially could trigger rebalancing decisions: cash inflows, cash outflows, and changes in the underlying parameters of the stock-return

model. Cash inflows and outflows require adjustments to the portfolio's positions, and the simplest way to adjust the portfolio in such cases is to buy or sell stocks using the current portfolio weights. (There are better ways to adjust the portfolio in the face of cash flows, but we will come back to that later.) Changes in the underlying parameters do not automatically require adjustments in the portfolio. The portfolio manager needs to decide whether a parameter change is big enough to warrant incurring transactions costs. Suppose that the transaction cost of making some adjustment to the portfolio is 5 basis points (i.e., 0.05% of the total portfolio value).² If, by making the adjustment, the portfolio manager can increase the expected return of the portfolio by only 3 basis points, then he or she has not recovered 2 basis points of transactions costs.

When the portfolio manager does decide to rebalance the portfolio, the transactions cost will affect the composition of the optimal portfolio. We ignored transactions costs in previous chapters partly because we wanted to focus on the main ideas of modeling and portfolio construction and partly because transactions costs do not greatly affect the initial creation of the portfolio. However, after the portfolio is in place and changes must be made midstream, transactions costs take on greater significance.

Transactions costs affect the returns on stocks within the portfolio. Take stock *A*, which has an exceptionally high transaction cost of 1% owing to thin trading but a relatively high expected return of 12%. Also consider stock *B*, which has a relatively low transaction cost of 10 basis points and an expected return of 10%. Not including transactions costs, the difference in the expected return between stocks *A* and *B* is 2%. However, including transactions costs, \$100 invested in stock *A* will return \$10.88, whereas \$100 invested in stock *B* will return \$9.89.³ Thus, after transactions costs, the difference in the expected return between the two securities is less than 1%. This certainly has an impact on the composition of the optimal portfolio.

We begin this chapter with a discussion of the rebalancing decision. Then we discuss how the optimal portfolio changes when we consider transactions costs. In the last section of the chapter we explain ways to control transactions costs.

10.2 THE REBALANCING DECISION

The rebalancing decision involves two questions: when to rebalance the portfolio and how to go about it. From an econometric point of view, the answer to the first question is clear. The factor model we used to pick stocks implies a certain rebalancing period. If we estimated the model on monthly returns, then our optimal portfolio is optimal only for one month. If we estimated the model on quarterly returns, then our optimal portfolio is optimal for one quarter. We must rebalance whenever the optimal portfolio expires, which is the model's periodicity. We discuss model periodicity in the first part of this section. The more challenging rebalancing question is how to change α or other parameters in the model in order to update the portfolio. We will discuss that aspect of rebalancing in the second part of this section.

10.2.1 Rebalancing and Model Periodicity

Financial economists have discovered that models of stock returns do not predict daily returns well. Models approximate patterns that emerge from returns over time, whereas daily returns are heavily influenced by specific news and events that a model often does not take into account. In the extreme case of hourly returns, a model's predictive powers will approach zero. Some financial economists have had moderate success in applying their models to weekly returns. On the whole, though, weekly returns are still too influenced by specific events. For this reason, the majority of financial research is conducted on monthly returns. On the flip side, the danger of trying to apply the model to longer periods is that the parameters may change during the period. For example, to estimate the β of annual returns, one would need a sample spanning at least 20 years. It is unlikely that the β of a firm would stay constant for 20 years. The same problem arises for quarterly returns. Monthly returns are an acceptable middle-of-the-road solution to the tradeoff between predictive power and the risk of unstable parameters.

Whether monthly rebalancing is really the best route depends on market conditions. Markets may be very stable for a month so that nothing fundamental changes. Or markets may fluctuate rapidly, forcing more frequent rebalancing. External restrictions on the portfolio manager, such as large cash inflows and outflows, also alter the rebalancing frequency.

Our suggestion is to “update” the estimation of the models at least once a month if the model uses monthly returns. Updating the estimation does not necessarily lead to rebalancing. The portfolio manager simply should make sure that his or her information is up to date, and from there, he or she can decide whether to rebalance.

10.2.2 Change in α and Other Parameters

Changes in the parameters of a stock-return model can trigger rebalancing. Recall from the previous discussion of the factor model that the stock return can be expressed as

$$r_i = \alpha_i + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i \quad (10.1)$$

If we used an economic factor model to predict stock returns, then we should be on the lookout for changes in α and the factor exposures ($\beta_{i1}, \dots, \beta_{iK}$). If the model is the fundamental factor model, then we should be concerned about α and the factor premiums ($f_1, \dots, f_K$).

Parameters can change at any time. Corporate actions and major changes in the business environment all affect the parameters of a model. If firm A merges with firm B , for instance, the α and other parameters of firm A are bound to change. If the government imposes a new restriction on the activity of firm A , its α and other parameters most likely will change. Even a CEO’s retirement can affect a stock’s parameters. Then there are changes that do not become public knowledge. If the portfolio manager bases α on private information, α could react to new private information.

Quantitative portfolio managers always should think in terms of the model. If a manager reads a piece of negative news about a stock, it warrants a little time considering whether the news lowers the value of α or another parameter.

When the manager suspects that a parameter has changed, it is time to reestimate the model. Updating the values of the parameters updates the expected return and risk of each stock, and it changes the optimal portfolio. After finding the new optimal portfolio, the manager can decide whether to rebalance the portfolio by comparing the benefit of rebalancing to the transactions costs.

10.3 UNDERSTANDING TRANSACTIONS COSTS

By *transactions costs*, we generally mean the commissions that brokerages charge to execute orders. The commissions vary among brokers and also depend on the trades themselves. The same broker may charge different commissions for trading the same stock depending on the number of shares of the trade or the value of the transaction.

There are also two kinds of hidden transactions costs that portfolio managers should be aware of. These are the *bid–ask spread* and the *price impact*. The bid–ask spread equals the difference between the price at which one can buy the stock (the ask price) and the price at which one can sell the stock (the bid price). The bid–ask spread may reflect the operation cost and profit of a market maker or designated market maker. The lower the liquidity of a stock, the bigger is its bid–ask spread.

Figuring out the bid–ask spread can be tricky.⁴ The bid–ask spread is not fixed over time, and, at any given moment, we may be able to observe either the bid price or the ask price. If you compare the ask price from one moment to the bid price from another moment, it is unclear whether the difference between these two numbers reflects the change in the equilibrium price or the bid–ask spread.⁵

Price impact is the effect that large orders have on the observed stock price. A large order will move the stock price, causing the portfolio manager to pay more when buying, or earn less when selling, than the price per share quoted before trading. Price impact increases as the transaction size becomes large relative to the market size or daily trading volume of the stock. For a large order, the entire order may not be executed at the quoted price, so the average transaction price for the trade may be quite different from the quoted price before trading. This difference is one measure of the costs of price impact.

According to one estimate, the average broker commission is about 4 basis points of the transaction value, while the price impact is about 21 basis points. If one includes the price effect of delayed trades, the transaction cost exceeds 0.4% of the transaction value. As mentioned earlier, trading small-cap stocks costs more than trading large-cap stocks. The average commissions for large-cap stocks and small-cap stocks are 2.4

and 5.9 basis points, respectively. The average price impact is about 21 basis points for large cap and 19.9 basis points for small cap. Including the price effect of delayed trades, the total transaction cost is around 44.6 basis points for small-cap stocks.⁶

In general, it is impossible to anticipate the exact cost of the transaction in advance of its execution. This is especially true of the price impact and the bid–ask spread. One can develop a model and make an educated guess, but that is still far from knowing the exact number.⁷

10.4 MODELING TRANSACTIONS COSTS

Given the complicated and unpredictable nature of transactions costs, it is conventional to model them as a fixed proportion of the total value of the transaction. That is, we choose a constant, perhaps 5 or 10 basis points, to approximate the transaction cost per dollar transaction. Let us call this constant c . If the transaction value is \$10, then the transaction cost is \$10 c .

The total transaction value (TV) is easy to determine. Let V_t denote the dollar value of the current portfolio. Let $\{w_1^b, \dots, w_N^b\}$ denote the weights of stocks in the current portfolio and $\{w_1^a, \dots, w_N^a\}$ denote the weights of stocks in the prospective portfolio. (Superscript b refers to “before,” and superscript a refers to “after.”) That is, w_i^b is the weight of stock i in the current portfolio, and w_i^a is the weight of stock i in the prospective portfolio. Then $V_t w_i^b$ is the current holding of stock i in dollar terms, and $V_t w_i^a$ is the prospective holding of stock i in dollar terms. If $V_t w_i^a$ is greater than $V_t w_i^b$, it suggests buying stock i . If $V_t w_i^a$ is smaller than $V_t w_i^b$, it suggests selling stock i . In both cases, the difference, $V_t w_i^a - V_t w_i^b$, shows how much to buy or sell, in dollar terms. Therefore, the transaction value is simply the sum of this difference across all stocks:

$$TV = \sum_{i=1}^N |V_t w_i^a - V_t w_i^b| \quad (10.2)$$

The transaction cost (TC) is a constant fraction of the transaction value; thus

$$TC = cV_t \sum_{i=1}^N |w_i^a - w_i^b| \quad (10.3)$$

Eventually, we want to include the transactions cost formula in our optimization problem. For that purpose, it is useful to write down TC as a linear function, that is,

$$TC = V_t \sum_{i=1}^N c_i (w_i^a - w_i^b) \quad (10.4)$$

where

$$c_i = \begin{cases} c & \text{if } w_i^a > w_i^b \\ -c & \text{if } w_i^a < w_i^b \end{cases} \quad (10.5)$$

Once formulated in this way, we can express the transaction cost as a vector product of the weight vector and the transactions cost vector. Let us denote the vector of the current weights as $\mathbf{w}^b$ and the vector of the prospective weights as $\mathbf{w}^a$. That is, $\mathbf{w}^b = \{w_1^b, \dots, w_N^b\}$, and $\mathbf{w}^a = \{w_1^a, \dots, w_N^a\}$. Also, we define the vector of transactions costs as $\mathbf{c}$. That is, $\mathbf{c} = \{c_1, \dots, c_N\}$. Then

$$TC = V_t (\mathbf{w}^a - \mathbf{w}^b)' \mathbf{c} \quad (10.6)$$

Note that the exact value of $\mathbf{c}$ depends on the values of $\mathbf{w}^a$ and $\mathbf{w}^b$. Thus $\mathbf{c}$ is not a constant; rather it is a function of $\mathbf{w}^a$ and $\mathbf{w}^b$ in the mathematical sense [see [Eq. \(10.5\)](#)].

We can generalize our discussion of transactions costs and allow for different stocks to have different transactions costs. Considering the bid–ask spread and, especially, the price impact, one may want to model the transactions cost as proportional to the liquidity of each stock. For example, one might calculate the average trading volume of each stock and assume that the trading cost is inversely proportional to the trading volume.

Elements of $\mathbf{c}$ then would have both different signs and different absolute values.⁸

10.5 PORTFOLIO CONSTRUCTION WITH TRANSACTIONS COSTS

No matter what conditions or costs arise, the principle of portfolio selection always remains the same. Given whatever constraints we may face, we always want the portfolio with the best combination of expected return and risk. (In the case of managing against a benchmark, we always want the best combination of expected excess return and tracking error.) Transactions costs introduce a new variable into the process of determining the optimal portfolio (or the optimal tracking portfolio), but they do not alter the selection principle itself.

10.5.1 The Optimal Portfolio with Transactions Costs

We may formulate the problem of optimizing the portfolio in the face of transactions costs as a problem of minimizing risk given a certain expected return. In the context of our discussion of rebalancing, however, we prefer to use an alternative but equivalent formulation. Specifically, we formulate the optimization problem as one of maximizing the risk-adjusted return on the portfolio, where the risk-adjusted return is defined as the expected return in excess of the variance.

We start by recalculating the expected return of the portfolio, taking into account the transactions costs. If the expected return of the portfolio is μ_p and V_t dollars are invested in it, the value of the portfolio at the end of the holding period equals $V_t(1 + \mu_p)$. However, the portfolio manager needs to spend $(V_t\mathbf{w}^a - V_t\mathbf{w}^b)'\mathbf{c}$ in transactions costs, and the ending value of the portfolio must take this into account. That is, we need to subtract $(V_t\mathbf{w}^a - V_t\mathbf{w}^b)'\mathbf{c} (1 + \mu_p)$ from the ending portfolio value. Similarly, we need to subtract the transactions costs from the expected return. That is,

$$\text{Effective expected return} = \mu_p - (\mathbf{w}^a - \mathbf{w}^b)'\mathbf{c} - (\mathbf{w}^a - \mathbf{w}^b)'\mathbf{c}\mu_p \quad (10.7)$$

The effective expected return has three components. The first component is the gross expected return of the portfolio. The second component, to be subtracted from the gross expected return, is the transactions cost expressed as a fraction of the portfolio value.⁹ The last component is the time value of the transactions cost. Since the transactions cost is paid up front rather than at the end of the period, the portfolio manager loses twice, once by paying the cost and once again by not being able to invest and create profit. The second and the third components of the expected return reflect these losses.

For realistic values of the transactions cost and the expected return, the time value of the transactions cost is very small. For example, if the transactions cost is about 0.1% of the transaction value and the expected return is 1%, then the time value of transactions cost is only 0.001% of the transaction value. Therefore, in the discussion that follows, we ignore the time-value term.¹⁰

To obtain the expression for the risk-adjusted return, let us introduce additional notation. Let $\boldsymbol{\mu} = \{\mu_1, \dots, \mu_N\}$ be the vector of the expected stock returns. That is, μ_i is the expected return of stock i . Let $\boldsymbol{\Sigma}$ be the variance-covariance matrix of stock returns. That is,

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & & & \\ \sigma_{N1} & \sigma_{N2} & \cdots & \sigma_N^2 \end{bmatrix} \quad (10.8)$$

where σ_i^2 denotes the variance of the return of stock i , and σ_{ij} denotes the covariance between the return of stock i and the return of stock j . Thus $\mu_P = \mathbf{w}^a \boldsymbol{\mu}$ is the expected return of the prospective portfolio, and $\sigma_P^2 = \mathbf{w}^a \boldsymbol{\Sigma} \mathbf{w}^a$ is the variance of the prospective portfolio return. Let us define the *risk-aversion parameter* as A . Given A , we define the risk-adjusted return as $\mu_P - A\sigma_P^2$. Thus $A = 2$ means that the portfolio manager would not mind if the variance goes up by 1% as long as the expected return goes up by 2%.

Using the preceding notation and considering the transactions costs, the risk-adjusted return becomes

$$\begin{aligned}\text{Effective risk-adjusted return} &= \mu_p - (\mathbf{w}^a - \mathbf{w}^b)' \mathbf{c} - A \mathbf{w}^{a'} \Sigma \mathbf{w}^a \\ &= \mathbf{w}^{a'} (\boldsymbol{\mu} - \mathbf{c}) + \mathbf{w}^{b'} \mathbf{c} - A \mathbf{w}^{a'} \Sigma \mathbf{w}^a\end{aligned}\quad (10.9)$$

The optimal portfolio can be found by maximizing [Eq. \(10.9\)](#) subject to

$$\mathbf{w}^{a'} \mathbf{1} = 1 \quad (10.10)$$

and any other relevant constraints.

While [Eq. \(10.9\)](#) looks like a typical quadratic equation, in fact, it is highly nonlinear. The transactions cost vector $\mathbf{c}$ depends on the weight vector $\mathbf{w}^a$ [see [Eq. \(10.5\)](#)]. Thus this problem cannot be solved by a conventional quadratic optimization technique. In [Appendix 10A](#) we present a simple technique that will produce an approximate solution.

10.5.2 The Tracking Portfolio with Transactions Costs

If the portfolio manager is more concerned about tracking error than overall risk, then he or she should maximize the effective tracking-error-adjusted return rather than the effective risk-adjusted return.

Recall that the tracking error (TE) of the portfolio is defined as the standard deviation of the difference between the portfolio return r_p and the benchmark return r_B :

$$\begin{aligned}TE^2 &= V(r_p) - 2C(r_p, r_B) + V(r_B) \\ &= \mathbf{w}' \Sigma \mathbf{w} - 2\mathbf{w}' \boldsymbol{\gamma} + V(r_B)\end{aligned}\quad (10.11)$$

where

$$\boldsymbol{\gamma} \equiv C(\mathbf{r}, r_B) \quad (10.12)$$

As explained in [Chapter 9](#), the last term does not depend on the weight vector, so only the first two terms will be used in the optimization problem.

The *tracking-error-aversion parameter* A measures the manager's aversion to squared tracking error in the portfolio. The effective tracking-error-adjusted return becomes

$$\text{Effective } TE - \text{adjusted return} = \mathbf{w}^{a'}(\boldsymbol{\mu} - \mathbf{c}) + \mathbf{w}^{b'}\mathbf{c} - A[\mathbf{w}^{a'}\boldsymbol{\Sigma}\mathbf{w}^a - 2\mathbf{w}^{a'}\boldsymbol{\gamma} + V(r_B)] \quad (10.13)$$

To find the optimal tracking-error portfolio, we maximize the effective tracking-error-adjusted return with certain constraints. For the purpose of solving this optimization problem, we can ignore the terms that do not include $\mathbf{w}^a$ or $\mathbf{c}$. Thus we solve

$$\max_{\mathbf{w}^a} \mathbf{w}^{a'}(\boldsymbol{\mu} - \mathbf{c} + 2A\boldsymbol{\gamma}) + \mathbf{w}^{b'}\mathbf{c} - A\mathbf{w}^{a'}\boldsymbol{\Sigma}\mathbf{w}^a \quad (10.14)$$

subject to the constraint that the sum of weights should be 1 and any other constraints. This problem cannot be solved by a conventional quadratic optimization technique because $\mathbf{c}$ is a function of $\mathbf{w}$. In [Appendix 10A](#) we present a simple technique that will produce an approximate solution and in [Appendix 10B](#) we present a more complicated optimization framework that produces an exact solution.

10.6 DEALING WITH CASH FLOWS

As we mentioned in the introduction to this chapter, a manager sometimes has to rebalance the portfolio, or at least make additional investments outside the regular rebalancing schedule. Cash inflows and outflows require this sort of extra rebalancing. In this section we discuss some common methods that the portfolio manager can use to reduce transactions costs in the presence of cash flows. The first method involves investing cash temporarily in index futures or exchange-traded funds (ETFs) until there is a better time (such as the next scheduled rebalancing period) to return it to the portfolio. The second method involves purchasing specific, perhaps more liquid, stocks so as to achieve target portfolio weights through fewer transactions.

10.6.1 Reducing Transactions Costs Using Futures and ETFs

For portfolio managers who have daily cash inflows and out-flows, adjusting the portfolio at each cash movement is just not practical.¹¹ For the typical fund, daily flows are small relative to the fund's assets, and new contributions frequently offset fund redemptions. For such a fund, net inflows can be invested temporarily in cash or money market instruments without adversely affecting the portfolio return. Conversely, the fund's cash buffer can handle small redemptions, avoiding the need to liquidate any stocks. However, for funds with large daily net flows, maintaining a large cash reserve will negatively affect the portfolio return, especially in an environment of low interest rates. If outflows are large, the manager cannot rely on the cash buffer to meet redemptions and may be forced to liquidate part of the portfolio. This, of course, can be particularly problematic if there is low liquidity or if spreads are wide in some of the stocks.

In such cases, the manager might be better off adjusting his or her exposure in one of two ways, either by equitizing cash with index futures or by purchasing an ETF that invests in the same sector as the fund or a sector very correlated with the fund. ETFs became a more feasible solution for many funds when the Securities and Exchange Commission (SEC) lifted some restrictions on ETF ownership in mutual fund portfolios. Even with the wide availability of sector-specific ETFs, however, a manager still might prefer to use futures because futures have three distinct advantages over ETFs: futures are more liquid in normal trading hours, they can be traded after hours on GLOBEX, and they have an added tax benefit in that gains are treated as partly long term and partly short term. On the other hand, if the horizon for using futures is longer than a few days, then the portfolio manager might have to deal with the challenges of rolling over futures contracts.

10.6.2 Rebalancing toward Optimal Target Weights

Whether in a period of cash flows or at the point of conducting regular rebalancing, a portfolio manager can make the necessary trades in such a way that the portfolio maintains an optimal mix of stock weights. Imagine a portfolio that is rebalanced monthly and experiences cash flows at the end of the month. When it comes time for the monthly rebalancing, the manager can either trade so that he or she maintains the original optimal portfolio weights or so that he or she achieves the new optimal weights determined

by updating/reestimating the model. At the end of the month, the manager faces cash flows but can trade in such a way as to redirect the portfolio toward either the original or new optimal target weights. In the following subsections we discuss the algorithm that enables the manager to conduct optimal trades first in the case of regular rebalancing and then in the case of cash flows. In the analysis, we do not include trading costs per se. Rather, we show that by reducing the amount of trading, we can reduce tax costs as well as direct trading costs.

Standard Rebalancing

Standard rebalancing assumes that the portfolio manager has a target set of stock weights to which he or she wants to rebalance the portfolio. If the portfolio tracks a benchmark, then the target weights are those of the benchmark. Unless a new optimal portfolio with new stock weights has been constructed since the last rebalancing date, standard rebalancing involves selling the “winners” and buying the “losers” of the period since the last rebalancing date. You may be wondering why anyone would want to buy the losers. The reason is that the portfolio manager is concerned about the weights of stocks within the portfolio, not about their individual returns during a given period. Stocks that have performed relatively well will have gained weight in the portfolio so that they now exceed their targets; stocks that have done less well will have lost weight so that they now fall short of their targets. Rebalancing means restoring the weights back to their target levels.

Some notation may be useful. Let's call w_i^b the weight of stock i in the portfolio before rebalancing, w_i^a the target weight of each stock i , p_i the price per share of stock i , s_i^b the number of shares owned in stock i of the portfolio, and s_i^a the number of shares implicitly held in the optimal portfolio. The weights of the target portfolio and the current portfolio at any time $t + 1$ are given by

$$w_{i,t+1}^b = \frac{p_{i,t+1} s_{i,t+1}^b}{V_{t+1}} \quad (10.15)$$

$$w_{i,t+1}^a = \frac{p_{i,t+1} s_{i,t+1}^a}{V_{t+1}} \quad (10.16)$$

At any point in time, one can calculate the difference between the target weights and the current portfolio weights and use this difference to rebalance the portfolio. The number of shares of each stock that must be bought or sold at time $t + 1$ is given by

$$x_{i,t+1} = s_{i,t+1}^a - s_{i,t+1}^b = \frac{w_{i,t+1}^a}{p_{i,t+1}} (V_{t+1} + C_{t+1}) - s_{i,t+1}^b \quad (10.17)$$

where V_{t+1} is the value of the portfolio at time $t + 1$ before rebalancing, and C_{t+1} is any monetary contribution to (or withdrawal from) the portfolio at time $t + 1$. The excess shares $x_{i,t+1}$ are negative if one needs to sell shares of the stock and positive if one needs to buy more of the stock. One should bear in mind that these formulas do not take into account the bid–ask spreads or the price impact, both of which slightly distort the rebalancing procedure and create costs to the portfolio.

An Example

This example of standard rebalancing involves a portfolio of seven stocks (A through G). [Table 10.1](#) considers the portfolio at times t and $t + 1$. At time t , the portfolio is valued at \$100,000; by time $t + 1$, it has increased to \$107,986.68. At time t , the portfolio contains the optimal target stock weights, but as stock prices change between times t and $t + 1$, the portfolio drifts out of alignment with the targets. The table shows the difference between the weights at times t (optimal target weights) and $t + 1$ (misaligned weights). The “Shares to buy/sell” row of the table gives the results of the formulas that we presented earlier for calculating the number of stock shares to sell or buy in order to return to the optimal scenario.

TABLE 10.1

Standard Rebalancing

		Stock Holdings						
Date		A	B	C	D	E	F	G
t	Price per share	\$137.45	\$30.46	\$115	\$75	\$85	\$15	\$42.09
	Shares owned	200	200	200	200	200	200	200
	Weight (w_i^a)	0.2749	0.0609	0.2300	0.1500	0.1700	0.0300	0.0842
$t + 1$	Price-per-share	\$154.88	\$32.53	\$125.35	\$76.50	\$89.47	\$15.37	\$45.83
	Shares owned	200	200	200	200	200	200	200
	Weight before rebalance (w_i^b)	0.2869	0.0602	0.2322	0.1417	0.1657	0.0285	0.0849
	Rebalance shares	191.66	202.23	198.14	211.73	205.19	210.76	198.34
	Shares to buy/sell	-8.34	2.23	-1.86	11.73	5.19	10.76	-1.66
	Rebalance weight (w_i^{RB})	0.2749	0.0609	0.2300	0.1500	0.1700	0.0300	0.0842
	Dollar value	\$29,686	\$6,579	\$24,837	\$16,198	\$18,358	\$3,240	\$9,090

Using Cash Flows to Rebalance without Selling

Often it is less costly to let a portfolio drift toward the optimal target portfolio rather than make trades that try to match it exactly. There are tax costs associated with rebalancing owing to the selling of securities with capital gains. There is also a transactions cost with each purchase or sale of a security. Thus the manager can choose to not sell any securities but only buy more shares of the securities whose weights have fallen below target. While it does not rebalance the portfolio immediately, this method gives the portfolio a push in the right direction. It is easy to give the portfolio a push by using any cash inflows to add, proportionally to the optimal target weights, to all the portfolio's existing positions. A slightly harder push, investing the cash inflows according to an algorithm, will send the portfolio a bit faster toward the optimal weights.

The easiest method is to divide the cash inflows among all the portfolio's securities in amounts proportional to the target weights. Thus, if one has $\$C_{t+1}$ in cash to inject into the portfolio at time $t + 1$, then one will purchase $x_{i,t+1}$ of each security:

$$x_{i,t+1} = \frac{C_{t+1} w_{i,t+1}^a}{p_{i,t+1}} \quad (10.18)$$

The new weights that result from this process will fall in between the weights before adjustment and the optimal target weights. Their proximity to the target weights will depend on how large C_{t+1} is relative to V_{t+1} .

An Example

We return to the example of stocks A through G . Now, instead of doing the standard rebalancing, let us rebalance by distributing the cash flow C across the entire portfolio in proportion to the target weights.

The cash injection $C_{t+1} = \$2,000$. One can see from [Table 10.2](#) that the weights *slowly* converge to the target weights.

TABLE 10.2

Rebalancing by Purchasing at Target Weights

Date		Stock Holdings						
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
t	Price per share	\$137.45	\$30.46	\$115	\$75	\$85	\$15	\$42.09
	Shares owned	200.00	200.00	200	200	200	200	200
	Weight (w_i^a)	0.2749	0.0609	0.2300	0.1500	0.1700	0.0300	0.0842
$t + 1$	Price per share	\$154.88	\$32.53	\$125.35	\$76.50	\$89.47	\$15.37	\$45.83
	Shares owned	200.00	200.00	200.00	200.00	200.00	200.00	200.00
	Weight before rebalance (w_i^b)	0.2869	0.0602	0.2322	0.1417	0.1657	0.0285	0.0849
	Shares to buy/sell	3.55	3.75	3.67	3.92	3.80	3.90	3.67
	New shares owned	203.55	203.75	203.67	203.92	203.80	203.90	203.67
	Rebalance weight (w_i^{RB})	0.2866	0.0603	0.2321	0.1418	0.1658	0.0285	0.0849
	Dollar value	\$31,527	\$6,628	\$25,530	\$15,601	\$18,233	\$3,134	\$9,335

Another method for helping the portfolio gravitate toward the benchmark weights more quickly is to invest the new cash only in stocks that are below their corresponding target weights. All the stocks in the portfolio are divided into two groups, one in which stocks exceed their target weights (i.e., $\tilde{w}^b > w^a$) and another in which stocks fall short of target weights (i.e., $\tilde{w}^b < w^a$). It is important to calculate these weights *after* the cash has entered the portfolio. We thus use the symbol $\tilde{w}^b$ to represent the weights of the portfolio considering the current value of the portfolio and the cash inflow. The weight of each stock with the cash inflow is

$$\tilde{w}_{i,t+1}^b = \frac{s_{i,t+1}^b p_{i,t+1}}{V_{t+1} + C_{t+1}} \quad (10.19)$$

All these portfolio weights should be compared with their corresponding target weights ($w_{i,t+1}^a$). We must figure out how much of the cash should be allocated to each of the stocks that are below target weight. Suppose that for the first n_u stocks ($i = 1, \dots, n_u$), the weights are below the target weights. Let us call the cash allocated to stock i , $c_{i,t+1}$. We can determine this amount for each stock:

$$c_{i,t+1} = w_{i,t+1}^a (V_{t+1} + C_{t+1}) - s_{i,t+1}^b p_{i,t+1} \quad (10.20)$$

After the $c_{i,t+1}$ have been calculated,¹² we need to calculate the sum of them: $\sum_{i=1}^{n_u} c_{i,t+1}$. We then take the difference between the sum and C_{t+1} . The difference, which represents the amount of slack funds, equals $C_{t+1} - \sum_{i=1}^{n_u} c_{i,t+1}$. This number always will be less than or equal to zero. It will equal zero when there are enough funds (enough cash) to perfectly rebalance the portfolio. It will be less than zero when there are too few funds (too little cash) to restore deficient stocks to their required weights. In that case, the amount of cash distributed to each stock will have to be scaled down proportionally. Whether the amount of slack funds is zero or negative, the number of shares $x_{i,t+1}$ that need to be purchased of each stock is¹³

$$x_{i,t+1} = \frac{c_{i,t+1}}{p_{i,t+1}} \frac{C_{t+1}}{\sum_{i'=1}^{n_u} c_{i',t+1}} \quad (10.21)$$

The benefit of this type of “buy” system for rebalancing is that it should make the portfolio converge toward the optimal target weights at a faster rate than if we simply invested in the stocks in proportion to the target weights.

A manager might want to know exactly how much cash he or she would need to rebalance the portfolio using this system of purchasing deficient stocks. There are two steps to figuring this out. The first step is to calculate

the ratio of the weight of each stock before rebalancing and *before* new money has been added ($w_{i,t+1}^b$) to the target weight ($w_{i,t+1}^a$) of each stock minus one. That is,

$$\left(\frac{w_{i,t+1}^b}{w_{i,t+1}^a} - 1 \right) \quad (10.22)$$

The second step is to find the maximum of those values and multiply it by the total value of the portfolio V_{t+1} at time $t + 1$. That is,

$$\max \left[\frac{w_{1,t+1}^b}{w_{1,t+1}^a} - 1, \frac{w_{2,t+1}^b}{w_{2,t+1}^a} - 1, \dots, \frac{w_{N,t+1}^b}{w_{N,t+1}^a} - 1 \right] \cdot V_{t+1} \quad (10.23)$$

One can easily verify that this number is the minimum amount of cash inflow required to perfectly rebalance the portfolio without selling.

An Example

This example considers the same portfolio as the one in the two preceding examples. In this case, however, the rebalancing method attempts to achieve the desired weights more quickly. The seventh row of [Table 10.3](#) shows the weights of the stocks in the portfolio after new cash arrives. These values are the result of [Eq. \(10.19\)](#). The next row provides the number of shares to purchase of each stock in order to rebalance as much as possible according to [Eq. \(10.21\)](#). Finally, for this example, the second-to-last row of the table computes the new weights of the portfolio given the cash injection. One can see how this method pushes the portfolio toward convergence with the target weights much more quickly than the preceding buy- only method.¹⁴ In this example, \$4,697.43 would have been needed to perfectly rebalance without selling.

TABLE 10.3

Rebalancing by Purchasing Only Deficient Stocks

		Stock Holdings						
Date		A	B	C	D	E	F	G
t	Price per share	\$137.45	\$30.46	\$115	\$75	\$85	\$15	\$42.09
	Shares owned	200.00	200.00	200	200	200	200	200
	Weight (w_i^a)	0.2749	0.0609	0.2300	0.1500	0.1700	0.0300	0.0842
$t + 1$	Price per share	\$154.88	\$32.53	\$125.35	\$76.50	\$89.47	\$15.37	\$45.83
	Shares owned	200.00	200.00	200.00	200.00	200.00	200.00	200.00
	Weight before rebalance (w_i^b)	0.2869	0.0602	0.2322	0.1417	0.1657	0.0285	0.0849
	Weight after cash flow ($\tilde{w}_i^b$)	0.2816	0.0592	0.2279	0.1391	0.1627	0.0280	0.0833
	Shares to buy/sell	0.00	4.36	1.32	11.42	6.56	10.70	1.47
	New shares owned	200.00	204.36	201.32	211.42	206.56	210.70	201.47
	Rebalance weight (w_i^{RB})	0.2816	0.0604	0.2294	0.1471	0.1680	0.0294	0.0840
	Dollar value	\$30,977	\$6,648	\$25,236	\$16,174	\$18,480	\$3,239	\$9,234

The process of rebalancing without selling should be used whenever there are net cash inflows into the portfolio. A corresponding symmetric algorithm can be used when there are net withdrawals, in which case the rebalancing would involve only selling (and no buying). The selling, too, can be done optimally while reducing transactions costs and tax costs.

10.7 CONCLUSION

Much of portfolio management theory talks about optimizing portfolios without considering transactions costs. Unfortunately, some of the best portfolio management strategies involve trading stocks that are either illiquid or have high trading costs. Thinking about transactions costs ahead of time therefore can save a portfolio a great deal of money.¹⁵ For a quantitative portfolio manager who closely tracks a benchmark, even a small drag on performance owing to transactions costs can cause significant underperformance of the benchmark. Thus we have discussed methods of factoring transactions costs into the optimization techniques for tracking portfolios. These methods help a manager to steer clear of stocks that seem to have high α 's before trading but in fact have low or even negative α 's once trading costs are considered.

In this chapter we also discussed some rather rudimentary ways to reduce transactions costs when there are frequent cash flows into or out of the portfolio. These methods, though simple to implement, work quite well at reducing the number of transactions involved in rebalancing owing to cash flows. They also allow the manager to spread trades out over time so as to avoid a sudden, immense price impact on the portfolio.

Portfolio managers can save a great deal of money by being aware of explicit transactions costs. They can save yet more by anticipating the “hidden” costs of capital gains and dividend taxes. We cover the critical issue of tax management in the next chapter.

¹ Due to volatility in the markets caused by the coronavirus, the S&P Dow Jones index committee made a decision to postpone their rebalancing date to April 24, 2020. Invesco simply forgot to rebalance on April 24, 2020, and did so on April 29, 2020. See Riquier (2020).

² It is more common to discuss transactions costs with respect to the amount traded. However, when comparing transactions costs to the expected returns of the portfolio, it is better to convert this amount into a number that can compare the cost as a fraction of the entire portfolio value.

³ The transactions costs reduce the \$100 invested in stock *A* to \$99, which will earn 12%, or \$11.88. So stock *A* will reach a total value of \$110.88. Transactions costs for stock *B* reduce the \$100 to \$99.9, which will earn 10%, or \$9.99. Stock *B* therefore will reach a total value of \$109.89.

⁴ See Roll (1984), Glosten (1987), and Glosten and Harris (1988).

⁵ It is somewhat helpful that NASDAQ Level II and III allow many users to see the national best bid and offer prices for given volumes at any given point in time.

⁶ The data were obtained from Abel Noser and reflect one-way transactions costs in 2020. See [Table 16.12](#) in [Chapter 16](#).

⁷ Many commercial software providers attempt to supply price impact models to help portfolio managers and traders estimate these costs.

⁸ If the transactions costs are not a constant fraction of transaction value, then c_i defined in this section becomes a nonlinear function of transaction value. If, for example, the portfolio manager gets charged a flat fee for trades, then c_i is inversely proportional to the transaction value. When c_i is nonlinear, the portfolio optimization problem becomes nonlinear as well. Some commercial software packages have nonlinear programming routines to deal with this, but the reliability of these routines varies. Market impact also can alter the relationship between the trade size and transactions costs. There is a recursive effect because the optimal trade size depends on the transactions costs, which, in turn, depend on the trade size. In [Appendix 10B](#), we describe a method for an exact solution to the optimization problem that resolves most of these problems. In [Appendix 10C](#), we describe a technique that allows the portfolio optimization to deal with a variety of market impact cost models.

- ⁹ Note that there is no adjustment for how long the portfolio is going to be held. We are assuming that the expected return is expressed for the investment horizon. That is, if the portfolio manager plans to rebalance the portfolio six months later, then μ_P represents the return for six months. As long as the expected return is expressed for the investment horizon, there is no need to adjust the transactions cost c . Some portfolio managers “amortize” the transactions cost to reflect the holding period, but a simpler, equivalent way is to restate μ_P .
- ¹⁰ If we do not ignore the time-value term, then the optimization problem in the next section becomes rather complicated; the effective expected return becomes quadratic, rather than linear, in weights. Fortunately, the time-value term is small in most cases.
- ¹¹ This section is a brief summary of Chincarini (2004).
- ¹² The formula for $c_{i,t+1}$ as presented is equivalent to $c_{i,t+1} = (V_{t+1} + C_{t+1}) (\tilde{w}_{i,t+1}^a - \tilde{w}_{i,t+1}^b)$, where the portfolio weights are computed including the cash injection C_{t+1} .
- ¹³ When slack funds equal zero, $\frac{C_{t+1}}{\sum_{i=1}^{n_u} c_{i,t+1}} = 1$.
- ¹⁴ In this particular example, one can compute the savings in transactions costs and taxes from using the algorithm. Although very small in percentage terms, the algorithm results in about half the costs of standard rebalancing.
- ¹⁵ In fact, a Yale University professor approached one of the coauthors about launching a fund based on the professor’s “phenomenal” portfolio strategy. Once the professor and the coauthor considered transactions costs, however, they realized that it actually underperformed the major indices.

Approximate Solution to the Optimal Portfolio Problem

Problems specified by Eqs. (10.9) and (10.14) are not quadratic in $\mathbf{w}^a$ and are difficult to solve with conventional quadratic optimizers. The mathematics to solve this more complicated problem are described in Appendix 10B. The current appendix presents a simple approximate solution that works very well for most situations.

Nonlinearity in Eqs. (10.9) and (10.14) arises because we do not know, before solving the problem, which stock to sell and which stock to buy. If we knew, the problem would be quadratic as usual. Thus, to avoid nonlinearity, we need to determine which stocks to buy and which stocks to sell before solving the problems in Eqs. (10.9) and (10.14).

We suggest the following shortcut. First, we can solve the problem ignoring the transactions costs. That is, solve

$$\max_{\mathbf{w}^a} \mathbf{w}^a \mu - A \mathbf{w}^{a'} \Sigma \mathbf{w}^a \quad (10A.1)$$

subject to

$$\mathbf{w}^{a'} \mathbf{1} = 1 \quad (10A.2)$$

and any other relevant constraints. Call the solution to this problem $\mathbf{w}^* = \{w_1^*, \dots, w_N^*\}$. Then define the transactions cost vector $\mathbf{c} = \{c_1,$

..., c_N using these weights:

$$c_i = \begin{cases} c & \text{if } w_i^* > w_i^b \\ -c & \text{if } w_i^* < w_i^b \end{cases} \quad (10A.3)$$

That is, we determine which stocks to buy and which stocks to sell based on the optimization without considering transactions costs. Once we have determined which stocks to buy and which stocks to sell, we can impose this as a constraint to the optimization problem in [Eqs. \(10.9\) and \(10.14\)](#). In the case of [Eq. \(10.9\)](#), the problem to be solved becomes

$$\max_{\mathbf{w}^a} \mathbf{w}^{a'} (\boldsymbol{\mu} - \mathbf{c}) - A \mathbf{w}^{a'} \boldsymbol{\Sigma} \mathbf{w}^a \quad (10A.4)$$

subject to

$$\mathbf{w}^{a'} \boldsymbol{\iota} = 1 \quad (10A.5)$$

$$\begin{aligned} w_i^a &> w_i^b \text{ for } i \text{ such that } w_i^* > w_i^b \\ w_j^a &< w_j^b \text{ for } j \text{ such that } w_j^* < w_j^b \end{aligned} \quad (10A.6)$$

and any other relevant constraints. Now $\mathbf{c}$ is a fixed constraint and does not depend on the value of $\mathbf{w}^a$ (as long as the additional constraints we imposed are satisfied), and we can solve this problem using a conventional quadratic optimizer.

An Exact Solution to the Optimal Portfolio Problem

Although more complicated, there is a way to construct the portfolio optimization problem so as to consider transactions costs without having to use an approximate method. The minimization problem is similar to our standard one, such that,

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} \quad \text{s.t.} \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (10B.1)$$

$$\mathbf{lb} \leq \mathbf{x} \leq \mathbf{ub} \quad (10B)$$

where $\mathbf{x}$ is the vector of unknowns in the problem, $\mathbf{Q}$ is a symmetric positive semi-definite matrix, $\mathbf{c}$ is a vector of coefficients, $\mathbf{A}$ is a matrix of coefficients for the equality and inequality constraints, $\mathbf{b}$ is a vector of constraint values, $\mathbf{lb}$ is a lower-bound vector, and $\mathbf{ub}$ is an upper-bound vector.¹

As explained in [Chapter 9](#), in the traditional mean-variance optimization problem, we substitute the following variables: $\mathbf{x} = \mathbf{w}$, the stock weights; $\mathbf{Q} = 2\mathbf{\Sigma}$, the variance-covariance matrix of stock returns; $\mathbf{c} = \mathbf{0}$; $\mathbf{A}$ is chosen typically to have a row of ones and a row of expected returns, and the other constraints. In many software packages, the $\mathbf{A}$ matrix is split into two matrices, one for equality constraints and one for inequality constraints.

In order to solve for the optimal weights of the portfolio given a current portfolio and transactions costs, we start by expanding the $\mathbf{x}$ vector to include six types of variables. In addition to the prospective weights of the portfolio (w_i^a), we create phantom positive weights, w_i^+ , which represent the $(N \times 1)$ “long” positions of the portfolio; then we create the phantom negative weights, w_i^- , which represent the “short” positions of the portfolio; w_i^b , which represent the current weights of the portfolio, and the binary weights v_i^+ and v_i^- . That is, the $\mathbf{x}$ variable will consist of six types of variables in the following way:

$$\mathbf{x} = \begin{bmatrix} \mathbf{w}_{N \times 1} \\ \mathbf{w}_{N \times 1}^+ \\ \mathbf{w}_{N \times 1}^- \\ \mathbf{w}_{N \times 1}^b \\ \mathbf{v}_{N \times 1}^+ \\ \mathbf{v}_{N \times 1}^- \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \\ w_1^+ \\ w_2^+ \\ \vdots \\ w_N^+ \\ w_1^- \\ w_2^- \\ \vdots \\ w_N^- \\ w_1^b \\ w_2^b \\ \vdots \\ w_N^b \\ v_1^+ \\ v_2^+ \\ \vdots \\ v_N^+ \\ v_1^- \\ v_2^- \\ \vdots \\ v_N^- \end{bmatrix} \tag{10B.3}$$

That is, rather than just N weights for N stocks, we have $6N$ weights that have specific importance in the optimization. Of course, in the end, the variables of the ultimate importance are the first N weights: the optimal stock weights.

Since we now have $6N$ weights, the $\mathbf{Q}$ matrix must also be modified to be $(6N \times 6N)$ by adding zeros all around the variance-covariance matrix so that,

$$\mathbf{Q} = \begin{bmatrix} 2\Sigma_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix}_{(6N) \times (6N)} \quad (10B.4)$$

We constrain the prospective weights of the optimized portfolio, w_i^a , to be related to these phantom weights and the prior portfolio weights as follows:

$$w_i^a = w_i^+ - w_i^- + w_i^b \quad (10B.5)$$

This essentially makes the phantom weights equal to the incremental buy or sell for each stock in the portfolio from the prior portfolio weights. We also need to constrain the phantom weights, w_i^+ and w_i^- , to be positive and orthogonal to each other (i.e., only one of them is positive for any given security in the portfolio). This is needed so that we do not unnecessarily buy and sell the same stock. Thus, we introduce the binary variables v_i^+ and v_i^- for each stock in the portfolio. We introduce the following constraints in order to ensure that the phantom weights are orthogonal:

$$v_i^+ \kappa_l \leq w_i^+ \leq v_i^+ \kappa_h \quad (10B.6)$$

$$v_i^- \gamma_l \leq w_i^- \leq v_i^- \gamma_h \quad (10B.7)$$

$$v_i^+ + v_i^- \leq 1 \quad (10B.8)$$

Since v_i^+ and v_i^- are binary variables, the inequality constraint in Eq. (10B.8) forces only one of them to be 1 and the other 0. This combined with the constraints in Eqs. (10B.6) and (10B.7) force the phantom weights to also be orthogonal to each other. For generality, we used the parameters κ_b , γ_b , κ_h , and γ_h , but later in this section we will assign them values of 0 and 1 to simplify this particular optimization.²

We then create a vector of transactions costs, $\mathbf{c}$, as before. Now that the general setup is in place, we show how to arrange the matrices (equality and inequality constraints) to complete the optimization. We will show the equality matrix constraints and inequality matrix constraints separately, but you can also combine them into one $\mathbf{A}$ matrix if desired. The equality matrix, $\mathbf{A}_{eq}$, should be a $(2N + 2)$ by $(6N)$ matrix, and the corresponding $\mathbf{b}_{eq}$ vector will be $(2N + 2)$ by (1) .

$$\mathbf{A}_{eq} = \begin{bmatrix} \boldsymbol{\iota}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \\ \boldsymbol{\mu}_{1 \times N} & -\mathbf{c}_{1 \times N} & -\mathbf{c}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \\ \mathbf{I}_{N \times N} & -\mathbf{1} \cdot \mathbf{I}_{N \times N} & \mathbf{I}_{N \times N} & -\mathbf{1} \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix}_{(2N+2) \times (6N)} \quad (10B.9)$$

$$\mathbf{b}_{eq} = \begin{bmatrix} 1 \\ \mu_p \\ \mathbf{0}_{N \times 1} \\ \mathbf{w}_{N \times 1}^b \end{bmatrix}_{(2N+2) \times (1)} \quad (10B.10)$$

The inequality matrix, $\mathbf{A}_{ineq}$, should be a $(5N)$ by $(6N)$ matrix and the corresponding $\mathbf{b}_{ineq}$ vector will be $(5N)$ by (1) .

$$\mathbf{A}_{ineq} = \begin{bmatrix} \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{I}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & -\kappa_h \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \kappa_l \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & -\gamma_h \cdot \mathbf{I}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \gamma_l \cdot \mathbf{I}_{N \times N} \end{bmatrix}_{(5N) \times (6N)} \quad (10B.11)$$

$$\mathbf{b}_{ineq} = \begin{bmatrix} \boldsymbol{\iota}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \end{bmatrix}_{(5N) \times (1)} \quad (10B.12)$$

where $\boldsymbol{\mu}_{1 \times N}$ is the row vector of the mean returns for each of the N stocks; μ_P is the desired after-tax portfolio return; $\mathbf{1}_{1 \times N}$ is a row of six 1s (that is, the first six entries of matrix $\mathbf{A}$'s row 1 are 1s); $\mathbf{0}_{1 \times N}$ is a row of N entries of 0; $\mathbf{I}_{N \times N}$ is the identity matrix (that is, an N by N matrix with 1s on the diagonal and 0s everywhere else); and so on and so forth. In order to aid the reader, we indicate the dimension of every matrix as a subscript.

The transactions costs are incorporated into the optimization via the second row of the equality matrix, which constrains the optimization so that $\boldsymbol{\mu}'\mathbf{w} - \mathbf{c}'\mathbf{w}^+ - \mathbf{c}'\mathbf{w}^- = \mu_P$. Since the phantom weights must be positive and orthogonal, this creates the appropriate after-tax average return.

As stated earlier, you can either put upper- and lower-bound constraints in the inequality, or $\mathbf{A}_{ineq}$, matrix or specify them directly as upper- and lower-bound vectors in many optimization packages. In this example, we chose to specify them directly; thus we constrain weights and phantom weights to be between 0 and 1, as well as the prior portfolio and the binary weights.³ Thus

[illegible]

$$\mathbf{ub} = [\mathbf{l}_{1 \times N} \quad \mathbf{l}_{1 \times N} \quad \mathbf{l}_{1 \times N} \quad \mathbf{l}_{1 \times N} \quad \mathbf{l}_{1 \times N} \quad \mathbf{l}_{1 \times N}] \quad (10B.14)$$

We can solve this problem with a mixed integer quadratic optimization (MIQO).⁴ If you do not have access to a MIQO optimizer, but have access to a mixed integer linear programming optimizer, then you can use an algorithm to find the solution.⁵

10B.1 A NUMERICAL EXAMPLE

We will continue the numerical example in [Appendix 9A](#). Thus we will seek to construct a portfolio with an after-tax average return of 8%, with no short sales allowed, and the weights of the portfolio must sum to 1. To make things simple, we will assume all stocks cost 1% to trade, except stock 1, which costs 20% to trade. That is, if one trades \$100 of stock 1, one will incur a cost of \$20, while a \$100 trade of any other stock will result in a cost of \$1. This will be represented by a $\mathbf{c}$ vector. Of course, this solution would work with any cost vector. We also use the following prior weights of the portfolio: $\mathbf{w}^b = [0.23 \ 0.1 \ 0.12 \ 0.14 \ 0.13 \ 0.28]$. We will construct an optimization that minimizes variance subject to a target after-tax return of 8% while considering the prior portfolio weights, w_i^b .

Since our focus is on transactions costs and not particular upper and lower limits to the phantom weights, we can set the parameters $\kappa_l = \gamma_l = 0$ and $\kappa_h = \gamma_h = 1$ so that we allow the weights to fluctuate between 0 and 1, but force them to be orthogonal to each other. The constraint matrices will be⁶

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\iota}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \boldsymbol{\mu}_{1 \times 6} & -\mathbf{c}_{1 \times 6} & -\mathbf{c}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \mathbf{I}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{I}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \end{bmatrix}_{44 \times 36} \quad (10B.21)$$

$$\mathbf{b} = \begin{bmatrix} \mathbf{b}_{eq} \\ \mathbf{b}_{ineq} \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ \mathbf{0}_{6 \times 1} \\ \mathbf{w}_{6 \times 1}^b \\ \boldsymbol{\iota}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \end{bmatrix}_{1 \times 44} \quad (10B.22)$$

where $\boldsymbol{\mu}_{1 \times 6} = [14.40 \ 10.19 \ 9.87 \ 7.52 \ 20.05 \ 2.66]$; $\mathbf{c}_{1 \times 6} = [20 \ 1 \ 1 \ 1 \ 1 \ 1]$; $\boldsymbol{\iota}_{1 \times 6}$ is a row of six 1s (that is, the first six entries of matrix $\mathbf{A}$'s row 1 are 1s); $\mathbf{0}_{1 \times 6}$ is a row of six entries of 0; $\mathbf{I}_{6 \times 6}$ is the identity matrix (that is, a 6-by-6 matrix with 1s on the diagonal and 0s everywhere else); and so on and so forth. In order to aid the reader, we indicate the dimension of every matrix as a subscript, including for the large $\mathbf{A}$ and $\mathbf{b}$ matrices.

The economic meaning of these constraints is the following: the first two rows of this $\mathbf{A}$ matrix are the equality constraints that the sum of weights equals 1 and that the target after-tax average return is 8%.

The third row through the eighth row of the equality matrix establishes the links between the actual portfolio weights that are chosen, the phantom weights, and the prior portfolio weights. Rows 9 through 14 constrain the

prior portfolio weights to equal $\mathbf{w}^b$. Rows 15 through 20 constrain the binary variables to be orthogonal; that is, only one of them can be positive, which consequently constrains the phantom weights to be orthogonal (i.e., we don't buy and sell the same security). In order to do this, we employ the constraints from rows 21 to 44. As discussed earlier, we accomplish this by setting $\kappa_l = \gamma_l = 0$ and $\kappa_h = \gamma_h = 1$.

The solution using an optimization routine with continuous and binary variables is:⁷

$$\mathbf{w} = \begin{bmatrix} 0.230 \\ 0.081 \\ 0.120 \\ 0.309 \\ 0.000 \\ 0.260 \end{bmatrix} \quad (10B.23)$$

Notice that the solution does not adjust the weight of stock 1 at all. This is quite understandable given the high transactions cost that we imposed on stock 1. In the absence of transactions costs, the optimal portfolio weights would have been [0.287 0.016 0.065 0.283 0 0.348].

¹ As we discussed in [Appendix 9A](#), you can also place these upper- and lower-bound constraints directly in the $\mathbf{A}$ matrix.

² It is essential that the software package allows for quadratic optimization with binary variables and continuous variables and that the user specifies which variables are binary variables. If the software only allows for integers, then upper and lower bounds of 0 and 1 on the integers will achieve similar results. The CPLEX suite has these and can be used as a library with MATLAB. For users who do not have the option to specify binary variables in a quadratic optimization, we provide an algorithm later in this appendix to solve for this problem with a linear programming integer optimizer.

³ Some of these weight boundaries were implicitly already constrained by prior constraints.

⁴ For example, IBM's CPLEX optimization system has a MIQO function called CPLEXMIQP. These are also referred to as mixed integer quadratic programs (MIQPs).

⁵ For example, MATLAB does not have an MIQP but does have an MILP called INTLINPROG. In order to use an MILP in place of an MIQP, certain adjustments to the matrices must be made and an iterative procedure followed, which converges to the MIQP solution [Kelley (1960)]. Rather than describe all the details for this particular optimization, we will discuss the general concepts and the principal changes required to do a similar transactions cost optimization using the iterative MILP to replicate an MIQP. The steps are as follows. Step 1: Introduce a slack variable called z to represent the quadratic term. The idea is that instead of minimizing $\lambda \mathbf{x}'\mathbf{Q}\mathbf{x}$, you minimize λz s.t. $\mathbf{x}'\mathbf{Q}\mathbf{x} - z \leq 0$ and $z \geq 0$. Then as you iteratively solve the MILP approximations, include new linear constraints, each of which approximates the nonlinear constraint locally near the current point. This is repeated until a satisfactory solution emerges. Practically, the inequality matrix $\mathbf{A}_{ineq}$ is altered to include a second column representing z (which can be thought of an extra choice parameter, like the $N + 1$ weight in a standard N -asset problem), with a lower bound of 0 and an upper bound of infinity. Also, you should specify that the v_i are integers and place upper and lower bounds on them of 0 and 1. Step 2: Transform the nonlinear quadratic objective function, which minimizes risk and maximizes return with a tradeoff of λ , into a linear optimization problem with non-linear (e.g., quadratic) constraints. That is, convert the optimization problem

$$\min_{\mathbf{x}} \lambda \mathbf{x}'\mathbf{Q}\mathbf{x} - \mathbf{x}'\mathbf{r} \quad (10B.15)$$

into this one:

$$\min_{\mathbf{x}, z} \lambda z - \mathbf{x}'\mathbf{r} \quad \text{s.t.} \quad \mathbf{x}'\mathbf{Q}\mathbf{x} - z \leq 0, \quad z \geq 0 \quad (10B.16)$$

where $\mathbf{r}$ is a vector of variables to maximize. For example, in the case of transactions costs, it would represent the mean returns of the stocks as well as the negative of the transactions costs for the phantom weights (i.e., it would consist of the second row of $\mathbf{A}_{eq}$ and that row would be removed from $\mathbf{A}_{eq}$). Since many linear programming softwares are set to minimize $\mathbf{f}'\mathbf{x}$, the $\mathbf{f}$ for this particular optimization would look something like

$$\mathbf{f} = \begin{bmatrix} -\mu_{N \times 1} \\ \lambda_{1 \times 1} \\ \mathbf{c}_{N \times 1} \\ \mathbf{c}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \end{bmatrix}_{(6N+1) \times (1)} \quad (10B.17)$$

Use a Taylor series expansion of the constraint around $\mathbf{x}_0$, ignoring squared terms and higher, such that

$$\mathbf{x}'\mathbf{Q}\mathbf{x} - z = \mathbf{x}_0'\mathbf{Q}\mathbf{x}_0 - z + 2\mathbf{x}_0'\mathbf{Q}(\mathbf{x} - \mathbf{x}_0) - O(\|\mathbf{x} - \mathbf{x}_0\|^2) \quad (10B.18)$$

$$\approx -\mathbf{x}_0' \mathbf{Q} \mathbf{x}_0 - z + 2\mathbf{x}_0' \mathbf{Q} \mathbf{x} \quad (10B.19)$$

Then add the following constraint to the constraint matrix for the next iteration:

$$2\mathbf{x}_k' \mathbf{Q} \mathbf{x} - z \leq \mathbf{x}_k' \mathbf{Q} \mathbf{x}_k \quad (10B.20)$$

Using this approximation, in every iteration of the MILP problem you add a new constraint to the inequality constraint matrix of the form above. In particular, add a row to $\mathbf{A}_{ineq}$ with the values of $2\mathbf{x}_k' \mathbf{Q}$ based on the prior optimized value of asset weights $\mathbf{x}_k$, $a-1$ for the z term, and then 0s for the other decision variables in the optimization problem (e.g., phantom weights, binary variables, etc). You would also add a value to the $\mathbf{b}$ vector equal to $\mathbf{x}_k' \mathbf{Q} \mathbf{x}_k$.

On every iteration, instead of using the actual value of $\mathbf{x}_k$ from the most recent optimization, one could also use a linear combination of the previous solution and the new solution. That is, use $\tilde{\mathbf{x}}_k = 0.5\mathbf{x}_k + 0.5\mathbf{x}_{k-1}$ instead of $\mathbf{x}_k$ when adding the constraint. Continue the process with the new constraints and using the MILP. On every iteration, compare the value of z to the value of the quadratic at the solution point (i.e., $z - \mathbf{x}_k' \mathbf{Q} \mathbf{x}_k$). If that is within the tolerance level, stop the process and you have the optimal value of your stock weights. For example, you could choose to stop when $z < \mathbf{x}_k' \mathbf{Q} \mathbf{x}_k (1 - \epsilon)$, where $\epsilon = 0.0005$.

⁶ The $\mathbf{Q}$ matrix containing the variance-covariance matrix must also be expanded to account for the additional five variables with 0s on the diagonals. See [Eq. \(10B.4\)](#).

⁷ Using the CPLEX function CPLEXMIQP, available through the CPLEX library in MATLAB, this optimization took 0.0170 seconds to reach a solution.

An Approximate Optimal Portfolio with Market Impact Costs

Estimating market impact costs can be very valuable for a portfolio manager. Unfortunately, there is a nonlinear element to the market impact of trades. The larger your portfolio, the larger the market impact of a trade. As a result, standard portfolio optimization techniques cannot be used because, as the optimal weight of each stock changes, so does the cost of trading. In addition to the optimization dilemma, the portfolio manager also needs a good description of how the size of a trade and market impact are related. This market impact data can either be obtained from a professional organization that has made detailed estimates of transactions costs or from a parametric model of transactions costs. For example, you might specify transactions costs from trading as

$$tc_{it} = C_{it} + \left| \frac{100s_{it}/2}{p_{it}} \right| + |c_{it}| \quad (10C.1)$$

where C_{it} is the percentage commission cost from the trade, s_{it} is the bid–ask spread of stock i at time t , p_{it} is the price of stock i at time t , and c_{it} is the percentage market impact costs for stock i at time t based on a market impact model. These transactions costs, tc_{it} , are in price percentage points

(i.e., a value of 0.20 would indicate that this particular trade would lead to costs of 0.20%).

Market impact models will vary but might take a form similar to this:

$$c_{it} = a + \psi \left(\frac{n_{it}}{V_{it}} \right)^\phi \quad (10C.2)$$

where n_{it} represents the number of shares of security i that need to be traded, V_{it} is the average daily trading volume of the stock (shares traded, not dollars traded), and a , ϕ , and ψ are parameters that need to be estimated.¹

As mentioned previously, market impact transactions cost models are difficult to use in a standard optimization framework.² In this appendix we are going to introduce an approximation technique to allow for portfolio optimizations when the user has a detailed model of market impact costs as described above. The first step is to apply an approximation method to approximate a variety of transactions cost models.³ The second step is to formulate the optimization problem so as to solve for market impact.

10C.1 APPROXIMATION OF TRANSACTIONS COSTS

The first step is to approximate the transactions cost model, which is usually based on the number of shares traded of a particular stock, and other parameters. The larger the trade of a stock, the larger the market impact. Thus, in order to make the market impact model usable in a standard optimization framework, we need to start with a given portfolio size, say \$100 million. We then vary the weights of each stock from 0% of the portfolio to a maximum allowable percentage based on the portfolio manager's needs. We can choose increments of 1% or smaller and for each weighting we compute the model's estimated market impact costs. Once these values of market impact (or total transactions costs) are calculated, a series of weights for each stock along with the market impact for each particular stock weight will be known for each period (e.g., monthly).⁴ For

example, for stock i , we might have a series of weights $\mathbf{w}_{it} = [0.0 \ 0.01 \ 0.02 \ 0.03 \ 0.04 \ 0.05 \ 0.06 \ 0.07 \ 0.08 \ 0.09 \ 0.10]$ along with the corresponding market impact costs $\mathbf{c}_{it} = [c_{it}^{0.00} \dots c_{it}^{0.10}]$. From this we compute the total transactions cost vector for each stock, which is given by Eq. (10C.1), and multiply this by $\mathbf{w}_{it}$ to obtain a vector $\tilde{\mathbf{tc}}_{it}$, which is the weight of stock i for every weight level multiplied by the actual transactions costs $\mathbf{tc}_{it}$, given a portfolio of value $\$V_t$.

The second step is to estimate for each stock a regression of the following form:

$$\tilde{\mathbf{tc}}_{it} = a_{it} \mathbf{w}_{it} + b_{it} \mathbf{w}_{it}^2 \quad (10C.3)$$

where $\tilde{\mathbf{tc}}_{it}$ is a vector of transactions costs from the transactions cost model multiplied by each stock's vector of incremental weights, and a_{it} and b_{it} are parameters estimated from the linear regression.⁵

This approximation model works extremely well for many transactions cost models.⁶ As an example, the actual transactions costs and approximate transactions costs for AT&T for December 2013 were 0.000232% and 0.000257% for a \$5 million trade (i.e., 1% position of a \$500 million portfolio).⁷

This approximation is key to allowing for market impact in the optimizations since we now have a clear link between weight choice and transactions costs impact. In the next section, we show how to use this approximation in a portfolio optimization problem.

10C.2 LONG-ONLY PORTFOLIO

In order to incorporate our approximate transactions costs into the portfolio optimization problem, we must modify the quadratic optimization program slightly. First, we must use a quadratic optimization routine that can accept quadratic constraints, in addition to linear constraints, as discussed in [Appendix 9B.3](#).⁸ Second, we must modify the traditional portfolio optimization setup to work with transactions costs.

The mathematical expression of the quadratic optimization with quadratic constraints is given as

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} \quad s.t. \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (10C.4)$$

$$\mathbf{l}' \mathbf{x} + \mathbf{x}' \mathbf{Q}^* \mathbf{x} \leq r \quad (10C.5)$$

$$\mathbf{l} \mathbf{b} \leq \mathbf{x} \leq \mathbf{u} \mathbf{b} \quad (10C.6)$$

where $\mathbf{x}$ is the vector of unknowns in the problem, $\mathbf{Q}$ is a symmetric positive semi-definite matrix, $\mathbf{c}$ is a vector of coefficients, $\mathbf{A}$ is a matrix of coefficients for the equality and inequality constraints, $\mathbf{b}$ is a vector of constraint values, $\mathbf{l}$ is a column vector, r is a scalar, $\mathbf{Q}^*$ is a symmetric matrix, $\mathbf{l} \mathbf{b}$ is a lower-bound vector, and $\mathbf{u} \mathbf{b}$ is an upper-bound vector.

The main difference in this optimization problem is the quadratic constraint in Eq. (10C.5). In the traditional mean-variance optimization problem, we substitute the following variables: $\mathbf{x} = \mathbf{w}$, the stock weights, $\mathbf{Q} = 2\mathbf{\Sigma}$, the variance-covariance matrix of stock returns, $\mathbf{c} = \mathbf{0}$, $\mathbf{l} = \mathbf{0}$, $r = 0$, $\mathbf{Q}^* = \mathbf{0}$, $\mathbf{A}$ is chosen typically to have a row of 1s and a row of expected returns, and the lower and upper bounds are set as desired.

In order to create an optimal portfolio that minimizes the risk of the portfolio and achieves a desired after-transactions cost return, which includes market impact costs, we must set up additional matrices. For this general discussion, we will not create all of the other constraints; we will just assume that the $\mathbf{A}$ and $\mathbf{b}$ matrices have been set up for the particular optimization of the portfolio manager. We will assume that the optimization has only N choice parameters, that is, that $\mathbf{x} = [w_1 \ w_2 \ \dots \ w_N]$. Thus, $\mathbf{Q} = 2\mathbf{\Sigma}$. We will need to specify, $\mathbf{l}$, $\mathbf{Q}^*$, and r as follows:

$$\mathbf{Q} = \begin{bmatrix} \hat{b}_1 & 0 & \dots & 0 \\ 0 & \hat{b}_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \hat{b}_N \end{bmatrix} \quad (10C.7)$$

and $\mathbf{l} = -\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - \hat{\mathbf{a}}$, $\mathbf{c} = \mathbf{0}$, and $r = -\mu_P$, where $\tilde{\boldsymbol{\mu}}$ is a vector of the expected returns of each stock minus the constant estimate in the transactions cost regression ($\hat{a}_i$); $\boldsymbol{\mu}$ is the expected return of each stock; μ_P is the desired target of after-transactions cost expected return for the portfolio to match; and $\hat{b}_i$ is the coefficient estimate from the transactions cost regression for stock i .⁹

10C.3 MARKET-NEUTRAL PORTFOLIO

The market-neutral problem is slightly more complicated. We need to create phantom weights and binary weights. Once again, we will assume the user has set up their desired constraints for the regular weights ($\mathbf{w}$), phantom weights (both $\mathbf{w}^+$ and $\mathbf{w}^-$), and the binary weights (both $\mathbf{v}^+$ and $\mathbf{v}^-$). We will also have to increase the size of the $\mathbf{Q}$ matrix to accommodate the other variables as

$$\mathbf{Q} = \begin{bmatrix} 2\boldsymbol{\Sigma}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix} \quad (10C.10)$$

where $\boldsymbol{\Sigma}$ is the variance-covariance matrix of stock returns [e.g., see [Eq. \(9A.13\)](#)]. We also need to specify, $\mathbf{l}$, $\mathbf{Q}^*$, and r as follows:

$$\mathbf{Q}^* = \begin{bmatrix} \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \boldsymbol{\Sigma}_{2, N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \boldsymbol{\Sigma}_{2, N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix} \quad (10C.11)$$

where

$$\Sigma_2 = \begin{bmatrix} \hat{b}_1 & 0 & \dots & 0 \\ 0 & \hat{b}_2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \hat{b}_N \end{bmatrix} \quad (10C.12)$$

$$\mathbf{Q} = \begin{bmatrix} 2\Sigma_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix} \quad (10C.10)$$

where $\boldsymbol{\mu}$ is the expected return of each stock; $\hat{\mathbf{a}}_{N \times 1}$ is the coefficient on w_i in the regression for stock i ; $\hat{b}_i$ is the coefficient estimate on w_i^2 from the transactions cost regression for stock i ; $\mathbf{c} = \mathbf{0}$, μ_P is the desired target of after-transactions cost expected return for the portfolio to match; and $r = -\mu_P$.¹⁰

10C.4 MARKET IMPACT DURING REBALANCING

In the previous two sections we showed how to use the market impact approximation on a long-only and a market-neutral optimization when building the portfolio from scratch. It may also be of use to model market impact when rebalancing a portfolio. The approach is very similar to the concepts discussed in [Appendix 10B](#). As before, we will assume the user has set up their desired constraints for the regular weights ($\mathbf{w}$), phantom weights (both $\mathbf{w}^+$ and $\mathbf{w}^-$), prior or current portfolio weights ($\mathbf{w}^b$), and the binary weights (both $\mathbf{v}^+$ and $\mathbf{v}^-$) as described in [Appendix 10B](#). Thus, we

will need to specify, $\mathbf{I}$, $\mathbf{Q}^*$, and $\mathbf{r}$ as before. We will also have to modify $\mathbf{Q}$ (i.e., expand the variance-covariance matrix).

For the market-neutral portfolio described in [Appendix 10C.3](#), we add a prior set of portfolio weights as discussed in [Appendix 10B](#). Thus, the only modification we need to make to $\mathbf{Q}$, $\mathbf{Q}^*$, and $\mathbf{I}$ is to add a column of six $\mathbf{0}_{N \times N}$ matrices between the third and fourth columns of $\mathbf{Q}$ and $\mathbf{Q}^*$ in [Eqs. \(10C.14\)](#) and [\(10C.15\)](#), and add $\mathbf{0}_{1 \times N}$ between the third and fourth columns of $\mathbf{I}$. We also need to add the constraints that come with prior weights, as described in [Appendix 10B](#). These modifications will account for the new prior portfolio weights, and the optimization setup is complete.

For the long-only portfolio, which needs to be set up as in [Appendix 10B](#), we need to remove the second rows of $\mathbf{A}_{eq}$ and $\mathbf{b}_{eq}$ that target an after-transactions cost return and instead alter $\mathbf{Q}$, $\mathbf{Q}^*$, and $\mathbf{I}$ very similarly to the way we did them for the market-neutral portfolio that accounts for market impact during rebalancing.

These changes will allow for optimizations that consider market impact during rebalancing.

10C.5 A NUMERICAL EXAMPLE

We will continue the previous numerical example in [Appendix 10B](#). Thus, we will seek a portfolio with an after-tax average return of 8%, with no short sales allowed, and the weights of the portfolio must sum to 1. In this case we will construct transactions costs for every stock based on the actual spread costs (i.e., bid–ask spread) and market impact costs based on a transaction model used in portfolio management. Although not shown here, we first acquired the relevant data to compute the market impact cost for all of our six stocks in a given period in our sample. We then computed the spread costs for this same period. We then computed the market impact costs for various weights for each of our stocks assuming a portfolio size of \$500 million and estimated the linear regression to obtain the $\hat{a}_{it}$ and $\hat{b}_{it}$ for every stock. The regression estimates for the stocks were $\hat{\mathbf{a}}'_t = [0.027 \ 0.019 \ 0.354 \ 0.119 \ 3.654 \ 2.844]$ and $\hat{\mathbf{b}}'_t = [0.511 \ 0.377 \ 58.531 \ 12.694 \ 5704.284 \ 3742.573]$. Thus, for stock 1, a 10% position in this stock would represent a position value of \$50

million ($0.10 \cdot 500$). Suppose this stock was trading at \$101.2, that would mean that buying 10% of this stock in the portfolio would require purchasing 494,071 shares of the stock. This stock is a quite liquid stock and trades about 12,410,000 shares per day, on average. Thus, this trade would represent only 4% of the average daily volume. The transaction cost for this stock is 0.07784%. That is, to trade \$50 million all at once would result in a cost of \$38,917. The approximation of transaction cost equals $0.07818\% \left(\frac{0.027 \cdot 0.10 + 0.511 \cdot 0.10^2}{0.10} \right)$.¹¹

Thus, our constraints are given as

$$\Sigma_2 = \begin{bmatrix} 0.511 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.377 & 0 & 0 & 0 & 0 \\ 0 & 0 & 58.531 & 0 & 0 & 0 \\ 0 & 0 & 0 & 12.694 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5704.284 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3742.573 \end{bmatrix} \quad (10C.16)$$

$$\mathbf{l} = \begin{bmatrix} -1 \cdot \boldsymbol{\mu}_{N \times 1} \\ 0.027 \\ 0.019 \\ 0.354 \\ 0.119 \\ 3.654 \\ 2.844 \\ \hat{\mathbf{a}}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \end{bmatrix} \quad (10C.17)$$

The solution to this optimization is

$$\mathbf{w} = \begin{bmatrix} 0.298 \\ 0.000 \\ 0.000 \\ 0.339 \\ 0.078 \\ 0.285 \end{bmatrix} \quad (10C.18)$$

In order to understand these results, remember that based on our estimations, stocks 1 and 2 are the most liquid (lowest transactions costs), stocks 3 and 4 are the next most liquid, and stocks 5 and 6 are relatively illiquid. Compared to the prior portfolio weights for stock 5 and 6 of 13% and 28%, the optimization barely altered the weight of stock 6 and sold only 5.2% of stock 5 as opposed to an optimization that ignored market impact costs—costs that are increasingly larger for less liquid stocks. We did the same optimization without considering market impact costs, and the optimal solution was $\mathbf{w}' = [0.293 \ 0.017 \ 0.067 \ 0.287 \ 0.000 \ 0.336]$.

QUESTIONS

- 10.1. Describe the events that might trigger a rebalancing of a portfolio.
- 10.2. “If the portfolio manager needs to sell a fraction of the portfolio to meet the cash outflow demand and if there is no change in any parameters of the model, then the portfolio manager does not have any reason to change the portfolio weights.” Is this statement correct?
- 10.3. Consider a portfolio consisting of 100 stocks with weights given by $w_1, \dots, w_{100}$.
 - (a) If the one-hundredth stock drops out of the portfolio and if the portfolio manager intends to keep the relative weights unchanged for the remaining 99 stocks, what would be the new weights of the remaining stocks?

- (b) If the portfolio manager needs to add one more stock to the portfolio and makes its weight to be w_{101} , what would be the new weights of the original 100 stocks?
- 10.4.** Suppose that we estimated an economic factor model using monthly data.
- (a) From the estimates of the model, it is possible to predict the annual stock returns by multiplying the predicted return by 12. What is the implicit assumption that allows us to make this inference?
- (b) Explain why it is preferable that the model periodicity be identical to the rebalancing frequency.
- 10.5.** Consider an economic factor model
- $$r_{it} = \alpha_i + \beta_i f_t + \epsilon_{it}$$
- Using the observations from $t = 1$ to $t = T$, we obtained $\hat{\alpha}_i^{(T)}$ and $\hat{\beta}_i^{(T)}$. Using one more observation, i.e., from $t = 1$ to $t = T + 1$, we obtained $\hat{\alpha}_i^{(T+1)}$ and $\hat{\beta}_i^{(T+1)}$. We would like to test whether the parameters changed.
- (a) Can we say that $\hat{\beta}_i^{(T)}$ and $\hat{\beta}_i^{(T+1)}$ are independent?
- (b) What is the covariance between $\hat{\beta}_i^{(T)}$ and $\hat{\beta}_i^{(T+1)}$?
- (c) Find the formula for the t -statistic when the null hypothesis is that β did not change.
- 10.6.** Evaluate the dividend reinvestment plan offered by certain companies in the context of transactions costs.
- 10.7.** What is the bid–ask spread? Why is it difficult to observe the bid–ask spread in reality?
- 10.8.** On the New York Stock Exchange, designated market makers (DMMs) offer their own bid and ask quotes. Explain whether DMMs would increase or decrease their bid–ask spreads in each of the following situations:
- (a) Transaction volume is decreasing.
- (b) More and more limit orders arrive.
- (c) The company is about to announce earnings.
- 10.9.** Explain the various components of transactions costs, including implicit costs as well as explicit costs.

- 10.10.** Explain why transactions costs may be higher for small stocks than for large stocks.
- 10.11.** One simple way to control transactions costs is to determine what percentage of the *average daily trading volume* (ADTV) is acceptable as a weight for any stock in the portfolio. Let's denote the vector of allowable ranges for all stocks as $\mathbf{w}_{\text{ADTV}}$. Write down a problem for finding an optimal portfolio considering these critical weights.
- 10.12.** To set up an optimal portfolio problem considering transactions costs, we need to adjust the expected return of stocks and calculate the transactions-cost-adjusted expected return. Suppose that a \$1 transaction creates 5 cents in transactions costs.
- (a) If stock A 's monthly expected return is 1%, what would be the monthly transactions-cost-adjusted expected return?
 - (b) If stock A 's annual expected return is 15%, what would be the annual transactions-cost-adjusted expected return?
 - (c) Explain why we would subtract the same percentage from the expected return to account for the transactions costs regardless of whether the return is monthly or annual.
- 10.13.** Let w_i^b denote the weight of stock i in the existing portfolio and w_i^a denote the weight of stock i in the new portfolio to be created. Let c_i denote the transactions cost of stock i .
- (a) If μ_i is the expected return of stock i , what is the transactions-cost-adjusted expected return of stock i ?
 - (b) What is the marginal effect of w_i^a on the transactions-cost-adjusted expected return? Explain why this marginal effect can be positive.
- 10.14.** If you hold a stock whose transactions cost is high, it is likely that your *future* transactions costs will be high because you will have to sell the stock eventually. This fact implies that accounting for the *current* transactions costs only is not enough. How should the optimal portfolio problem be modified to account for the future transactions costs?
- 10.15.** Explain the advantage of futures and ETFs in controlling transactions costs.

¹ One such model is the structural model estimated from U.S. equity data by Almgren, Thum,

$$\text{Haphtmann, and Li (2005) } c_{it} \approx \frac{I}{2} + \text{sgn}(n_{it})\eta\sigma_{it}\left|\frac{n_{it}}{V_{it}T}\right|^{3/5}, \text{ where } I = \gamma\sigma_{it}\frac{n_{it}}{V_{it}}\left(\frac{N_{it}}{V_{it}}\right)^{1/4},$$

$\gamma = 0.314$, $\eta = 0.142$, σ_{it} is the daily volatility of the returns of stock i at the beginning of month t ; N_{it} is the total amount of shares outstanding in the security; V_{it} (a.k.a. ADTV) is the average daily trading volume of the stock (shares traded, not dollars traded); T is the time interval in which the trade takes place in number of days (in the example that follows, we use $T = 1$); and n_{it} represents the number of shares of security i that the portfolio is trading. Another model is the market impact model of Northfield. The model is of the form $c_{it} = B_{it}|n_{it}| +$

$C_{it}|n_{it}|^{0.5}$, where B_{it} and C_{it} are parameters estimated by Northfield, n_{it} is the number of shares to be purchased for security i in month t , and c_{it} is expressed in terms of percentage price movement. To give an example of how the first cost model works, let's look at the data from two stocks in December 2013: AT&T (ticker symbol: T), a very liquid stock, and AGL Resources (ticker symbol: GAS), a less liquid stock. AT&T for this particular period had a market capitalization of \$183 billion, a stock price of \$35.16, and a 10-day average daily trading volume of 18,930,000 shares. The spread was 1 cent, or a 0.0284% spread. The trading cost in percentage terms for a 1% position in a \$500 million portfolio was 0.0232%. That is, a \$5 million trade of AT&T representing 142,000 shares would cost the trader \$1,160. This does not represent commissions; it is simply the market impact and spread costs. AGL Resources for this particular period had a market capitalization of \$5.6 billion, a stock price of \$47.23, and a 10-day average daily trading volume of 490,000 shares. The spread was 2 cents, or a 0.0423% spread. The trading cost in percentage terms for a 1% position in a \$500 million portfolio was 0.1621%. That is, a \$5 million trade of AGL Resources representing 105,865 shares would cost the trader \$8,105.

² Furthermore, in the case of our example, transactions cost model 1 was not usable as of 2018 even in the leading software provider platforms for portfolio optimization, such as Axioma, Northfield, and BARRA.

³ This technique was proposed by Chincarini (2017).

⁴ This is done by computing the dollar amount of a trade for a given weight and portfolio size. Thus, for a 1% position with a portfolio size of \$100 million, the trade size would be \$1 million. Then, dividing by the price of the stock in that period, we obtain the number of shares required to be traded in that stock. This is used in the transactions cost model to obtain the market impact costs in percentage terms. Most transactions cost models require some sort of number of shares to be traded to estimate the transactions costs.

⁵ There is a different a_{it} and b_{it} for every net asset level since the transactions costs of each stock vary with assets under management. $\tilde{tc}_{it}$ can also be thought of as net transactions cost as it represents the percentage transactions cost of each stock multiplied by the stock's weight, w_{it} , reflecting the net transactions cost impact of each stock at each weight to the entire portfolio.

⁶ The Weierstrass approximation theorem states that every continuous function defined on a closed interval $[a, b]$ can be uniformly approximated as closely as desired by a polynomial function.

In the case of many common transactions cost models, a quadratic function is sufficient for a very good approximation. The $\bar{R}^2$ for stocks is typically very close to 1.

⁷ The regression estimates for this particular stock in this particular month were $\hat{a} = 0.0206$, $\hat{b} = 0.5096$, and $\bar{R}^2 = 0.9999$.

⁸ For example, CPLEX's CPLEXQCP.

⁹ Under certain circumstances, a portfolio manager may wish to create a portfolio that maximizes after-transactions cost returns while achieving a certain target variance. The market impact optimization for a long-only portfolio can be modified to achieve this. In this particular case, we assume that the portfolio manager has chosen his other constraints for $\mathbf{A}$ and $\mathbf{b}$ and modifies the remaining matrices to achieve his goal as follows:

$$\mathbf{Q} = \begin{bmatrix} \hat{b}_1 & 0 & \dots & 0 \\ 0 & \hat{b}_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \hat{b}_N \end{bmatrix} \quad (10C.8)$$

$$\mathbf{Q}^* = 2\mathbf{\Sigma} \quad (10C.9)$$

where $\mathbf{\Sigma}$ is the variance-covariance matrix of stock returns; $\mathbf{l} = \mathbf{0}$, $\mathbf{c} = -\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - \hat{\mathbf{a}}$, and $r = \sigma_P$, where $\tilde{\boldsymbol{\mu}}$ is a vector of the expected returns of each stock minus the constant estimate in the transactions cost regression; $\boldsymbol{\mu}$ is the expected return of each stock; σ_P is the target volatility for the portfolio to match; and $\hat{b}_i$ is the coefficient estimate from the transactions cost regression for stock i .

¹⁰ Under certain circumstances, a portfolio manager may wish to create a portfolio that maximizes after-transactions cost returns while achieving a certain target variance. The market impact optimization for a market-neutral portfolio can be modified to achieve this. In this particular case we assume that the portfolio manager has chosen his other constraints for $\mathbf{A}$ and $\mathbf{b}$ and modifies the remaining matrices to achieve his goal as follows:

$$\mathbf{Q}^* = \begin{bmatrix} 2\mathbf{\Sigma}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix} \quad (10C.14)$$

$$\mathbf{Q} = \begin{bmatrix} \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{\Sigma}_{2, N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{\Sigma}_{2, N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix} \quad (10C.15)$$

where $\mathbf{\Sigma}$ is the variance-covariance matrix of stock returns, $\mathbf{\Sigma}_2$ is defined in Eq. (10C.12), $\mathbf{l} = \mathbf{0}$, $\mathbf{c} = -\tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\mu}}' = [\boldsymbol{\mu}_{1 \times N} \mathbf{1} \cdot \hat{\mathbf{a}}_{1 \times N} \mathbf{1} \cdot \hat{\mathbf{a}}_{1 \times N} \mathbf{0}_{1 \times N} \mathbf{0}_{1 \times N}]$, and $\mathbf{r} = -\sigma_P$, where $\tilde{\boldsymbol{\mu}}$ is a $(1 \times 5N)$ vector of the expected returns of each stock, the constant estimates in the transactions cost regression, and zeros, $\boldsymbol{\mu}$ is the expected return vector, σ_P is the after-transactions cost risk for the portfolio to match, and $\hat{\mathbf{a}}_{1 \times N}$ is the vector of coefficients on w_i in the regression for stock i . The main adjustment is that we switch the matrices $\mathbf{Q}$ and $\mathbf{Q}^*$.

¹¹ We must divide by 0.10 since the formula is for $\tilde{tc}_i = tc_i \cdot w_i$.

CHAPTER 11

Tax Management

It is the mark of an educated man to be able to entertain a thought without accepting it.

—Aristotle

11.1 INTRODUCTION

A study by James Garland on taxable investing concluded that “Those who stand to lose the least—to taxes and fees—stand to win the most when the game’s all over.”¹ In the game of investing, the winners are the ones who know not only how to earn a lot but also how to hold onto what they earn. Many investment managers have forgotten to apply this seemingly obvious lesson. From 1996 to 2020, the typical mutual fund lost approximately 24% of its gross return to the government in taxes.² The typical fund manager undoubtedly put plenty of effort into managing his or her fund, but he or she got to keep only a little over half of what he or she earned for his or her efforts.

By devising a solid tax management strategy, the typical manager could have kept more earnings in the portfolio and out of Uncle Sam’s reach. Taxes are by no means a constant in the investment world. A manager’s investment decisions can alleviate or exacerbate the tax burden on the portfolio, with dramatic consequences for the investment return. The manager can take advantage of two types of tax management techniques, passive and active. *Passive* techniques focus on minimizing the tax burden

when the portfolio manager must generate cash outflows. Three major passive techniques exist: dividend-income management, tax-lot management, and capital gain and loss management. *Active* techniques aim to create extra tax advantages whether the portfolio manager needs to generate cash outflows or not. [Table 11.1](#) provides an overview of tax management techniques.

TABLE 11.1

Overview of Tax Management Techniques

Category	Name of Technique	Description
Passive	Dividend management	How to create the portfolio to have the right dividend yield desired by the investor
	Tax-lot management	Which tax lots to sell given that a withdrawal of cash is necessary and a specific stock is chosen
	Capital gain/loss management	Which stocks to sell when it is necessary for a cash withdrawal
Active	Loss harvesting	How to rebalance the portfolio when the amount to be withdrawn is zero

We start this chapter by describing the principles underlying all tax management techniques. It is necessary to study these principles carefully to understand tax management tools properly. After studying the principles of tax management, we discuss each of the tax management techniques listed in [Table 11.1](#). The techniques are most relevant to U.S.-taxable investors, but investors in international equities and investors not subject to U.S. tax laws still will find the discussion useful because the same basic tax structure exists in many countries, and the principles of tax management apply everywhere.³

11.2 DIVIDENDS, CAPITAL GAINS, AND CAPITAL LOSSES

To understand the principles of tax management, let us briefly review how investment income is taxed by the Internal Revenue Service (IRS). As far as

taxes are concerned, there are three important types of investment income: dividends, long-term capital gains/losses, and short-term capital gains/losses. Capital gains and capital losses are treated asymmetrically, so there are effectively five types of investment income.

Individual taxpayers used to pay taxes on dividends at the ordinary personal income tax rate.⁴ This meant that the government essentially double taxed dividends—once through the corporate income tax on the company distributing the dividends and again through the personal income taxes of the people receiving them. Congress tried to rectify the double-taxation problem in 2003.⁵ Under the new rule, the majority of dividends that individuals receive (called *qualified dividends*) are taxed at rates lower than their ordinary income tax rates. In 2020, dividend tax rates ranged from 0% to 23.8% depending on the recipient's income level. The new rule reduces the double-taxation effect, but it does not eliminate it completely. Financial economists have warned that corporations still have a tax incentive to retain earnings rather than pay dividends, so investors should watch out for further changes in the dividend tax structure.

Unlike individuals, corporations that receive dividends can deduct some of them from taxable income.⁶ For the 2020 tax year, corporations generally could deduct up to 50%, and, in some cases, up to 65%, of dividends received.⁷ The remainder of the dividend is taxed at the corporate income tax rate. Given the average corporate income tax rate of 26% (including state and federal taxes), the effective tax rate on dividends to most corporations is about 13%. The reason for allowing a dividend deduction is to avoid triple taxation of corporate income. Dividend-paying corporations pay corporate income tax, and individuals who receive dividends pay personal income tax. If dividend-receiving corporations—which redistribute the dividends they receive to their own shareholders—had to pay full taxes, then the dividend income would be taxed three times!

An important thing to remember about dividend taxation is that the same taxes are levied whether dividends are reinvested or not. This is one rather unfair element of the dividend tax system. Dividend recipients who immediately reinvest in the same stock end up paying taxes for nothing. If the corporation had not paid the dividend and had instead kept it as retained earnings, the stock would have grown as much, and because no

dividend tax would have been owed, investors' after-tax returns would have been higher.

As far as taxes are concerned, individual investors do not gain from dividends. Those who reinvest the dividends pay taxes for nothing. Those who keep the dividends could have generated the same amount of cash without owing dividend taxes simply by selling part of their holdings in the stock. Since any investor can create this sort of "synthetic dividend" when he or she wants cash, he or she really has no need for a dividend payment from the corporation.

When an individual investor sells shares in a stock, he or she avoids dividend taxes but does potentially incur capital gains taxes. Capital gains taxes apply to profit realized from the sale of a stock.⁸ Capital gains on paper, or "paper profit," incur no taxes. For example, if an investor bought 100 shares of XYZ stock at \$10 per share and still holds the stock at the current price of \$20, then the investor has an unrealized capital gain of \$1,000 [= $100 \cdot (\$20 - \$10)$]. This gain is not taxed until the investor sells the stock.

The tax system distinguishes between capital gains that are long term (holdings over a year) and short term (holdings under a year). Short-term capital gains are taxed at the ordinary personal income tax rate. Long-term capital gains receive preferential treatment. For the 2020 tax year, the maximum tax rate applicable to long-term capital gains was 20%. For low-income individuals, the rate was 0%. Thus, for individuals who paid 37% in personal income tax, the short-term capital gains tax was almost twice the long-term capital gains tax. Why does the government give preferential treatment to long-term gains? It may be an attempt to discourage "speculative investing" by rewarding investors who leave their money in one place for a while.

Capital losses also affect taxes. Like capital gains, they are classified as long term or short term. If an individual taxpayer has a long-term capital loss, it is deducted first from long-term capital gains and then from any short-term capital gains. If the individual's capital loss is short term, it is deducted first from short-term capital gains and then from any long-term capital gains. All capital losses, whether long term or short term, that remain after these deductions are deducted from other types of income.

Corporations' capital losses are treated similarly except that they cannot be deducted from other types of income even if they exceed capital gains. If

capital losses exceed capital gains, then the difference is deducted from past capital gains and then from future capital gains. In all cases, capital losses can be carried into the future to offset future capital gains.

11.3 PRINCIPLES OF TAX MANAGEMENT

An investor who knows all the different places that taxes could hit him or her can figure out ways to lessen the impact of the blows. There are four principles of tax management to keep in mind when devising such strategies:

1. Individual taxpayers should avoid dividend income, but corporations can handle it more easily.
2. It is better to realize a long-term capital gain than a shortterm capital gain.
3. It is better to realize a capital loss than a long-term capital gain.
4. All investors should delay the realization of a capital gain and expedite the realization of a capital loss.

The first principle draws on what we know about dividend taxes on individual investors versus corporations. Dividend taxes are especially burdensome to individuals who do not plan to withdraw cash from their portfolios, but even individuals who want the cash could generate it on their own and avoid the taxes. Individual investors therefore should avoid dividend payments. Corporations, on the other hand, should prefer them to capital gains (as long as they need cash). Corporations pay lower taxes on dividends than on capital gains, so dividends provide a relatively cheap stream of cash.

The second principle simply reflects the fact that the long-term capital gains tax rate is lower than the short-term capital gains tax rate. The third principle is based on the fact that no matter whether they are short term or long term, capital gains increase taxable income, whereas capital losses reduce it.

The fourth principle relies on the concept of the time value of money, which is that a dollar today is worth more than a dollar tomorrow. As long as the interest rate is positive, you can invest a dollar today and end up with a dollar plus interest tomorrow. By expediting the realization of a capital

loss, you are decreasing your taxes and giving yourself extra dollars to invest today. By delaying the realization of a capital gain, you are postponing your taxes and giving yourself an extra day to invest your dollars.

How much money could an investor save by following the principles of tax management? Table 11.2 shows the potential savings. Realizing a long-term capital gain of \$1 rather than a short-term gain of \$1 saves 17 cents (for an investor in the 37% income tax/20% long-term capital gains tax bracket). Replacing the long-term capital gain of \$1 with a long-term capital loss of \$1 saves 40 cents.⁹ Delaying a \$1 long-term capital gain for one year postpones the payment of 20 cents in taxes for one year. At an average investment return of 10%, this equals 2 cents in savings. However, the extra 2 cents also will be taxed at the long-term capital gains rate, thus leaving the investor with only 1.6 of his or her 2 cents. However, this savings is measured in tomorrow's terms. Thus we need to discount this savings to the present day using the after-tax return of 8%. This leaves us with 1.48 cents. Thus, the total savings is really 1.48 cents. Similarly, realizing \$1 of capital loss today rather than next year saves about 1.48 cents.¹⁰

TABLE 11.2

Savings from the Principles of Tax Management

Investment Income	Taxes	Savings
Short-term capital gain of \$1	37 cents	
Long-term capital gain of \$1	20 cents	17 cents
Long-term capital gain of \$1	20 cents	
Long-term capital loss of \$1	−20 cents	40 cents
Long-term capital gain of \$1 now	20 cents	
Long-term capital gain of \$1 a year from now	18.52 cents	1.48 cents
Long-term capital loss of \$1 a year from now	−18.52 cents	
Long-term capital loss of \$1 now	−20 cents	1.48 cents

Note: We assumed a personal income taxpayer whose ordinary income is taxed at the rate of 37% and long-term capital gain is taxed at the rate of 20%. We also assumed that the average investment return is approximately 10%. Most of the calculations are straightforward. For the final four rows, recognize that the gains on the accrued earnings also will be taxed, which results in the difference. Taxes on the long-term capital gain of \$1 a year from now is the present value of the taxes imposed on the investor if the investor delayed paying taxes on the \$1 current capital gain by one year.

11.4 DIVIDEND MANAGEMENT

The first principle of tax management states that individual investors taxed at personal income tax rates should buy low-dividend stocks. Corporations, on the other hand, should prefer high-dividend stocks rather than stocks with high capital appreciation and low dividends. Creating a portfolio with the right level of dividend yield is called *dividend management* or, sometimes, *yield management*.

As we explained earlier, personal-taxable investors do not benefit at all from dividends. If they need to withdraw cash from a portfolio, they can create synthetic dividends for themselves by selling a part of the portfolio. If they do not need to withdraw cash from the portfolio, then dividends only create an unnecessary tax burden.

Avoiding dividend-paying stocks completely is one approach that a portfolio manager may adopt. To do so, the portfolio manager simply needs to add the “no-dividend constraint” to the optimization problem. Given the dividend rates of stocks $\mathbf{d} = (d_1, \dots, d_N)$ and the weights of stocks in the portfolio $\mathbf{w} = (w_1, \dots, w_N)$, the dividend rate of the portfolio is the weighted average of the dividend rate of stocks $\sum_{i=1}^N w_i d_i$ or, in vector notation, $\mathbf{w}'\mathbf{d}$. Then the no-dividend constraint can be written as

$$\mathbf{w}'\mathbf{d} = 0 \quad (11.1)$$

Let $\boldsymbol{\mu}$ be the vector of the expected stock returns, Σ the variance-covariance matrix of the stock returns, and A the risk aversion parameter. Then, to find an optimal portfolio, we solve

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} - A\mathbf{w}'\Sigma\mathbf{w} \quad (11.2)$$

subject to the constraint that the weights should sum to 1, that is,

$$\mathbf{w}'\mathbf{1} = 1 \quad (11.3)$$

and the no-dividend constraint in [Eq. \(11.1\)](#) and other constraints.

While finding an optimal portfolio of non-dividend-paying stocks is feasible, it is not the best we can do. We might end up missing stocks that

would lead to the true optimal portfolio. For instance, there may be dividend-yielding stocks that the quantitative model predicts will have very high α 's. A better approach than excluding dividend-paying stocks is to assign them some penalty. Given the annual dividend rate d_i of stock i , investing \$1 in stock i will generate an annual dividend of d_i . If the dividend tax rate is τ_d , then the tax burden becomes $\tau_d d_i$. If the weight of stock i in the portfolio is w_i , then the portfolio return will be reduced by $\tau_d w_i d_i$. Considering all the stocks, the portfolio return will be reduced by $\sum_{i=1}^N \tau_d w_i d_i$ or, in vector notation, $\tau_d \mathbf{w}' \mathbf{d}$. This reduction in the expected return is the appropriate penalty for including dividend-paying stocks.¹¹ Thus we may subtract this amount from the risk-adjusted return of the optimization problem in Eq. (11.2):

$$\begin{aligned} \text{Effective risk-adjusted return} &= \mathbf{w}' \boldsymbol{\mu} - A \mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} - \tau_d \mathbf{w}' \mathbf{d} \\ &= \mathbf{w}' (\boldsymbol{\mu} - \tau_d \mathbf{d}) - A \mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \end{aligned} \quad (11.4)$$

The new optimization problem is to maximize the effective risk-adjusted return in Eq. (11.4) subject to Eq. (11.3).

Now let us consider corporations, or corporate-taxable investors. Corporate-taxable investors would prefer dividends as long as the dividend income does not exceed the amount of the required cash needs.¹² If no cash withdrawal is required, dividends are actually as bad for corporate-taxable investors as they are for personal-taxable investors. Thus it makes sense to match the dividend-income schedule to the corporation's cash-withdrawal schedule. Suppose that the dividend rate of d_p would create just enough dividends to cover the cash-outflow requirement. Given the dividend rates of stocks $\mathbf{d} = (d_1, \dots, d_N)$ and the weights of stocks in the portfolio $\mathbf{w} = (w_1, \dots, w_N)$, the dividend rate of the portfolio is a weighted average of the dividend rate of the stocks $\sum_{i=1}^N w_i d_i$ or, in vector notation, $\mathbf{w}' \mathbf{d}$. To achieve the desired dividend rate, we may add the following "target dividend constraint" to the optimization problem in Eq. (11.2):

$$\mathbf{w}' \mathbf{d} = d_p \quad (11.5)$$

That is, the new optimization problem is to solve Eq. (11.2) subject to Eqs. (11.3) and (11.5).

Depending on the target dividend rate, the preceding problem may not have a solution. If the target dividend rate is too high and the investment universe does not include enough high-yield stocks, then the problem will not have a solution. And even if the problem has a solution, the resulting portfolio may not be truly optimal; we could have created a better portfolio with a different dividend rate. We therefore generalize the target dividend constraint into a penalty for deviating from the target.

If the dividend income is less than the cash-withdrawal requirement, then a part of the portfolio should be sold to generate cash. If the required cash-withdrawal rate is d_p and the portfolio dividend rate is $\mathbf{w}'\mathbf{d}$, then $(d_p - \mathbf{w}'\mathbf{d})$ dollars should be sold for each current dollar. Although we cannot determine the exact tax consequences of this sale, we can make an approximation. If the long-term capital gains tax rate of τ_l applies, then the tax burden will be $\tau_l(d_p - \mathbf{w}'\mathbf{d})$ dollars for each dollar invested now. Given that dividend income is taxed at the rate of τ_d , the deviation from the target dividend rate increases taxes by $(\tau_l - \tau_d)(d_p - \mathbf{w}'\mathbf{d})$ dollars for each dollar invested. This is in fact the reduction in the expected return and can be considered the proper penalty for the deviation from the target dividend. Thus we may subtract this amount from the risk-adjusted return of the optimization problem in Eq. (11.2):

$$\begin{aligned} \text{Effective risk-adjusted return} &= \mathbf{w}'\boldsymbol{\mu} - A\mathbf{w}'\Sigma\mathbf{w} - (\tau_l - \tau_d) \\ &\quad (d_p - \mathbf{w}'\mathbf{d}) \\ &= \mathbf{w}' [\boldsymbol{\mu} + (\tau_l - \tau_d)\mathbf{d}] - A\mathbf{w}'\Sigma\mathbf{w} \\ &\quad - (\tau_l - \tau_d)d_p \end{aligned} \tag{11.6}$$

This effective risk-adjusted return is valid only if the dividend rate is less than or equal to the cash-withdrawal rate. Too many dividends are as bad for corporate-taxable investors as they are for personal-taxable investors. Thus we may add the restriction that the portfolio dividend rate $\mathbf{w}'\mathbf{d}$ is less than or equal to the cash-withdrawal rate d_p , that is,

$$\mathbf{w}'\mathbf{d} \leq d_p \tag{11.7}$$

The new optimization problem is to maximize the effective risk-adjusted return in Eq. (11.6) subject to Eqs. (11.3) and (11.7).

11.5 TAX-LOT MANAGEMENT

The second and third principles of tax management imply that investors should realize all capital losses before any capital gains, and then they should realize all long-term capital gains before any short-term ones. The method for prioritizing the realization of losses and gains is called *tax-lot management* (or *tax-lot accounting*). For the discussion of tax-lot management, we assume that the portfolio manager already has determined the stocks and quantity of shares to sell from the optimization.

Let's walk through a simple example to understand the mechanics of tax-lot management. Suppose that you purchased 30 shares of XYZ stock on six different occasions, as shown in Table 11.3, and that you want to sell 10 shares of this stock today. Assume that today's date is March 1, 2020, and that the stock currently trades at \$20. Does it matter which shares out of the 30 shares you sell? From the IRS's point of view, it does. Different shares came at different purchase prices on different dates. If you sell shares that you bought on June 5, 2018, then you will realize capital gains and incur long-term capital gains taxes. If you sell shares that were bought on January 10, 2018, then you will realize long-term capital losses and reduce your taxable income. On other shares, you may realize a short-term capital gain or a short-term capital loss. A different tax bill accompanies each type of gain or loss.

TABLE 11.3

Tax Lots of Stock XYZ

Date of Purchase	Number of Shares	Purchase Price	Gain/Loss	Long Term/Short Term	Tax Burden
Jan 10, 2018	5	\$22	−\$2	Long	−\$0.40
Jun 5, 2018	5	\$18	\$2	Long	\$0.40
Feb 12, 2019	5	\$24	−\$4	Long	−\$0.80
Mar 16, 2019	5	\$23	−\$3	Short	−\$1.11
Dec 19, 2019	5	\$19	\$1	Short	\$0.37
Jan 24, 2020	5	\$21	−\$1	Short	−\$0.37

Note: The current price is assumed to be \$20 and the current date March 1, 2020. For the purpose of calculating the tax burden, we used a 20% long-term capital gains tax rate and a 37% short-term capital gains tax rate.

For purposes of calculating the tax bill, each purchase history is called a *tax lot*. You have six tax lots to choose from when you want to sell the 10 shares of XYZ stock. There are two rules for minimizing the tax burden¹³:

1. Sell the shares of the tax lot with the maximum loss first. Continue to sell the tax lots in descending order of tax loss until all tax lots with losses are sold.
2. For each of the remaining tax lots, calculate the tax burden by applying the appropriate capital gains tax rate (long term or short term). Sell the shares of the tax lot with the minimum tax burden first.

In [Table 11.3](#) we have calculated the potential tax burden for each tax lot assuming a 20% long-term capital gains tax rate and 37% short-term capital gains tax rate. According to our two rules, we would sell the tax lot of February 12, 2019, and the tax lot of March 16, 2019, in order to sell the 10 shares. If further sales were necessary, we would then proceed to the tax lots of January 10, 2018, January 24, 2020, December 19, 2019, and June 5, 2018.

More and more brokerages practice tax-lot management automatically for clients, so portfolio managers may not have to worry about this too much.¹⁴ Nonetheless, the portfolio manager needs to be aware of it and understand the logic behind it.

11.6 TAX-LOT MATHEMATICS

Delving into the specifics of tax-lot accounting generally requires use of an elaborate mathematical notation. The notation must keep track of three elements: the stock, the time when a tax lot was purchased, and the current time. To avoid triple subscripts, we have developed a new, streamlined notation for tax-lot accounting.

Instead of denoting the current time using a subscript, we distinguish only between the existing portfolio and the new portfolio using a superscript. We use superscript b (“before”) to indicate the existing portfolio and superscript a (“after”) to indicate the new portfolio. Thus w_i^b is the weight of stock i in the existing portfolio, whereas w_i^a is the weight of stock i in the new portfolio. Similarly, s_i^b is the number of shares of stock i in the existing portfolio, whereas s_i^a is the number of shares of stock i in the new portfolio. V^b indicates the dollar value of the existing portfolio, and V^a indicates the dollar value of the new portfolio. $V^a - V^b$ gives the net cash inflow. V^a and V^b will differ only when there is cash inflow or outflow.

In order to distinguish between tax lots, we refer to each lot by a combination of a stock indicator i and a time indicator t . That is, tax lot (i, t) refers to the shares of stock i purchased at time t . Stock indicator i varies from 1 to N , where N is the total number of stocks in the investment universe. The time indicator t varies from 1 to T , where T denotes the current time.

Quantities of a particular tax lot are now denoted by adding two subscripts. We will also place a tilde ($\sim$) above a letter to clarify that the notation refers to a tax lot. For instance, $\tilde{w}_{i,t}^b$ is the weight of tax lot (i, t) in the existing portfolio, and $\tilde{w}_{i,t}^a$ is the weight of tax lot (i, t) in the new portfolio. $\tilde{s}_{i,t}^b$ is the number of shares in tax lot (i, t) in the existing portfolio, and $\tilde{s}_{i,t}^a$ is the number of shares in tax lot (i, t) in the new portfolio. Time T indicates the current time, so $\tilde{s}_{i,T}^a$ is the number of shares of stock i to buy for the new portfolio, and $\tilde{w}_{i,T}^a$ is the corresponding weight ($\tilde{s}_{i,T}^b$ and $\tilde{w}_{i,T}^b$ should equal zero). Vectors are defined in a similar manner. For example, $\mathbf{w}^a = (w_1^a, \dots, w_N^a)$ is the vector of stock weights in the new portfolio,

whereas $\tilde{\mathbf{W}}^a = (\tilde{w}_{1,1}^a, \dots, \tilde{w}_{1,T}^a, \dots, \tilde{w}_{N,1}^a, \dots, \tilde{w}_{N,T}^a)$ is the vector of tax-lot weights in the new portfolio.

There is a simple relationship between tax-lot quantities and stock quantities. The sum of the tax-lot weights of one stock equals the stock's weight in the portfolio, and the sum of the tax-lot shares of one stock equals the total number of shares of that stock in the portfolio. That is,

$$w_i^b = \sum_{t=1}^T \tilde{w}_{i,t}^b, \quad w_i^a = \sum_{t=1}^T \tilde{w}_{i,t}^a \quad (11.8)$$

and

$$s_i^b = \sum_{t=1}^T \tilde{s}_{i,t}^b, \quad s_i^a = \sum_{t=1}^T \tilde{s}_{i,t}^a \quad (11.9)$$

11.7 CAPITAL GAIN AND LOSS MANAGEMENT

The third principle of tax management is the driving force behind capital gain and loss management. Realizing a capital loss reduces the tax burden, whereas realizing a capital gain increases it. Therefore, to minimize the tax burden, the portfolio manager should sell stocks that have experienced the greatest capital losses or the smallest capital gains. Capital gain and loss management is a systematic approach to identifying the stocks to sell.

In a sense, capital gain and loss management generalizes the concept of tax-lot management. In the tax-lot management section, we did not consider which stocks to sell because we assumed that the manager had already chosen them. Capital gain and loss management provides a method for first determining which stocks to sell and then zeroing in on which specific tax lots should go.

To make the problem tractable, we take the portfolio's cash outflows net of dividend income as a given. We also restrict ourselves to selling and do not allow any buying. Using the notation explained in the preceding section, the portfolio manager needs to generate $V^b - V^a$ dollars of cash by selling

some of the current holdings. What the portfolio manager can do is to sort the tax lots of all stocks according to potential capital loss and select the tax lots with the highest potential capital loss first until the cash-outflow requirement is met. The potential capital gain of tax lot (i, t) is $\tilde{S}_{i,t}^b (p_{i,T} - p_{i,t})$, where $p_{i,T}$ is the current price of stock i , $p_{i,t}$ is the purchase price, and $\tilde{S}_{i,t}^b$ is the number of shares in the tax lot in the existing portfolio. We can sort all tax lots in descending order of potential capital gains (i_1, t_1) , (i_2, t_2) , ... so that

$$p_{i_1,T} - p_{i_1,t_1} \leq p_{i_2,T} - p_{i_2,t_2} \leq \dots \quad (11.10)$$

Then we choose M tax lots (i_1, t_1) , (i_2, t_2) , ..., (i_M, t_M) from the beginning of the list until the cash-outflow requirement is met so that

$$\tilde{S}_{i_1,t_1}^b p_{i_1,T} + \dots + \tilde{S}_{i_M,t_M}^b p_{i_M,T} = V^b - V^a \quad (11.11)$$

While this approach certainly will minimize the tax gain, the cash outflow will alter the portfolio's risk profile and even may throw the portfolio out of optimal balance. A better approach would be to include the tax consequence as an additional constraint in the optimization problem. To do so, let us calculate the tax burden more explicitly. Suppose that the long-term capital gains tax rate is τ_l and that the short-term capital gains tax rate is τ_s . Suppose also that we sell $\tilde{S}_{i,t}^b - \tilde{S}_{i,t}^a$ shares from tax lot (i, t) . Then the tax burden for tax lot (i, t) is calculated as follows:

$$\tau_{i,t} (p_{i,T} - p_{i,t}) (\tilde{S}_{i,t}^a - \tilde{S}_{i,t}^b) \quad (11.12)$$

where $\tau_{i,t} = \tau_l$ if $T - t$ is greater than one year and $\tau_{i,t} = \tau_s$ if $T - t$ is less than one year. We can reformulate the above in terms of the weight as follows:

$$\begin{aligned}
& \tau_{i,t} (p_{i,T} - p_{i,t}) \left(\frac{V^b \tilde{w}_{i,t}^b}{p_{i,T}} - \frac{V^a \tilde{w}_{i,t}^a}{p_{i,T}} \right) \\
&= \tau_{i,t} \left(\frac{p_{i,T} - p_{i,t}}{p_{i,T}} \right) (V^b \tilde{w}_{i,t}^b - V^a \tilde{w}_{i,t}^a)
\end{aligned} \tag{11.13}$$

In words, this formula shows that the tax burden is the product of the tax rate ($\tau_{i,t}$), the capital appreciation rate $[(p_{i,T} - p_{i,t})/p_{i,T}]$, and the amount of the sale ($V^b \tilde{w}_{i,t}^b - V^a \tilde{w}_{i,t}^a$). The total tax burden is the sum of the tax burden of each tax lot, and the total tax rate is the total tax burden divided by the portfolio value:

$$\text{Total tax rate} = \sum_{i=1}^N \sum_{t=1}^{T-1} \tau_{i,t} \left(\frac{p_{i,T} - p_{i,t}}{p_{i,T}} \right) \left(\frac{V^b}{V^a} \tilde{w}_{i,t}^b - \tilde{w}_{i,t}^a \right) \tag{11.14}$$

In vector notation,

$$\text{Total tax rate} = \boldsymbol{\tau}' \left(\frac{V^b}{V^a} \tilde{\mathbf{w}}^b - \tilde{\mathbf{w}}^a \right) \tag{11.15}$$

where $\boldsymbol{\tau}$, $\mathbf{w}^a$, and $\mathbf{w}^b$ are the vectors made of $\tau_{i,t} [(p_{i,T} - p_{i,t})/p_{i,T}]$, $w_{i,t}^a$, and $w_{i,t}^b$, respectively. Given the tax rate for creating the new portfolio, we can include the rate as a penalty in the objective function of the optimization problem in [Eq. \(11.2\)](#). That is, the effective risk-adjusted return can be modified:

$$\begin{aligned}
\text{Effective risk-adjusted return} &= \tilde{\mathbf{w}}^{a'} \tilde{\boldsymbol{\mu}} - A \tilde{\mathbf{w}}^{a'} \tilde{\boldsymbol{\Sigma}} \tilde{\mathbf{w}}^a - \boldsymbol{\tau}' \left(\frac{V^b}{V^a} \tilde{\mathbf{w}}^b - \tilde{\mathbf{w}}^a \right) \\
&= \tilde{\mathbf{w}}^{a'} (\tilde{\boldsymbol{\mu}} + \boldsymbol{\tau}) - A \tilde{\mathbf{w}}^{a'} \tilde{\boldsymbol{\Sigma}} \tilde{\mathbf{w}}^a - \frac{V^b}{V^a} \tilde{\mathbf{w}}^{b'} \boldsymbol{\tau}
\end{aligned} \tag{11.16}$$

Note that all the vectors and matrices including $\tilde{\mu}$ and $\tilde{\Sigma}$ are defined with respect to the *tax lots* rather than the stocks. To find an optimal portfolio, we have to maximize Eq. (11.16) with the constraint that the weights should sum to 1, that is,

$$\tilde{\mathbf{w}}' \mathbf{1} = 1 \quad (11.17)$$

and with the constraint that stock purchases are not allowed,¹⁵ that is,

$$\tilde{\mathbf{w}}^a \leq \frac{V^b}{V^a} \tilde{\mathbf{w}}^b \quad (11.18)$$

11.8 LOSS HARVESTING

The three tax management methods that we have discussed so far fall into the passive tax management category. The aim of these methods is to protect the portfolio's gains from taxes by minimizing the tax burden. A savvy investor or portfolio manager can go a step further and actually generate added gains from tax management. The active approach for doing this is called *loss harvesting*. It is based on the fourth principle of tax management, that realizing capital losses early produces potential tax savings. Investors should try to “harvest” as many capital losses as soon as possible.

Loss harvesting may generate an extra return for non-tax-related reasons as well. Loss harvesting is a kind of momentum investment strategy. The portfolio manager sells losing stocks and holds onto winning stocks. If stock prices exhibit positive autocorrelation over time, the winning stocks will do well again in the upcoming period, and the portfolio will earn higher returns. There is some empirical evidence that stock prices are positively autocorrelated over time, especially for periods of less than three months.

There are two loss-harvesting approaches that invest the proceeds from the sale of stock differently. Typical loss harvesting involves creating a new optimal portfolio. This method may not work properly if the *wash-sale rule* is in effect. The wash-sale rule prohibits an investor from purchasing a stock that he or she sold within the past 30 days from a tax perspective. The rule therefore would prevent an optimization solution that requires buying

back a stock that just was sold. In a more sophisticated approach, which we call *loss harvesting and characteristic matching*, the stocks sold are replaced with similar stocks without solving the optimization problem again. This avoids conflicts with the wash-sale rule and still maintains the portfolio at an approximately optimal level.

11.8.1 Loss Harvesting and Reoptimizing

To simplify the setup, let us suppose that there is no cash-outflow requirement and no cash inflow either. (Or suppose that we have already taken care of cash outflows and inflows.) Then the loss harvesting and reoptimizing strategy can be carried out in the following manner:

1. Sell all tax lots that have capital losses.
2. Reoptimize the portfolio without selling, and reinvest the proceeds of the sale from the first step.

The first step is straightforward. For each tax lot (i, t) , compare the purchase price $p_{i,t}$ with the current price $p_{i,T}$. If the purchase price is higher than the current price, then we sell this tax lot. That is, we will set

$$\tilde{w}_{i,t}^a = 0, \quad \text{if } p_{i,t} > p_{i,T} \quad (11.19)$$

where $\tilde{w}_{i,t}^a$ is the weight of tax lot (i, t) in the new portfolio. If we sell all tax lots with capital losses, then the proceeds from the sale will be $V \sum_{(i,t): p_{i,t} > p_{i,T}} \tilde{w}_{i,t}^b$ dollars, where $\tilde{w}_{i,t}^b$ is the weight of tax lot (i, t) in the original portfolio.

The second step is to reoptimize the portfolio with the proceeds of the sale, $V \sum_{(i,t): p_{i,t} > p_{i,T}} \tilde{w}_{i,t}^b$ dollars. To avoid reversing the tax consequences, we need to impose a “no sale” restriction. This restriction can be written as

$$\tilde{w}_{i,t}^a = \tilde{w}_{i,t'}^b \quad \text{if } p_{i,t} \leq p_{i,T} \quad (11.20)$$

Also, to avoid violating the wash-sale rule, we need to make sure not to buy the stocks that were just sold:

$$\tilde{w}_{i,T}^a = 0, \quad \text{if } p_{i,t} > p_{i,T}, \text{ for any } t = 1, \dots, T-1 \quad (11.21)$$

Two conditions, Eqs. (11.19) and (11.21), together completely determine the weights of the stocks that we sold. The only “free” weights we need to determine at this point are $\tilde{w}_{i,T}^a$ for which $p_{i,t} \leq p_{i,T}$ for all t .

Now let us express the optimization problem in terms of the stock weights. For those stocks affected by the loss harvesting, we have already determined the weights:

$$w_i^a = \sum_{t: p_{i,t} \leq p_{i,T}} \tilde{w}_{i,t}^a \quad (11.22)$$

For the stocks not affected by the loss harvesting, the only restriction imposed is

$$w_i^a \geq w_i^b \quad (11.23)$$

if no tax lot of stock i has a capital loss. Then, to find the optimal portfolio, we need to solve the optimization problem in Eq. (11.2) subject to Eqs. (11.3), (11.22), and (11.23).

11.8.2 Loss Harvesting and Characteristic Matching

In the preceding approach we reoptimized the portfolio after harvesting losses. However, this reoptimized portfolio may not really be the optimal portfolio because we had to abide by the no-wash-sale and no-realized-gain constraints. We actually may find a better portfolio if we drop the optimization completely and instead use the *characteristic-matching approach*. The characteristic-matching idea plays a role in the creation of factor models when we estimate factor exposures. Characteristic matching is based on the assumption that stocks with similar characteristics, such as size and book-to-market ratio, have similar expected returns and risk levels. If we sell stock *XYZ* to harvest losses and if we cannot buy stock *XYZ* again because of the wash-sale rule, then the best thing we can do is purchase a stock that is similar to *XYZ*.¹⁶

The loss-harvesting and characteristic-matching strategy can be implemented with the following steps:

1. Sell all tax lots that have capital losses.
2. Reinvest the proceeds of the sale into *similar* stocks.

The first step here is identical to the first step of regular loss harvesting. We sell all tax lots for which the current price $p_{i,T}$ is lower than the purchase price $p_{i,t}$. If stock j is affected by this loss harvesting, then the proceeds from the sale of stock j are

$$\text{Proceeds from the sale of stock } j = \sum_{t: p_{j,T} \leq p_{j,t}} p_{j,T} \tilde{s}_{j,t}^b \quad (11.24)$$

In the second step, we need to identify stocks similar to the ones we sold. Suppose that we want to find similar firms using L characteristics. We can calculate the Z-score for each of the L characteristics for all stocks. Given the Z-scores of stock i , $\mathbf{z}_i = (z_{i1}, \dots, z_{iL})$, and stock j , $\mathbf{z}_j = (z_{j1}, \dots, z_{jL})$, we can define the “distance” between two stocks by¹⁷

$$\text{Distance} = \sum_{k=1}^L |z_{ik} - z_{jk}| \quad (11.25)$$

The stock most similar to stock j is the one with the smallest distance from stock j . That is the stock to purchase using the proceeds from the sale of stock j .

11.8.3 Loss Harvesting with a Benchmark

Loss harvesting can be a powerful tool for the benchmark portfolio manager. In the first part of this section we showed how to create optimal portfolios through loss harvesting. The portfolio manager with a benchmark can create his or her tracking portfolio in the same way. He or she can add the two constraints in Eqs. (11.22) and (11.23) to the tracking-error-minimization problem.

Characteristic matching also can be adjusted for the presence of a benchmark. In picking stocks to replace the losers, the benchmark portfolio manager must consider whether characteristic matching would increase the portfolio's tracking error. To prevent the portfolio from straying further from the benchmark, the replacement stocks should have α 's and β 's as close to the benchmarks α and β as possible.

For each stock we run the regression of the stock return on the benchmark return:

$$r_i = \alpha_i + \beta_i r_B + \epsilon_i \quad (11.26)$$

The α and β of this equation are the benchmark α and benchmark β defined in [Chapter 2](#). Once we have obtained the benchmark α and benchmark β for each stock, we add them to each stock's Z-score vector. The last two elements of the characteristic vector $\mathbf{z}_i = (z_{i1}, \dots, z_{iL})$ are now the benchmarks α and β . We then determine the similarity between any pair of stocks by [Eq. \(11.25\)](#) as before.

This method preserves the relationship between the portfolio and the benchmark reasonably well, although it may not maintain the absolute lowest tracking error.

11.9 GAINS FROM TAX MANAGEMENT

We ran a simulation based on real data to show just how big the gains from tax management can be.¹⁸ In the simulation we compare two hypothetical investors. We call the first investor, the *naive* investor, and the second, the *t-smart* investor. The naive investor does not manage taxes efficiently; he simply invests in every stock in a sector index. He rebalances the portfolio every month to adjust for index entries and exits. The t-smart investor chooses the same portfolio as the naive investor initially. However, every month, the t-smart investor sells “loser” stocks (i.e., stocks that have capital losses) and replaces them with characteristically matched stocks from outside the index universe to minimize realized capital gains. We tested each investor's results for the investment period from July 1990 to June 2000.¹⁹ We repeated the simulation for each of the 18 different sector indexes.

Table 11.4 reports the before-tax annualized return, the aftertax annualized return, and the effective tax rate for each investor.²⁰ Each entry in the table is an average across 18 different sector simulations. The effective tax rate for the two investors differs by 4.82%. This means that the t-smart investor gains an extra 4.82% per year in after-tax portfolio return simply by practicing the tax management techniques we have discussed in this chapter.

TABLE 11.4

Comparison of Naive and T-Smart Investors

Investor	Pretax Return (%)	After-Tax Return (%)	Effective Tax Rate (%)
Naive investor	11.99	9.93	26.40
T-smart investor	16.68	14.41	21.58

As a second illustration of the power of tax management, we consider an alternative situation. This time, the two investors regularly withdraw cash from their portfolios by cashing out dividends. The naive investor is a buy-and-hold investor. He buys a portfolio at the beginning of the simulation period and holds it until the end of the simulation period. Each month, he cashes out dividends and pays taxes on those dividends. The t-smart investor buys the same portfolio as the naive investor at the beginning of the simulation period. Each month, she also withdraws the same amount of cash from her portfolio as the naive investor does. The difference, though, is that the t-smart investor tries to offset dividends with realized capital losses. She does so by selling losers and replacing them with characteristically matched stocks.

Table 11.5 reports the pretax and the after-tax returns, as well as the effective tax rates, for each investor. Each entry in the table is an average across 18 sector simulations with different cash withdrawal rates. The effective tax rates for the two investors differ by 4.69%. The t-smart investor gains 4.69% extra annual investment income from her tax management strategy.

TABLE 11.5

Comparison of Naive and T-Smart Investors, Regular Cash Withdrawal

Investor	Pretax Return (%)	After-Tax Return (%)	Effective Tax Rate (%)
Naive investor	13.44	11.33	24.06
T-smart investor	13.79	12.04	19.37

11.10 CONCLUSION

Portfolio managers who do not want to give away the fruits of their labor must practice effective tax management. In this chapter we discussed passive techniques for protecting the after-tax return of the portfolio and active techniques for boosting the return with additional tax-free gains. For passive tax management, we described three very useful methods: dividend management, tax-lot accounting, and capital gains and loss management. On the active management side, we discussed the strategy of loss harvesting. We showed how the portfolio manager can incorporate these strategies directly into the stock-return model without significantly altering the before-tax α of the portfolio.

As we saw from a simulation using real data, intelligently tax-managed portfolios can earn as much as 4% per year more than portfolios that overlook the effect of taxes. Why turn down an extra return of 4%? In the past, the benefits of tax management techniques might not have always justified the costs and difficulties of implementing them. Recent advances in technology, however, have greatly simplified and lowered the costs of implementation. A portfolio stands to gain only from strategies that keep taxes to a minimum.

QUESTIONS

- 11.1. Explain the concept of double taxation of corporate income.
- 11.2. You are a research analyst interested in how dividends affect stock prices. Assume that there is only one income tax rate (τ_i) and a capital gains rate (τ_c). Also assume that the owner has held the stock for more than one year already.
 - (a) Denoting the price of the stock before and after the ex-dividend date as p^b and p^a , what should be the theoretical drop

- in the stock price ex-dividend?
- (b) Suppose that you further study the data since 1984 and find that the typical stock drops by 90% of the dividend on the ex-dividend date. Could you profit from this information if you managed a tax-exempt pension fund? How?
- 11.3.** Explain why individual investors may have preferred capital gains to dividends in the U.S. tax system before 2002.
- 11.4.** Behind the principles of tax management is the concept of the time value of money.
- (a) If the economy is experiencing very high inflation, how would the time value of money be affected?
- (b) If the economy is not growing at all, can the time value of money be negative?
- 11.5.** Explain why realizing a capital loss as early as possible may reduce the tax burden.
- 11.6.** If the value effect (value stocks perform better) dominates, should the dividend management technique be modified? Does your answer change if the dividend rate is one of the factors in the stock return model?
- 11.7.** One assumption behind dividend management is that we can *predict* the dividend amount of each stock fairly accurately. Explain why predicting the dividend amount is easier than predicting earnings or the stock price. How would you actually predict the dividend amount?
- 11.8.** A popular tax-lot management technique, called *highest-in, first-out*, entails selling the tax lot with the highest purchase price first. Is this rule consistent with the fact that short-term capital gains are taxed at a higher rate than long-term capital gains? How should this rule be modified?
- 11.9.** Explain why the first-in, first-out rule of tax-lot management may create unnecessary tax burdens.
- 11.10.** Let $\mathbf{w} = (w_1, \dots, w_N)$ be the portfolio weights of stocks at the beginning of the month and

$$\tilde{\mathbf{w}} = (\tilde{w}_{11}, \dots, \tilde{w}_{1T}, \dots, \tilde{w}_{N1}, \dots, \tilde{w}_{NT})$$

be the portfolio weights of tax lots at the beginning of the month.

- (a) If $\mathbf{r} = (r_1, \dots, r_N)$ is the stock return of the month, what would be the weights of the stocks at the end of the month? What would be the weights of the tax lots at the end of the month?
 - (b) It is useful to construct matrix $\mathbf{D}$ such that $\mathbf{w} = \mathbf{D}\tilde{\mathbf{w}}$. Specify the typical elements of $\mathbf{D}$.
 - (c) Answer part (a) again using matrix $\mathbf{D}$ from part (b).
- 11.11.** If the loss-harvesting strategy is followed for many years, the portfolio manager may get into the situation of not having any capital losses to realize. How should the loss-harvesting strategy be modified to avoid such a situation?
- 11.12.** Explain why loss harvesting may have a negative influence on the portfolio return when returns exhibit negative serial correlation.
- 11.13.** Explain why loss harvesting may increase the tracking error of the portfolio.
- 11.14.** What is the wash-sale rule? Explain how to comply with the wash-sale rule in the portfolio optimization problem.
- 11.15.** The characteristic-matching technique introduced in this chapter is based on ideas from the fundamental factor model. In practice, replacing stocks by characteristic matching affects the return and the risk of the portfolio. For the characteristic matching to be perfect, what would the error term in the fundamental factor model have to be?
- 11.16.** 401(k) accounts are not taxable in principle. However, some people withdraw funds from their 401(k) accounts before they retire, and they pay taxes on the withdrawals. Given the possibility of early withdrawal, how would the portfolio manager of a 401(k) account modify the portfolio?
- 11.17.** Suppose that a taxable investor and a nontaxable investor pool money and create one portfolio.
- (a) Which tax management techniques can be adopted? Which techniques should be avoided?
 - (b) Does it really make sense to pool funds from these two investors?

¹ See Garland (1997).

- ² We used the same methodology as the original Garland study, but we changed the parameters to reflect the 2020 values of the average expense ratio (1.16%), average turnover rate (67%), and highest value for long-term capital gains (20%) and for qualified dividends (20%).
- ³ Since we wrote the first edition of *QEPM*, many robo-advisor companies have been created that employ the techniques we discuss in this chapter. These portfolio techniques are not relevant for nontaxable accounts, such as IRAs and 401(k) accounts.
- ⁴ The tax rates for individuals for capital gains and dividends have changed substantially over time. Overall, the concepts of tax management still remain valid. For a portfolio manager, however, with a diverse array of clients, the decision of which techniques to use for tax management may vary.
- ⁵ Jobs and Growth Tax Relief Reconciliation Act of 2003 and the Tax Cuts and Jobs Act of 2017. The new rules lowered the tax on qualified dividends and also lowered the corporate and individual income tax rates.
- ⁶ Hedge funds that manage equities usually design their legal structure in such a way as to allow for pass-through taxation. That is, the fund itself does not pay taxes; the income, gains, and losses all “pass through” to the investors, who take them directly on their own tax bills. Mutual funds that manage equities are required by the Investment Company Act of 1940 to distribute 98% of their gains and dividends to avoid being taxed as well. Thus investors in the mutual fund ultimately bear the burden of the taxes. In the case of both institutions, tax-managed techniques could greatly improve the welfare of their investors.
- ⁷ In order for a corporation to receive the 65% deduction, the corporation must own between 20% and 80% of the distributing corporation.
- ⁸ Capital gain taxes are also due when a short position is closed out and capital gains are realized.
- ⁹ Of course, it is not easy to replace a long-term capital gain of \$1 with a long-term capital loss of \$1. This calculation is for the purpose of illustration.
- ¹⁰ Another way to think about this is that delaying a \$1 long-term capital gain for one year generates a capital gain of \$1.1 one year later at the (pretax) investment return of 10%. Thus, the tax payment due is 22 cents ($1.1 \cdot 0.2$), and the after-tax income is 88 cents ($1.1 \cdot 0.8$). We can then discount the 88 cents by the after-tax return of 8% to obtain the present value of 81.48 cents. Compare this to the no-delay after-tax income of 80 cents. Thus the total savings is 1.48 cents. Similarly, realizing \$1 of capital loss today rather than next year saves about 1.48 cents.
- ¹¹ It is easy to adjust the penalty for dividend-paying stocks if one plans to withdraw cash. If the dividend amount exceeds the amount of cash one plans to withdraw, then the penalty is the tax on the excess dividend. If the dividend does not exceed the withdrawal amount, it may not increase the tax burden at all. Nonetheless, if the existence of dividend income makes the other tax management techniques hard to implement, it still might be best to avoid dividend-paying stocks.
- ¹² Even for a corporation, it is still better to generate cash by selling loser stocks, but in many cases this is less feasible due to ulterior reasons.
- ¹³ These tax-lot selection rules might not be optimal if we consider the net capital gains and losses of the entire portfolio. With sufficient net short-term realized gains as well as net long-term gains in the rest of the portfolio, it might be wise to consider short-term losses before long-term losses. Of course, this will then affect total net gains and losses, which will affect other stocks’ optimal tax-lot selling decisions. Thus, these simple rules should be considered when selling the shares of a particular stock in isolation of the rest of the portfolio. A more general tax-lot

selection approach considering net capital gains and losses of the entire portfolio is discussed in Section 11.7.

- 14 Many brokerages, however, use the IRS's default accounting method of first in, first out, which in many situations produces higher taxes than the simple rules we propose in this chapter.
- 15 The reader might notice that the variance-covariance matrix $\tilde{\Sigma}$ is not of full rank. This may cause a problem in applying the quadratic programming approach. To avoid this problem, we can restate the maximization problem in terms of $\tilde{\mathbf{w}}^d$ and $\mathbf{w}^a$ and express the variance-covariance matrix in terms of $\mathbf{w}^a$. Then the problem becomes a standard problem that quadratic programming can handle easily. That is, the effective risk-adjusted return in Eq. (11.16) should be modified so that it is minimized with respect to $\tilde{\mathbf{w}}^d$, but the variance-covariance term is used as $\mathbf{w}^a \Sigma \mathbf{w}^a$, with a constraint that specifies the relationship between $\tilde{\mathbf{w}}^d$ and $\mathbf{w}^a$.
- 16 Ideally, the stock would possess both similar risk characteristics and a similar expected α or expected return.
- 17 One could also calculate a weighted distance function that weights each factor differently.
- 18 This section is based on Chincarini and Kim (2001). Since this study was written, there has been much more interest in tax-managed strategies (see Chaudhuri et al. (2020)).
- 19 For the purpose of calculation, the entire portfolio holding was assumed to be liquidated at the end of June 2000. If the portfolio were not liquidated, future tax liabilities would be erroneously ignored in the calculation.
- 20 The pretax return is higher for the t-smart investor than for the naive investor. This is due to the momentum in price movement. When there is momentum in price movement, selling loser stocks and buying something else produces superior returns.

α Mojo

The factor model of stock returns gives the quantitative equity portfolio manager a good idea of what he or she can expect his or her portfolio to earn with a particular investment strategy. The manager can depend on the model because of the reliability of the statistical relationship between stock returns and the model's factors. The effectiveness not only of the model but also of quantitative equity portfolio management (QEPM) as a whole depends greatly on the quality of statistical results.

Yet all great investment strategies do not operate strictly within the boundaries of the factor–return relationship. In fact, to focus only on factor exposures and factor premiums is to ignore the model's greater potential. The data-derived factor model says that higher returns come only from more factor exposure or higher factor premiums. But look at the beginning of the model, and you will find α , a deceptively small term that represents the radical possibility of earning much higher returns *without* absorbing much more risk. How can a portfolio manager activate α and boost the returns he or she gets for the same amount of risk? By tapping into “alpha mojo.”

Alpha mojo (a.k.a. *α mojo*) produces outsize returns that defy the presumption of some universal, incontrovertible ratio between risk and reward. There are three major sources of α mojo, discussed in the next three chapters: leverage, which multiplies the portfolio's return; the market-neutral strategy, which insulates the portfolio against the volatile winds of

market trends and risks; and Bayesian α , which introduces the portfolio to new sources of return. Alpha mojo generates results that look like pure investment magic, but just as the magician knows the logic behind his or her tricks, the quantitative equity portfolio manager knows that the sources of mojo are actually quantitative methods of investing that exploit the laws of investing to phenomenal effect—while still upholding the tenets of QEPM.

CHAPTER 12

Leverage

Give me a lever long enough and a fulcrum on which to place it, and I shall move the world.

—Archimedes

12.1 INTRODUCTION

Given a benchmark, the portfolio return can be broken down into two parts: the part related to the benchmark and the part not related to the benchmark. The first part is measured by the benchmark β multiplied by the benchmark return, and the second part is measured by α^B plus a random component.¹ That is,

$$r_P = \alpha^B + \beta r_B + \epsilon \quad (12.1)$$

Part of the portfolio's return is beyond the manager's control. To some extent the portfolio goes the way the benchmark goes (represented by βr_B), and occasionally, it does the proverbial "random walk" (represented by ϵ). Yet the portfolio's success or failure is also up to the manager. He or she has the power to boost the β of the overall portfolio. Better yet, he or she has the power to boost the α^B . We call this power *α mojo*. Leverage is a potent source of *α mojo* and also can be used to boost the β of the overall portfolio.

Leverage is perhaps the ultimate source of α mojo. In physics, a lever multiplies the force that is applied to it; in investing, leverage multiplies the force of the portfolio. With leverage in the form of borrowed capital, a portfolio manager can invest double, triple, or even more of the amount of available equity. If he or she has \$10 million and can find a lender to match that amount, he or she will have a leveraged portfolio of \$20 million, and every dollar that the leveraged portfolio gains will be twice as much as the unleveraged portfolio would have garnered. Leverage applied in the correct way also can generate major α mojo and amplify the benchmark return. Leverage easily alters the risk-return profile of any portfolio. If the portfolio is managed well but its risk-return ratio does not satisfy the investor's appetite for risk, leverage will pump up both the expected return and the risk level. If the risk is too high, leverage through shorting can decrease the overall return and risk of the portfolio. Investors who like to time the market also can leverage the portfolio to enhance the impact of market-timing trades.²

The sort of power that leverage confers can be dangerous, though. A massive leveraged portfolio can earn the unleveraged portfolio's profits three times over, but it also will get hit three times as hard by any losses.³ Borrowed capital increases the portfolio's exposure to market swings. It opens up the portfolio manager to margin-call risk, which threatens the portfolio's liquidity by raising the possibility of having to meet daily margin requirements in adverse market conditions. It exacerbates market-timing mistakes. And if it isn't handled carefully, leverage can turn into an evil mojo, inflicting severe damage on the portfolio manager's α . A number of hedge funds have gone bankrupt when the portfolio was heavily leveraged and the α went negative. Long-Term Capital Management (LTCM) may be the most notorious case, but there have been others, such as the Niederhoffer Investments Fund.

For those who dare to use it, leverage comes in different forms. The manager can borrow additional funds to invest more than the equity capital. He or she can leverage the portfolio through the margin account, since most equities require only 50% initial margin. He or she can use financial instruments and derivatives, such as REPOs, futures, forwards, equity swaps, and option contracts, which require smaller margin requirements than equities and allow for even higher leverage of the portfolio for a given amount of equity capital. Finally, a portfolio manager can achieve leverage

by shorting securities. In this chapter we discuss how to leverage skillfully. We explain practical methods for increasing the leverage of a portfolio primarily through the use of index futures and single-stock futures. We cover several portfolio situations, including that of an index portfolio manager and that of an active portfolio manager attempting to leverage an equity portfolio with a positive α . We also discuss issues related to rebalancing a levered portfolio and protecting against unlimited losses.

12.2 CASH AND INDEX FUTURES

The easiest way to leverage an equity portfolio is to use futures contracts on equity indices such as the Standard & Poor’s (S&P) 500 futures, the NASDAQ 100 futures, the Russell 2000 futures, and the S&P 400 futures.⁴ Table 12.2 shows the relative monthly trading volume of these and other types of futures contracts, which gives a sense of their relative degree of liquidity. The S&P 500 futures and the NASDAQ 100 futures are the most liquid futures contracts available.

TABLE 12.1
The Advantages and Disadvantages of Leverage in Quantitative Equity Portfolio Management (QEPM)

Advantages	Disadvantages
1. Easily alter the risk-return profile of a portfolio	1. May increase overall risk of the portfolio if direction of leverage is the same as underlying exposure and introduces margin risk
2. May increase the overall expected return of the portfolio	2. May reduce the relative contribution of the portfolio manager’s overall α to the total return
3. May increase α to $l \cdot \alpha$, where l is the leverage ratio	3. Will magnify negative α ’s as well as $l \cdot \alpha$
4. Enhances the market returns of market timers	4. May cause huge losses in the portfolios of market timers

TABLE 12.2
Commonly Used Domestic Equity Futures Contracts

Futures Name	Ticker	Exchange	Contract Size	Index Value	Initial Margin (\$)	Maintenance Margin (\$)	Margin (%)	Volume	Relative Liquidity (% of Total)
S&P 500	SP1	CME	250	3,756.07	60,500	55,000	6.44	2,062	0.39
S&P 500 E-mini	ES1	CME	50	3,756.07	12,100	11,000	6.44	993,343	61.00
NASDAQ 100 E-mini	NQ1	CME	20	12,888.28	17,600	16,000	6.83	280,573	28.06
NASDAQ Comp E-mini	NL1	CME	20	12,888.28	17,600	16,000	6.83	0	0.00
S&P 400 Mid-Cap E-mini	FA1	CME	100	2,306.62	14,850	13,500	6.44	16,031	0.72
S&P 600 Small-Cap E-mini	GN1	CME	100	1,118.93	9,350	8,500	8.36	0	0.00
Russell 2000 T.R.	RTB1	CME	10	10,122.27	7,810	7,100	7.72	0	0.00
Russell 2000 E-mini	RTY1	CME	50	1,974.86	7,150	6,500	7.24	144,363	3.52
Russell 1000 T.R.	RTD1	CME	10	12,531.09	8,580	7,800	6.85	0	0.00
Russell 1000 E-mini	RVY1	CME	50	1,349.62	5,170	4,700	7.66	125	0.01
DJIA T.R.	RDR1	CBOT	2	70,778.10	9,900	9,000	6.99	0	0.00
DJIA E-mini	DM1	CBOT	5	30,606.48	9,900	9,000	6.47	81,740	5.76
DJIA Micro E-mini	HW1	CBOT	0.5	30,606.48	990	900	6.47	81,092	0.37
S&P 500 Growth	SG1	CME	250	2,577.22	38,500	35,000	5.98	0	0.00
S&P 500 Value	SU1	CME	250	1,267.18	22,000	20,000	6.94	0	0.00
S&P 500 Financial SS E-mini	IXA1	CME	250	363.52	8,250	7,500	9.08	144	0.03
S&P 500 Technology SS E-mini	IXT1	CME	100	1,311.92	10,450	9,500	7.97	97	0.02
S&P 500 Communication Services SS E-mini	XAS1	CME	250	353.47	5,500	5,000	6.22	20	0.00
S&P 500 Consumer Discretionary SS E-mini	IXY1	CME	100	1,625.07	10,450	9,500	6.43	105	0.02
S&P 500 Consumer Staples SS E-mini	IXR1	CME	100	681.07	3,850	3,500	5.65	71	0.01
S&P 500 Energy SS E-mini	IXP1	CME	100	397.55	4,400	4,000	11.07	195	0.02
S&P 500 Health Care SS E-mini	IXC1	CME	100	1,148.03	7,810	7,100	6.80	251	0.02
S&P 500 Industrial SS E-mini	IXI1	CME	100	893.21	7,370	6,700	8.25	72	0.02
S&P 500 Materials SS E-mini	IXD1	CME	100	767.22	5,720	5,200	7.46	126	0.01
S&P 500 Real Estate SS E-mini	XAR1	CME	250	178.33	3,575	3,250	8.02	142	0.00
S&P 500 Utilities SS E-mini	IXS1	CME	100	636.17	5,280	4,800	8.30	291	0.01

Note: Margin requirements differ for speculators, for exchange members, and for hedging purposes. We listed the requirements for speculators, which are the highest margin rates as of December 31, 2020. For speculators, initial margin is 1.10 times the maintenance margin at CME. The futures data were obtained from Bloomberg, and all data are as of the close of business on December 31, 2020. The tickers are Bloomberg tickers for the nearest-term generic futures contracts. The relative liquidity measures the percentage of 30-day ADDTV (average daily dollar trading volume) leading up to December 31, 2020 of that particular contract relative to the other contracts in the table. For instance, 61% of the total trading volume of those listed contracts occurred with the S&P 500 E-mini contract. The margin percentage was computed by dividing the initial margin requirements by the size of one contract using closing prices of the index for December 31, 2020. T.R. is for total return and SS is for Select Sector.

In this section we discuss leveraging with equity index futures. The discussion applies mainly to managers whose benchmarks are common equity indices with actively traded futures contracts. We first discuss the magnitude of possible leverage and then examine the practical mechanics.

12.2.1 Theoretical Limits of Leverage

The amount of leverage to add to a portfolio varies with the goals of the manager. Leverage is conventionally measured as the total absolute value of all investment positions divided by the equity capital. Thus, if a manager were long \$100 M of futures and short another \$100 M of index futures, the conventional measure of leverage would be 2. In this chapter, we use an alternative definition of leverage which could also be thought of as net dollar exposure. That is, in our definition, the manager in the preceding example would have a leverage or exposure of 0. In this chapter, we use leverage mainly to refer to futures positions that increase the net exposure of the underlying portfolio position. In these cases, our formula for leverage coincides with the conventional formula for leverage. When futures are shorted with an underlying long position, our definition of leverage is no longer the conventional definition of leverage. Thus when considering long positions combined with short positions, our definition of leverage is more

commonly known as the net dollar exposure. We will use the symbol l to illustrate the amount of leverage with respect to the underlying equity capital of the portfolio. Leverage at time t will be given by the expression $l = V_t^f / V_t$, where V_t^f is the total notional value of the futures positions, and V_t is the total equity capital. Thus $l = 1$ represents an exposure equal to 100% of the invested capital and hence may be described as “no leverage” (though we write it as the leverage of one), while $l = 2$ represents a 200% exposure to the equity capital.⁵

In the simple case of cash and index futures, leverage affects the portfolio manager’s return primarily by increasing its exposure to the benchmark, its benchmark β . To achieve a certain benchmark β for the portfolio, one needs to compute the appropriate number of futures contracts to purchase. The portfolio manager is limited by the extent of the margin requirements of the underlying contract. [Table 12.2](#) gives a summary of the percentage requirements of some major futures contracts as of December 31, 2020.

Let us use m_f to denote the margin percentage requirement for every dollar of futures positions acquired. Theoretically, the highest leverage that could be obtained is

$$l^{\max} = \frac{1}{m_f} \quad (12.2)$$

The β of the portfolio is simply the β of the futures contract multiplied by the leverage ratio.⁶ Thus the highest β that can be achieved is

$$\beta^{\max} = \frac{\beta_f}{m_f} \quad (12.3)$$

where β_f is the β of the futures contract.⁷

Suppose that the benchmark is the S&P 500. If the margin requirement on the S&P 500 futures is 5%, then the maximum leverage that can be achieved is 20 ($= 1/m_f$). This is also equal to the maximum β that can be achieved because the S&P 500 futures have a $\beta_f = 1$ when the benchmark is

the S&P 500. At that rather high exposure to the S&P 500, if the S&P 500 had a -4% return, the portfolio would lose approximately -80% ($= -4\beta^{\max} = -4l^{\max}$).

12.2.2 Leverage Mechanics

Although very high levels of leverage theoretically can be achieved in an equity portfolio, rarely do portfolio managers maintain a leverage ratio higher than 2 or 3.⁸ There are many reasons for this. Higher amounts of leverage put the portfolio at greater risk of vanishing entirely with a significant adverse market movement. For mutual funds and ETFs, the Securities and Exchange Commission (SEC) has to approve the leverage amount in the prospectus, and it may be very uncomfortable with leverage ratios above 2 or 3. For these practical reasons, we will focus on leverage ratios within a moderate range, but the formulas apply to any sort of leverage up to the theoretical maximum.

In order to construct a portfolio of a given β with cash and futures, we can use a formula that describes the β of the overall portfolio:

$$\beta_P = \sum_{i=1}^N w_i \beta_i \quad (12.4)$$

where β_P is the β of the overall portfolio, β_i represents the β of instrument i , and w_i represents the weight of instrument i in the portfolio. The weights are calculated relative to equity capital V so that the sum of the weights of noncash positions is l . The preceding equation just states the relationship that the β of the overall portfolio is a weighted average of component portfolio β 's. The β of cash is equal to 0 and drops out of the equation. Since we are using just cash and one futures instrument, we are left with only one term, the product of the weight of the futures contracts and the β of the futures. We denote the β of the futures as β_f .

In order to find the number of contracts that we need to purchase to achieve a given target β , denoted by β^* , we must satisfy the following equation:

$$\begin{aligned}\beta^* &= w_f \beta_f \\ &= \frac{N_f q S_t}{V_t} \beta_f\end{aligned}\tag{12.5}$$

where N_f is the number of futures contracts purchased, q is the futures contract multiplier (e.g., $q = 250$ for the S&P 500 futures contract), and S_t is the value of the index underlying the futures contract at time t (e.g., the S&P 500 value).⁹ By rearranging this equation, we obtain the standard formula to determine how many futures contracts should be purchased to achieve a given β^* :

$$N_f = \frac{\beta^* V_t}{\beta_f q S_t}\tag{12.6}$$

Suppose that the benchmark is the S&P 500 and that the portfolio manager is purchasing S&P 500 futures. To achieve a portfolio of $\beta^* = 2$, given an S&P 500 trading at 1,000, a multiplier for the large S&P 500 contracts of 250, and a sum of cash equal to \$100 million, the appropriate number of futures contracts to purchase would be 800. (The β of the S&P 500 futures with respect to the S&P 500 is 1.)

When the portfolio manager purchases N_f contracts of the futures, it “costs” him or her $qN_f F_t$ dollars, where F_t is the price of a futures contract. Then the portfolio manager needs to keep $(qN_f F_t)m_f$ dollars in a margin balance and hold the rest of the equity capital in cash earning interest. The general formula for the amount of cash held is $V_t - (qN_f F_t)m_f$.

In our example, the margin amount is \$10,075,282 (the product of $q = 250$, $N_f = 800$, $F_t = 1,007.53$, $m_f = 0.05$), and the cash portion of the portfolio is \$89,924,718 ($= \$100,000,000 - \$10,075,282$).

12.2.3 Expected Return and Risk

When the leverage ratio is 1, the return of the futures contracts r_f can be expressed as $\alpha_f + \beta_f r_B + \epsilon_f$ where r_B is the benchmark return. Given the

leverage ratio of l , the expected return and risk of the total portfolio become

$$E(r_p) = l(\alpha_f + \beta_f \mu_B) \quad (12.7)$$

$$V(r_p) = l^2(\beta_f^2 \sigma_B^2 + \omega_f^2) \quad (12.8)$$

where ω_f^2 is the residual risk of the futures contract, μ_B is the mean return of the benchmark, and σ_B^2 is the variance of the benchmark return. This clearly shows that although leverage can increase the expected return of the portfolio, it also increases proportionately the risk of the portfolio. In the case of leveraging by purchasing futures contracts on the benchmark itself, $\beta_f = 1$ and $\alpha_f = 0$ (or close to it), and we can ignore the ϵ_f because it will be very small. In this particular case, if we assume that the portfolio and futures returns already have subtracted the risk-free rate, the Sharpe ratio (SR) of the overall portfolio is equal to

$$SR_p = \frac{\mu_B}{\sigma_B} \quad (12.9)$$

In this particular case, the Sharpe ratio is independent of the degree of leverage, and the β of the total portfolio equals the leverage ratio.¹⁰

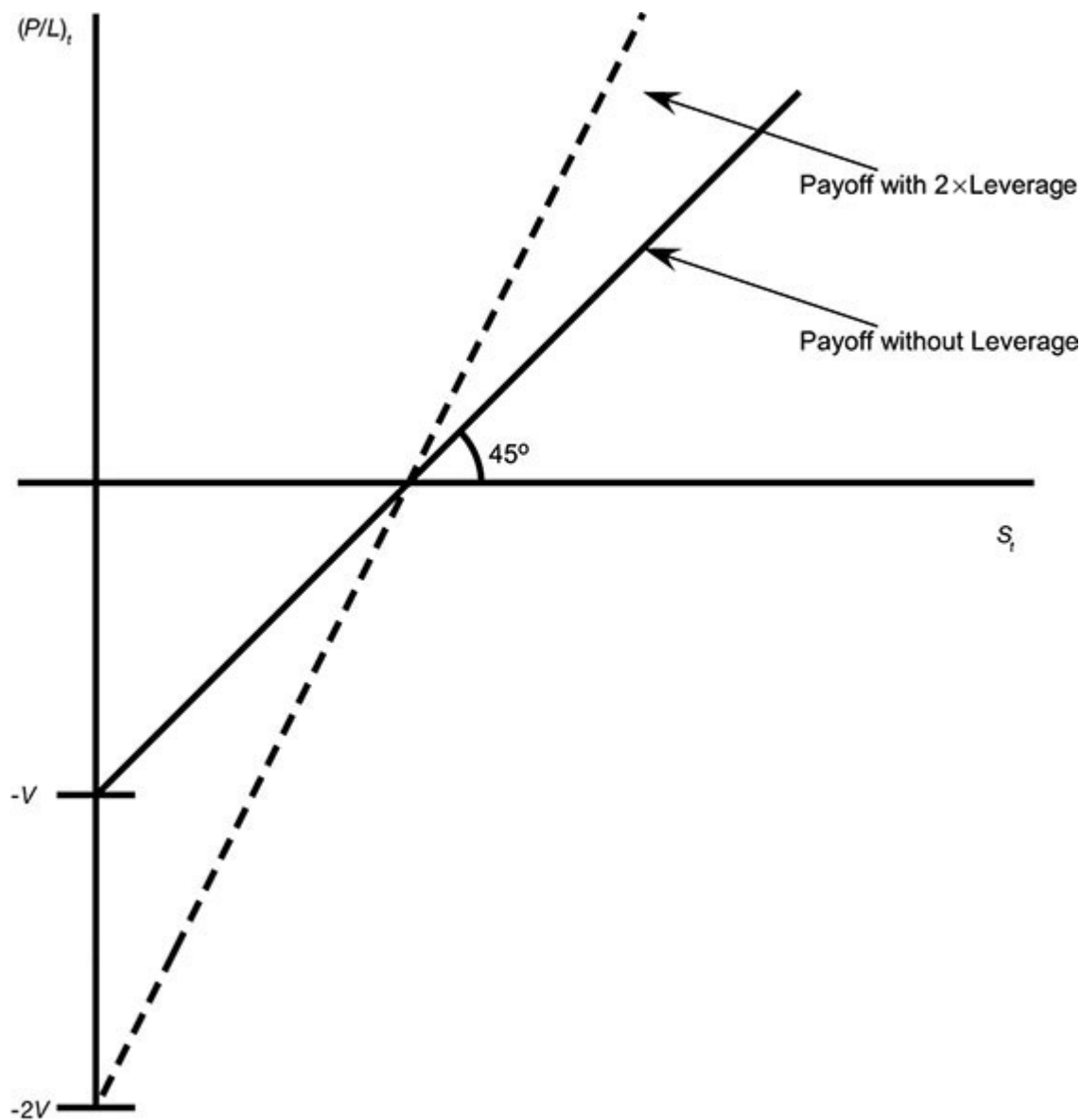
Most portfolio managers also earn interest on the margined amount minus a small “haircut”. On the margined sum, one can assume that the manager will receive something like 98% of the short-term interest rate for cash. Throughout this chapter we will use i to denote the interest rate on the cash position and i' to denote the interest rate on the margin. We expect that $i > i'$ owing to the haircut. In all our calculations for the expected return of the portfolio, we shall ignore the cash returns of the position for simplicity. However, cash returns are not trivial, especially when the portfolio manager has a substantial amount of cash in his or her overall portfolio.¹¹

Figure 12.1 show how leverage changes the payoff profile of the portfolio. The horizontal axis represents the value of the index, whereas the vertical axis represents profit and loss of the portfolio. If the index were to go to zero, the portfolio would lose the entire value, V . Without leverage, the index value and the payoff move one for one along the 45-degree line.

With leverage, the payoff becomes steeper versus the underlying index. This diagram depicts the payoff when the leverage is equal to 2. With leverage equal to 2, if the index drops by 50%, the portfolio would lose more than double its assets, or $2V$.

FIGURE 12.1

Portfolio payoffs with and without leverage.



12.3 STOCKS, CASH, AND INDEX FUTURES

The preceding section dealt with leverage in the case of an index manager. Leverage works somewhat differently for the portfolio manager who does not follow an index and instead selects stocks that he or she believes will outperform the benchmark. This manager will calculate leverage and the number of futures contracts in a similar way, but the underlying portfolio will be a portfolio of stocks with weights that already have been chosen according to the optimization method.¹² In this section we discuss leverage for the portfolio manager who owns a portfolio of stocks that he or she picked. This kind of portfolio most likely will have a β not equal to 1. We will call the β of this portfolio β_s .

12.3.1 Theoretical Limits to Leverage

The portfolio manager may wish to hold cash in addition to stocks. Let's use the term ξ to refer to the percentage of the equity capital that the portfolio manager holds in cash. Thus $1 - \xi$ represents the amount of the portfolio's equity invested in stocks. The maximum l and β the manager can achieve is

$$l^{\max} = \frac{\xi + (1 - \xi) m_s}{m_f} + (1 - \xi) \quad (12.10)$$

$$\beta^{\max} = [l^{\max} + (\xi - 1)] \beta_f + (1 - \xi) \beta_s \quad (12.11)$$

where m_s is the initial margin required on stocks (which is also used in determining the collateral value of stocks), and m_f is the required futures margin.¹³ Thus, with $m_f = 0.05$ and $m_s = 0.50$ if the manager wants to hold 50% of the equity in cash ($\xi = 0.50$), the maximum achievable leverage ratio is 15.5. The $\beta^{\max}$, for the case of the S&P 500 assuming that $\beta_f = \beta_s = 1$, is 15.5.

12.3.2 Leverage Mechanics

The portfolio manager may decide to use index futures to alter the overall beta of his or her total portfolio. To achieve the target β^* of the overall portfolio, the manager would have to satisfy

$$\begin{aligned}\beta^* &= w_f \beta_f + (1 - \xi) \beta_s + \xi \underbrace{\beta_{\text{cash}}}_{=0} \\ &= \frac{N_f q S_t}{V_t} \beta_f + (1 - \xi) \beta_s\end{aligned}\tag{12.12}$$

By rearranging this equation, we can obtain the number of contracts that the portfolio manager will have to purchase or sell to achieve the desired β^* :

$$N_f = \frac{[\beta^* + (\xi - 1) \beta_s] V_t}{\beta_f q S_t}\tag{12.13}$$

The portfolio manager can achieve any target β^* provided that $\beta^* \leq \beta^{\max}$. The portfolio manager need not hold any cash for most β ranges because the stock portfolio can be used as margin.¹⁴

Returning to our example from the preceding section, the manager now needs to purchase 600 contracts in order to achieve $\beta^* = 2$ with 50% of the equity in cash and a $\beta_s = 1$. This results in a portfolio that has $(qN_f F_t)$, or \$151,129,500, invested in future contracts when the price of a futures contract F_t is \$1,007.53. The purchase of the contracts requires margin collateral of $(qN_f F_t)m_f$, or \$7,556,462.¹⁵

12.3.3 Expected Returns and Risk

Given the benchmark return r_B , the return of the stock portfolio can be expressed as

$$r_s = \alpha_s + \beta_s r_B + \epsilon_s\tag{12.14}$$

Similarly, the return of the futures contracts can be written as

$$r_f = \alpha_f + \beta_f r_B + \epsilon_f \quad (12.15)$$

Noting that the weight of the stock portfolio is $(1 - \xi)$ and the weight of the futures contracts is $l + (\xi - 1)$, the return of the overall portfolio is¹⁶

$$\begin{aligned} r_p &= (1 - \xi)r_s + [l - (1 - \xi)] r_f \\ &= (1 - \xi)\alpha_s + [l - (1 - \xi)] \alpha_f \\ &\quad + \{(1 - \xi)\beta_s + [l - (1 - \xi)] \beta_f\} r_B + (1 - \xi)\epsilon_s + [l - (1 - \xi)] \epsilon_f \end{aligned} \quad (12.16)$$

If the underlying futures index is also the benchmark of the portfolio, then we may assume that $\alpha_f = 0$ and that $\beta_f = 1$. We also will ignore ϵ_f because it is very small. Under these assumptions, the expected return and variance of the overall portfolio are¹⁷

$$E(r_p) = (1 - \xi)\alpha_s + [(1 - \xi)(\beta_s - 1) + l] \mu_B \quad (12.17)$$

$$V(r_p) = [(1 - \xi)(\beta_s - 1) + l]^2 \sigma_B^2 + (1 - \xi)^2 \omega_s^2 \quad (12.18)$$

where μ_B and σ_B denote the expected return and the standard deviation of the benchmark, and ω_s denotes the standard deviation of ϵ_s . It is clear that leverage increases the market risk of the portfolio and reduces the relative contribution of stock picking. As leverage moves from $l = 1$ (no leverage) to $l = 2$ (leverage two times equity), the expected return of the total portfolio increases by $E(r_B)$, effectively reducing the value that the portfolio manager adds by stock picking.¹⁸ Depending on the goals of the quantitative manager, this may or may not be a good thing.

Let's assume for a moment that we already have subtracted the risk-free rate from all the return figures; the Sharpe ratio (SR) is defined as the ratio of the expected return to the standard deviation. Thus the squared Sharpe ratio would be

$$SR_p^2 = \frac{\left[(1-\xi)\alpha_s + (1-\xi)(\beta_s - 1)\mu_B + l\mu_B \right]^2}{\left[(1-\xi)(\beta_s - 1) + l \right]^2 \sigma_B^2 + (1-\xi)^2 \omega_s^2} \quad (12.19)$$

Rearranging the expression in terms of l , we get

$$SR_p^2 = \frac{\mu_B^2 l^2 + 2\mu_B \left[(1-\xi)\alpha_s + (1-\xi)(\beta_s - 1)\mu_B \right] l + \left[(1-\xi)\alpha_s + (1-\xi)(\beta_s - 1)\mu_B \right]^2}{\sigma_B^2 l^2 + 2(1-\xi)(\beta_s - 1)\sigma_B^2 l + (1-\xi)^2 (\beta_s - 1)^2 \sigma_B^2 + (1-\xi)^2 \omega_s^2} \quad (12.20)$$

This expression suggests that as the value of l increases, the Sharpe ratio of the overall portfolio approaches the Sharpe ratio of the benchmark.¹⁹ If the Sharpe ratio of the unleveraged portfolio were higher than the Sharpe ratio of the benchmark, then the leverage would reduce the Sharpe ratio eventually.

12.4 STOCKS, CASH, AND SINGLE-STOCK FUTURES

One of the drawbacks to using index futures to leverage a portfolio is that the relative contribution of the α is diminished in the returns of the overall portfolio. We saw in the preceding example that the contribution of stock picking can decrease by as much as 50%. Depending on the goals of QEPM, this may not be a concern, but there is a better way to leverage—through the use of single-stock futures. Leverage created with single-stock futures results in a *leveraged* α rather than just a leveraged portfolio. In other words, it generates α mojo.

Unfortunately, as of December 2020, single-stock futures do not exist in the United States; however, they do exist in other countries, including on Euronext, the European stock exchange.²⁰ Part of the reason that the U.S. stock futures exchange closed was a lack of investor interest, which subsequently created a set of practical problems for investors wanting to use single-stock futures. First of all, there are typically only a limited number of single-stock futures available for trading.²¹ Thus, a manager may not find futures for all the stocks in his or her portfolio.

Second, single-stock futures that do exist have less trading liquidity than index futures.²² Table 12.3 lists some of the major single-stock futures and their volumes on August 7, 2020, shortly before the exchange closed. The table also includes the percentage of a \$100 million portfolio that the daily volume represents. Even the third most-traded security has a volume that represents only 2.69% of a \$100 million portfolio. For reasonable holdings of around 1% or 2% in a stock, the portfolio manager would have difficulties using single-stock futures contracts to create a leveraged position. This problem becomes more pronounced as one moves down the list of most-traded securities.²³

TABLE 12.3

Highest-Volume Single-Stock Futures Contracts

Company Name	Ticker	Volume	Price	Percent of \$100 M Portfolio
Itau Unibanco Holding SA ADR	ITUB	5749	4.69	2.696
Ambev SA ADR	ABEV	1685	2.46	0.415
Mosaic Co	MOS	306	17.19	0.526
Gap Inc.	GPS	298	14.08	0.420
Cabot Oil & Gas Corporation	COG	234	20.59	0.482
CME Group Inc	CME	214	167.15	3.577
Walmart Inc	WMT	191	129.98	2.483
Bristol-Myers Squibb Co	BMJ	179	61.02	1.092
NRG Energy Inc	NRG	156	34.28	0.535
The AES Corporation	AES	153	17.16	0.263

Note: These statistics were for August 7, 2020. Price data were obtained from Bloomberg, and futures volume data were obtained from www.onechicago.com (now not available). The last column in the table is the percentage of the daily trading volume that would be represented in a \$100 million portfolio. For Gap Inc., for example, the daily trading volume would represent less than 0.5% of the portfolio. Thus, if a manager would like a 1% or greater holding of GPS, his or her trade would take up the entire day's trading volume. The formula is $\text{percentage} = \frac{100 \cdot \text{volume} \cdot \text{price}}{100 \text{ M}}$.

12.4.1 Theoretical Limits of Leverage

The theoretical limits to leveraging with single-stock futures are very similar to the limits using index futures. There are three essential differences between the two cases, however. The first is that individual stock futures have initial margin requirements that are higher than index

futures.²⁴ The second difference is that single stock futures allow the manager to increase the maximum β by specifically leveraging higher- β stocks or decrease it by specifically leveraging lower- β stocks. Leveraging single stocks according to their β 's is also a sensible way to leverage one's α if one already has chosen the optimal stock weights. A third concern that is unique to leveraging with single-stock futures is that if the futures are not actively traded, then it will be possible to achieve only partial leverage. For the time being, we will assume that the futures are traded actively enough for the portfolio to achieve maximum leverage.

For leveraging a single security, the formulas for the maximum leverage and maximum β for single-stock futures are identical to Eqs. (12.10) and (12.11), but with the single-stock futures margin requirement replacing the index futures margin requirement. For these leverage formulas to be the same for the entire portfolio when leveraging single-stock futures, all stocks must be leveraged in relative proportion to their weights in the portfolio.

12.4.2 Leverage Mechanics

There are numerous ways to weight single-stock futures in order to achieve the desired β level. In this subsection we consider three specific scenarios that a portfolio manager might encounter, each requiring a different weighting scheme.

In one scenario, all single-stock futures contracts are sufficiently liquid to use for leverage. In this case, to achieve his or her target β^* for the overall portfolio, the manager must purchase the following amount of futures contracts²⁵:

$$N_i = \frac{[\beta^* + (\xi - 1)\beta_s]V_t w_i}{\beta_s q_s p_{i,t}} \quad (12.21)$$

In this equation, q_s is the contract multiplier for single-stock futures, which is equal to 100, $p_{i,t}$ is the price of stock i , β_s is the β of the stock portfolio with respect to the benchmark, and w_i is the weight of stock i in the equity portfolio. The preceding formula indicates that the portfolio manager may

purchase single-stock futures for each stock in proportion to the weight of the stock in the equity portfolio.

The second scenario is that only a subset N_1 of the N stocks in the portfolio is available for single-stock futures trading. The manager therefore uses only the stocks with futures to achieve the desired β . To achieve his or her target β^* for the overall portfolio, he or she must purchase the following amount of futures contracts:

$$N_i = \frac{[\beta^* + (\xi - 1)\beta_s]V_t \tilde{w}_i}{\tilde{\beta}_s q_s p_{i,t}} \quad (12.22)$$

where $w_{\sim i}$ is the relative weight of the subset of stocks whose futures contracts are bought from or sold to each other (that is, $w_{\sim i} = w_i / \sigma_{j=1}^{N_1} w_j$), $\beta_{\sim s}$ is the weighted average β of the subset of stocks whose futures contracts are bought and sold (that is, $\beta_{\sim s} = \sigma_{j=1}^{N_1} w_{\sim i} \beta_j$), and all other variables are as defined previously.

In the third scenario, there are futures for some of the stocks in the portfolio, and the manager decides to combine single-stock futures with index futures to achieve the desired leverage. He or she purchases the following amounts of single-stock futures contracts and index futures, respectively:

$$N_i = \frac{[\beta^* + (\xi - 1)\beta_s]V_t \tilde{w}_i}{q_s p_{i,t} \left[\tilde{\beta}_s + \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \beta_f \right]} \quad (12.23)$$

$$N_f = \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \frac{\sum_{i=1}^{N_1} N_i q_s p_{i,t}}{q S_t} \quad (12.24)$$

An example might help to further illustrate these three scenarios. Suppose that the benchmark is the S&P 500 and that we have a simple five-stock portfolio consisting of the Mosaic Company (MOS), the Gap (GPS), Cabot Oil & Gas (COG), CME Group (CME), and Walmart (WMT). [Table 12.4](#) contains the portfolio and the weights of each stock in the portfolio. Using the preceding formulas, we can compute the number of futures contracts required to achieve our desired β^* . As in previous examples, let's assume that our desired $\beta^* = 2$ and that there is \$100 million in equity capital. The actual portfolio β of the five stocks in this example is 0.83, and the β_i for each of the stock's futures contract is listed in the table (we assume that it is the same as the underlying stock's β). The prices of each stock are also listed in the table. For this example, we do not round the futures contracts to the nearest whole number. Of course, in reality, you would have to do this.

TABLE 12.4

Example of Single-Stock Leverage in a Simple Portfolio

Method 1					
Ticker	w_i	β_i	p_i	$N_{f,i}$	$V_{f,i}$
MOS	0.200	1.73	17.19	16,412.71	\$28,213,445.19
GPS	0.200	1.56	14.08	20,037.96	\$28,213,445.19
COG	0.300	0.27	20.59	20,553.75	\$42,320,167.78
CME	0.100	0.30	167.15	843.96	\$14,106,722.59
WMT	0.200	0.31	129.98	2,170.60	\$28,213,445.19
Total		0.83			\$141,067,225.94
Method 2					
Ticker	$\tilde{w}_i$	β_i	p_i	$N_{f,i}$	$V_{f,i}$
MOS	0.286	1.73	17.19	18,448.72	\$31,713,355.02
GPS	0.286	1.56	14.08	22,523.69	\$31,713,355.02
COG	0.429	0.27	20.59	23,103.46	\$47,570,032.53
CME	0.000	0.30	167.15	0.00	\$0.00
WMT	0.000	0.31	129.98	0.00	\$0.00
		1.05			\$110,996,742.57
Method 3					
Ticker	$\tilde{w}_i$	β_i	p_i	$N_{f,i}$	$V_{f,i}$
MOS	0.286	1.73	17.19	13,117.15	\$22,548,386.65
GPS	0.286	1.56	14.08	16,014.48	\$22,548,386.65
COG	0.429	0.27	20.59	16,426.70	\$33,822,579.98
CME	0.000	0.30	167.15	0.00	\$0.00
WMT	0.000	0.31	129.98	0.00	\$0.00
S&P 500		1.00	3351	40.37	\$33,822,579.98
Total		1.05			\$112,741,933.27

Note: The variables in the table represent attributes of the stocks. $\tilde{w}_i$ represents the stock weight in the underlying equity portfolio; β_i represents the stock's β ; p_i represents the stock's actual price as of August 7, 2020; $N_{f,i}$ represents the number of single-stock futures contracts purchased or sold; $V_{f,i}$ represents the notional value of the futures contracts purchased; and $\tilde{w}_i$ represents the relative weight of the subset of securities whose single-stock futures are purchased. The total β 's represent the weighted-average β of the subset of stocks whose futures are purchased. In method 1, this corresponds to the β of the underlying stock portfolio.

Let's walk through the example for each of the three scenarios just outlined. Under the first scenario, in which we can use all the stocks' futures contracts to create leverage, we purchase a total futures value of \$141,067,225.94 to achieve our desired $\beta^* = 2$. Since the underlying

portfolio has a β that is lower than 1 with respect to the benchmark, we need more than an equal amount in dollars to achieve our desired leverage. The number of contracts of each single-stock futures is listed in the table.

Under the second scenario, we assume that some of the singlestock futures—say, CME and WMT—are not liquid enough to trade. Thus we can use only the other three securities' futures to achieve our desired leverage. In this case we purchase a total value of \$110,996,742.57 of futures. This consists of 18,449, 22,524, and 23,103 contracts of MOS, GPS, and COG, respectively. Although in this scenario we have again achieved our desired $\beta^* = 2$, we have leveraged the portfolio disproportionately. The resulting portfolio α is tilted toward the first three stocks.

Under the third scenario, we continue to use the first three single-stock futures, but we also use the index futures of the S&P 500 to achieve the desired β^* . In this case we purchase a total value of \$78,919,353 of the single-stock futures and \$33,822,580 of the index futures (40 contracts). This also achieves our desired goal, but we have diluted the overall α of the portfolio by using index futures.

12.4.3 Expected Returns, Risk, and α Mojo

In the first scenario, the total portfolio is composed of the stock portfolio with the weight of $1 - \xi$ and the portfolio of single-stock futures with the weight of $l - \xi + 1$. Recall that given the benchmark return r_B , the return of the stock portfolio can be expressed as

$$r_s = \alpha_s + \beta_s r_B + \epsilon_s \quad (12.25)$$

The return of the single-stock futures portfolio is essentially identical to the return of the stock portfolio. For simplicity, we assume²⁶

$$r_{sf} = \alpha_s + \beta_s r_B + \epsilon_s \quad (12.26)$$

Calculation of the expected return and the variance of the total portfolio is straightforward in this case. Given the weight of the stock portfolio and the single-stock futures portfolio, the total portfolio return is

$$\begin{aligned}
r_P &= (1 - \xi)r_s + (l - 1 + \xi)r_{sf} \\
&= l\alpha_s + l\beta_s r_B + l\epsilon_s
\end{aligned} \tag{12.27}$$

The expected return and the variance of the total portfolio are

$$E(r_P) = l\alpha_s + l\beta_s \mu_B = lE(r_s) \tag{12.28}$$

$$V(r_P) = l^2\beta_s^2\sigma_B^2 + l^2\omega_s^2 = l^2V(r_s) \tag{12.29}$$

where μ_B and σ_B denote the expected return and the standard deviation of the benchmark, and ω_s denotes the standard deviation of ϵ_s . We can easily calculate the Sharpe ratio (SR) of the total portfolio as well. Assuming that the risk-free rate has been subtracted already, the Sharpe ratio is simply the ratio of $E(r_P)$ to $\sqrt{V(r_P)}$:

$$SR_P = \frac{l\alpha_s + l\beta_s\mu_B}{\sqrt{l^2\beta_s^2\sigma_B^2 + l^2\omega_s^2}} = SR_s \tag{12.30}$$

Thus the Sharpe ratio of the total portfolio is identical to the Sharpe ratio of the stock portfolio, regardless of the leverage. That is, the leverage does not change the Sharpe ratio in the first scenario.

In the second scenario, the total portfolio is again composed of the stock portfolio and the single-stock futures portfolio. The weights of these portfolios are identical to the first scenario. The difference is the composition of the single-stock futures portfolio. The single-stock futures portfolio is N_1 stock futures that are liquid enough. The weights of these futures are given by $\tilde{w}_i = w_i / \sum_{j=1}^{N_1} w_j, i = 1, \dots, N_1$.

It will be useful to think of the stock portfolio as composed of two subportfolios: the portfolio of stocks with liquid futures and the portfolio of stocks without liquid futures. Let us call the first portfolio $s1$ and the second $s2$. Then $s1$ is the portfolio of N_1 stocks with the weights $\tilde{w}_i$ defined earlier. $s2$ is the portfolio of the remaining stocks with the weights

$\tilde{w} = w_i / \sum_{j=N_1+1}^N w_j$, $i = N_1 + 1, \dots, N$. Defining $\psi = \sum_{j=1}^{N_1} w_j$, we may express the return of the stock portfolio as

$$\begin{aligned} r_s &= \psi r_{s1} + (1 - \psi) r_{s2} \\ &= [\psi \alpha_{s1} + (1 - \psi) \alpha_{s2}] + [\psi \beta_{s1} + (1 - \psi) \beta_{s2}] r_B \\ &\quad + \psi \epsilon_{s1} + (1 - \psi) \epsilon_{s2} \end{aligned} \quad (12.31)$$

Assuming that the return of the single-stock futures is identical to the return of the underlying stocks, the return of the single-stock futures portfolio is identical to the return of portfolio $s1$.²⁷ Thus we may express the return of the single-stock futures portfolio as

$$r_{sf} = r_{s1} = \alpha_{s1} + \beta_{s1} r_B + \epsilon_{s1} \quad (12.32)$$

Now, calculation of the expected return and the variance of the total portfolio is straightforward. Given the weight of the stock portfolio and the single-stock futures portfolio, the total portfolio return is

$$\begin{aligned} r_p &= (1 - \xi) r_s + [l - (1 - \xi)] r_{sf} \\ &= [l - (1 - \xi)(1 - \psi)] r_{s1} + (1 - \xi)(1 - \psi) r_{s2} \\ &= [l - (1 - \xi)(1 - \psi)] \alpha_{s1} + (1 - \xi)(1 - \psi) \alpha_{s2} \\ &\quad + \{[l - (1 - \xi)(1 - \psi)] \beta_{s1} + (1 - \xi)(1 - \psi) \beta_{s2}\} r_B \\ &\quad + [l - (1 - \xi)(1 - \psi)] \epsilon_{s1} + (1 - \xi)(1 - \psi) \epsilon_{s2} \end{aligned} \quad (12.33)$$

The expected return and the variance of the total portfolio are

$$\begin{aligned} E(r_p) &= [l - (1 - \xi)(1 - \psi)] \alpha_{s1} + (1 - \xi)(1 - \psi) \alpha_{s2} \\ &\quad + \{[l - (1 - \xi)(1 - \psi)] \beta_{s1} + (1 - \xi)(1 - \psi) \beta_{s2}\} \mu_B \end{aligned} \quad (12.34)$$

$$\begin{aligned} V(r_p) &= \{[l - (1 - \xi)(1 - \psi)] \beta_{s1} + (1 - \xi)(1 - \psi) \beta_{s2}\}^2 \sigma_B^2 \\ &\quad + [l - (1 - \xi)(1 - \psi)]^2 \omega_{s1}^2 + [(1 - \xi)(1 - \psi)]^2 \omega_{s2}^2 \\ &\quad + 2[l - (1 - \xi)(1 - \psi)][(1 - \xi)(1 - \psi)] \omega_{s12} \end{aligned} \quad (12.35)$$

where ω_{s12} is the covariance between ϵ_{s1} and ϵ_{s2} .

Assuming that the risk-free rate already has been subtracted, the Sharpe ratio is simply the ratio of $E(r_P)$ to $\sqrt{V(r_P)}$, and the squared Sharpe ratio is the ratio of $E(r_P)^2$ to $V(r_P)$. Expressing the squared Sharpe ratio as a function of the leverage l , one can find the relationship:

$$SR_P^2 \rightarrow \frac{(\alpha_{s1} + \beta_{s1}\mu_B)^2}{\beta_{s1}^2 \sigma_B^2 + \omega_{s1}^2} = SR_{s1}^2 \quad (12.36)$$

as the leverage becomes large.²⁸ That is, as the leverage gets larger, the risk-return relationship of the total portfolio becomes very similar to that of portfolio $s1$, the portfolio of stocks with liquid futures.

In the third scenario, the total portfolio is composed of the stock portfolio with weight $1 - \xi$, the index future with weight $(1 - \psi)(l - 1 + \xi)$, and the single-stock futures portfolio with weight $\psi(l - 1 + \xi)$. The single-stock futures portfolio is the same as the one in the second scenario, that is, the portfolio of N_1 stock futures that are liquid enough, with weights $\tilde{w}_i = w_i / \sum_{j=1}^{N_1} w_j$, $i = 1, \dots, N_1$. We will assume that the futures index return is identical to the benchmark return²⁹ (i.e., $r_f = r_B$). The returns for the stock portfolio and for the single-stock futures portfolio are identical to those of the second scenario, as described in Eqs. (12.31) and (12.32).

The return of the total portfolio can be calculated using the weights of the three subportfolios:

$$\begin{aligned} r_P &= (1 - \xi)r_s + (1 - \psi)(l - 1 + \xi)r_f + \psi(l - 1 + \xi)r_{sf} \\ &= \psi l r_{s1} + (1 - \xi)(1 - \psi)r_{s2} + (1 - \psi)(l - 1 + \xi)r_B \\ &= \psi l \alpha_{s1} + (1 - \xi)(1 - \psi)\alpha_{s2} \\ &\quad + [\psi l \beta_{s1} + (1 - \xi)(1 - \psi)\beta_{s2} + (1 - \psi)(l - 1 + \xi)]r_B \\ &\quad + \psi l \epsilon_{s1} + (1 - \xi)(1 - \psi)\epsilon_{s2} \end{aligned} \quad (12.37)$$

The expected return and the variance of the total portfolio are

$$E(r_p) = \psi l \alpha_{s1} + (1 - \xi)(1 - \psi) \alpha_{s2} \\ [\psi l \beta_{s1} + (1 - \xi)(1 - \psi) \beta_{s2} + (1 - \psi)(l - 1 + \xi)] \mu_B \quad (12.38)$$

$$V(r_p) = [\psi l \beta_{s1} + (1 - \xi)(1 - \psi) \beta_{s2} + (1 - \psi)(l - 1 + \xi)]^2 \sigma_B^2 \\ + \psi^2 l^2 \omega_{s1}^2 + (1 - \xi)^2 (l - \psi)^2 \omega_{s2}^2 \\ + 2\psi l (1 - \xi)(1 - \psi) \omega_{s12} \quad (12.39)$$

where ω_{s12} is the covariance between ϵ_{s1} and ϵ_{s2} as defined before.

Assuming that the risk-free rate already has been subtracted, the squared Sharpe ratio is the ratio of $E(r_p)^2$ to $V(r_p)$. Expressing the squared Sharpe ratio as a function of the leverage l , one can find the following relationship:

$$SR_p^2 \rightarrow \frac{\left\{ \psi \alpha_{s1} + [\psi \beta_{s1} + (1 - \psi)] \mu_B \right\}^2}{[\psi \beta_{s1} + (1 - \psi)]^2 \sigma_B^2 + \psi^2 \omega_{s1}^2} \quad (12.40)$$

as the leverage increases. Note that the right-hand side of this equation can be interpreted as the Sharpe ratio of a portfolio made of $s1$ (the portfolio of stocks for which liquid futures are available) and the benchmark, with the respective weights of ψ and $1 - \psi$. That is, the Sharpe ratio of the total portfolio approaches the Sharpe ratio of $\psi r_{s1} + (1 - \psi) r_B$ as the leverage increases. This is not surprising. As the leverage increases, the portfolio is dominated by the stocks with liquid futures available and the benchmark.

From these formulas, we find that the portfolio's α gets the biggest boost from leverage that uses all single-stock futures (scenario 1). When futures are available for all the portfolio's stocks and all the futures have enough trading liquidity, the best approach is to leverage only with single-stock futures. It is the ultimate form of α mojo. Leveraging with a subset of futures provides the next-best α boost, but it may be suboptimal.

12.5 STOCKS, CASH, INDIVIDUAL STOCKS, AND SINGLE-STOCK AND BASKET SWAPS

A portfolio manager who produces positive α may wish to leverage his or her portfolio for a variety of reasons. As we have seen, one of the drawbacks to using index futures for leverage is that it saps the relative α contribution. Although single-stock futures would resolve this problem, they may not have enough breadth or liquidity to serve as a realistic solution. In this section we discuss two more related alternatives to leveraging. The first alternative is to buy individual stocks on margin; the second is to create a series of individual stock swap contracts with a broker-dealer.

12.5.1 Margining Individual Stocks

Buying stocks on margin requires an initial margin of m_s , which according to Regulation T is currently 50%.³⁰ A manager technically could purchase his or her entire basket of securities on margin, effectively creating a leveraged position as high as $l = 2$. Thus, margining securities allows for any amount of leverage between $l = 1$ and $l = 2$. The benefit of this approach is that the leverage does not diminish the relative α contribution—in fact, margining securities leverages the α of the portfolio manager. The drawback of this form of leverage is that it is more costly than leveraging with index futures. The broker-dealer charges the broker call rate as interest on the margined loans, a rate that is typically higher than the futures carry rate (i.e., the implied repo rate).³¹ Thus, for an index manager who wants to leverage the portfolio, index futures are clearly the better choice. However, a manager who creates α by picking stocks should weigh the extra interest costs of the margin against the strength of his or her α .

12.5.2 Single-Stock and Basket Swaps

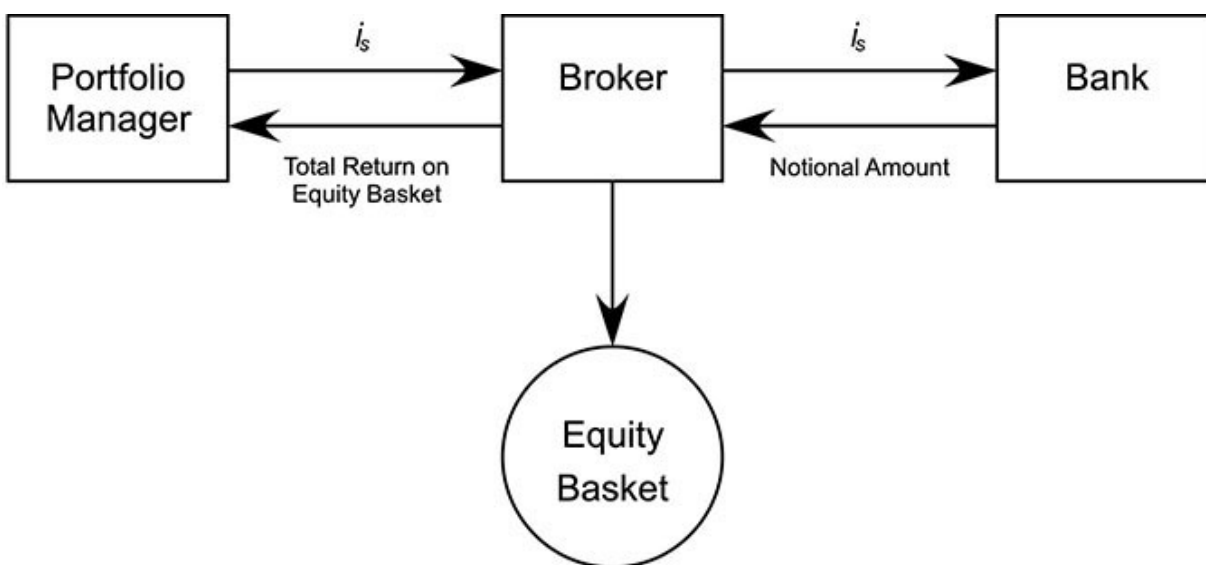
In common parlance, a *swap* is an exchange of one thing for another. In financial markets, a swap is an agreement between two parties to exchange payments with each other. The most common swaps are interest-rate and currency swaps. With interest-rate swaps, it is usually the case that one party pays fixed interest and the other party pays floating interest. With currency swaps, one party makes payment in one currency, and the other party makes payment in another.³²

Equity swaps are usually set up so that one party pays the return of an equity index, whereas the other party pays fixed or floating interest on the notional principal. Equity swaps are a convenient way for portfolio managers to increase or decrease their exposure to the equity index without actually buying or selling equities. For example, a typical equity swap might have a broker-dealer pay fund X the total return on the S&P 500 in exchange for fund X paying the broker-dealer some reference interest rate (say, the 6-month LIBOR rate plus a spread).³³ The swap can be customized many different ways.

Equity swaps can be structured for equity indices or for individual equity securities.³⁴ The portfolio manager can construct a customized swap with the broker-dealer for a basket of securities (Fig. 12.2.) This basket or portfolio of securities in the swap is the same as the basket of securities that the portfolio manager believes will outperform the market. The broker-dealer will pay the portfolio manager the return of the basket of securities during the swap period, whereas the manager will pay the broker-dealer some interest rate i_s typically specified as LIBOR + x . (Alternatively, it might be SOFR + x .) The broker-dealer also will charge the portfolio manager some additional fees for setting up the trades because he or she must hedge his or her position. The broker-dealer typically will hedge the position by buying the actual equity basket of securities and taking on a little risk or by buying a futures contract in proportion to the equity basket and taking on slightly more risk. Collateral for such a trade can be arranged through the custodian of the portfolio manager or directly through the broker-dealer.

FIGURE 12.2

Example of flows from an equity basket swap.



The swap periods should match the portfolio's rebalancing frequency. If a portfolio manager rebalances his or her portfolio monthly, the swap should be reset monthly. With a monthly swap, the broker-dealer will pay the portfolio manager the return of the basket at the end of the month, and the portfolio manager will pay the interest cost i_s . The portfolio manager then will reconstruct his or her optimal portfolio, send a new basket list to the broker-dealer, and construct a new swap for the following month. In this arrangement, the portfolio manager can invest directly in the equities and construct a swap for leverage, or he or she just can hold cash and leverage the entire position with individual stock or basket swaps.

Equity swaps have several advantages over other means of leveraging a portfolio. With swaps on a basket of securities, the portfolio manager can alter his or her α directly by choosing a basket of his or her actual optimal portfolio of securities. This is another kind of α mojo that index futures do not provide. Also, leveraging with equity swaps is usually cheaper than margining stocks, and it is an alternative for securities that do not have liquid single-stock futures contracts.

12.6 STOCKS, CASH, AND OPTIONS

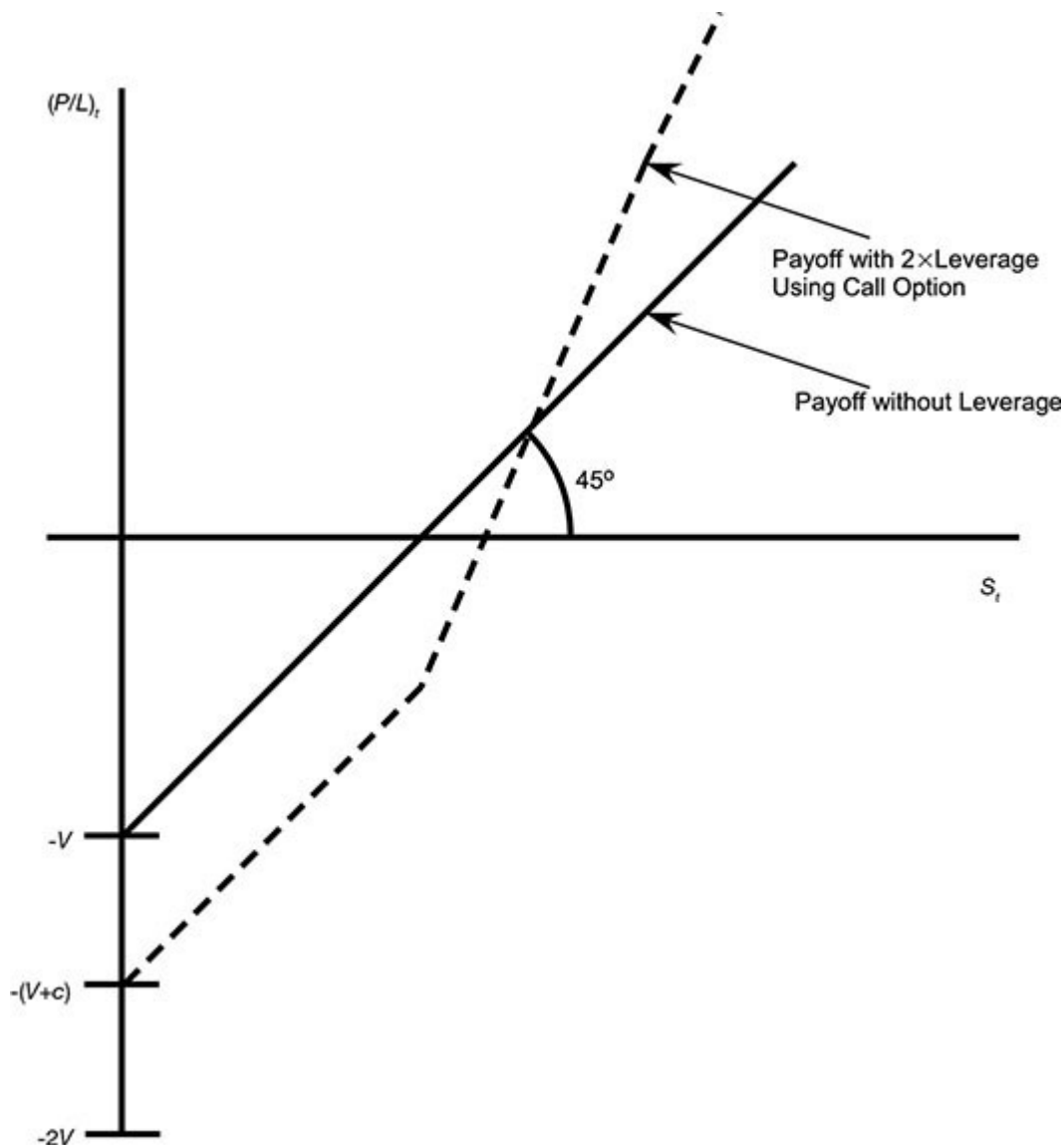
A manager who has built a portfolio may wish to leverage the portfolio yet another way. Rather than using index futures to increase the market exposure of the entire portfolio, the manager may wish to leverage the

portfolio with index options. The reason to do this is twofold. First of all, options have the potential for a nonlinear payoff. They can leverage returns on the upside without leveraging them on the downside. The second reason is that options offer a greater magnitude of leverage than do index futures. This kind of flexibility comes at a cost, though, because options are pricey.

Figure 12.3 shows how the payoff is altered with a call option on an underlying portfolio. The payoff of the overall portfolio is leveraged two times for upward movements in the index but continues to be unlevered for downward movements.

FIGURE 12.3

Portfolio payoffs with and without leverage using call options.



The number of call option contracts required on a given index to achieve a given level of β^* for the entire portfolio given a current β_s for the stock portion of the portfolio will be given by

$$N_O = \frac{[\beta^* + (\xi - 1)\beta_s]V_t}{\beta_O q_O S_t} = \frac{[\beta^* + (\xi - 1)\beta_s]V_t}{\Delta_O \beta_I q_O S_t} \quad (12.41)$$

where N_O represents the number of call option contracts to purchase, q_O represents the option contract multiplier, β^* equals the desired level of exposure, β_s is the β of the stock portfolio, Δ_O is the delta of the call option, β_I is the β of the underlying index with respect to the benchmark, and $\beta_O = \Delta_O \beta_I$ is just the delta approximation for the β of the option contract.³⁵

For example, suppose that we use the same numbers as in the preceding example. Thus the S&P 500 is trading at 1,000, the portfolio manager would like to hold 50% cash, and the desired $\beta^* = 2$. Normally, a portfolio manager simply will look up the prices of traded call options and either use a standard Black-Scholes Δ or compute his or her own. Suppose that the price of an at-the-money call option (i.e., $K = 1,000$) was $c = 43.58$. Suppose that the Δ of this call equaled 0.55.³⁶ Using our formula, we find that we should have

$$N_O = \frac{[2 + (0.5 - 1) \cdot 1]100 \text{ M}}{0.55 \cdot 1 \cdot 100 \cdot 1,000} = 1819.05 \quad (12.42)$$

Thus the portfolio manager should purchase 1,819 calls on the S&P 500. This will provide the desired leverage of 2.

Leveraging through options offers the major advantage of magnifying only the upside. The chief danger of leverage, as explained at the outset of this chapter, is that it creates the possibility of a very hard fall when the portfolio runs into negative returns. With options leverage, though, the portfolio loses only what it would lose anyway. Perhaps this sounds too

good to be true. Indeed, the cost of this kind of insurance is steep. The options in this example have a three-month duration. They would cost a total of $100 \cdot 1,819 \cdot 43.58 = \$7,927,202$. Leverage with downside protection costs \$8 million dollars for a \$100 million portfolio. This is clearly an expensive method of leveraging, feasible only in specific cases in which the risk of the portfolio calls for downside protection.

12.7 REBALANCING

The portfolio manager has to rebalance the portfolio regularly to maintain the desired leverage position. The frequency of the rebalancing depends on how closely and how long he or she wants to track the β^* level. If he or she is leveraging only over the short term, he or she may just close the position at the end of his or her short-term time horizon without rebalancing. Portfolio managers offering the leveraged portfolio to clients through a mutual fund, however, may wish to rebalance daily.³⁷ Other portfolio managers may believe that leverage will offer higher long-run returns and might choose to rebalance monthly. In this section we describe methods for rebalancing at intervals between t and $t + k$, where k can be any period the portfolio manager chooses from $k = 1$ (1 day) to $k = 20$ (1 month, in business days) or longer.

12.7.1 Cash and Futures

For the simplest case of using cash and futures, we know that the β_P of the portfolio at time t will be

$$\beta_{P,t} = \frac{N_{f,t} q S_t}{V_t} \beta_f \quad (12.43)$$

where subscript t indicates the time period t of each variable. At $t + k$, some of the variables will have changed. In particular, the total portfolio value V_{t+k} and the value of the index underlying the futures S_{t+k} will have changed.

On day $t + k$, the portfolio manager will have to compute the desired number of futures contracts required to maintain his or her desired

exposure. To satisfy

$$\beta^* = \frac{N_{f,t+k} q S_{t+k}}{V_{t+k}} \beta_f \quad (12.44)$$

the number of future contracts required is

$$N_{f,t+k} = \frac{\beta^* V_{t+k}}{\beta_f q S_{t+k}} \quad (12.45)$$

The portfolio manager needs to purchase $N_{f,t+k} - N_{f,t}$ contracts, which can be expressed as

$$\begin{aligned} N_{f,t+k} - N_{f,t} &= \frac{\beta^* V_{t+k}}{\beta_f q S_{t+k}} - \frac{\beta^* V_t}{\beta_f q S_t} \\ &= \frac{\beta^*}{\beta_f q} \left(\frac{V_{t+k}}{S_{t+k}} - \frac{V_t}{S_t} \right) \\ &= \frac{\beta^*}{\beta_f q} \frac{V_t}{S_{t+k}} \left(\frac{V_{t+k} - V_t}{V_t} - \frac{S_{t+k} - S_t}{S_t} \right) \end{aligned} \quad (12.46)$$

This equation says that whether the portfolio manager will have to purchase or sell futures contracts (i.e., whether $N_{f,t+k} - N_{f,t}$ is positive or negative) depends on the total return of the portfolio $(V_{t+k} - V_t)/V_t$ and the return of the underlying futures index $(S_{t+k} - S_t)/S_t$. The value of the total portfolio at time $t + k$ is given as

$$V_{t+k} = N_{f,t} q (F_{t+k} - F_t) + I_{t,t+k} V_t \quad (12.47)$$

where F_{t+k} is the price of the futures contract at time $t + k$, F_t is the price of the futures contract at time t , and $I_{t,t+k} V_t$ represents the interest earned on

the margin cash and the other cash of the portfolio at rates i' and i , respectively.³⁸ Thus the return on the total portfolio is given as

$$\frac{V_{t+k} - V_t}{V_t} = \frac{N_{f,t} q S_t}{V_t} \frac{F_{t+k} - F_t}{S_t} + (I_{t,t+k} - 1) \quad (12.48)$$

The first part of the right-hand side ($N_{f,t} q S_t / V_t$) is simply the leverage ratio. Assuming no basis risk, we know that $[(F_{t+k} - F_t) / S_t] = [(S_{t+k} - S_t) / S_t]$, and we can make this substitution into our equation.³⁹ Thus

$$\frac{V_{t+k} - V_t}{V_t} = l \frac{S_{t+k} - S_t}{S_t} + (I_{t,t+k} - 1) \quad (12.49)$$

Now we may rewrite Eq. (12.46) as follows⁴⁰:

$$N_{f,t+k} - N_{f,t} = \frac{\beta^*}{\beta_f q} \frac{V_t}{S_{t+k}} \left[(l - 1) \frac{S_{t+k} - S_t}{S_t} + (I_{t,t+k} - 1) \right] \quad (12.50)$$

From Eq. (12.50) we can make the following general statement: For a leveraged position ($l > 1$) and a positive return on the index $[(S_{t+k} - S_t) / S_t > 0]$, the portfolio manager must purchase more futures contracts to maintain β^* or the desired leverage. On the other hand, the portfolio manager will have to sell futures contracts if the index return is sufficiently negative. Since we have a leveraged position, any time the target β^* is greater than the β of the futures contract, we may state the following: If the target β^* is greater than the β of the futures contract, a positive return on the index will lower β over time, and the portfolio manager will have to purchase more futures contracts. A negative index return will increase β over time, and the portfolio manager will have to sell futures contracts. Table 12.5 illustrates the various general rules.

TABLE 12.5

Rebalancing Actions to Achieve Desired β^*

	$r_{t,t+k} > 0$	$r_{t,t+k} < 0$
$\beta^* > \beta_t$	Buy futures	Sell futures
$\beta^* < \beta_t$	Sell futures	Buy futures

Suppose that there is a portfolio manager whose benchmark is the S&P 500 and who would like to use S&P 500 futures to leverage his or her portfolio. The following conditions exist: $\beta_f = 1$ for the S&P 500 futures contracts; the multiplier for the contracts is $q = 250$; the beta the manager would like to achieve is $\beta^* = 2$; the value of the equity capital is \$100 million; the S&P 500 and the futures contract are trading at 1,000 and 1,007.53, respectively, on day t ; and the continuously compounded annual interest rates on margin and cash are $i' = 0.02$ and $i = 0.03$, respectively. To achieve the desired β of the total portfolio on day t , the manager must purchase exactly $N_{f,t} = \beta^* V_t / \beta_f q S_t = 2 \cdot 100,000,000 / (1 \cdot 250 \cdot 1,000) = 800$ contracts. This will create a portfolio with a $\beta_P = 2$ and a leverage of 2 as well.

Suppose that one day has passed; it is now day $t + 1$. The S&P 500 and futures values have increased by 5% in one day, so the β exposure has changed. The manager wants to readjust it back to 2. Using our formula, we know that the portfolio manager would have to purchase a net amount of futures contracts. This value is given by

$$\begin{aligned}
 N_{f,t+k} - N_{f,t} &= \frac{\beta^*}{\beta_f q} \left(\frac{V_{t+k}}{S_{t+k}} - \frac{V_t}{S_t} \right) \\
 &= \frac{2}{1 \cdot 250} \left(\frac{110,083,335.75}{1,050} - \frac{100,000,000}{1,000} \right) = 38.73
 \end{aligned} \tag{12.51}$$

Now, in reality, the portfolio manager will have to buy either 38 or 39 contracts, unless he or she uses S&P 500 E-minis to fine-tune the purchase. It is better to round to the nearest integer, so he or she probably will purchase 39 contracts.

12.7.2 Stocks, Cash, and Futures

A very similar change in the β of the overall portfolio will occur with the portfolio of stocks, cash, and futures.⁴¹ At $t + k$, the futures prices and portfolio values will have changed. In this case, the portfolio manager actually may change the composition of the underlying portfolio; thus β_s may change from time t to $t + k$. The net change in futures contracts becomes

$$N_{f,t+k} - N_{f,t} = \frac{\beta^*}{\beta_f q} \left(\frac{V_{t+k}}{S_{t+k}} - \frac{V_t}{S_t} \right) + \frac{1}{\beta_f q} \left(\frac{(\xi_{t+k} - 1)\beta_{s,t+k} V_{t+k}}{S_{t+k}} - \frac{(\xi_t - 1)\beta_{s,t} V_t}{S_t} \right) \quad (12.52)$$

where $\beta_{s,t}$ and $\beta_{s,t+k}$ are the stock portfolio β at times t and $t + k$, respectively. This is more complicated to analyze. However, if we assume that the stock portfolio β is the same or similar between time t and time $t + k$ and that the percentage held in cash is the same, then the net purchases of futures contracts simplifies to

$$N_{f,t+k} - N_{f,t} = \frac{\beta^* + (\xi_{t+k} - 1)\beta_s}{\beta_f q} \left(\frac{V_{t+k}}{S_{t+k}} - \frac{V_t}{S_t} \right) \quad (12.53)$$

In this case, the conditions that determine whether there are net purchases or net sales are exactly the same as they were in the preceding subsection with only cash and futures.

12.8 LIQUIDITY BUFFERING

Leveraging a portfolio increases the risk of the portfolio. In fact, except in the case of index options that amplify only gains, leveraging itself can create unrecoverable losses—ones that run in excess of the equity in the portfolio. In order to maintain the portfolio's liquidity, a manager must purchase some form of portfolio insurance. Although there are various ways to achieve this downside protection, we shall focus on how to use *out-of-the-money put options* to prevent the levered portfolio's losses from exceeding the amount of equity in the portfolio. This method is simple to implement.

First, we need to compute the index loss that would exactly use up the equity value of the portfolio. Once we know this value, we can purchase the appropriate put options in terms of strike price and quantity. Liquidity buffering is applicable to all the leveraging methods discussed in this chapter, but we will focus on applying it to the cash and futures leverage case.

When we create leverage with cash and futures, we know that in a period of time between t and $t + k$, the value of the portfolio's equity equals

$$\begin{aligned}
 V_{t+k} &= N_{f,t} q (F_{t+k} - F_t) + I_{t,t+k} V_t \\
 &= N_{f,t} q S_t r_{t,t+k}^f + I_{t,t+k} V_t \\
 &= \left(\frac{\beta^* V_t}{\beta_f q S_t} \right) q S_t r_{t,t+k}^f + I_{t,t+k} V_t
 \end{aligned} \tag{12.54}$$

where $r_{t,t+k}^f \equiv (F_{t+k} - F_t) / S_t$, and all other variables have been defined previously. The goal is to find the strike price of the put options at which all the portfolio's equity would be consumed. This is the point at which closing the futures position would wipe out the portfolio's equity. By purchasing put options at this strike price, any further deterioration in the returns of the index and futures would be completely offset by the put options. There would be no further loss in portfolio value, making the leveraged portfolio feasible.

Assuming for simplicity that there is no basis risk (i.e., that we can substitute $r_{t,t+k}$ for $r_{t,t+k}^f$, since they are equal), we must find the index return at which all the account equity would be dissolved.⁴² This would be the value that equates the loss on the futures position with the current account equity multiplied by the interest factor, that is,

$$\left(\frac{\beta^* V_t}{\beta_f q S_t} \right) q S_t r_{t,t+k} = -I_{t,t+k} V_t \tag{12.55}$$

which implies

$$r_{t,t+k}^* = \frac{-\beta_f I_{t,t+k}}{\beta^*} \quad (12.56)$$

Let us return to the example we used in the preceding subsection. Suppose that the portfolio manager's target $\beta^* = 2$, $\beta_f = 1$, the index value on day t is 1,000, and the futures' value on day t is 1,007.53. We also will assume for simplicity that interest rates are 0 or that $I_{t,t+k} \approx 1$. The return that would consume the account equity over the k -period horizon would be -0.5 , or a negative return of 50%.

We have derived this return on day t , so we need to find the actual strike price of the put options to purchase. In particular, if the index is trading at S_t , then we will want to purchase options with a strike price $K = (1 + r_{t,t+k}^*)S_t$. Thus, in our particular example, we would purchase options with a strike price of 500 (or the closest value to that).

The final step is determining how many options contracts to purchase. The simplest method is to use a dollar hedge adjusted for β . Thus we compute the number of options contracts to purchase, given the β of the underlying index. Thus

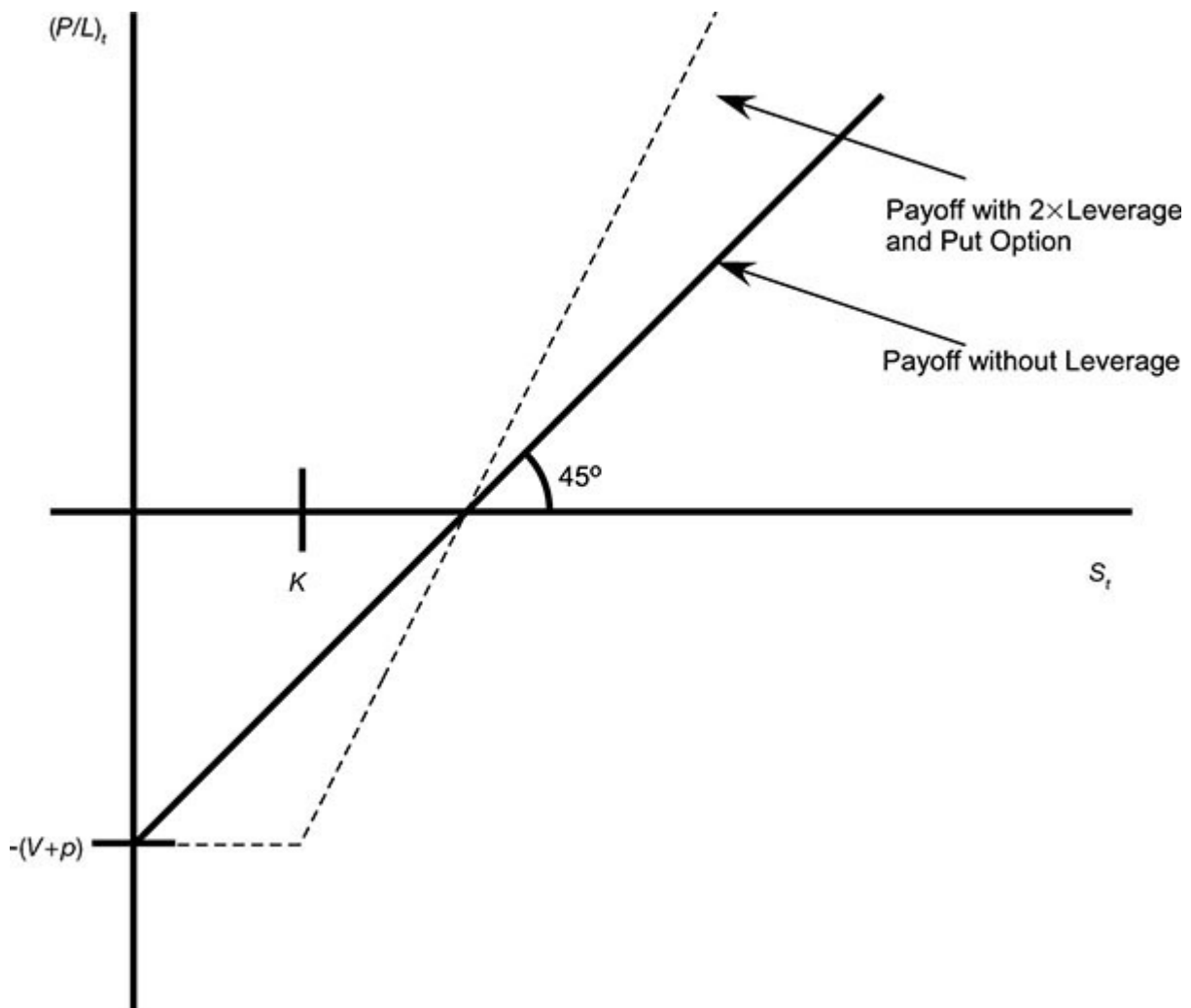
$$N_O = \frac{\beta^* V_t}{\beta_I q_O S_t} \quad (12.57)$$

where β_I is the β of the underlying index with respect to the benchmark, q_O is the contract multiplier, and N_O is the number of option contracts to purchase.⁴³ Thus, in our previous example with $\beta^* = 2$, $\beta_I = 1$, $S_t = 1,000$, and $F_t = 1,007.53$, q_O for the S&P 500 being equal to 100, the number of options contracts to purchase is 2,000.

An illustration of the overall payoff to the portfolio when out-of-the-money options are used to limit the losses of the portfolio to the equity capital is shown in Fig. 12.4. The resulting portfolio looks similar to an in-the-money call option purchased at strike price K and purchased so as to have a twofold exposure to the index in this particular case.

FIGURE 12.4

Portfolio payoffs with and without leverage and liquidity buffering.



We have closed the matter for day t . We have created a leveraged portfolio and purchased out-of-the-money put options to protect the liquidity of the account. We should note that these options are extremely out of the money, and index options are not typically available extremely out of the money. It is rare, even on a major index, to find any active trading in options even 20% out of the money. When it does happen, the bid-ask spreads are very wide.

Therefore, practically speaking, creating this type of leverage requires the portfolio manager to form some kind of over-the-counter (OTC) trade with one of the major broker-dealers. The portfolio manager can have the structured desk of a broker-dealer create these, or the broker-dealer can create them with *flex options*. Flex options (flexible options) are customized equity or index options contracts available on a number of exchanges, including the Chicago Board Options Exchange (CBOE). The flexible

elements of the options include the strike price, contracts terms, exercise styles, and expiration dates. Flex options eliminate counterparty risk because they are guaranteed by the Options Clearing Corporation, and they offer more liquidity than a typical OTC contract because they trade on a secondary market. They can be traded in size, expanding or even canceling entire positions.

These options might cost slightly more than the Black-Scholes or standard option-pricing model would predict, but being extremely out of the money, they still would be very cheap.⁴⁴

Continuing with the preceding example, we need to purchase 2,000 put option contracts at a strike price of 500. Using the Black-Scholes formula with an interest rate of 3%, a volatility of the S&P 500 of 20%, and a time to maturity of three months, the Black-Scholes price for one option is $\$1.16 \cdot 10^{-11}$. The cost of 2,000 contracts is thus $\$2.32 \cdot 10^{-6}$. As a percentage of the equity capital, this is $2.32 \cdot 10^{-14}\%$. Clearly, the cost of this insurance, at least theoretically, is insignificant. In reality, the broker-dealer might charge as much as 5 cents per options contract.

To absolutely protect the equity value of the portfolio, one would need to rebalance these options daily. Some mutual funds require daily rebalancing. From a practical perspective, though, these out-of-the-money options contracts could be set up for a three-month duration and not rebalanced until they expire. Rebalancing would take place every three months for the entire portfolio. The risk of losing the entire equity value of the portfolio is minute, especially considering the fact that all exchanges implement trading halts during severe declines in equity.⁴⁵ Halts in trading give the portfolio manager time to reposition his or her portfolio if his or her downside potential is not perfectly hedged.

We have just illustrated liquidity buffering in the case of cash and futures. Liquidity buffering works just as well for a portfolio of stocks, cash, and futures. In that case, as long as the portfolio beta is well estimated and the portfolio well diversified, the manager can be comfortable that using index options as buffers will prevent catastrophe.

12.9 LEVERAGED SHORT

So far we have discussed the idea of leveraging a long-only portfolio. What if the portfolio manager has a negative view of the market as a whole? There is no reason that the manager cannot apply these leveraging concepts to short positions as well, whether he or she is an index manager or a stock picker. There are just some minor differences in technique for leveraging a short portfolio.

When an index manager shorts the market, he or she can bypass trading individual stocks altogether, keep his or her equity capital in cash to serve as margin, and short index futures contracts. Shorting is a little more complicated for a portfolio manager who picks stocks to outperform the market. He or she must selectively short low- α stocks. We discuss some of the issues related to shorting individual stocks in the next chapter on market-neutral portfolios. For now, we can say that the stock-picking portfolio manager can short stocks a number of ways: by shorting the stocks combined with shorting index futures, by shorting the stocks combined with single-stock or basket swap contracts (in which, as opposed to the long-position swaps, the portfolio manager pays the stock returns), or by shorting the stocks and/or shorting singlestock futures. The concepts involved are very similar to those that govern the long side; they just operate in reverse.

Consider the case of the index portfolio manager who wishes to short the entire index. We already showed how to find, for the long portfolio, the number of contracts to buy to reach a desired overall portfolio β . With shorting, the only difference is that rather than *buying* the futures, the manager will *short* them. The equations for long-only portfolios apply to short-only portfolios, the only difference being that the number of contracts will be negative (reflecting the fact that the β^* is negative).

Given an S&P 500 trading at 1,000, a multiplier for the large S&P 500 contracts of 250, a sum of cash equal to \$100 million, and a $\beta^* = -2$, the appropriate number of futures contracts to sell would be 800. (Remember that $\beta_f = 1$ when the benchmark is the S&P 500.) The portfolio manager should sell the 800 futures contracts at a cost of $(qN_f F_t)m_f$ in margin balance. The remainder is held in cash, earning the interest rate.

As for rebalancing the position, the same concepts apply that apply to long-only portfolios. The triggers for rebalancing simply change. A market rally, for instance, would be cause to short futures contracts. A market decline would call for the purchase of futures contracts.

As for liquidity buffering, the portfolio manager also should follow the guidelines of liquidity buffering for long-only portfolios, except that now he or she should buy out-of-the-money call options rather than out-of-the-money put options.

12.10 CONCLUSION

Leverage is powerful. It introduces plenty of risk into the portfolio and has caused the collapse of entire funds. Yet it is the ultimate source of α mojo, and it can boost the portfolio's overall return. In this chapter we explained leveraging for index managers and stock pickers for long-only portfolios and ones that are short on the market. We looked at leveraging with equity index futures, single-stock futures, equity swaps, and options. We explained methods of fine-tuning the amount of leverage in a portfolio so as to control the amount of associated risk. These methods included calculating the appropriate number of futures contracts to achieve any desired β or leverage level, rebalancing the portfolio back to the desired leverage ratio, and creating buffers to ensure that the portfolio does not lose more than its equity capital. We have seen α mojo at work in the leveraged portfolio. In the next chapter we find it again in the market-neutral portfolio.

¹ See [Chapter 2](#) for further explanation of α^B and other variations of α .

² See [Table 12.1](#) for a list of the advantages and disadvantages of leverage.

³ Some forms of leverage, such as a long-short portfolio, actually can reduce the market risk of the portfolio. Throughout this chapter, when we speak of leverage, we are implicitly referring to an increase in exposure or amplification of the underlying position. Some people call this *net leverage*, which is different than *gross leverage*, which considers the absolute sum of all positions divided by equity capital. We discuss long-short types of portfolios in [Chapter 13](#).

⁴ Futures capital gains and losses are treated differently from other investments for individual investors. In particular, all futures gains and losses are treated as 60% long-term gains and 40% short-term gains regardless of the time of purchase and sale. This can amount to quite a bit for short-term trades. For example, a short-term trade with futures would result in a tax burden of 26.8% with futures compared with a tax burden of 37% with stocks for the top marginal tax rate individual using the individual tax rates of 2020.

⁵ Note that we are not including the cash position in the numerator when we calculate the leverage ratio as described above, which is standard, since cash is not really considered a risky investment.

- ⁶ Throughout this chapter, when we refer to β , we are referring to the β of the respective instrument versus the benchmark. For portfolio managers who are more accustomed to using β with respect to the market (i.e., the S&P 500), one simply need interpret all our β 's as the β versus the S&P 500. If one does this, in all cases the overall β of the portfolio will be with respect to the S&P 500, irrespective of the benchmark. Of course, if the benchmark is the S&P 500, the resulting β also will mean a similar thing. The key is that all the measured β 's are, with respect to one underlying instrument, either a benchmark or a market index.
- ⁷ Since $l \equiv V_f^f/V_e$, the maximum leverage is achieved when all the equity capital is just sufficient to cover the margin requirements, that is, when $mV_f^f = V_e$. Substituting this into the equation results in the maximum leverage ratio. The β of the portfolio is given by $\beta_P = w\beta_f$, where w is the weight on the futures position (i.e., the futures position over the equity capital). Since we want the maximum β , we will create the maximum leverage on that particular futures contract of $1/m_f$, which by substitution leads to the $\beta^{\max}$.
- ⁸ Rydex pioneered the idea of levered mutual funds and ETFs, which was then copied by other institutions. The early days of creating mutual funds of leverage equal to 2 were challenging. At the time, the CFTC wanted to limit the size of futures positions held by mutual funds and did not allow a mutual fund to hold more than 5% of its NAV in initial margin on futures positions. This made a leverage of 2 or higher untenable with futures contracts. In order to get around this margin restriction, these mutual funds sometimes used options and other loopholes to leverage the portfolio to a level of 2 and higher. The CFTC eventually dismantled most of its margin regulations, and leveraging is less complicated today.
- ⁹ For those unfamiliar with trading futures contracts, these terms might look odd. Every futures contract represents the value of an underlying portfolio. Thus, when buying one futures contract, it is contract to buy $1 \cdot q \cdot S_t$ of the underlying index portfolio at a future date. Thus, if one buys N_f contracts, one is agreeing to buy $N_f \cdot q \cdot S_t$ worth of a portfolio with equal composition to the underlying index at a future date.
- ¹⁰ In the case of cash and futures only, $l = V_f^f/V_e$. Thus, to achieve a desired leverage ratio l^* , the portfolio manager solves $l^* = (N_{f,t} q S_t)/V_e$ for the number of contracts to purchase. Thus, $N_{f,t} = (l^* V_e)/q S_t$. This is equivalent to the β^* formula when $\beta_f = 1$. The leverage ratio in this chapter uses the translation of $V_f^f = N_{f,t} q S_t$, which is known as *equal-position matching*. Some investment professionals prefer to define the leverage ratio in terms of *dollar matching*, where the spot price of the underlying contract would be replaced with the actual price of the futures contract.
- ¹¹ For those interested, all the expected-return calculations in this chapter can be modified to include the amount held in cash by adding the term $\xi \cdot r_{\text{cash}}$, where ξ is the proportion of equity held in cash. In the case of cash and index futures, $\xi = 1$; it changes in other sections of this chapter. Using notation employed later in this chapter and continuous compounding, the additional cash term can be expressed as $\xi (I_{t,t+k} - 1)$, where

$$I_{t,t+k} = \left[\frac{m_f q N_{f,t} F_t}{V_e} \left(e^{i' \frac{k}{360}} - e^{i \frac{k}{360}} \right) + e^{i \frac{k}{360}} \right]$$

If there is no haircut on margin or if margin is covered by other means, let $i' = i$ or let $m = 0$.

- ¹² This section also applies to the case of a portfolio manager who owns an exchange-traded fund (ETF) as the main portfolio. ETFs are not preferable for index leverage because they cost more than futures. The size of the trade may be too large for the ETF to handle because ETFs are generally less liquid than futures, and ETFs do not trade after market close (futures contracts do, on GLOBEX). Investors in ETFs essentially pay two intermediaries, the broker and the specialist on the exchange, who will most likely hedge the position with futures. ETFs are clearly less efficient, and only certain circumstances call for them. For example, if the index manager is a sector portfolio manager, his or her best bet is to use sector ETFs combined with futures to get some leverage because there are no liquid sector futures contracts. For the best combination of liquid futures to leverage with, see Chincarini (2004).
- ¹³ Since the portfolio manager can use the cash and the equity portfolio as margin for the futures positions, both need to be considered. However, the equity portfolio is worth only m_S , or 50% of cash as collateral. Thus, given that a portion ξ of the equity is held in cash, the amount of cash equivalents available for margining equals $V_t [\xi + (1 - \xi)m_S]$. This should be equal to $m_f V_t^f$ for maximum margin. Rearranging this equation for V_t and plugging it into the equation for leverage results in the equation for maximum leverage when holding a cash-and-equity portfolio. The broker-dealer may require an additional haircut on the equity as margin. Typically, a portfolio management company can have a triparty agreement with its custodian and its broker-dealer such that the custodian sets up a subaccount with securities to be posted for margin for the broker-dealer.
- ¹⁴ In fact, for $\xi = 0$, $\beta^{\max} = 11$ when $m_S = 0.5$, $m_f = 0.05$, and $\beta_S = 1$ as in our example.
- ¹⁵ In the case of futures, cash, and an underlying stock portfolio, leverage is defined as $l = [V_t^f + (1 - \xi)V_t]/V_t$. One can rearrange this equation in order to find the number of futures contracts required for a given target leverage l^* . It is $N_{f,t} = (V_t/qS_t) [l^* + (\xi - 1)]$. For futures indices with $\beta_f = 1$ (e.g., when the benchmark is the S&P 500 and the portfolio manager is using S&P 500 futures), the number of futures contracts required to achieve a certain β^* and l^* is the same.
- ¹⁶ We did not include the cash return in the expected-return calculation; see footnote 10.
- ¹⁷ The expected return of the portfolio and futures is given by $E(r_S) = \alpha_S + \beta_S E(r_B)$ and $E(r_f) = \alpha_f + \beta_f E(r_B)$.
- ¹⁸ For example, if $E(r_B) = 0.13$, $\alpha_S = 0.02$, and $\xi = 0$, moving from a leverage ratio of 1 to a leverage ratio of 2 reduces the contribution of α_S as a percentage of the total portfolio return from 13% to 7%.
- ¹⁹ Applying l'Hospital's rule, $\lim_{l \rightarrow \infty} SR_P^2 = \mu_B^2 / \sigma_B^2 = SR_B^2$
- ²⁰ Single-stock futures trading existed from November 2002 to September 2020 in the United States through the exchange OneChicago. OneChicago, LLC, was a joint venture created by the CME, CBOE, and CBOT for the primary purpose of trading single-stock futures (former website: www.onechicago.com). Contract size was 100 shares of the underlying security. Regular trading hours were 8:30 a.m. to 3:00 p.m. Central time. The initial and maintenance margin requirements for single-stock futures were 20% of the cash value of the contract. This was higher than the margin requirement for index futures but still better than buying stocks outright. Also, there was no need to borrow the stock for shorting, nor was there an "uptick" rule.

- 21 On January 31, 2005, only 122 individual stock futures were available for trading, although growth in the early years was promising. By April 2006, the number available had grown to 200 securities. Shortly before the exchange closed, on August 7, 2020, there were 1,600 listed securities available for trading, but this also included non-stock securities, such as ETFs. On Euronext, about 350 securities are available as of 2021.
- 22 On August 7, 2020, the total volume of all single-security futures was 11,699 contracts. Furthermore, only 84 contracts had a positive volume on that particular day. The open interest was 149,953.
- 23 One method for a portfolio manager to deal with the low trading volume is to place a series of limit orders on individual stock futures and have them executed.
- 24 Before OneChicago closed, the single-stock futures requirement was 20%. ICE Futures Europe has a margin requirement of 5%–10%, and the Euronext exchange has other margin requirements.
- 25 See [Appendix 12B](#) for the derivation of these equations.
- 26 This assumption is true as long as we ignore basis risk.
- 27 Note that the weights of futures in the futures portfolio are identical to the weights of stocks in portfolio $s1$.
- 28 Use L'Hospital's rule to show this.
- 29 We are assuming that the underlying security of the futures contract is the benchmark and that there is no basis risk.
- 30 Regulation T is the rule established by the Federal Reserve Board that limits the amount that an investor can borrow to establish a new position in a security.
- 31 The cost of carry for a futures contract is the implied rate in the *fair-value equation*, $F_t = S_t e^{r(T-t)}$.
- 32 For a more detailed discussion of swaps, the reader is referred to Hull (2005), Cox and Rubenstein (1985), Wilmott et al. (1993), and McDonald (2003).
- 33 LIBOR is the London Inter-Bank Offer Rate, the short-term interest rate at which many international banks borrow from each other. Due to the highly publicized LIBOR scandal that broke in 2012, it might be more common to see the new overnight benchmark called the Secured Overnight Financing Rate (SOFR) being used.
- 34 Many broker-dealers, including Goldman Sachs, Citibank, and Credit Suisse, structure such equity products. Jefferies, Thomas Weisel Partners, and Raymond James are better for mid-cap or small-cap securities.
- 35 Index option contracts are 100 times the actual index. Thus $q_O = 100$. All the major index options are European-style options, except the S&P 100, which is an American-style option. For more details, please consult the Chicago Board Options Exchange (www.cboe.com) and/or a derivatives textbook.
- 36 In reality, it will be very unlikely to find a call option that is perfectly at the money (i.e., strike price equal to index value); thus a portfolio manager will have to find the option with the closest moneyness.
- 37 Since we wrote the first edition of this book, many mutual fund and ETF companies offering daily leverage were sued due to a discrepancy between what investors may have expected and the

actual returns of the levered portfolios. For more info, see Cheng and Madhavan (2009), Avellaneda and Zhang (2010), Shum and Kang (2012), Tang and Zu (2013), and Bruno et al. (2014).

- 38 The actual formula for $I_{t,t+k}$ is $I_{t,t+k} = \left[\left(\frac{m_f q_l N_{f,t} F_t}{V_t} \right) \left(e^{i' \frac{k}{360}} - e^{i \frac{k}{360}} \right) + e^{i \frac{k}{360}} \right]$. This is simply 1 plus the continuously compounded interest from both the cash position and the margin position, where margin interest i' is typically lower than cash interest i .
- 39 Basis risk is the risk that the basis will change over time. The *basis* is defined as the futures rate of the index minus the spot rate of the index. That is, $B_t = F_t - S_t$. If the futures are held to expiration, the futures price will converge to the spot price, and there is no basis risk. However, at any time prior to futures expiration, the basis can change, and this introduces basis risk because the futures movement and spot movement do not offset each other. This can be an issue for hedgers or investors who use leverage. Basis risk can also affect the effectiveness of rolling over a hedge. In reality, the basis moves even if just by a little bit owing to contraction of the time until maturity, but this is very insignificant. When the basis risk is 0, it implies that $r_{t,t+k}^f = r_{t,t+k}$.
- 40 Since we are dealing with only cash and futures, the leverage ratio reduces to just l .
- 41 The analysis in this subsection follows also for a portfolio manager who invests in ETFs, cash, and futures.
- 42 With no basis risk, $F_{t+k} - F_t = S_{t+k} - S_t$; thus $(S_{t+k} - S_t)/S_t = (F_{t+k} - F_t)/F_t$. The former is the return on the underlying equity index, which we call $r_{t,t+k}$, and is what we are interested in because the options depend on the value of the underlying index, not the futures. Some people might wish to define the return on the futures contracts as $(F_{t+k} - F_t)/F_t$, but it really does not matter for this analysis. We are merely transforming the equation into the item we are interested in.
- 43 In the daily risk management of option contracts, the β of the option contract is actually not the same as the underlying index. That is, $N_o = \frac{\beta^* V_t}{\beta_o q_o S_t}$, where $\beta_o = \beta_l |\Delta|$, and Δ is the option's Δ . In order to determine the number of options to purchase, we use β_l , since our purpose for creating this option is not for daily risk management but rather to protect against the catastrophic scenario of a huge negative return. In that scenario, the Δ of the option will converge to -1 very quickly; thus, for the purposes of liquidity protection, we assume that it is -1 from the onset.
- 44 The theoretical Black-Scholes price of these options is infinitesimal, but the broker-dealer typically charges more because, at "fair value," he or she earns next to nothing for taking on a risk.
- 45 The New York Stock Exchange (NYSE) halts trading for 15 minutes if the S&P 500 declines by more than 7% on a given day, for another 15 minutes if it declines by 13%, and for the entire day if it drops by 20% or more. For more information, see www.nyse.com.

Fair-Value Computations

Fair value is a concept used commonly in futures markets. Since a futures contract trades on the basis of an underlying spot security, there should be some relationship between the underlying security price and the futures price. This theoretical price of the futures contract is known as *fair value*.

There is no perfect formula for computing the fair value of the futures contract. It will depend on the dividends paid by the underlying index, the timing of those dividends, the available interest rates in the market, the number of days until the expiration of the futures contract, and the current index level. Different banks have different ways of computing fair value; however, all fair values are based on a similar fundamental concept known as *present discounted value*. If one is using a continuously compounded interest rate to compute fair value, then the fair-value formula is expressed as

$$F_t = S_t e^{(i-q)(T-t)} \quad (12A.1)$$

where F_t is the fair-value futures price, S_t is the current value of the underlying index, i is the prevailing interest rate over the time until expiration of the futures contract, q is the continuously compounded dividend yield of the index, and $T - t$ is the time until the futures contract expires.

Most people in the market do not deal with continuous compounded interest; thus the more popular formula is a formula that involves money

market interest conventions. Thus

$$F_t = (S_t - d_t) \left(1 + i \frac{k}{360} \right) \quad (12A.2)$$

where i is the money market interest rate, k is the number of days until the futures contract expires, and d_t is the present discounted value of dividend payments occurring on the index during the life of the futures contract. Ideally, one should know in advance the payment of all dividends and discount each one appropriately; unfortunately, this is not an easy task. Thus most people who compute fair value just use some estimated value of dividends to be paid on the index and present discount value them by some average time in which they are expected to be received. It really does not have a large effect on the fair value unless the dividend yield of the index is high and/or a lot of dividends are expected to be paid out shortly.

An example might help to illustrate the point. Suppose that the S&P 500 is trading at 1,000, that a 90-day futures contract exists, that the S&P 500 will pay 5 points of dividends at the end of the first month, and that the money market interest rate is 4%. The fair value of the futures contract using [Eq. \(12A.2\)](#) will be

$$\left[1,000 - \frac{5}{\left(1 + 0.04 \cdot \frac{30}{360} \right)} \right] \left(1 + 0.04 \cdot \frac{90}{360} \right) = 1,004.97 \quad (12A.3)$$

APPENDIX 12B

Derivation of Equations (12.21), (12.22), and (12.23)

The derivation of Eq. (12.21) is as follows:

$$\beta^* = \sum_{i=1}^N \left(\frac{N_i q_s p_{i,t}}{V_t} \right) \beta_i + (1 - \xi) \beta_s \quad (12B.1)$$

where q_s is the contract multiplier for single-stock futures, which is equal to 100, and $p_{i,t}$ is the price of stock i . For Eq. (12.21), we assumed that the portfolio manager purchases single-stock futures contracts in proportion to the weighting of the underlying equity portfolio. Thus $w_i = N_i q_s p_{i,t} / \sigma_{j=1}^N N_j q_s p_{j,t}$. Thus we substitute in the preceding expression and obtain

$$\begin{aligned}
\beta^* &= \sum_{i=1}^N \left(\frac{w_i \sum_{j=1}^N N_j q_s p_{j,t}}{V_t} \right) \beta_i + (1 - \xi) \beta_s \\
&= \left(\frac{\sum_{j=1}^N N_j q_s p_{j,t}}{V_t} \right) \sum_{i=1}^N w_i \beta_i + (1 - \xi) \beta_s \\
&= \left(\frac{N_i q_s p_{i,t}}{w_i V_t} \right) \beta_s + (1 - \xi) \beta_s
\end{aligned} \tag{12B.2}$$

Rearranging the last term, we obtain

$$N_i = \frac{[\beta^* + (\xi - 1) \beta_s] V_t w_i}{\beta_s q_s p_{i,t}} \tag{12B.3}$$

The derivation of [Eq. \(12.22\)](#) is very similar to the preceding derivation. However, we make the assumptions that $\tilde{w}_i = N_i q_s p_{i,t} / \sum_{i=1}^{N_1} N_i q_s p_{i,t}$ and that all the arguments are similar to those above, except that we replace w_i with $\tilde{\beta}_s = \sum_{i=1}^{N_1} \tilde{w}_i \beta_i$ and replace β_s with $\tilde{w}_i$, where $\tilde{\beta}_s$. This leads to the expression $N_i = \{[\beta^* + (\xi - 1) \beta_s] V_t \tilde{w}_i\} / \beta_s q_s p_{i,t}$.

The derivation of [Eq. \(12.23\)](#) is also similar to the preceding derivation. However, we need to make an additional assumption about the relative weight of the index futures and the single-stock futures. In particular, we assume that $(1 - \sum_{i=1}^{N_1} w_i) / \sum_{i=1}^{N_1} w_i = N_f q S_t / \sum_{i=1}^{N_1} N_i q_s p_{i,t}$. Thus

$$\begin{aligned}
\beta^* &= \sum_{i=1}^{N_1} \frac{N_i q_s p_{i,t}}{V_t} \beta_i + \frac{N_f q S_t}{V_t} \beta_f + (1-\xi) \beta_s \\
&= \sum_{i=1}^{N_1} \frac{N_i q_s p_{i,t}}{V_t} \beta_i + \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \left(\sum_{i=1}^{N_1} N_i q_s p_{i,t} \right) \frac{\beta_f}{V_t} + (1-\xi) \beta_s \\
&= \sum_{i=1}^{N_1} \frac{\tilde{w}_i \left(\sum_{j=1}^{N_1} N_j q_s p_{j,t} \right)}{V_t} \beta_i + \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \left(\sum_{i=1}^{N_1} N_i q_s p_{i,t} \right) \frac{\beta_f}{V_t} + (1-\xi) \beta_s \\
&= \left(\sum_{i=1}^{N_1} N_i q_s p_{i,t} \right) \left[\frac{\tilde{\beta}_s}{V_t} + \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \frac{\beta_f}{V_t} \right] + (1-\xi) \beta_s
\end{aligned} \tag{12B.4}$$

Rearranging the last term, we obtain

$$N_i = \frac{\left[\beta^* + (\xi - 1) \beta_s \right] V_t \tilde{w}_i}{q_s p_{i,t} \left[\tilde{\beta}_s + \left(\frac{1 - \sum_{i=1}^{N_1} w_i}{\sum_{i=1}^{N_1} w_i} \right) \beta_f \right]} \tag{12B.5}$$

Once this is known, we already have the relationship between the index futures and the single-stock futures; thus

$$N_f = [(1 - \sum_{i=1}^{N_1} w_i) / \sum_{i=1}^{N_1} w_i] (\sum_{i=1}^{N_1} N_i q_s p_{i,t} / q S_t).$$

Tables of Futures Leverage Multipliers Needed to Achieve Various Degrees of Leverage

In this appendix we compiled a series of tables with various baseline β_s values of a portfolio manager and the *futures leverage multiplier* $\{[\beta^* + (\xi - 1)\beta_s]/\beta_f q\}$ which when multiplied by the ratio of the portfolio's value to the index's value will determine the number of appropriate futures contracts to purchase to alter the β of the total portfolio to the desired level. For these tables, we used S&P 500 E-mini and NASDAQ 100 E-mini futures because they are the most liquid. In both cases we assumed that the benchmark for the portfolio manager was the S&P 500; hence all β 's are with respect to the S&P 500. However, the leverage ratio, l , is always with respect to the actual portfolio. As we stated earlier, our measure of leverage is not the conventional measure of leverage. However, they are the same when the futures positions are taken to enhance the exposure of the portfolio. Thus, in the tables, anytime the futures leverage multiplier is positive, the leverage ratio, l , will coincide with the conventional measure of leverage. However, when the futures leverage multiplier is negative (indicating to sell futures), the leverage ratio, l , will not coincide with the conventional measure of leverage. To obtain the conventional measure of leverage when the futures multiplier is negative, use the formula $l^c = (1 - l) + 1$, where l^c is the symbol for the conventional leverage measure. Our leverage ratio can also

be interpreted as the net dollar exposure of the entire position. We also assume that the portfolio manager does not hold cash ($\zeta = 0$).

Table 12C.1 presents the S&P 500 E-mini leverage multiplier assuming $\beta_f = 1$. Table 12C.2 presents the NASDAQ 100 E-mini leverage multiplier needed to achieve the desired β^* for a given portfolio β_s . For the NASDAQ 100 E-mini, we used $\beta_f = 1.10$ (since the benchmark is the S&P 500). The portfolio manager can use these tables as rough guides to adjust either the β of the portfolio or the overall leverage of the portfolio.

TABLE 12C.1

S&P 500 E-mini Futures' Leverage Multipliers to Obtain Desired β^* or Leverage (l)

β_s	β^*											
	0.2	0.4	0.6	0.8	1	1.2	1.4	1.6	1.8	2	3	4
0.20	0.0000	0.0040	0.0080	0.0120	0.0160	0.0200	0.0240	0.0280	0.0320	0.0360	0.0560	0.0760
	1.00	1.20	1.40	1.60	1.80	2.00	2.20	2.40	2.60	2.80	3.80	4.80
0.40	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0160	0.0200	0.0240	0.0280	0.0320	0.0520	0.0720
	0.80	1.00	1.20	1.40	1.60	1.80	2.00	2.20	2.40	2.60	3.60	4.60
0.60	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0160	0.0200	0.0240	0.0280	0.0480	0.0680
	0.60	0.80	1.00	1.20	1.40	1.60	1.80	2.00	2.20	2.40	3.40	4.40
0.80	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0160	0.0200	0.0240	0.0440	0.0640
	0.40	0.60	0.80	1.00	1.20	1.40	1.60	1.80	2.00	2.20	3.20	4.20
1.00	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0160	0.0200	0.0400	0.0600
	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.60	1.80	2.00	3.00	4.00
1.20	-0.0200	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0160	0.0360	0.0560
	0.00	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.60	1.80	2.80	3.80
1.40	-0.0240	-0.0200	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0120	0.0320	0.0520
	-0.20	0.00	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.60	2.60	3.60
1.60	-0.0280	-0.0240	-0.0200	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0080	0.0280	0.0480
	-0.40	-0.20	0.00	0.20	0.40	0.60	0.80	1.00	1.20	1.40	2.40	3.40
1.80	-0.0320	-0.0280	-0.0240	-0.0200	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0040	0.0240	0.0440
	-0.60	-0.40	-0.20	0.00	0.20	0.40	0.60	0.80	1.00	1.20	2.20	3.20
2.00	-0.0360	-0.0320	-0.0280	-0.0240	-0.0200	-0.0160	-0.0120	-0.0080	-0.0040	0.0000	0.0200	0.0400
	-0.80	-0.60	-0.40	-0.20	0.00	0.20	0.40	0.60	0.80	1.00	2.00	3.00
3.00	-0.0560	-0.0520	-0.0480	-0.0440	-0.0400	-0.0360	-0.0320	-0.0280	-0.0240	-0.0200	0.0000	0.0200
	-1.80	-1.60	-1.40	-1.20	-1.00	-0.80	-0.60	-0.40	-0.20	0.00	1.00	2.00
4.00	-0.0760	-0.0720	-0.0680	-0.0640	-0.0600	-0.0560	-0.0520	-0.0480	-0.0440	-0.0400	-0.0200	0.0000
	-2.80	-2.60	-2.40	-2.20	-2.00	-1.80	-1.60	-1.40	-1.20	-1.00	0.00	1.00

Note: For these examples, we use $\beta_f = 1$, $q = 50$, and $\zeta_f = 0$. The first number is the futures leverage multiplier $\frac{[\beta^* + (\zeta - 1)\beta_s]}{\beta_f q}$, and the number below it is the effective leverage l of the portfolio from this action. Our leverage ratio is not always equal to the conventional leverage ratio. The text describes how to convert our leverage ratio (a.k.a. net leverage) to the conventional leverage ratio (a.k.a. gross leverage). The futures multiplier is rounded to four decimal places.

TABLE 12C.2

NASDAQ 100 E-mini Futures' Leverage Multipliers to Obtain Desired β^* or Leverage (l)

β^*												
β_s	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.60	1.80	2.00	3.00	4.00
0.20	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0455 1.91	0.0545 2.09	0.0636 2.27	0.0727 2.45	0.0818 2.64	0.1273 3.55	0.1727 4.45
0.40	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0455 1.91	0.0545 2.09	0.0636 2.27	0.0727 2.45	0.1182 3.36	0.1636 4.27
0.60	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0455 1.91	0.0545 2.09	0.0636 2.27	0.1091 3.18	0.1545 4.09
0.80	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0455 1.91	0.0545 2.09	0.1000 3.00	0.1455 3.91
1.00	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0455 1.91	0.0909 2.82	0.1364 3.73
1.20	-0.0455 0.09	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0364 1.73	0.0818 2.64	0.1273 3.55
1.40	-0.0545 -0.09	-0.0455 0.09	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0273 1.55	0.0727 2.45	0.1182 3.36
1.60	-0.0636 -0.27	-0.0545 -0.09	-0.0455 0.09	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0182 1.36	0.0636 2.27	0.1091 3.18
1.80	-0.0727 -0.45	-0.0636 -0.27	-0.0545 -0.09	-0.0455 0.09	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0091 1.18	0.0545 2.09	0.1000 3.00
2.00	-0.0818 -0.64	-0.0727 -0.45	-0.0636 -0.27	-0.0545 -0.09	-0.0455 0.09	-0.0364 0.27	-0.0273 0.45	-0.0182 0.64	-0.0091 0.82	0.0000 1.00	0.0455 1.91	0.0909 2.82
3.00	-0.1273 -1.55	-0.1182 -1.36	-0.1091 -1.18	-0.1000 -1.00	-0.0909 -0.82	-0.0818 -0.64	-0.0727 -0.45	-0.0636 -0.27	-0.0545 -0.09	-0.0455 0.09	0.0000 1.00	0.0455 1.91
4.00	-0.1727 -2.45	-0.1636 -2.27	-0.1545 -2.09	-0.1455 -1.91	-0.1364 -1.73	-0.1273 -1.55	-0.1182 -1.36	-0.1091 -1.18	-0.1000 -1.00	-0.0909 -0.82	-0.0455 0.09	0.0000 1.00

Note: For these examples, we use $\beta_f = 1.10$, $q = 20$, and $\xi_f = 0$. The first number is the futures leverage multiplier $\frac{[\beta^* + (\xi - 1)\beta_s]}{\beta_f q}$, and the number below it is the effective leverage l of the portfolio from this action. Our leverage ratio is not always equal to the conventional leverage ratio. The text describes how to convert our leverage ratio (a.k.a. net leverage) to the conventional leverage ratio (a.k.a. gross leverage). The futures multiplier is rounded to four decimal places.

Thus suppose that a portfolio manager is managing a portfolio with a $\beta_s = 1.6$ with respect to the benchmark, which we have assumed is the S&P 500, and he or she would like to increase the overall β of the portfolio to 3. His or her portfolio has a value of \$100 million, and the current level of the S&P 500 is 3,756. In [Table 12C.1](#), we find the 1.6 value under the β_s column and then slide over until we reach the column under $\beta^* = 3$, which tells the portfolio manager that the futures leverage multiplier is 0.0280. The ratio of the portfolio's value to the index's value is 26,624.07 (\$100 M/3,756). Thus, the appropriate number of futures contracts to purchase to achieve the desired position is 745.47 ($= 0.0280 \cdot 26,624.07$). Thus, in practice, we would round to the nearest integer and actually purchase 745 futures contracts. This will achieve his or her desired β^* and a leverage ratio of 2.4 (the number below the futures multiplier in the table). To accomplish the same sort of β^* with respect to the benchmark using NASDAQ 100 futures would require purchasing 493.48 ($= 0.0636 \cdot 7,759.16$) futures contracts but result in a lowered leveraged position of 2.27 owing to the higher β of the NASDAQ 100 versus the benchmark (for this example we assumed the NASDAQ was at 12,888). In practice, we would round to the nearest integer and purchase 493 NASDAQ 100 futures contracts.

In the two preceding examples, our leverage ratio, l , is equal to the conventional measure of leverage. Suppose instead, we wish to use S&P 500 futures to reduce our overall β to 1. In this particular case we find the value of 1.6 under the β_s column and slide over until we reach the column under $\beta^* = 1$, which tells the portfolio manager that the futures leverage multiplier is -0.0120 . The appropriate number of futures to short will be $319.49 (= -0.0120 \cdot 26,624.07)$. This will achieve the desired β^* and a leverage ratio or net dollar exposure of 0.40. The conventional leverage measure in this example would be $1.6[(1 - 0.4) + 1]$. If we were instead to use NASDAQ 100 futures to achieve the same position, we would sell $211.83 (= -0.0273 \cdot 7,759.16)$ futures contracts resulting in a leverage ratio or net dollar exposure of 0.60. The conventional leverage measure in this example would be $1.4[(1 - 0.6) + 1]$.

QUESTIONS

- 12.1. Name four ways an equity manager can leverage a portfolio.
- 12.2. Name three advantages of using leverage in QEPM.
- 12.3. Name three disadvantages of using leverage in QEPM.
- 12.4. What type of portfolio manager is most likely to use leverage with only cash and index futures?
- 12.5. In the case of a portfolio manager using only cash and S&P 500 futures, what is theoretically the largest leverage he or she could achieve? The largest portfolio β he or she could achieve? Assume that the margin futures requirement is 6.5% and that the β of the S&P 500 futures is 1.
- 12.6. Which of the equity index futures contracts that portfolio managers use are the most liquid? What evidence supports your answer?
- 12.7. Suppose that you are a portfolio manager with \$150 million in cash. You would like to create a portfolio with an exposure to the S&P 500 of 2. Use the index values in [Table 12.2](#) for each futures contract.
 - (a) How many contracts of the S&P 500 futures must you purchase?
 - (b) How many contracts of the S&P 500 E-mini futures would you have to purchase?
 - (c) Which one would you probably use and why?

- 12.8.** Suppose that you are a portfolio manager with \$50 million in cash. You are interested in using futures to obtain an overall β exposure of 2 to the S&P 500. Use the index values in [Table 12.2](#) for each futures contract.
- How many futures contracts would you need if you used S&P 500 futures to achieve this exposure?
 - How many futures contracts would you need if you used Russell 2000 E-mini futures to do this (assume the Russell 2000 futures have a $\beta_f = 1.25$ versus the S&P 500)?
 - Which futures contract probably would be the better one to use and why?
- 12.9.** Suppose that a portfolio manager has \$10 million in cash and \$90 million in a portfolio of equities with $\beta_s = 0.8$. Suppose that the portfolio manager would like to use the S&P 500 futures contracts for leverage (that is, $\beta_f = 1$).
- What is the largest leverage he can achieve?
 - What is the largest overall portfolio β he can achieve theoretically?
- 12.10.** Suppose that a small-cap quantitative equity portfolio manager wishes to leverage his underlying stock portfolio using index futures. Which index futures contract would you probably recommend and why?
- 12.11.** Suppose that a mid-cap quantitative equity portfolio manager wishes to leverage her underlying stock portfolio using index futures. She uses S&P 400 Mid-Cap E-mini contracts. How many futures contracts will she be required to purchase?
- 12.12.** Suppose that the portfolio manager holds no cash and a portfolio of equities with an exposure versus the benchmark of $\beta_s = 0.9$. The α_s of the portfolio is 2% per annum, the expected return of the benchmark is 10%, the benchmark risk or standard deviation is 16%, and the residual risk of the portfolio versus the benchmark is 4%. The leverage of the portfolio is 2 (that is, $l = 2$) using future contracts on the underlying benchmark.
- What is the expected return and risk of the overall portfolio?
 - How does it compare with that of the benchmark?

- 12.13.** Name an advantage and a disadvantage of using single-stock futures versus index futures for leverage.
- 12.14.** Is the theoretical maximum leverage larger or smaller using single-stock futures versus index futures? Why?
- 12.15.** Suppose that a portfolio manager has the following \$10 million portfolio. He has three stocks A , B , and C . Each stock has a weight of 0.30, 0.40, and 0.30, respectively. The price of each stock is \$20, \$50, and \$100, respectively. The β of each stock with respect to the benchmark is 0.5, 0.7, and 0.9, respectively.
- What is the overall β of the portfolio β_s ?
 - How many futures contracts would have to be purchased if the portfolio manager would like to increase the overall portfolio β to 2.5? (Specify for N_A , N_B , and N_C .) Assume that all single-stock futures are available and liquid and that the portfolio manager uses scenario 1 in the text.
- 12.16.** True or false: leveraging with index futures has no effect on the Sharpe ratio.
- 12.17.** Using the relative contribution of the α^B to the overall expected portfolio return, show whether index leverage or single-stock leverage results in better leverage for a portfolio manager who is good at stock picking. Show your work.
- 12.18.** Suppose that a portfolio manager has a stock portfolio and cash and wants to use single-stock futures and index futures to increase her leverage to a value of $l = 2$ using scenario 3. Given the following parameters: $\xi = 0.3$, $\psi = 0.60$, $\alpha_{s1} = 0.02$, $\alpha_{s2} = 0.015$, $\beta_{s1} = 0.95$, $\beta_{s2} = 0.95$, $\beta_f = 1$, $\mu_B = 0.15$, $\sigma_B = 0.40$, $\omega_{s1} = 0.10$, $\omega_{s2} = 0.05$, and $\omega_{s1,s2} = 0.02$.
- What is the expected return of the portfolio?
 - What is the expected risk?
- 12.19.** What could a quantitative equity portfolio manager use to boost his α^B in the absence of liquid single-stock futures contracts?
- 12.20.** What are the advantages of using options to leverage? What is the practical disadvantage?
- 12.21.** Suppose that a portfolio manager started with a portfolio of stocks and cash. She used index futures to leverage the portfolio. The original stock portfolio had a $\beta_s = 1.2$, the futures $\beta_f = 2$ with

respect to the benchmark, and the value of the portfolio was \$100 million at date t , of which \$40 million was in cash ($\xi = 0.40$). The target β^* versus the benchmark was set to be 3. The contract multiplier of the futures contract is 20. After 15 days, the market collapses, with the market going down by 15%. In particular, the underlying index of the futures contract has dropped from $S_t = 1,240$ to $S_{t+k} = 1,054$. The value of the overall portfolio has dropped from $V_t = \$100$ million to $V_{t+k} = \$77.5$ million. [Note: For simplicity, assume that the return on cash over this period is negligible (that is, $I_{t,t+k} = 1$)].

- (a) How many futures contracts does the portfolio manager need to purchase on date t to establish $\beta^* = 3$ for the portfolio?
- (b) What was the return on the underlying benchmark over the period t to $t + k$? (*Hint: The benchmark is not the same as the index underlying the futures contract.*)
- (c) What was the return on the overall leveraged portfolio?
- (d) Considering your answer in part (b), what does this say about leverage?
- (e) What is the number of futures contracts the portfolio should purchase to reestablish an overall $\beta^* = 3$ at time $t + k$?

12.22. Suppose that we assume that there is no basis risk, and there is a portfolio manager who uses only cash and index futures to manager his equity exposure. He uses S&P 500 futures, not the E-minis. His benchmark is the S&P 500. Suppose that from time t to time $t + k$, the S&P 500 moves from 3,584 to 4,480. Suppose also that over the period the interest on cash is a total of 1% (that is, $I_{t,t+k} = 1.01$). The portfolio manager's original $\beta^* = 1.50$, and the original value of the portfolio is \$500 million.

- (a) Does the portfolio manager need to purchase or sell futures contracts to reestablish the original β^* ?
- (b) How many futures contracts does he need to buy or sell? (*Hint: You need to determine the contract multiplier associated with this contract and the original leverage l .*)

12.23. Explain what is meant by *liquidity buffering*? Why is it important?

12.24. Is leveraging short any different from leveraging long? If so, how?

CHAPTER 13

Market Neutral

Not being able to govern events, I govern myself.

—Michel de Montaigne

13.1 INTRODUCTION

Market-neutral strategies have gained popularity over the last few years. From 2000 to 2020, the equity market-neutral strategy declined from 2.57% of the hedge-fund universe to 1.51%.¹ The first known hedge fund was market neutral. It was founded in 1949 by Alfred W. Jones, an Australian-born Harvard graduate who believed that he had an ability to predict individual stock returns but no ability to time the market or predict its general direction. Jones realized that if he could, in the right proportion, short the securities that he thought would lose value and buy the securities that he thought would gain value, he would effectively eliminate the impact of the overall market return on his portfolio. His portfolio's return would equal the difference in returns between the winners and losers, reflecting only his ability to pick the stocks. Jones's track record was quite impressive. In fact, a 1966 *Fortune* article entitled, "The Jones That Nobody Can Keep Up With," reported that his fund beat the best-performing mutual fund of that year by 44%.

Jones's market-neutral strategy took hold in the investment community, and although people today apply a variety of techniques and designs to it, the goal remains the same: to reduce the sensitivity of the portfolio to

market movements by taking long and short positions in individual securities or market indices that offset the effects of the market. One of those effects is market risk. Table 13.1 shows that over the last 25 years, a market-neutral portfolio would have earned a return a little less than half of the Standard & Poor's (S&P) 500 but with about half of the volatility. In fact, the market-neutral portfolio had less risk than any other major index except for the money market index. By taking a big slice of risk out of the portfolio, market-neutral strategies clear the path for a manager who is good at stock picking to focus on his or her strength. This can boost α above the α of the traditional long portfolio. In other words, market neutral is a good source of α mojo.

TABLE 13.1

Statistics of a Market-Neutral Portfolio versus Other Major Indices

Index	Average Return (%)	SD (%)	Correlation with Market Neutral
Market neutral	4.49	9.08	1
S&P 500	11.19	15.09	0.28
NASDAQ 100	16.43	24.58	0.23
Government bond	8.27	10.22	-0.24
Commodity	2.19	22.53	0.27
Money market	2.41	0.65	0.14
Real estate	10.91	19.08	0.35

Note: All index statistics were measured over the period 12/31/1994 to 12/31/2020 using monthly return data. Market neutral is the Credit Suisse Equity Market Neutral Index; the S&P 500 represents S&P 500 total returns; NASDAQ 100 represents the NASDAQ 100 price returns; the government bond index is the Bloomberg U.S. Government Bond Index; the commodity index is the total return S&P GSCI Commodity Index; the money market index is the S&P Treasury Bill Total Return Index; and the real estate index is the Dow Jones U.S. Real Estate Total Return Index. Data were obtained from Bloomberg. Average return and standard deviation (SD) have been annualized.

Market-neutral strategies can work for index managers too. Combined with a long exposure to the equity index, buying favorable stocks and shorting unfavorable ones adds the α from stock picking to the benchmark return. Market-neutral portfolios also can help to diversify an investor's overall portfolio because they have very low correlations with the market. Table 13.1 shows that the correlations between the market-neutral portfolio and various indices are all less than 0.36.²

In this chapter we discuss the basics of equity market-neutral strategies of portfolio management. We show the various methods for constructing a market-neutral portfolio, go over the basic mathematics, and discuss the advantages of the strategy.

13.2 MARKET-NEUTRAL CONSTRUCTION

The market-neutral concept applies to a variety of investment strategies, including convertible arbitrage, fixed-income arbitrage, and risk arbitrage. Here we focus on applying it to an equity portfolio run by a stock-picking manager.

A manager can construct a market-neutral position by purchasing stocks that he or she thinks are “buys” and shorting ones that he or she thinks are “sells.” He or she can separate the buys from the sells with an aggregate Z-score model, a fundamental factor model, or an economic factor model—any of these will help to rank the stocks by expected return. He or she also can sort these stocks using other methods, including directly through an optimization. Then, to achieve market neutrality, the manager must use some mechanism for controlling risk so that the entire portfolio has *zero net exposure* to the market even as it captures the excess return from stock picking. Risk-control mechanisms run the gamut from very simple to quite complex. We discuss a number of them in this section.

13.2.1 Security Selection

In order to construct the portfolio, the manager must identify the buys and sells. With a quantitative model, he or she can estimate the expected returns or α 's of the stocks in his or her investment universe. Stocks with low expected returns or α 's go in the “sell” pile; the rest are “buys.” We will assume that the manager already has forecasted the expected returns of stocks, or their α 's, from either an aggregate Z-score model or a factor model.

13.2.2 Dollar Neutrality

The simplest market-neutral strategy is *dollar neutrality*, which means that the dollar amount of the long positions equals the dollar amount of the short

positions. Suppose that the portfolio manager has identified 10 stocks that are buys and 10 stocks that are sells and that he or she has \$100 million with which to trade. He or she can weight the group of buys and sells in any manner he or she chooses as long as the ratio of the weights of the longs to the weights of the shorts is 1:1. The short positions also will require margin collateral from the broker, but we take up this issue in a later section. If V_L represents the notional amount invested in the long positions and V_S represents the notional amount invested in the short positions, then dollar neutrality implies that

$$V_L = V_S \quad (13.1)$$

The weights of the long and short stocks each total 1. Thus $\sum_{i=1}^{N_L} w_i^L = 1$ and $\sum_{i=1}^{N_S} w_i^S = 1$, where w_i^L represents the relative weights of stocks in the long portfolio and w_i^S represents the relative weights of stocks in the short portfolio prior to shorting, and N_L and N_S represent the number of stocks in the long and short portfolios, respectively. It is convenient to think of the short portfolio as simply a long portfolio in which the stocks are sold in certain proportions rather than purchased.³

Suppose that we have an equity value of \$90 million. In [Table 13.2](#) we identify 10 stocks we would like to buy and 10 stocks that we would like to short.

TABLE 13.2

Example of a Dollar-Neutral Portfolio

Good/Long Portfolio					
Ticker	w_i^L	β_i	Dollar Amount	Share Price	Shares
TSLA	0.264	2.18	\$23,715,735.07	705.67	33,607
ETSY	0.107	1.66	\$9,616,896.10	177.91	54,055
NVDA	0.102	1.46	\$9,161,536.05	522.2	17,544
PYPL	0.094	1.12	\$8,482,215.28	234.2	36,218
LB	0.092	1.66	\$8,300,605.96	37.19	223,195
ALB	0.087	1.60	\$7,861,523.59	147.52	53,291
AMD	0.087	2.28	\$7,790,410.83	91.71	84,946
FCX	0.084	2.28	\$7,603,621.21	26.02	292,222
CDNS	0.083	1.05	\$7,467,455.90	136.43	54,735
NOW	0.080	1.14	\$7,195,380.82	550.43	13,072
Total	1.000	1.78	\$90,000,000.00		
Bad/Short Portfolio					
Ticker	w_i^S	β_i	Dollar Amount	Share Price	Shares
SLB	0.089	2.28	\$7,982,029.72	21.83	365,645
AAL	0.091	1.82	\$8,161,269.58	15.77	517,519
NOV	0.091	2.22	\$8,171,115.07	13.73	595,129
FANG	0.093	2.59	\$8,376,817.73	48.4	173,075
HFC	0.094	1.83	\$8,464,580.37	25.85	327,450
MRO	0.102	3.36	\$9,172,852.41	6.67	1,375,240
UAL	0.103	1.62	\$9,268,717.94	43.25	214,306
OXY	0.112	2.35	\$10,060,697.52	17.31	581,207
FTI	0.112	2.50	\$10,060,713.91	9.4	1,070,289
NCLH	0.114	2.86	\$10,281,205.74	25.43	404,294
Total	1.000	2.36	\$90,000,000.00		

Note: The data are from December 2020. The total portfolio β represents the weighted average of the β 's of each portfolio (long and short) independent of the other. The weights presented in the table may not add up to 1 due to rounding.

The weights in each portfolio were chosen according to some relative Z-score weighting. The sum of the weights in both portfolios equals 1. The β 's are reported with respect to the S&P 500. The weighted-average β for the long portfolio is 1.78, whereas the weighted-average β of the short portfolio is 2.36. The number of shares to go long and short, also listed in the table, are a function of the value of the portfolio and the current price of the

stocks. This is a dollar-neutral portfolio, but it is not neutral to market risk. In fact, the β of the overall portfolio is roughly

$$\beta_P = \beta_L + (-1)\beta_S = 1.78 - 2.36 = -0.58 \quad (13.2)$$

A portfolio neutral to market risk would display a β_P of 0. This dollar-neutral portfolio's β of -0.58 exhibits some market risk.

13.2.3 Beta Neutrality (a.k.a. Risk-Factor Neutrality)

The dollar-neutral portfolio is relatively uncorrelated with the market. Whenever one side of the portfolio (long or short) rises, the other side drops. However, dollar neutrality does not guarantee that the overall portfolio is neutral with respect to many of the risk factors in the economy. A dollar-neutral portfolio may have very low exposure to risk factors, but there is no guarantee that the exposure will be zero. Generally, it could even be quite high or negative. Thus it is better to create a market-neutral portfolio that is not only dollar neutral but also neutral with respect to market risk factors according to some risk model of equities.

The market-neutral portfolio should be, at the very minimum, both dollar neutral and neutral to market risk. For the portfolio to be neutral to market risk, the weighted-average capital asset pricing model (CAPM) β of its long and short sides must equal 0.⁴ Thus

$$\sum_{i=1}^{N_L} w_i^L \beta_i = \sum_{i=1}^{N_S} w_i^S \beta_i \quad (13.3)$$

where N_L and N_S represent the number of stocks in the long and short portfolios, respectively, and β_i is the β of stock i with respect to the market index return.⁵ One should remember that in the short portfolio, the stocks will be shorted; thus the actual exposure will be the negative of the overall β of the short portfolio.

With this method, the manager achieves a simple yet effective market-neutral portfolio. Some managers may want to make the portfolio neutral to many more factors—perhaps all the factors—contained in some risk model. This can be done by building the entire portfolio with a constraint that sets

all the factor exposures to zero. More generally, the optimization would be run such that

$$\sum_{i=1}^{N_L} w_i^L \beta_{i,k} = \sum_{i=1}^{N_S} w_i^S \beta_{i,k}, \quad \text{for all } k \quad (13.4)$$

where k represents the k th factor. With no exposure to any of the risk factors, the portfolio theoretically should return the risk-free rate and have an α of 0. However, to the extent that markets are not efficient, the portfolio may have a positive α .

Continuing with our example from the preceding section, we use an optimizer⁶ to construct a dollar-neutral and β -neutral portfolio. [Table 13.3](#) contains the portfolios, which are both dollar neutral and β neutral. Notice that we have allowed a relatively high β of 2.0 for the long and short portfolios because if we had tried to keep their β 's lower, it would have been difficult to achieve a β -neutral portfolio without excluding many stocks. Even with such a high β , on the long side we end up having to altogether forgo purchasing two of our preferred stocks, AMD and CDNS. On the short side, in order to achieve the high β , we have to assign a 65.5% weight to UAL. This means shorting \$59 million of UAL. Here we run into some practical limitations of the optimization. Shorting so much of one stock might not be feasible from a liquidity point of view, and it would reduce the overall diversification of the portfolio. We will discuss these sorts of practical issues in more detail in the last part of this book ([Chapters 16 and 17](#)), which covers empirical examples of quantitative equity portfolio management (QEPM). In this case, a better way to achieve the β neutrality might have been to relax the dollar-neutrality constraint.

TABLE 13.3

Example of a Dollar-Neutral and β -Neutral Portfolio

Good/Long Portfolio					
Ticker	w_i^L	β_i	Dollar Amount	Share Price	Shares
TSLA	0.186	2.18	\$16,758,853.37	540.73	30,993
ETSY	0.089	1.66	\$8,013,791.20	305.34	26,245
NVDA	0.057	1.46	\$5,129,344.51	1386.71	3,699
PYPL	0.019	1.12	\$1,736,408.45	472.27	3,677
LB	0.090	1.66	\$8,059,242.39	269.05	29,954
ALB	0.083	1.60	\$7,435,420.35	582.94	12,755
AMD	0.000	2.28	\$0	92.85	—
FCX	0.455	2.28	\$40,912,457.48	117.4	348,488
CDNS	0.000	1.05	\$0	195.18	—
NOW	0.022	1.14	\$1,954,572.25	387.83	5,040
Total	1.000	2.00	\$90,000,000.00		
Bad/Short Portfolio					
Ticker	w_i^L	β_i	Dollar Amount	Share Price	Shares
SLB	0.030	2.28	\$2,716,195.44	570.56	4,761
AAL	0.009	1.82	\$822,773.14	384.33	2,141
NOV	0.027	2.22	\$2,465,736.48	168.4	14,642
FANG	0.045	2.59	\$4,008,451.59	143.5	27,933
HFC	0.009	1.83	\$842,532.27	258.91	3,254
MRO	0.094	3.36	\$8,456,794.48	88.65	95,395
UAL	0.655	1.62	\$58,937,517.68	141.77	415,726
OXY	0.034	2.35	\$3,016,089.98	721.54	4,180
FTI	0.040	2.50	\$3,610,344.11	379.85	9,505
NCLH	0.057	2.86	\$5,123,564.84	92.48	55,402
Total	1.000	2.00	\$90,000,000.00		

Note: The data are from December 2020. The total portfolio β represents the weighted average of the β 's of each portfolio (long and short) independent of the other. The weights presented in the table may not add up to 1 due to rounding.

13.2.4 Market-Neutral Portfolio Out of a Long-Only Portfolio

An alternative way to create a dollar-neutral portfolio or a β -neutral portfolio is to create one out of two existing portfolios. A dollar-neutral portfolio can be created easily as long as we have two portfolios with different expected returns. Suppose that we have two portfolios, A and B . Portfolio A 's expected return is higher than portfolio B 's expected return. A simple way to create a dollar-neutral portfolio with a positive return is to

take a long position in portfolio A and a short position of the same size in portfolio B .

For a β -neutral portfolio, we need two portfolios with identical factor exposures.⁷ Suppose that portfolio A has factor exposures identical to benchmark B 's factor exposures. The portfolio manager could have constructed portfolio A using the technique of factor exposure targeting described in [Chapter 9](#). Then, taking a long position in portfolio A and a short position of the same size in benchmark B results in a β -neutral portfolio. If the benchmark is also traded as a futures contract, this market neutrality can be achieved easily by shorting the futures contract.⁸

13.3 MARKET NEUTRAL'S MOJO

Suppose that a portfolio manager creates a market-neutral portfolio hedged against all risk factors. It is then possible to compute the expected return of this strategy. Assuming that stock returns are driven by some multifactor model, the excess return of stock i can be represented as

$$r_i = \alpha_i + \beta_{i,1}f_1 + \cdots + \beta_{i,K}f_K + \epsilon_i \quad (13.5)$$

where there are K factors representing security returns, $\beta_{i,k}$ is the sensitivity of stock i to factor k , f_k is the factor excess return, and ϵ_i is the residual return of security i . The excess return to a marketneutral portfolio when all risk factors are set to be neutral is

$$\begin{aligned} r_p &= \sum_{i=1}^{N_L} w_i^L r_i^L - \sum_{j=1}^{N_S} w_j^S r_j^S \\ &= \sum_{i=1}^{N_L} w_i^L \alpha_i - \sum_{j=1}^{N_S} w_j^S \alpha_j + \sum_{i=1}^{N_L} w_i^L \epsilon_i - \sum_{j=1}^{N_S} w_j^S \epsilon_j \\ &\equiv \alpha_L - \alpha_S + \epsilon_L - \epsilon_S \end{aligned} \quad (13.6)$$

Thus the expected excess return of the market-neutral portfolio is

$$E(r_p) = \sum_{i=1}^{N_L} w_i^L \alpha_i - \sum_{j=1}^{N_S} w_j^S \alpha_j \quad (13.7)$$

According to the arbitrage pricing theory (APT), the market-neutral portfolio should have no expected excess return, but we are assuming that the market is not completely efficient and/or that the portfolio manager is good at picking outperforming and underperforming stocks. If you make a further assumption that $\alpha_L = -\alpha_S = \alpha$ (i.e., that the long portfolio's α and the short portfolio's α have the same absolute value but opposite signs), then the market-neutral portfolio achieves its 2α , that is,

$$E(r_p) = 2\alpha \quad (13.8)$$

This is the market-neutral strategy's hidden source of α mojo. It is one of the advantages of the market-neutral portfolio over a long-only portfolio, which can achieve only 1α . It is important to point out, though, that this α mojo comes purely from the leverage in the market-neutral portfolio, in which the full capital is invested both on the long side and on the short side.⁹ Without the leverage, the α would be no greater than the long-only portfolio's α unless very big, unexploited opportunities existed on the short side.

In addition to the expected return, it is possible to calculate the variance of the market-neutral portfolio:

$$\begin{aligned} V(r_p) &= V(\alpha_L - \alpha_S + \epsilon_L - \epsilon_S) \\ &= V(\epsilon_L - \epsilon_S) \\ &= V(\epsilon_L) + V(\epsilon_S) - 2C(\epsilon_L, \epsilon_S) \end{aligned} \quad (13.9)$$

If we make the assumption that $V(\epsilon_L) = V(\epsilon_S) = \omega^2$, then the variance of the market-neutral portfolio reduces to

$$V(r_p) = 2\omega^2(1 - \rho) \quad (13.10)$$

where ρ is the correlation coefficient between ϵ_L and ϵ_S .

Though α mojo is by itself a benefit, extra α comes at a price, and this should matter to the portfolio manager. The manager should be concerned with attaining a higher *information ratio* than the long-only portfolio.¹⁰ The information ratio (IR) of a long-only portfolio is

$$IR_L = \frac{\alpha}{\sqrt{\omega^2}} \quad (13.11)$$

whereas the information ratio of the market-neutral portfolio we considered earlier is

$$IR_N = \frac{2\alpha}{\sqrt{2\omega^2(1-\rho)}} \quad (13.12)$$

Thus the ratio of information ratios of a market-neutral to a long-only portfolio is

$$\begin{aligned} \frac{IR_N}{IR_L} &= \frac{\frac{2\alpha}{\sqrt{2\omega^2(1-\rho)}}}{\frac{\alpha}{\sqrt{\omega^2}}} \\ &= \sqrt{\frac{2}{1-\rho}} \end{aligned} \quad (13.13)$$

One can immediately see the benefits of a market-neutral strategy in the information ratio. Since the correlation between ϵ_L and ϵ_S is always less than or equal to 1, the market-neutral strategy improves on the long-only portfolio. In the extreme case of perfect correlation, the ratio of information ratios tends toward infinity.¹¹ In this case, the long and short portfolios (after shorting) move in opposite directions, reducing the overall portfolio variance.

Market neutral's advantage, however, does not come only from the correlation between the long and short portfolios. The mere ability to short

securities itself offers a distinct advantage. A long-only manager cannot take any particular action on a stock that he or she expects will crash; all he or she can do is not buy it. These constraints may actually cause greater mispricings and make $\alpha_S > \alpha_L$. With a long-short portfolio, the portfolio manager can have an outright negative weight on the stock and take advantage of this.

13.4 THE MECHANICS OF MARKET NEUTRAL

This section guides you through the mechanics of creating long and short positions in the securities markets.

13.4.1 Margin and Shorting

Equity securities in the United States can be purchased on margin. This allows an investor to invest more in securities than he or she has in cash. The Federal Reserve Board regulates margin and margin requirements through Regulation T.¹² For some time now, Regulation T has required that a customer deposit at least 50% of the current market value (CMV) of a marginable security. This is known as the *initial margin*, which is the margin required on first purchase of the equity security. For example, if a portfolio manager purchases 20,000 shares at \$10 per share, the customer technically could borrow up to \$100,000 from the broker for this \$200,000 purchase of securities. Of course, the broker-dealer will charge interest on the margin, often the broker *call rate* (the rate the bank charges the broker-dealer) plus some markup. We will call this margin rate i_M .¹³ There is margin for both going long and shorting stocks. However, because of the interest cost of trading on margin, the portfolio manager should trade on margin only with a specific objective in mind.

Suppose that a portfolio manager buys 200 shares of *ABC* at a price per share of \$50 and on 50% margin. This would be \$10,000 of security *ABC*. His or her CMV is \$10,000, his or her debit balance D is \$5,000, and his or her equity E is \$5,000 ($E = \text{CMV} - D$). These are broker-dealer terms. Clearly, as each day passes and the brokerage firm marks to market, the CMV, and hence the equity, will change. Marking to market means the

broker-dealer obtains the current value of the security each day and monitors whether the funds deposited by the portfolio manager are sufficient to cover the risks of a margined position. For example, if the price of *ABC* drops to \$30 the next day, the account equity will drop to \$1,000. This is worrisome because the manager has only \$1,000 in equity and a loan of \$5,000. To address this sort of problem, the Financial Industry Regulatory Authority (FINRA) and the stock exchanges have established rules that require a *minimum maintenance margin*. If a customer's equity value falls below some threshold, the broker-dealer makes a *margin call* that requires the customer to deposit extra securities or cash. The current threshold value is 25% of the CMV.¹⁴ If a customer receives a margin call, he or she must deposit enough funds to reach the minimum maintenance margin, or the broker-dealer may sell securities to achieve this minimum.

When a portfolio manager shorts a stock, he or she is effectively borrowing the stock from someone (the broker-dealer arranges this) and then selling it. Since this is like trading on margin, the manager must deposit margin for the stock. Regulation T therefore applies to short sales of equity securities as well. With a short account, the initial margin is also 50%, but the minimum maintenance margin is 30%.¹⁵ Accounting for a short position differs slightly from accounting for long margin trades, but the concepts are basically the same. On shorting the security, the broker-dealer receives the value of the short sale in cash plus additional cash (or some other deposit) representing the 50% margin. We call this initial cash balance *C*. The CMV of the securities is computed daily. The equity *E* of the account is equal to the cash deposit less the CMV, that is, $C - \text{CMV}$. For example, to short 200 shares of a stock at a price per share of \$50, using a 50% margin, the portfolio manager deposits \$5,000 as cash. The proceeds from the short sale are \$10,000. The cash in the account is therefore \$15,000. The CMV is \$10,000, so the equity in the account is \$5,000. As with long positions, if the stock price rises, the broker-dealer may require additional deposits as the minimum maintenance margin is reached.¹⁶

When a portfolio manager has both long positions and short positions with a broker-dealer, the broker-dealer computes the margin requirements of the long and short accounts separately and then nets them to determine the total equity in the account.

13.4.2 The Margin and Market Neutral

With Regulation T in mind, a market-neutral portfolio manager with $\$V$ in capital can invest $\$V$ in a long portfolio and then short $\$V$ of the short portfolio. A margin account will be required for every security sold short in the short portfolio, so the manager will have to post margin. In this case, though, the long positions can be used as margin collateral. Given the 50% margin requirement, the portfolio manager either could deposit $\$V/2$ in cash with the broker to cover the shorts or could just deposit $\$V$ of long securities.¹⁷ Theoretically, the portfolio manager could purchase $\$V$ of the long portfolio and short $\$V$ of the short portfolio and satisfy the Regulation T margin requirements.

In practice, however, a broker or prime broker usually requires an additional *liquidity buffer* to meet mark to markets on the short portfolio and to satisfy dividend payments on the short portfolio.¹⁸ Thus the broker may require an additional 10% of the value of the capital. In our example, this would be an additional $0.10V$. Let's call this extra liquidity buffer requirement m_{lb} . For any amount of capital V , the maximum that can be invested on one side in a fully leveraged market-neutral portfolio is $V^* = V(1 - m_{lb})$. Thus the total maximum amount invested would be $2V^*$. Of course, the portfolio manager can choose to leverage to a lesser degree. He or she could invest $V/2$ in the long portfolio and $V/2$ in the short portfolio, and this would achieve the same degree of gross leverage as a long-only portfolio, but zero net leverage.¹⁹

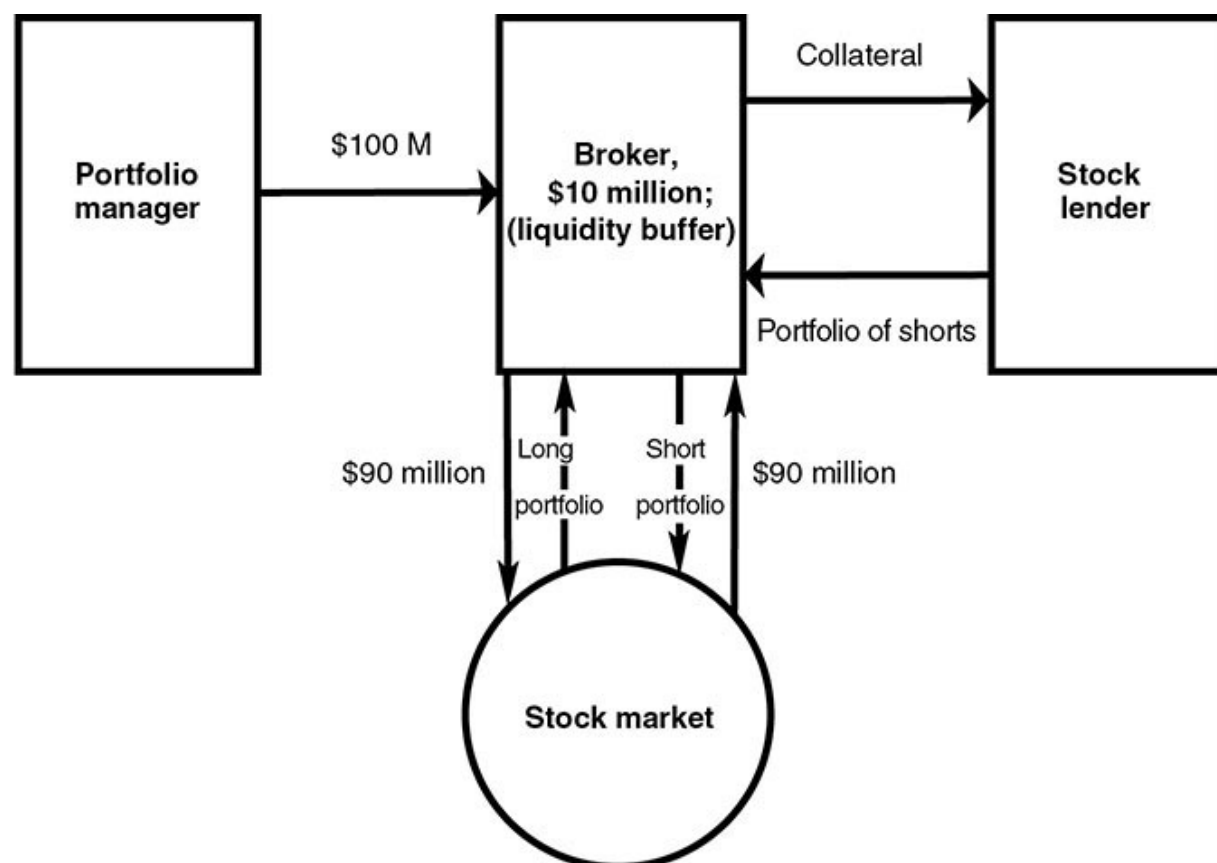
13.4.3 Sources of the Return

Suppose that a portfolio manager has identified a basket of securities to go long and a basket of securities to sell short. Call these portfolios L and S . A quantitative model has predicted excess returns for the long portfolio and negative or relatively lower excess returns for the short portfolio. The manager's equity capital is V , and the additional margin required by the broker is 10% ($m_{lb} = 0.10$). Let's assume that the manager has \$100 million in equity capital ($V = \$100$ million). Given the margin requirement, he or she will purchase \$90 million of the long portfolio and sell short \$90 million of the short portfolio. Thus the portfolio manager has chosen to fully leverage the market-neutral portfolio (i.e., V^* invested in the long and

short sides). He or she also has chosen to be dollar neutral. [Figure 13.1](#) illustrates the pattern of flows to create this market-neutral portfolio.

FIGURE 13.1

Flows in creating a market-neutral portfolio.



Ignoring any margin calls, we can dissect the returns that will accrue to the market-neutral portfolio. The first source of return is the difference in total return between the long portfolio and the short portfolio, $r_L - r_S$, where r_S is the return *before shorting*, the actual return of the weighted short portfolio.²⁰ The expected α of the overall portfolio, if the hedging is done perfectly, equals $\alpha_L - \alpha_S$. In practice, it may differ slightly from this value, but it can be measured exactly.

The second source of return is the interest on the proceeds of the short sale that are used as collateral. It usually will be a haircut to competitive short-term investment opportunities available in the marketplace. One

should expect to receive, over a period of k days and using continuous compounding, $V^*(e^{i \frac{k}{360}} - 1)$.²¹

The third source of return is the interest paid by the broker on the liquidity buffer. We shall assume that the liquidity buffer rate is i' . Thus, over a period of k days, one would receive $m_{lb} V(e^{i' \frac{k}{360}} - 1)$.

The value of the portfolio after k days (assuming we close all positions) would be²²

$$\begin{aligned} V_{t+k} &= V_t + V_t^* r_L - V_t^* r_S + (V_t - V_t^*) \left(e^{i \frac{k}{360}} - 1 \right) + V_t^* \left(e^{i \frac{k}{360}} - 1 \right) \\ &= V_t (1 - m_{lb}) + V_t (1 - m_{lb}) (r_L - r_S) + V_t m_{lb} \left(e^{i' \frac{k}{360}} - 1 \right) + V_t (1 - m_{lb}) \left(e^{i \frac{k}{360}} - 1 \right) \end{aligned} \quad (13.14)$$

Suppose that we hold the market-neutral portfolio for one month ($k=30$). Portfolio L increases by 7%, whereas portfolio S decreases by 4%. Suppose that the interest on the collateral and liquidity buffer is 2% per annum at a continuously compounded rate. In our example, the new value of the portfolio would be \$110,066,805.63. This is composed of the long P/L of \$6,300,000.00 plus the short P/L of \$3,600,000.00 plus the interest on the collateral of \$150,125.07 and finally the interest on the liquidity buffer of \$16,680.56. The return is 10.07% for the one-month period. This calculation is summarized in [Table 13.4](#).

TABLE 13.4

Sources of Return from Example Market-Neutral Portfolio

Number Source	Rate of Return (%)	Formula	P/L (\$)
1. Total return from long portfolio	7	$r_L V_t^*$	6,300,000
2. Total return from short portfolio	-4	$-r_S V_t^*$	3,600,000
3. Interest on collateral	2	$V_t^* (e^{i \frac{k}{360}} - 1)$	150,125.07
4. Interest on liquidity buffer	2	$V_t m_{lb} (e^{i' \frac{k}{360}} - 1)$	16,680.56
5. Total portfolio	10.07	Sum rows 1–4	10,066,805.63

Note: The total return for the long and short portfolios includes dividends. Of the \$100 million equity capital, only \$90 million is invested long and \$90 million is invested short owing to the liquidity buffer needed on the shorts by the broker-dealer equal to \$10 million.

13.5 THE BENEFITS AND DRAWBACKS OF MARKET NEUTRAL

Market-neutral portfolios offer a host of benefits for quantitative portfolio managers who wish to select specific stocks with their quantitative models. They allow the portfolio manager to focus his or her ability on stock selection with few disruptions from movements of the overall market. There are also some minor drawbacks to a market-neutral portfolio, including, of course, the transactions costs of creating it. [Table 13.5](#) lists more of the benefits and drawbacks.

TABLE 13.5

Benefits and Drawbacks of a Market-Neutral Portfolio

Benefits	Drawbacks
1. There may be more inefficiencies on the <i>short</i> side of the market. Analysts tend to be biased toward buys, there are institutional constraints on short selling, and several behavioral biases (e.g., "It's un-American to short") also prevent it.	Shorting stocks is subject to the <i>uptick rule</i> given certain market conditions. Borrowing certain stocks is sometimes difficult.
2. Bad stocks can be weighted less than 0%, whereas 0% is the minimum weight in long-only portfolios.	The short side of the portfolio requires an extra liquidity buffer of cash, which handles the short margin maintenance. The rebalancing of the short side can be costly, especially for large upside moves in the marketplace.
3. A truly market-neutral portfolio has no benchmark and hence has fewer weight constraints than a long-only portfolio, allowing for more opportunity to exploit inefficiencies.	The interest on the proceeds from the short sale of securities is up to the discretion of the broker. There is usually a haircut from available interest rates in the market for institutional investors, while retail investors usually receive zero interest.
4. There are extra benefits of diversification to the extent that the short portfolio and long portfolio are less than perfectly correlated. The information ratio may be higher for a market-neutral portfolio (see Section 13.3).	Even if scaled down appropriately, a short position theoretically has the potential for unlimited losses, especially for concentrated portfolios.*
5. The portfolio is generally uncorrelated with stock market movements and thus may be a suitable equity investment for those not interested in having market exposure. An investor can have positive returns whether the market goes up or down.	The market-neutral portfolio generally will have lower performance than a long-only portfolio in a strong bull market.
6. It allows the portfolio manager to focus on stock picking and not worry about market timing.	A bad quantitative model to predict will be amplified.

* Leverage risk is often cited as a drawback to the market-neutral portfolio. However, this is not entirely correct. A market-neutral portfolio manager does not have to create a fully leveraged portfolio. He or she can create a portfolio with 50% of the capital invested in the long portfolio and 50% of the capital invested in the short portfolio that would have a leverage risk very similar to that of a long-only portfolio. It should

be remembered, though, that the market-neutral portfolio still will have slightly higher risk, because a short position theoretically can have an unlimited loss.

Perhaps the most return-enhancing benefit of a market-neutral portfolio is the ability to short stocks. There tends to be a long bias in the market. Many money managers, including mutual fund managers, are restricted to long-only positions by institutional rules, and a majority of investors are either not aware of or just plain wary of shorting strategies. A wide variety of stocks may be overpriced owing to the lack of adequate shorting in the market. Thus the very act of shorting may exploit market inefficiencies.

A closely related advantage of a market-neutral portfolio is that in a long-only portfolio the worst view of a stock results in only a 0% weighting of that stock, whereas in a market-neutral portfolio, the manager can express a negative view with an outright negative weighting.

A third benefit is that the market-neutral portfolio is more flexible than the long-only portfolio. The long-only portfolio cannot stray too far from the benchmark owing to risk constraints, so its stock weightings must match more closely those of the benchmark. The market-neutral portfolio is not bound to the benchmark, so its stock weightings can follow many different patterns.

Extra diversification is another big benefit of the market-neutral portfolio when the stocks in the long and short portfolios are not very correlated. We know from basic portfolio theory that combining two securities with less than perfect correlation results in diversification. That is, by combining the two securities, the portfolio manager obtains a portfolio with returns similar to but with risk less than what either security in isolation would achieve. Since most stocks tend to be somewhat positively correlated, a long-only manager can get only so much of this diversification kick. However, since the stocks in the short portfolio are sold short, the manager effectively creates a negative correlation between the short and long portfolios. Depending on the extent of this negative correlation, the market-neutral portfolio can achieve greater diversification than can the long-only portfolio.

The benefit of diversification is truly amazing. Consider [Table 13.6](#), which describes two portfolios if they are both held long. One is the “good” portfolio because its expected return is high, and the other is the “bad”

portfolio because its expected return is low. Both portfolios have similar risk.

TABLE 13.6

Return and Risk Characteristics of Good and Bad Stock Portfolios

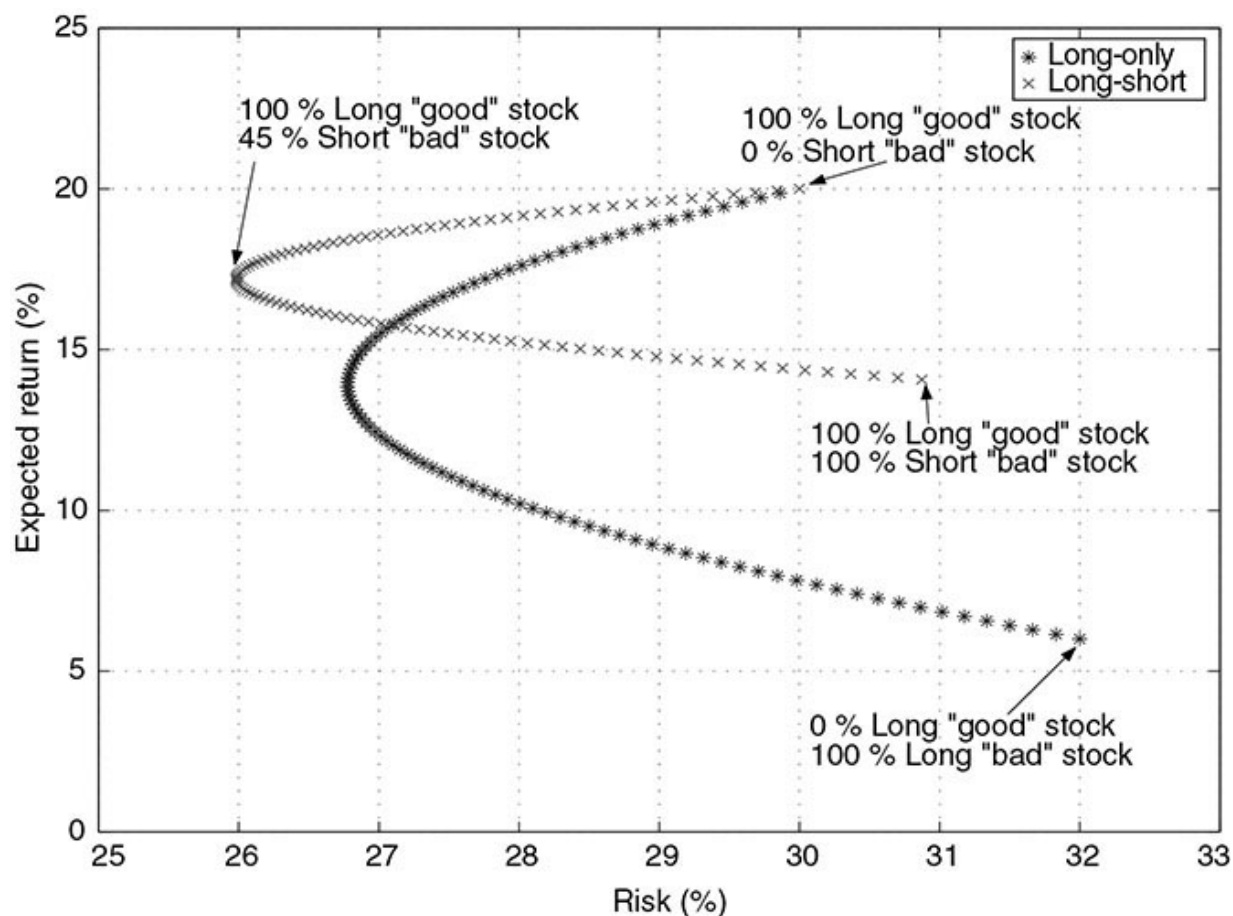
	μ	σ
Good portfolio	0.20	0.30
Bad portfolio	0.06	0.32
$\rho=0.5$		

Note: μ is the expected return of each portfolio, σ is the standard deviation or risk of each portfolio, and ρ is the correlation between the good and bad portfolios (before shorting).

Now consider the possible ways of owning the two portfolios. For an overall portfolio that is long only, the weightings of the long and short portfolios must sum to 1. That is, we could own 100% of the good portfolio and 0% of the bad portfolio or 0% of the good portfolio and 100% of the bad portfolio or some combination in between those two extremes. From the perspective of diversification, it is better to own a bit of both portfolios. [Figure 13.2](#) illustrates the idea. As we move from the point at which 100% is invested in the bad portfolio to the point at which 55% is invested in the good and 45% in the bad, we improve our risk-return combination. The frontier between 100% in the good portfolio and 55%–45% represents the efficient portfolios for a long-only manager, and we would pick an investment point along this frontier.

FIGURE 13.2

Diversification enhancement from a long-short portfolio.



However, if we could short stocks, we could do better. By allowing for shorting the bad portfolio, the efficient frontier actually moves to the left. There is a whole area, ranging from 100% long the good portfolio to 100% long the good portfolio and 71% short the bad portfolio, where we actually achieve lower risk for the same return that we could earn with the long-only portfolio. This risk reduction demonstrates the power of market-neutral investing.

In addition to the low correlation between the long and short sides of the portfolio, the market-neutral portfolio is relatively uncorrelated with market movements. This offers a benefit to investors who wish to invest in equities but do not wish to be exposed to overall market risk.

A final benefit of market-neutral portfolios is that they allow the portfolio manager to focus on stock picking without having to worry about market timing or exposure to a poorly performing market. During the internet bubble, a market-neutral manager would not have been outright long internet stocks. He or she would have been long certain stocks while

shorting others so that his or her portfolio probably would have fared well even when the bubble burst.

Most of the drawbacks of the market-neutral portfolio are not terribly detrimental. Shorting stocks is subject to certain restrictions that make shorting stocks sometimes difficult.²³ This means that one cannot always short quite as easily as one can go long. Interest rates on the proceeds from the short sale may be less than current market rates, causing a slight loss in potential return. Prime brokers typically require an additional cash liquidity buffer to handle the short portfolio for things such as dividend payments and margin calls. This takes some of the portfolio out of the market, but it is not a big problem because the market-neutral portfolio is usually leveraged anyway.²⁴ In a bull market, market-neutral portfolios will usually underperform all equity portfolios. This is the price of avoiding market collapses. Finally, a market-neutral portfolio focuses the returns on stock picking. Thus, while it aids a manager who is good at stock picking, it only magnifies the losses of a manager who, either for lack of a good model or simply because of bad luck, does not pick stocks well. There is no market-neutral strategy to disguise a poor model.

13.6 REBALANCING

Market-neutral portfolios need to be rebalanced as time passes to maintain market neutrality. At the same time, the market-neutral portfolio must meet margin requirements. We shall briefly discuss these two rebalancing needs.

Let's start with the case of dollar neutrality. Suppose that we create a dollar-neutral portfolio at time t . At time $t + 1$, the weights of the long portfolio and short portfolio will have changed. In particular, dollar neutrality at time t implies that

$$\frac{V_t^L}{V_t^S} = \frac{\sum_{i=1}^{N_L} w_{i,t}^L V_t^L}{\sum_{j=1}^{N_S} w_{j,t}^S V_t^S} = 1 \quad (13.15)$$

At time $t + 1$, this will no longer be true because

$$\frac{V_{t+1}^L}{V_{t+1}^S} = \frac{\sum_{i=1}^{N_L} w_{i,t}^L (1+r_{i,t+1}) V_t^L}{\sum_{j=1}^{N_S} w_{j,t}^S (1+r_{j,t+1}) V_t^S} \neq 1 \quad (13.16)$$

Only if the long portfolio and short portfolio had performed identically during the period would the overall portfolio have remained perfectly balanced. In all but the rarest of cases, therefore, the dollar neutrality must be reestablished. If the long portfolio has grown larger than the short portfolio, then selling stocks (in amounts proportional to their weights) from the long portfolio and keeping the funds in cash will reestablish the dollar neutrality. If the short portfolio has grown larger than the long portfolio, then buying back or covering some of the shorts will reestablish the dollar neutrality.

For the case of β neutrality, at time t , we have

$$\frac{\beta_t^L}{\beta_t^S} = \frac{\sum_{i=1}^{N_L} w_{i,t}^L \beta_i}{\sum_{j=1}^{N_S} w_{j,t}^S \beta_j} = 1 \quad (13.17)$$

However, at time $t + 1$, we have

$$\frac{\beta_{t+1}^L}{\beta_{t+1}^S} = \frac{\sum_{j=1}^{N_S} w_{i,t}^S (1+r_{j,t+1})}{\sum_{i=1}^{N_L} w_{i,t}^L (1+r_{i,t+1})} \frac{\sum_{i=1}^{N_L} w_{i,t}^L (1+r_{i,t+1}) \beta_i}{\sum_{j=1}^{N_S} w_{j,t}^S (1+r_{j,t+1}) \beta_j} \neq 1 \quad (13.18)$$

At time $t + 1$, rebalancing reestablishes β neutrality. The rebalancing process is simpler for β neutrality than it is for dollar neutrality. You can restore β neutrality by reducing the dollar value of the portfolio that now has the larger β , as in the case of dollar neutrality. You also can adjust the weights of the stocks within one of the portfolios, either the long or the short, so that the average weighted β is the same as it was originally. Or, in lieu of either of these methods, the portfolio manager simply can use index futures to alter the entire portfolio β . If the overall β has become slightly

positive (or slightly negative), the portfolio manager can use concepts from [Chapter 12](#) on leverage to reduce (or increase) the β to 0.²⁵

13.7 GENERAL LONG-SHORT

This chapter has focused on the construction of market-neutral portfolios in which the securities that the portfolio manager goes long exactly offset the securities that he or she shorts. Some portfolio managers bias the portfolio toward either the long or short sides either by design or as a consequence of their investment processes. This style of managing the portfolio not to be perfectly market neutral is known as *long-short*. In this section we discuss the general concept of partially hedging the portfolio this way. We also discuss the concept of *equitization*, or using index futures contracts to alter the portfolio's exposure to the overall market. Finally, we discuss *pair trading*, which is a bit like market neutrality on a pair of stocks.

13.7.1 Long-Short

Up to this point in the chapter we have looked at market-neutral portfolios that are either dollar neutral or risk factor neutral. Some portfolio managers want to construct their portfolios so that they are not entirely market neutral but retain some bias toward either the long or the short side. All of the concepts that we have explored thus far still apply, except that in the *long-short* portfolio there is some difference between either the dollar values or the risk exposures of the long and short sides. For example, the portfolio may have an overall β of 0.4 with respect to the market portfolio. It is not really market neutral, although it certainly has less risk than an all-equity index such as the S&P 500.

Long-short is a common portfolio style. It represents about 5% of the hedge fund world.²⁶ A long-short manager who is bearish on the equity market might be slightly short, whereas another who is optimistic might be slightly long. Along with the additional risk, the long-short strategy lets a manager benefit from overall market movements and α mojo.

13.7.2 Equitization

Equitization is the process of converting cash into a synthetic equity position with equity futures, forwards, or options. Through equitization, a portfolio manager with a market-neutral portfolio can keep his or her α and also have full exposure to the equity market. To alter the portfolio's exposure to the market, the manager can use equity index futures to go long the index an amount equal to 100% of the equity capital.²⁷ The return on the new portfolio will equal the return from the market-neutral side plus the return on the index futures. The market risk of the portfolio will be 1 (that is, $\beta_P = 1$). The portfolio will have all the attributes of a long-only portfolio, as well as the advantages of a market-neutral portfolio.

Equitization is especially useful to a portfolio manager who must maintain a significant exposure to a specific equity benchmark but has views on the relative performance of stocks. To take one example, if the manager feels that certain sectors of the economy are the most mispriced, he or she can focus on making many trades in those sectors. If he or she finds great opportunities in, say, the consumer staples sector, he or she can create a market-neutral position within consumer staples, leaving other sector exposures long-only, as dictated by the long index futures position. The resulting portfolio will still behave very much like the benchmark, but benefit from the hidden α of the market-neutral position in consumer staples.

13.7.3 Portable α

Equitization can be taken a step further via the concept of *portable α* .²⁸ Portable α is α that can be transported from the portfolio to another asset class or subclass entirely. This allows plan sponsors, financial advisors, pension fund managers, and other investors the flexibility to separate decisions relating to α from the asset-allocation decision. For portfolio managers, portable α makes it possible to separate the α from the β of the portfolio. The construction of portable α combines concepts from leverage (Chapter 12) and market neutrality (this chapter).

For example, suppose that a portfolio manager is very good at managing small-cap stocks but thinks that large-cap stocks will outperform small-cap stocks. He or she always has had a positive α^B when it comes to managing small-cap stocks against a small-cap benchmark such as the Russell 2000. He or she believes that his or her success is due to many inefficiencies in

the small-cap universe that he or she is able to exploit with quantitative models. His or her models are less successful in the large-cap universe, perhaps because there are fewer inefficiencies there. He or she tends to manage mainly small-cap stocks, but given that his or her benchmark is the S&P 500 (not the Russell 2000), he or she realizes that if he or she is correct about large-cap stocks performing better than small-cap stocks, this will hurt his or her relative performance tremendously. This is a perfect situation for the manager to transport his or her small-cap α . The manager can continue to manage his or her portfolio as he or she wishes, choosing small-cap stocks that he or she feels are relatively mispriced, but he or she also should short the Russell 2000 futures in order to achieve β neutrality vis-à-vis the Russell 2000 and purchase S&P 500 futures so as to achieve a β equal to 1 versus the S&P 500. This leaves the manager with a portfolio that has the overall market risk of the S&P 500 plus the α^B of his or her portfolio over the Russell 2000.²⁹

Below we describe the steps of transporting α in this example and the resulting overall portfolio:

Step 1: Find the β of the small-cap portfolio versus the Russell 2000:

$$\tilde{r}_P = \alpha_{P,R} + \beta_{P,R} \tilde{r}_R + \epsilon_{P,R}.^{30}$$

Step 2: Short the required amount of Russell 2000 futures to achieve a $\beta^* = 0$ for the overall portfolio. Then the excess return of the new portfolio becomes $\tilde{r}_P - \hat{\beta}_{P,R} \tilde{r}_R$.

Step 3: Find the β of the new portfolio versus the S&P 500. This is done by regressing $\tilde{r}_P - \hat{\beta}_{P,R} \tilde{r}_R$ against the excess returns of the S&P 500 ($\tilde{r}_S$). The new beta estimate will be given by

$$\frac{\hat{C}[\tilde{r}_P - \hat{\beta}_{P,R} \tilde{r}_R, \tilde{r}_S]}{\hat{V}(\tilde{r}_S)} = \frac{\hat{C}[\tilde{r}_P, \tilde{r}_S]}{\hat{V}(\tilde{r}_S)} - \hat{\beta}_{P,R} \frac{\hat{C}[\tilde{r}_R, \tilde{r}_S]}{\hat{V}(\tilde{r}_S)} = \hat{\beta}_{P,S} - \hat{\beta}_{P,R} \hat{\beta}_{R,S}$$

. The constant term, or alpha intercept, will be given as

$$E[\tilde{r}_P - \hat{\beta}_{P,R} \tilde{r}_R] - (\hat{\beta}_{P,S} - \hat{\beta}_{P,R} \hat{\beta}_{R,S}) E[\tilde{r}_S] = \hat{\alpha}_{P,R} - (\hat{\beta}_{P,S} - \hat{\beta}_{P,R} \hat{\beta}_{R,S}) \bar{r}_S$$

Step 4: Go long the required amount of S&P 500 futures to achieve a $\beta^* = 1$ (i.e., $1 - [\hat{\beta}_{P,S} - \hat{\beta}_{P,R} \hat{\beta}_{R,S}]$). The new excess

return of the portfolio will be

$$\tilde{r}_P - \hat{\beta}_{P,R}\tilde{r}_R + (1 - \hat{\beta}_{P,S} + \hat{\beta}_{P,R}\hat{\beta}_{R,S})\tilde{r}_S.$$

Step 5: The α of the new portfolio with respect to the S&P 500 is

$$E[\tilde{r}_P - \hat{\beta}_{P,R}\tilde{r}_S - (\hat{\beta}_{P,S} - \hat{\beta}_{P,R}\hat{\beta}_{R,S})\tilde{r}_S] = \hat{\alpha}_{P,R} - (\hat{\beta}_{P,S} - \hat{\beta}_{P,R}\hat{\beta}_{R,S})\bar{r}_S.$$

. Thus, this method gives the portfolio manager a β exposure of 1 to the S&P 500 with an α of $\hat{\alpha}_{P,R} - (\hat{\beta}_{P,S} - \hat{\beta}_{P,R}\hat{\beta}_{R,S})\bar{r}_S$.

This means that the portfolio manager achieves a perfect α transport when $\hat{\beta}_{P,S} = \hat{\beta}_{P,R}\hat{\beta}_{R,S}$. Depending on other values, his transported α might be larger or smaller than his original Russell 2000 α .

Portable α creates a host of possibilities for transporting α in the manager's portfolio. Given that there is no strict mandate on the portfolio's base exposure, the manager even could transport α from an equity universe to a bond universe. For example, the portfolio manager in the preceding example has a positive α versus the Russell 2000 equity benchmark by managing a portfolio of smallcap stocks. He or she could short the Russell 2000 futures in the same proportion so as to neutralize his or her overall β . He or she then could use the proceeds to invest in either 10-year bond futures or bond exchange-traded funds (ETFs) to acquire a market exposure to the bond markets.

The concept of portable α is quite useful, although there are some problems associated with it. First, derivatives, swaps, and even liquid futures do cost money to buy and sell. Second, index futures may not perfectly track the underlying portfolio, even if the β versus the index has been measured. The resulting residual risk may be larger than the α that the portfolio manager is trying to transport. Third, futures may not exist for a particular portfolio or asset class. In other cases, the futures may exist but be insufficiently liquid. For example, there are some futures for certain sector indices, but they are extremely illiquid and thus not practical. Fourth, while quantitative portfolio managers have the expertise to transport α , many pension fund committees and other investors simply do not have the expertise to handle α mojo. Fifth, margin may need to be posted for the futures trading, so a slightly lower interest rate is received than with alternative investment vehicles. Sixth, index futures have roll-over risk and

basis risk, and swaps have counterparty risk. For the most part, these risks are small.

Despite the minor problems associated with portable α , it is a low-cost mechanism to use capital efficiently and separate the α from the β of the portfolio so as to achieve the best of both worlds for the investor.

13.7.4 Pair Trading

Pair trading means creating a market-neutral balance between two securities that are similar except that one is undervalued and one is overvalued (according to a quantitative model of stock returns). The pair trade involves buying the undervalued security and shorting the overvalued security. Just as a market-neutral portfolio removes exposure to market risk and solidifies the connection between the portfolio's return and its α , a pair trade removes market risks from the pair and brings out the α 's of the individual securities. Pair trades can be very useful for quantitative portfolio managers for a variety of reasons. They allow the portfolio manager to make trades that focus on α without the complications of market or industry risk. Second, they expand the investment universe to stocks in otherwise overlooked sectors. Even if the manager considers the consumer discretionary sector extremely overvalued as a whole, he or she may find good picks within that sector. Pair trading takes advantage of picking these stocks while still avoiding exposure to the overvalued sector. A long-only portfolio simply would avoid the sector altogether. Another useful aspect of pair trading is that it uses equity capital efficiently. The pair trade involves using margin on the short side, so every dollar invested in the pair trade is equivalent to \$2 invested in a long-only portfolio without leverage. This is the same efficient use of capital that a leveraged long-only portfolio would provide.

A pair trade can occur when the portfolio manager believes that there is a difference between either the *absolute values* or the *relative values* of the two securities. It could be that the securities are both expected to have a positive α , but one of them is expected to have a much higher α . In this case, buying the security with the higher α and shorting the one with the lower α is still a reasonable pair trade because there is a difference in the relative expected movements of the securities. Or it could be the case that a portfolio manager finds two securities that are similar, but one is expected

to have a negative α and the other a positive α . This is an absolute pair trade, the ideal case.

The first step in pair trading is to identify similar stocks, and *similar* has more than one meaning. One is that the stocks are in the same industry. The manager can apply his or her quantitative model to stocks within an industry and then rank them by aggregate Z-score, or α , or expected return. The pair could be two stocks at opposite ends of the ranked list.

Similar stocks also can be found with *characteristic matching*, as discussed in [Chapter 7](#). The manager looks for stocks with close fundamental ratios. One should be careful to look at fundamental ratios that are not factors in the quantitative model of stock returns because those factors already drive the expected α 's.

Statistical analysis can help to identify stocks that have behaved similarly over time. One test is take the historical returns of stocks and estimate their correlations with each other (their β coefficients).³¹ Two stocks whose returns are highly correlated, or whose β with respect to one another is close to 1 and statistically significant, have behaved very similarly in the past and may tend to in the future. The problem with this sort of analysis is that it applies a relationship gleaned from past data to future expected returns. The value of the analysis, though, comes from the fact that the daily relative volatility of a pair of stocks will be under control if the stocks are highly correlated.

Another way to find similar stocks is to run a regression of stock A 's returns against stock B 's returns using daily data for a period of one year. The manager runs a series of regressions for every month of the past year and stores the α 's of the regression. He or she looks for pairs in which the β is close to 1 and the α of one stock versus the other is significant and related over time. This kind of similarity would show up in the autoregressive behavior of the α of two stocks relative to each other. If the pattern appears, there is a potential pair trade. The risk of one stock offsets the risk of the other, but one stock has an excess return over the other that seems to persist.

It is also possible to take a contrarian view of pair trades. Historical statistical data show whether or not two stocks that are similar have diverged in return movement. If they have, one might believe that they eventually must converge. In the contrarian pair trade, one would short the stock that has had the higher return over some period and purchase the one that has had the lower return.

Many portfolio managers spend a lot of time researching the fundamentals of the companies before enacting the pair trade. A quantitative portfolio manager, however, relies on his or her quantitative models and statistical techniques. Once a pair-trade candidate has been identified, the next step is to determine how much to trade. There are a number of methods for this, as well. Some managers want to be a bit long or a bit short on one side of a pair trade. We will focus on the case in which the portfolio manager wants to eliminate as much risk as possible and focus on the relative α 's of the stocks. As with a market-neutral portfolio, this sort of pair trade can be dollar-neutral or risk-factor-neutral—although, unlike the market-neutral portfolio, it cannot be both at once. Dollar-neutral pair trades are common, although they may be flawed if one stock is much more volatile than the other. Dollar neutrality implies that $w_A = w_B$, where the same dollar amount is placed in stock A and stock B . Risk-factor neutrality adjusts for the risk-factor β . The β of each stock is measured against some index, $w_B = w_A(\beta_A/\beta_B)$, where β_A and β_B are the measured β 's of each stock over the same time horizon with respect to some industry index or the market index. You can use $w_A = 1$ and get the appropriate risk-adjusted weight for stock B for the pair trade. Since a pair trade is ever undiversified, a portfolio manager may wish to use volatility instead of β to determine the relative sizes of the stock trades. In this case, the weights would be determined as $w_B = w_A(\sigma_A/\sigma_B)$.

Suppose that a quantitative portfolio manager ran his or her quantitative model on the pharmaceutical industry. From the model, he or she has determined that Johnson & Johnson is relatively more attractive than Merck. Since the companies are similar in that they come from the same industry and specialize in drug manufacturing, the manager wants to make a pair trade on these two stocks from January 2021 onward.

The manager already has completed the first step of identifying the stocks to pair trade. The next step is to determine the weights to hold in each security. Using historical data from December 31, 2014, to December 31, 2020, the portfolio manager estimates $\beta_{JNJ} = 0.717$ and $\beta_{MRK} = 0.513$ versus the S&P 500. For β neutrality, he or she will short \$1.40 times Merck for every \$1.00 exposure to Johnson & Johnson. Another way of saying this is that the relative normalized weights (weights that sum to 1) should be 0.417 and 0.582 for JNJ and MRK, respectively. If, instead, the portfolio

manager wants to be dollar neutral, he or she should short \$1 of Merck for every \$1 of Johnson & Johnson so that the relative weights are 0.50 and 0.50.

The performance of such a pair trade from December 31, 2020, to August 31, 2021, appears in [Table 13.7](#). From this table we can see that owning JNJ outright would have provided a decent return over the period of 10.01%, but with an annualized volatility of 10.79%. Shorting Merck also would have produced a positive 2.19% return but with higher annualized volatility of 17.42%. Together, though, the dollar-neutral pair trade produced an unleveraged return of 6.10% with a lower volatility of 9.77%. Thus, for an average return similar to that of either trade on its own, the pair trade produced a volatility that was less than the volatility of either of the individual securities. The correlation with the S&P 500 was -0.41 compared to -0.38 and 0.23 . The β -neutral pair trade did not perform quite as well as the dollar-neutral one, but both accomplished the desired goal of maintaining return while reducing risk.

TABLE 13.7

Example of a Pair Trade

	Holding Period Return (%)	SD (%)	Correlation with S&P 500
JNJ	10.01	10.79	-0.38
MRK	-2.19	17.42	0.23
Pair (dollar-neutral)	6.10	9.77	-0.41
Pair (beta-neutral)	5.45	9.59	-0.43

Note: Data were obtained from Bloomberg. The holding-period returns and other statistics were computed, over the period of December 31, 2020, to August 31, 2021. The position initiated on December 31, 2020, was either β neutral or dollar neutral and not rebalanced for the entire duration of the trade.

13.8 CONCLUSION

Where leverage multiplies a portfolio's exposure to the market, the market-neutral strategy insulates the portfolio from market winds. Market-neutral strategies reduce or eliminate an investor's exposure to the volatility of the marketplace while still producing substantial returns. In fact, these strategies even can boost the relative contribution of a portfolio's α and be

used to transport α across asset classes. In this chapter we discussed the two main variants of market neutrality: dollar neutrality, in which the long and short positions equal each other in dollar value, and β neutrality, in which the long and short positions have similar β 's with respect to certain risk factors. We showed that market-neutral investing is a great source of α mojo because, by letting the portfolio manager sell bad stocks short, it boosts α and the information ratio. We also discussed cousins of the market-neutral strategy, including long-short strategies, equitization of a market-neutral strategy, portable α , and pair trading.

We have not exhausted the wellsprings of α mojo quite yet. In the next chapter we uncover the third major source of investing mojo, a rigorous statistical theory that boosts the value of α by quantifying qualitative information.

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- ¹ *Source:* BarclayHedge. The actual assets under management grew from \$5.5 billion to \$57.8 billion. The broader category of long-short funds and market-neutral funds declined from 19.45% to 6.07% over the same period.
 - ² For more details, see Patton (2004). He uses various definitions of market neutrality and finds that while some market-neutral funds do not behave neutrally with respect to the market, most do. He also finds that market-neutral funds are clearly less affected by the market than are other types of hedge funds.
 - ³ In terms of writing a computer program, however, it may be more convenient to remove the short-sale constraint in the optimization and treat the weights of stocks in the short portfolio as the stocks that produce negative weights. An additional constraint can be added that the negative weights sum to -1 and the positive weights sum to positive 1 to maintain the dollar neutrality.
 - ⁴ To leave some exposure to market risk, the portfolio can be set with a β not exactly equal to 0 . Even if β_P is not exactly equal to 0 , it is sometimes possible to achieve β neutrality with a futures overlay. At this point, though, we are not considering the use of futures.
 - ⁵ Of course, the portfolio manager can construct the market-neutral portfolio to be neutral with respect to the market index of his or her choice (e.g., S&P 500, Russell 1000, and NASDAQ 100).
 - ⁶ We discuss optimizers in [Chapter 9](#).
 - ⁷ In general, we can create a β -neutral portfolio if we have two portfolios whose factor exposure profiles are a multiple of one another.
 - ⁸ In Part V of this book we implement this technique as well as the more direct construction of market-neutral portfolios, as explained earlier.
 - ⁹ We will see in the margin section of this chapter that in practice the maximum α mojo that can be achieved is 1.8 owing to additional cash requirements by the brokers. Theoretically, though, a market-neutral portfolio can achieve two times the α of a long-only portfolio. Hedge funds that

use prime brokers can increase their leverage beyond a leverage of two, depending on the relationship with the prime broker.

- 10 The information ratio is a common performance-measurement criterion for active managers. It is the ratio of active return to active risk. A higher ratio implies a better portfolio manager, one who achieves more return for the same amount of risk. This is described in more detail in [Chapter 15](#) on performance measurement.
- 11 These arguments were presented in Michaud (1993). Michaud argues, however, that additional costs associated with the short side of the strategy actually may reduce this efficiency of the portfolio. He also worries about the true ability to perfectly hedge all the additional risk factors. It should be mentioned that the ratio in [Eq. \(13.13\)](#) holds even without any additional leverage in the market-neutral over the long-only portfolio.
- 12 The Securities Exchange Act of 1934 granted the Federal Reserve Board this authority.
- 13 Broker-dealers make a large amount of their profits from margin lending.
- 14 There are other rules as well. The customer must have a minimum equity of \$2,000. (This is mostly a concern for individual investors, not usually for managers who run portfolios with millions of dollars in assets.) Some firms also have higher maintenance margins than the minimum requirement. For example, during the internet bubble of 2000, many brokerages placed especially high maintenance margins on certain internet stocks.
- 15 The minimum maintenance margin is only 5%, though, if the portfolio manager shorts against the box (i.e., shorts the same security that he or she is long).
- 16 The value of the stock position that will result in a margin call can be figured out by dividing the cash balance by $(1 + m_{MM})$, where m_{MM} is the minimum maintenance margin requirement.
- 17 Unmargined securities are worth half the equity of cash. Many hedge funds achieve greater leverage than described here due to their relationships with prime brokers.
- 18 When someone shorts a stock, all dividend payments from the shorted stocks must be paid to the stock's owner.
- 19 In [Chapter 12](#), we defined leverage to be net leverage or leverage with respect to market exposure. Gross leverage is also used by practitioners and is the sum of the absolute positions of a trade or portfolio divided by equity capital.
- 20 While the differences in dividend payments for stocks in the long portfolio and the short portfolio can be considered as a source of return as well, this is already reflected in the return formula.
- 21 Interest rates from the broker are not continuously compounded rates, but they can be converted to continuously compounded rates, and this makes the math more presentable. Also, one may wish to alter the day count convention from $k/360$ to something else. For a discussion on conversions between rates and general conventions, see Steiner (1998).
- 22 For the actual computation of historical returns of a market-neutral portfolio, we briefly discuss these calculations in [Chapter 15](#) on performance measurement. The formula is $r_P = r_L - r_S + r_{\text{cash}}$.
- 23 For many years, stocks were subject to the uptick rule, which stated that a listed security could only be sold short at a price above the price of the preceding sale (plus tick) or at the last sales price if higher than the previous sale. Short sales were not permitted on minus ticks or zero-minus ticks. Since many practitioners believed these rules did very little for markets, they were removed in 2007. However, in 2010, after the great financial crisis of 2008, the SEC modified

the rule, which is now known as the *alternative uptick rule* or the *short-sale restriction*. The rule only prohibits the execution of a short sale order at a price that is less than or equal to the current national best bid if the price of that covered security decreases by 10% from the prior day. This prohibition stays in effect the rest of that trading day plus the next trading day.

- 24 Many investors believe that market-neutral portfolios must be leveraged. This is not true. Leverage is the choice of the portfolio manager. One can create a market-neutral portfolio that has the same leverage as a long-only portfolio.
- 25 In addition to rebalancing needs for dollar and β neutrality, there are also rebalancing needs owing to the management of short positions.
- 26 Source: BarclayHedge and author's calculations.
- 27 This example assumes that the β of the market-neutral portfolio is equal to 0. While there will be a requirement by the futures broker for margin on the index futures, this amount can be supplied through the liquidity buffer used for the short positions. Since the long futures contract will implicitly hedge the short positions, less is required on both sides of the deal to meet margin requirements.
- 28 Some people refer to this as *α transport*.
- 29 The portfolio manager will achieve perfect α transport only under certain conditions discussed in the detailed steps to create portable α .
- 30 We use double subscripts (P, R) to indicate the regression of the portfolio excess return (" P ") on the Russell 2000 excess return (" R "), where $\tilde{r}_P = r_P - r_f$, and $\tilde{r}_R = r_R - r_f$ (i.e., the excess return of the portfolio or of the Russell 2000 over the risk-free rate). We also use (" S ") to represent the S&P 500. The user can do the entire analysis with or without the risk-free rate. A bar over a variable indicates the in-sample average of the variable. For example, $\bar{\tilde{r}}_S$ is the average return of the S&P 500 minus the risk-free rate over the estimation period.
- 31 One also could use a multifactor model to infer the implied correlations among stocks.

Market-Neutral Portfolio Construction Techniques

As discussed in [Appendix 9B](#), phantom weights and binary variables can be used to construct a market-neutral portfolio optimization that allows for more general portfolios, including leverage constraints, number of long and short positions, and other possibilities. In this appendix, we will review the ideas in [Appendix 9B](#) and illustrate an example of constraining leverage.

In order to constrain the total leverage of the portfolio, we will want the sum of the long weights plus the sum of the short weights to equal some amount, say l_p . For example, if we want a portfolio with a leverage of 2, we would specify that $l_p = 2$. If we want a portfolio of leverage 4, we would specify that $l_p = 4$, and so on. Some investors might want to constrain leverage further, such that the long and short leverages are different. In this case, the user might specify that the sum of the long weights equals l_l (long exposure) and the sum of the short weights equal l_s (short exposure). For example, if a user wanted to create a 130–30 portfolio, then $l_l = 1.30$ and $l_s = 0.30$. When using phantom weights and binary variables, these constraints are quite easy to add to the market-neutral optimization.

We start by expanding the $\mathbf{x}$ vector to include five types of variables. They include the N actual portfolio weights, w_i ; the phantom positive weights w_i^+ , which represent the N “long” positions of the portfolio; the phantom negative weights, w_i^- , which represent the N “short” positions of

the portfolio; and the binary variables v_i^+ and v_i^- for each stock in the portfolio. We then introduce the following constraints in order to ensure that the phantom weights are orthogonal to each other and are also related to the actual weights (w_i) of the portfolio:

$$w_i = w_i^+ - w_i^- \quad (13A.1)$$

and

$$v_i^+ \kappa_l \leq w_i^+ \leq v_i^+ \kappa_h \quad (13A.2)$$

$$v_i^- \gamma_l \leq w_i^- \leq v_i^- \gamma_h \quad (13A.3)$$

$$v_i^+ + v_i^- \leq 1 \quad (13A.4)$$

Since v_i^+ and v_i^- are binary variables, the inequality constraint in [Eq. \(13A.4\)](#) forces only one of them to be 1 and the other 0. This constraint combined with the constraints in [Eq. \(13A.2\)](#) and (13A.3) forces the phantom weights to also be orthogonal to each other. For generality, we used the parameters κ_l , γ_l , κ_h , and γ_h , but in most market-neutral applications, it will be sufficient to set these equal to 0 and 1. When constructing the optimization, it is important to specify that v_i^+ and v_i^- are binary variables.¹

The minimization problem is similar to our standard one:

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \mathbf{Q} \mathbf{x} + \mathbf{x}' \mathbf{c} \quad s.t. \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (13A.5)$$

$$\mathbf{lb} \leq \mathbf{x} \leq \mathbf{ub} \quad (13A.6)$$

where $\mathbf{x}$ is the vector of unknowns in the problem, $\mathbf{Q}$ is a symmetric positive semi-definite matrix, $\mathbf{c}$ is a vector of coefficients, $\mathbf{A}$ is a matrix of coefficients for the equality and inequality constraints, $\mathbf{b}$ is a vector of constraint values, $\mathbf{lb}$ is a lower-bound vector, and $\mathbf{ub}$ is an upper-bound vector.²

We will show the equality matrix constraints and inequality matrix constraints separately, but you can also combine them into one $\mathbf{A}$ matrix if desired. The equality matrix, $\mathbf{A}_{eq}$, will be an $(N + 4)$ by $(5N)$ matrix and the $\mathbf{b}_{eq}$ vector will be $(N + 4)$ by (1) .

$$\mathbf{A}_{eq} = \begin{bmatrix} \boldsymbol{\iota}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \\ \boldsymbol{\mu}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \\ \mathbf{I}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{1 \times N} & \boldsymbol{\iota}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \\ \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} & \boldsymbol{\iota}_{1 \times N} & \mathbf{0}_{1 \times N} & \mathbf{0}_{1 \times N} \end{bmatrix}_{(N+4) \times (5N)} \quad (13A.7)$$

$$\mathbf{b}_{eq} = \begin{bmatrix} 0 \\ \mu_p \\ \mathbf{0}_{N \times 1} \\ l_l \\ l_s \end{bmatrix}_{(N+4) \times (1)} \quad (13A.8)$$

The inequality matrix, $\mathbf{A}_{ineq}$, would be set up as follows:

$$\mathbf{A}_{ineq} = \begin{bmatrix} \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{I}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & -1 \cdot \mathbf{I}_{N \times N} & \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} \end{bmatrix}_{(5N) \times (5N)} \quad (13A.9)$$

$$\mathbf{b} = \begin{bmatrix} \boldsymbol{\mu}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \\ \mathbf{0}_{N \times 1} \end{bmatrix}_{(5N) \times (1)} \quad (13A.10)$$

where $\boldsymbol{\mu}_{1 \times N}$ is the row vector of the mean returns for each of the N stocks, μ_p is the desired annualized portfolio return, $\boldsymbol{\iota}_{1 \times N}$ is a row of six 1s (that is, the first six entries of matrix $\mathbf{A}$'s row 1 are 1s), $\mathbf{0}_{1 \times N}$ is a row of N entries of zero, $\mathbf{I}_{N \times N}$ is the identity matrix (that is, an N -by- N matrix with 1s on the diagonal and 0s everywhere else), and so on and so forth. In order to aid the reader, we indicate the dimension of every matrix as a subscript.

As stated earlier, you can either put upper- and lower-bound constraints in the inequality, or $\mathbf{A}_{ineq}$ matrix, or specify them directly as upper- and lower-bound vectors in many optimization packages. In this example, we chose to specify them directly; thus we constrain weights, phantom weights, and binary weights to be between 0 and 1.³ Thus,

$$\mathbf{lb} = [\mathbf{0}_{1 \times N} \quad \mathbf{0}_{1 \times N} \quad \mathbf{0}_{1 \times N} \quad \mathbf{0}_{1 \times N} \quad \mathbf{0}_{1 \times N} \quad \mathbf{0}_{1 \times N}] \quad (13A.11)$$

$$\mathbf{ub} = [\boldsymbol{\iota}_{1 \times N} \quad \boldsymbol{\iota}_{1 \times N} \quad \boldsymbol{\iota}_{1 \times N} \quad \boldsymbol{\iota}_{1 \times N} \quad \boldsymbol{\iota}_{1 \times N} \quad \boldsymbol{\iota}_{1 \times N}] \quad (13A.12)$$

The $\mathbf{x}$ solution variables are

$$\mathbf{x} = \begin{bmatrix} \mathbf{w}_{N \times 1} \\ \mathbf{w}_{N \times 1}^+ \\ \mathbf{w}_{N \times 1}^- \\ \mathbf{v}_{N \times 1}^+ \\ \mathbf{v}_{N \times 1}^- \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \\ w_1^+ \\ w_2^+ \\ \vdots \\ w_N^+ \\ w_1^- \\ w_2^- \\ \vdots \\ w_N^- \\ v_1^+ \\ v_2^+ \\ \vdots \\ v_N^+ \\ v_1^- \\ v_2^- \\ \vdots \\ v_N^- \end{bmatrix} \quad (13A.13)$$

We can solve this problem with a mixed integer quadratic programming (MIQP) optimizer.⁴ If an MIQP optimizer is not available, but a mixed integer linear programming (MILP) optimizer is, then use an algorithm similar to the one discussed in [Appendix 10B](#) to find the solution.

13A.1 A NUMERICAL EXAMPLE

We will continue the numerical example in [Appendix 9A](#). However, we will change the target return. Typically, with market-neutral portfolios, a lower return target should be sought, since the net position of the portfolio is zero. For this example, we chose a target return of 2% and the leverage of the long and short portfolios equal to 1 (i.e., a total leverage of 2, where $l_l = l_s = 1$).

The quadratic programming problem can be expressed as

$$\min_{\mathbf{x}} \frac{1}{2} \mathbf{x}' \Sigma \mathbf{x} \quad s.t. \quad \mathbf{A} \mathbf{x} \leq \mathbf{b} \quad (13A.14)$$

where

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{eq} \\ \mathbf{A}_{ineq} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\iota}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \boldsymbol{\mu}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \mathbf{I}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{1 \times 6} & \boldsymbol{\iota}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \boldsymbol{\iota}_{1 \times 6} & \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} \\ \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} & -1 \cdot \mathbf{I}_{6 \times 6} & \mathbf{0}_{6 \times 6} & \mathbf{0}_{6 \times 6} \end{bmatrix}_{40 \times 30} \quad (13A.15)$$

$$\mathbf{b} = \begin{bmatrix} \mathbf{b}_{eq} \\ \mathbf{b}_{ineq} \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ \mathbf{0}_{6 \times 1} \\ 1 \\ 1 \\ \mathbf{1}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \\ \mathbf{0}_{6 \times 1} \end{bmatrix}_{40 \times 1} \quad (13A.16)$$

where $\boldsymbol{\mu}_{1 \times 6} = [14.40 \ 10.19 \ 9.87 \ 7.52 \ 20.05 \ 2.66]$, $\mathbf{1}_{1 \times 6}$ is a row of six 1s (that is, the first six entries of matrix $\mathbf{A}$'s row 1 are 1s), $\mathbf{0}_{1 \times 6}$ is a row of 6 entries of 0, $\mathbf{I}_{6 \times 6}$ is the identity matrix (that is, a 6-by-6 matrix with 1s on the diagonal and 0s everywhere else), and so on and so forth, and $\mathbf{x} = [\mathbf{w}_{1 \times 6} \ \mathbf{w}_{1 \times 6}^+ \ \mathbf{w}_{1 \times 6}^- \ \mathbf{v}_{1 \times 6}^+ \ \mathbf{v}_{1 \times 6}^-]'$. In order to aid the reader, we indicate the dimension of every matrix as a subscript, including for the large $\mathbf{A}$ and $\mathbf{b}$ matrices.

The economic meaning of these constraints is the following. The first five rows of this $\mathbf{A}$ matrix are the equality constraints that the sum of weights equals 0, that the average return of the portfolio equals 2%, that $w_i = w_i^+ - w_i^-$, that the sum of the long weights equals 1 ($\sum_{i=1}^N w_i^+ = 1$), and that the sum of the short weights equals 1 ($\sum_{i=1}^N w_i^- = 1$). For the binary weights, a zero is placed in the matrix since they are not relevant for these constraints. The sixth row in the matrix (which is really the 11th row through the 16th row) constrains the binary variables to be orthogonal—that is, only one of them can be positive—which consequently constrains the phantom weights to be orthogonal (i.e., we don't buy and sell the same security). To establish this, we need the last four rows of constraints (which are really rows 17 to 40).

The solution using an optimization routine with continuous and binary variables is⁵:

$$\mathbf{w} = \begin{bmatrix} -0.213 \\ 0.198 \\ 0.357 \\ 0.445 \\ -0.099 \\ -0.687 \end{bmatrix} \quad (13A.17)$$

You will notice that the sum of the portfolio weights equals 0 (i.e., a dollar-neutral portfolio) and that stocks 1, 5, and 6 are sold short and stocks 2, 3, and 4 are held long. If you check the results for w_i^+ and w_i^- , you will note that they have the same value as the actual solution (i.e., there is no buying or selling of the same stock). The mean return of this portfolio is 2% with a standard deviation of 14.62%.

QUESTIONS

- 13.1. Who is said to have formed the first market-neutral portfolio? In what year was it formed?
- 13.2. True or false: Market-neutral portfolios provide a useful addition to an investor's portfolio because they have a low correlation with other asset classes.
- 13.3. Is there a combination of the risk-free asset and the S&P 500 that also would achieve a low correlation with the market (i.e., the S&P 500)? What about a low β ?
- 13.4. (a) What is dollar neutrality?
(b) Suppose that we take a simple two-stock portfolio. The portfolio manager shorts \$1 million of T with a $\beta = 0.70$ and goes long \$1 million of MGM with a $\beta = 2.42$. Is the portfolio market neutral?
- 13.5. (a) What is beta neutrality?

- (b) What is the minimum level of neutrality that a marketneutral portfolio should have?

- 13.6. Suppose that a portfolio manager would like to construct a market-neutral portfolio from the four stocks below. He already has identified TGT and BA as stocks for the short portfolio and WYNN and POOL as stocks for the long portfolio. Also suppose that the sum of the weights in the long portfolio must equal 1.

Stock	Ticker	β_i	w_i
1	POOL	0.72	
2	BA	1.65	
3	TGT	0.97	0.670
4	WYNN	2.38	

- (a) Suppose that he wishes to build a dollar-neutral portfolio. What are the weights of the other securities?
- (b) Suppose that he wishes to build a dollar-neutral and beta-neutral portfolio. What are the weights of the other stocks in the portfolio?

- 13.7. Suppose that a portfolio manager wants to construct a market-neutral portfolio from the four stocks below. She already has identified FIS and KO as stocks for the short portfolio and MMM and ABC as stocks for the long portfolio. She has found the factor exposures of the stocks to two factors that she believes drive stock returns. Also, suppose that the sum of the weights in the long and short portfolios must equal 1.

Stock	Ticker	$\beta_{i,1}$	$\beta_{i,2}$
1	MMM	0.93	-0.49
2	FIS	0.81	-0.68
3	KO	0.58	0.71
4	ABC	0.57	-0.46

- (a) Suppose that she wishes to build a dollar-neutral portfolio. What should be the weights of the securities in the portfolio?

- (b) Suppose that she wishes to build a portfolio that is dollar neutral and beta neutral to all factors. What should be the weights of the stocks in the portfolio? (*Note:* Round to two decimal places.)
- 13.8.** Suppose that we construct a simple market-neutral portfolio. For each value of ρ below (ρ being the correlation between the residual returns of the long portfolio and the short portfolio), what is the ratio of the information ratio of the market-neutral portfolio to that of the long-only portfolio? (*Note:* Assume that the market-neutral portfolio is fully leveraged.)
- (a) $\rho = 0$
 - (b) $\rho = 0.5$
 - (c) $\rho = 1$
- 13.9.** Suppose that a portfolio manager shorts 1,000 shares at a price of \$50 of stock ZZZ. The initial margin is 50%, and the maintenance margin is 25%. He deposits enough to satisfy the initial margin. At what price will the portfolio manager receive a margin call? (*Note:* Assume unrealistically that the portfolio manager does not have other collateral to satisfy the margin account credit balance.)
- 13.10.** Shortly after it became public that Martha Stewart was being investigated for insider trading, the stock of her company (MSO) dropped from \$19.40 in May 2002 to \$7.00 by September 2002. A portfolio manager believed that the stock price would not continue to decrease. Thus she wanted to buy the maximum number of shares possible using her margin account.
- (a) Given that initial margin is 50%, how many shares could she buy if she deposited \$125,000 into her brokerage account?
 - (b) If the maintenance margin of her account is 25%, what price can the stock reach before she will receive a margin call?
- 13.11.** True or false: To create a market-neutral portfolio, a portfolio manager needs to use cash as collateral on the shorts.
- 13.12.** In a typical market-neutral portfolio, what are the four sources of return when a liquidity buffer is required?
- 13.13.** Suppose that a performance analyst has access to the returns of a market-neutral portfolio. He knows the returns were 2.4% for the month of April 2020. He also knows that the interest rate on both the collateral and liquidity buffer is 5% per annum (that is, $i = i' =$

0.05). The liquidity buffer is 6% (that is, $m_{lb} = 0.06$). The total value of the portfolio is \$100 million. Assume that he is fully leveraged up to a liquidity buffer.

- (a) What must have been the return difference between the long and short portfolios?
 - (b) Does the market-neutral manager know how to separate “good” stocks from “bad” stocks?
- 13.14.** Name four advantages of a market-neutral portfolio over a long-only portfolio.
- 13.15.** Name four disadvantages of a market-neutral portfolio over a long-only portfolio.
- 13.16.** Why is rebalancing necessary when managing a market-neutral portfolio?
- 13.17.** Briefly describe each of the following portfolio techniques and their relation to the market-neutral strategy.
- (a) Long-short
 - (b) Equitization
 - (c) Portable α
 - (d) Pair trading
- 13.18.** A quantitative portfolio manager is very good at managing against the Russell 2000. In fact, she is expected to achieve an α^B of 0.5% per month. Unfortunately, her benchmark is the S&P 500. She thus shorts the required amount of Russell 2000 futures and purchases an equivalent amount of S&P 500 futures contracts. Suppose that the S&P 500 is expected to have a return of 6% in the next month.
- (a) What would be the expected return of her new portfolio?
 - (b) Would the realized return be similar? Explain.
- 13.19.** In 1999, many stocks were believed to be overvalued by quantitative portfolio managers. One of those stocks was Amazon (AMZN). This stock was particularly overvalued relative to other companies in the same line of business, such as Barnes & Noble (BKS). Suppose that a portfolio manager wanted to construct a pair trade consisting of AMZN and BKS. Use the table below to answer the questions. [Note: The β and σ for each stock were measured using monthly historical data from May 1997 to April 1999. The σ is presented in annualized percentage terms. Assume that the long and short weights (before shorting) sum to 1.]

Ticker	β_i	σ_i
AMZN	0.07	120
BKS	0.28	50

- How would you weight each stock if you wanted a dollar-neutral pair trade?
- How would you weight each stock if you wanted a beta-neutral pair trade?
- How would you weight each stock if you wanted a volatility-neutral pair trade?
- Which of the three methods above would you actually use, and why?
- Below are the results from the three types of pair trades from April 1999 to December 2001. Did the pair trade work for all three types of constructions? Which one worked the best? [Note: The standard deviations (SD) are annualized.]

	Holding Period Return (%)	SD (%)
AMZN	-87	88
BKS	-15	58
Pair (dollar-neutral)	36	63
Pair (beta-neutral)	67	78
Pair (volatility-neutral)	15	53

- Does risk make sense in this setting? Does β ?
- 13.20.** If a portfolio manager constructs a portfolio consisting of 50 pair trades, has he created a market-neutral portfolio? Explain.
- 13.21.** A good friend of yours is a portfolio manager at a leading bank and is very good at creating a positive α^B versus the NASDAQ 100. Unfortunately, her benchmark is the S&P 500, which she is not very good at beating. Thus year after year she has received bad bonuses. What would you advise her to do?

- ¹ It is essential that the software package allows for quadratic optimization with binary variables and continuous variables. The CPLEX suite has these and can be used through the CPLEX library in MATLAB. For users who do not have this option, we provide an algorithm in [Appendix 10B](#) that can be adapted for a market-neutral portfolio optimization.
- ² As we discussed in [Appendix 9A](#), one can also place these upper- and lower-bound constraints directly in the **A** matrix.
- ³ Some of these weight boundaries were implicitly already constrained by prior constraints.
- ⁴ For example, IBM's CPLEX optimization software has a MIQP function called CPLEXMIQP.
- ⁵ Using the CPLEX function CPLEXMIQP, available through the CPLEX library in MATLAB, this optimization took 0.0484 seconds to reach a solution.

CHAPTER 14

Bayesian α

Fool me once, shame on you. Fool me twice ... well the point is we can't get fooled again.

—George W. Bush

14.1 INTRODUCTION

Whether from a qualitative research report or from knowledge passed along by word of mouth, portfolio managers usually have useful investing information that does not originate in a data set. Managers often attempt to build such information into an existing quantitative factor model by transforming it into the constant term α , but these attempts can be awkward. Assigning values to qualitative ideas can be somewhat arbitrary, and the new information may not combine efficiently with the estimation in the model. In such cases, *Bayesian theory* is extremely helpful. The theory, which provides a rigorous way to combine heterogeneous sets of information, is a good guide for how best to combine extra qualitative information with the estimation from a factor model.

We start this chapter with a quick review of Bayesian theory and define what we call the *Bayesian α* . We lay out the two stages of Bayesian analysis, determining the prior and determining the posterior. We walk through a number of typical examples to illustrate how to determine the prior, which summarizes all the extra qualitative information that the manager wants to add to the model. We then discuss the posterior, which

combines the extra information with the standard estimation from the factor model. At the end of the chapter we warn managers to avoid mistakes in applying the theory that violate the information criterion.¹

14.2 THE BASICS OF BAYESIAN THEORY

Bayesian theory is a systematic application of a statistical theorem called *Bayes' rule*. To lay the foundation for Bayes' rule, let us review the concept of conditional probability. Conditional probability is the probability of an event (b) happening when we know that another event (a) will happen. For example, event a is, "It will rain tomorrow," and event b is, "The temperature will be below 100 degrees tomorrow." We may ask, "What is the probability of the temperature being below 100 degrees tomorrow if it rains tomorrow?" We can express the question as a conditional probability $P(b|a)$. If the answer to this question is 99% (i.e., when it rains, the temperature is unlikely to reach 100 degrees), it is the same as saying that the probability of b conditional on a is 99%, or $P(b|a) = 99\%$.

Bayes' rule relates probabilities to each other. It says that the probability of b conditional on a , $P(b|a)$, is proportional to two probabilities: the probability of a conditional on b , $P(a|b)$, and the probability of b , $P(b)$. Mathematically, we may write

$$P(b|a) \propto P(a|b)P(b) \quad (14.1)$$

where $\propto$ means "is proportional to." Continuing the example, if rain is more likely when the temperature is below 100 degrees, then a temperature below 100 degrees is more likely when it rains. In other words, when $P(a|b)$ has a high value, $P(b|a)$ has a better chance of having a high value as well. Also, if a temperature below 100 degrees is likely whether it rains or not, then it is more likely to be under 100 degrees if it rains. Expressed in terms of probabilities, the idea is that if $P(b)$ has a high value, then $P(b|a)$ probably also will be quite high.

We can interpret Bayes' rule in [Eq. \(14.1\)](#) in a more general way. That is, we can take a and b to be random variables rather than events and $P(\cdot)$ to be the probability density of a random variable rather than the probability of an event. *Random* variables are any variables (such as stock returns) that

can take on different values, with different levels of likelihood for each potential value. We can revise our example about tomorrow's weather so that we are discussing random variables rather than events. Suppose now that b is a random variable that equals the temperature at noon tomorrow, and a is a random variable that equals the amount of rainfall tomorrow. For discrete random variables (i.e., variables with only a limited set of potential values), probability is the same thing as probability density. For continuous random variables, the probability of attaining a certain value must be recovered by integrating the probability density over an interval. For example, if b is the temperature at noon tomorrow, then the probability that the temperature will be between 90 and 100 degrees can be found by integrating $P(b)$ over b between 90 and 100. Whether it is for a discrete variable or for a continuous variable, the conditional probability density is the probability density of a random variable conditional on a specific value of another random variable. If a is the amount of rainfall tomorrow and b is the temperature at noon tomorrow, then $P(b|a)$ is the probability density of temperature b given a specific amount of rainfall a .

Let us apply Bayes' rule to a specific kind of conditional probability density, the probability density of parameters conditional on data. We will indicate parameters as θ (potentially a vector) and data as $\mathbf{x}$ (probably a very long vector). For factor models, the coefficients on explanatory variables and the error variances are the parameters. In the fundamental factor model, the factor premiums and the error variances are the parameters, whereas in the economic factor model, the factor exposures and the error variances are the parameters. For both factor models, the values of dependent and explanatory variables are data.

To apply Bayes' rule to the probability density of parameters conditional on data, we simply can replace b and a in [Eq. \(14.1\)](#) with θ and $\mathbf{x}$:

$$p(\theta | \mathbf{x}) \propto p(\mathbf{x} | \theta)p(\theta) \quad (14.2)$$

This formula takes the central place in Bayesian theory, and each term deserves an explanation. The last term, $p(\theta)$, is called the *prior*.² This is the probability density of the parameters *prior to* data analysis. The prior may be based on qualitative or nondata information, intuition, or logic.

The probability density of the data conditional on the parameters, $p(\mathbf{x}|\theta)$, is called the *likelihood function*, or the *likelihood* for short. It shows how

likely it is that the data are drawn from the given parameters. For example, if the data are -100 and -101 , then we know that they are very unlikely to be from the normal distribution with a mean of 0 and a variance of 1 . On the other hand, if the data are -0.1 and 0.1 , then we can say that they are quite likely to be from the normal distribution with a mean of 0 and a variance of 1 . The likelihood function expresses this. To put it another way, the likelihood function measures the distance between the data and the parameters.

The probability density of the parameters conditional on the data $p(\theta|\mathbf{x})$ is called the *posterior*. It is the probability density of the parameters *after* we analyze the data. The posterior combines the information we had before analyzing the data (the prior) and the information we obtained from the data (the likelihood). Thus the posterior summarizes all we know about the parameters. The formula in [Eq. \(14.2\)](#) is a simple way to combine nondata or qualitative information about the parameters into the data analysis. We simply need to multiply the prior and the likelihood. Not only is it simple, but it turns out to be the best way of adding nondata information to the data analysis.

Since Bayesian theory seems quite straightforward, one might wonder why it deserves to be called a theory. Although it is indeed straightforward mathematically, Bayesian theory represents a huge philosophical jump from “classical” statistical theories. In classical theory, there is a clear distinction between what is a random variable and what is not a random variable. The parameters of a model are not considered random variables. While we do not know the true values of the parameters, the classical thinking goes, true values must be somewhere “out there,” so we cannot treat them as random variables. Thus the parameters cannot have probability distributions, and we can think neither of the prior nor of the posterior. Bayesian theory, on the other hand, makes no distinction between random and nonrandom variables. Even if we do not know the exact value of a variable, we still can assign a probability to each possible value and treat the variable as random. Another important feature of Bayesian theory is that the probabilities do not have to be objective. In fact, the prior is a subjective probability of the parameters. If you have special information, the prior you calculate will be different from the prior another person calculates. In classical theory, the probability of a value is taken almost as a law of nature, and there is no room to add one’s subjective opinion to it.

While the philosophical difference between Bayesian and classical is great, the practical differences should not be exaggerated. When both the classical theory and the Bayesian theory are applicable, their conclusions are identical more often than not. In this chapter we are mostly interested in situations in which only the Bayesian theory is applicable. Classical theory has only limited use (or almost no use) when one attempts to combine qualitative information with data analysis. In such cases, we do not have any choice but to adopt Bayesian theory.

14.3 BAYESIAN α MOJO

Instead of adopting the Bayesian approach from beginning to end, we will focus on where Bayesian theory clearly adds value to quantitative equity portfolio management (QEPM), which is in incorporating qualitative information on α into the QEPM process in a systematic way. Portfolio managers often have nondata information on individual stocks, and this information is often qualitative rather than quantitative. Maybe a research report appears saying that stock XYZ has an exceptionally positive outlook but that its current stock price does not reflect the positive outlook. Maybe an analyst, who has information not available to the portfolio manager, comes up with a selection or a ranking of stocks, and the portfolio manager would like to incorporate that into the analysis. Or perhaps the manager has a data set that he or she finds informative but not suitable as the basis for a factor model.

In such cases, a portfolio manager typically attempts to set the value of α in the factor model in some ad hoc way that reflects the extra information. The portfolio manager may use the following economic factor model to estimate the return of stock i , r_i :

$$r_i = \alpha_i + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i \quad (14.3)$$

where $f_1, \dots, f_K$ are factor premiums, $\beta_{i1}, \dots, \beta_{iK}$ are factor exposures of stock i , and ϵ_i is the error component of stock i 's return. The alpha of stock i , α_i , is the part of the average stock return that the model cannot explain. The portfolio manager bases the factor premiums and factor exposure on his or her original data set. Since this data-based model cannot explain α_i , the

manager attempts to set the value of α_i according to his or her nondata information. However, since the nondata information is qualitative, assigning values could be arbitrary.

Bayesian theory becomes very useful at this stage. Rather than setting the value of α in an arbitrary way, the portfolio manager can adopt the Bayesian approach and assign a distribution to the α that is consistent with the nondata information. This assigned distribution is what Bayesian theorists call the *prior*. Once the prior is assigned, the portfolio manager can follow the standard procedure of the Bayesian approach to obtain the best estimates of [Eq. \(14.3\)](#) that reflect not only the information in the data but also the useful nondata information. Calculating the *Bayesian α* , as we call it, is consistent with the information criterion that we introduced in [Chapter 2](#). Attempts to incorporate nondata information into the model without calculating the Bayesian α most likely will fail to exploit the information completely and violate the information criterion, seriously distorting the portfolio. The Bayesian α , on the other hand, is entirely a gain to the portfolio manager and another strong source of α mojo.

14.4 QUANTIFYING QUALITATIVE INFORMATION

The first step of Bayesian analysis is to decide on the prior of the α from the nondata information, a somewhat challenging step because nondata information is often qualitative. In this section we will go through three relatively simple but commonly faced situations involving calculating the prior. More complicated cases will be dealt with in the next two sections. First, we will start with the situation in which the portfolio manager receives a list of stocks screened by other analysts. Then we will consider the case in which the portfolio manager receives a ranking of selected stocks. Lastly, we will consider the case in which the portfolio manager wants to incorporate buy and sell recommendations for selected stocks.

14.4.1 Quantifying a Stock Screen

Suppose that we have many lists of analyst-recommended stocks, each list made by an analyst through some screening process. How can we convert

these lists into the prior for the α of each stock? When a stock is included in a list, it means that an analyst believes that the stock is likely to outperform the stocks not included in the list. That is, a listed stock is likely to have a higher α than that of an unlisted stock. If stock A is included in a list but stock B is not, we can say that

$$P(\alpha_A > \alpha_B) > 0.5 \quad (14.4)$$

If stock A is included in many analysts' lists and stock B in none, then we can be more confident that α_A is greater than α_B because more analysts agree on the superiority of stock A . Specifically, if we consider each list (or analyst) equally reliable, then we can say that

$$P(\alpha_A > \alpha_B) = \frac{\text{number of lists including stock } A}{\text{number of all lists}} \quad (14.5)$$

That is, if half the analysts say that α_A is greater than α_B , then we may be 50% sure that α_A is greater than α_B . If every analyst says that α_A is greater than α_B , then we may be (almost) 100% sure that α_A is greater than α_B .

If stock B is not included in any of the lists, the lists do not say anything about stock B . Thus, as far as α_B is concerned, we do not need to do anything special. We can estimate α_B in the usual way and obtain the estimate $\hat{\alpha}_B$ and its standard error $S(\hat{\alpha}_B)$. From the Bayesian perspective, $\hat{\alpha}_B$ and its standard error are considered to be the mean and the standard deviation of α_B . That is, assuming a normal distribution, we can write

$$\alpha_B \sim N[\hat{\alpha}_B, S(\hat{\alpha}_B)^2] \quad (14.6)$$

Once Eqs. (14.5) and (14.6) are obtained, it is straightforward to find the prior distribution for α_A . The inclusion of stock A into the list does not imply anything about the relative amount of uncertainty regarding α_A and α_B . Thus it is reasonable to take $S(\hat{\alpha}_B)$ as the standard deviation of α_A as well. Now we only have to determine the mean of α_A . Let μ_A be the mean of α_A . Assume that α_A has a normal distribution and is independent of α_B

(which is reasonable given that we do not have any information indicating otherwise). Then

$$\alpha_A - \alpha_B \sim N[\mu_A - \hat{\alpha}_B, 2S(\hat{\alpha}_B)^2] \quad (14.7)$$

It follows immediately that

$$p_A = P(\alpha_A - \alpha_B > 0) = 1 - \Phi\left[\frac{\hat{\alpha}_B - \mu_A}{\sqrt{2S(\hat{\alpha}_B)^2}}\right] \quad (14.8)$$

where Φ is the standard normal cumulative distribution function. Considering Eqs. (14.5) and (14.8),

$$\mu_A = \hat{\alpha}_B - 2 S(\hat{\alpha}_B) \Phi^{-1}(1 - p_A) \quad (14.9)$$

where Φ^{-1} is the inverse standard normal cumulative distribution function that takes a probability as the argument. Thus we have determined the prior distribution for α_A . We can generalize this analysis in a number of directions. First of all, we can find the distribution of α_B using all the stocks that are not included in any of the lists. If we cannot find such stocks, then we can use an arbitrary set of stocks and modify Eq. (14.5). Also, if different analysts are not equally reliable, then we may assign different weights to different analysts and lists.

A more important generalization of the analysis is the case in which there is only one list of stocks. It may be that all analysts work as a team and produce a single list, or it may even be the case that the portfolio manager is the one who produces the list. For one stock list, the preceding analysis is still valid except for Eq. (14.5). For Eq. (14.5), the portfolio manager himself or herself must decide on the probability that α_A is greater than α_B . While this may seem quite subjective, this is not necessarily a problem. The prior distribution is meant to represent subjective opinion. Unless the portfolio manager chooses different probabilities for different stocks, Eq. (14.9) is common for all stocks included in the list. That is, all stocks in the list have the same mean as expressed in Eq. (14.9).

14.4.2 Quantifying a Stock Ranking

Suppose that instead of having a simple list of stocks, we have analyst-prepared rankings of stocks. We can easily extend the idea developed in the preceding subsection to this case. If stock A is ranked higher than stock B , it means that an analyst believes that the α of stock A is higher than the α of stock B . The probability of the α of stock A being higher than the α of stock B can be computed by comparing the number of analysts who believe this to be true to the total number of analysts who do not. That is,

$$P(\alpha_A > \alpha_B) = \frac{\text{number of analysts ranking stock } A \text{ higher than stock } B}{\text{number of all analysts}} \quad (14.10)$$

Note that [Eq. \(14.10\)](#) is essentially identical to [Eq. \(14.5\)](#). Thus we can follow the same procedure for dealing with stock screens.

Once stock B is chosen, this same stock should be compared with all the other stocks to exploit all the information in the ranking. One possibility is to choose for stock B a stock that is ranked most frequently at the bottom of ranked lists, and for each of the remaining stocks, calculate [Eq. \(14.10\)](#) relative to stock B . In any case, how stock B is selected does not really matter as long as it is compared with all other stocks. The procedure is identical whether all stocks are ranked by every analyst or different stocks are ranked by different analysts. If there is only one ranking (perhaps made by the portfolio manager himself or herself), then the portfolio manager assigns a number to $P(\alpha_A > \alpha_B)$ in [Eq. \(14.10\)](#) and follows the same procedure.

14.4.3 Quantifying the Buy and Sell Recommendations

Qualitative information also may be in the form of analysts' buy and sell recommendations. Instead of a simple "buy" or "sell," analysts often give stocks one of five recommendations: strong buy, buy, neutral, sell, and strong sell. These recommendations can be converted easily into the probability of one stock having a higher α than another stock. One way to do this is to use the *buy ratio*, which is the number of buy recommendations divided by the total number of buy and sell recommendations. Once the buy ratio is calculated for every stock, we can find the probability of the α of stock A being greater than the α of stock B by subtracting the buy ratio of stock B from the buy ratio of stock A . That is,

$$P(\alpha_A > \alpha_B) = \frac{\text{number of buy recommendations for } A}{\left(\text{number of buy recommendations for } A + \text{number of sell recommendations for } A \right)} - \frac{\text{number of buy recommendations for } B}{\left(\text{number of buy recommendations for } B + \text{number of sell recommendations for } B \right)} \quad (14.11)$$

The buy ratio of stock A shows what fraction of analysts believes that stock A is superior, and the buy ratio of stock B shows the fraction of analysts that believes that stock B is superior. The difference between the two buy ratios shows the fraction of analysts that believes that stock A is superior to stock B . This is exactly true if the same analysts make recommendations for stocks A and B and approximately true in other cases. After [Eq. \(14.11\)](#) is determined, we can proceed as we did in the previous cases.

14.5 THE Z-SCORE-BASED PRIOR

The Z-score approach explained in [Chapter 5](#) produces rankings of stocks based on certain factors. It is therefore possible to generate the prior of the α from the Z-score and then estimate a factor model using other unused information. This approach is superior to using the Z-score alone because it employs all the information available in the data. Suppose that we have L factors and a calculated Z-score for each stock. If we have not aggregated the Z-score, then the Z-score for each stock is a vector of L numbers. Let $\mathbf{z}_i = (z_{i1}, \dots, z_{iL})$ be the Z-score of stock i . The Z-score for each factor implies a certain ranking of stocks. Since there are L factors, we have L implied rankings of stocks. Once we interpret the Z-score as the ranking of stocks, the prior of the α can be generated according to the method for stock rankings described earlier. That is, given stock A and stock B , the probability that the α of stock A is greater than the α of stock B is

$$P(\alpha_A > \alpha_B) = \frac{\sum_{l=1}^L I(z_{Al} > z_{Bl})}{L} \quad (14.12)$$

where $I(\cdot)$ is an indicator function returning the value 1 if the expression inside the parentheses is true and 0 if not.

The preceding formula is based on the idea that each factor is equally informative. However, as we discussed in [Chapter 5](#), we may have reason to assign different weights to different factors. Suppose that for factor l we want to assign weight v_l so that $v_1 + \dots + v_L = 1$. Then the probability that the α of stock A is greater than the α of stock B changes to

$$p_A = P(\alpha_A > \alpha_B) = \sum_{l=1}^L v_l I(z_{Al} > z_{Bl}) \quad (14.13)$$

After the probability is calculated, we can easily determine the prior of the α of stock A as explained in the preceding section. We choose an arbitrary stock B and obtain the estimate $\hat{\alpha}_B$ and its standard error $S(\hat{\alpha}_B)$ from a factor model. Then the prior for the α of stock A becomes

$$\alpha_A \sim N[\mu_A, S(\hat{\alpha}_B)^2] \quad (14.14)$$

where

$$\mu_A = \hat{\alpha}_B - \sqrt{2} S(\hat{\alpha}_B) \Phi^{-1}(1 - p_A) \quad (14.15)$$

The derivation of this formula is the same as the derivation explained in the preceding section. That is, p_A is calculated from [Eq. \(14.8\)](#), and we can substitute p_A into [Eq. \(14.15\)](#).

14.6 SCENARIO-BASED PRIORS

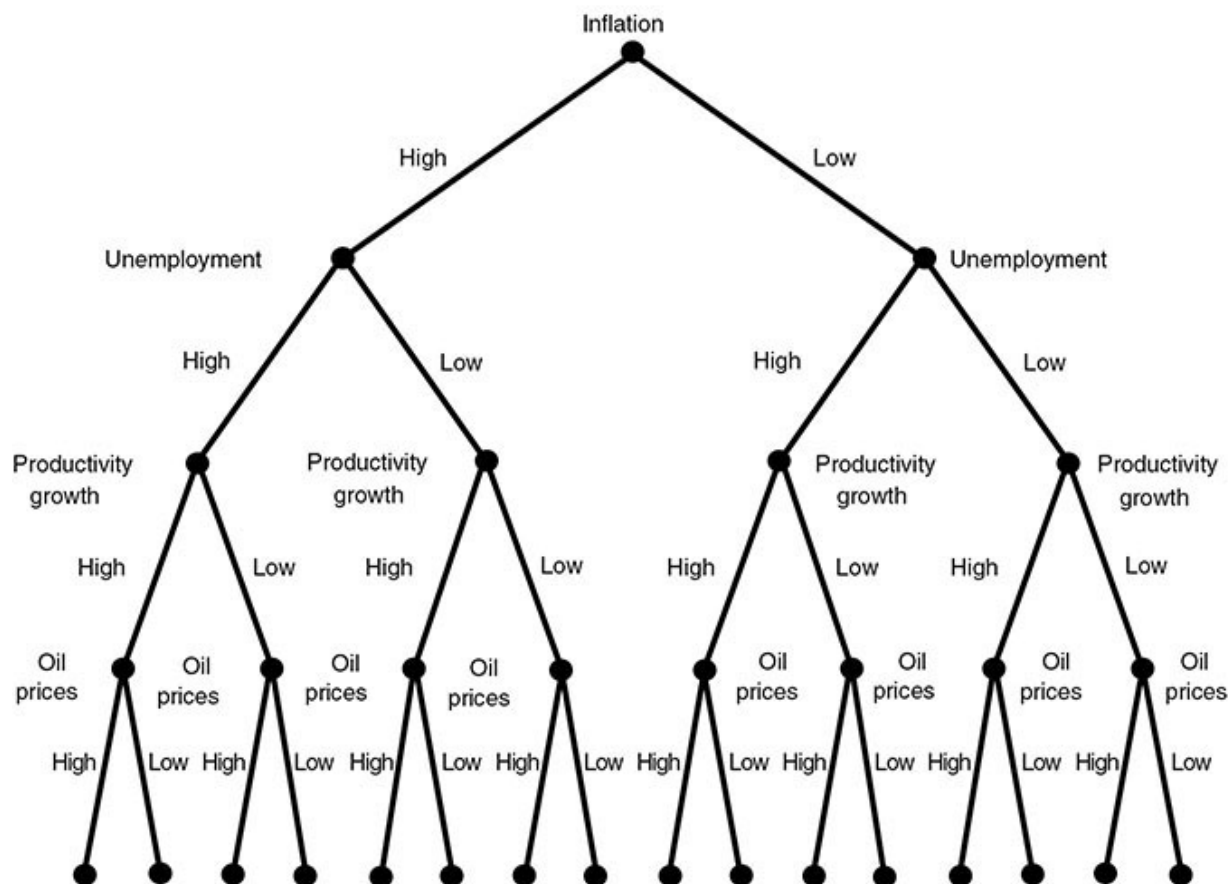
The prior of the α can be generated systematically from what is known as *scenario analysis*. Scenario analysis is suitable when the portfolio manager has strong opinions about how individual stocks will perform under specific situations. Different variables in the economic or natural environments present opportunities for scenario analysis. For instance, investors use scenario analysis before presidential elections to guess how different election outcomes will affect stocks. Investors in the agricultural and certain

manufacturing industries use it to try to anticipate the effect of weather patterns on future stock returns. The portfolio manager can systematically incorporate his or her views about stock returns under different conditions into the prior of the α through scenario analysis. For each scenario, the portfolio manager determines—whether based on his or her own beliefs or on some outside evidence—two quantities: the probability of the scenario being realized and the α of the stock when the scenario is realized. Given the probability of each scenario and the value of the α , he or she can construct a distribution for the stock's α , which becomes the prior of the α .

The first step in scenario analysis is to identify all possible scenarios. Typically, this can be done with an *event tree*. Suppose that the portfolio manager has certain beliefs about future stock returns, which are conditional on the following events: high or low inflation, high or low unemployment, high or low productivity growth, and high or low oil prices. An event tree organizes these events, as illustrated in [Figure 14.1](#), producing 16 possible scenarios. In general, if E pairs of events are considered, then 2^E scenarios are possible.

FIGURE 14.1

Scenario analysis event tree.



The second step in scenario analysis is to assign probabilities to each scenario. If each pair of events is independent of the others, then the probability of a scenario is simply the product of the probabilities of the events that happen in the scenario. For example, the probability of scenario 1 is given by³

$$P(\text{high inflation, high unemployment, high productivity growth, high oil price}) = P(\text{high inflation}) \cdot P(\text{high unemployment}) \cdot P(\text{high productivity growth}) \cdot P(\text{high oil price}) \quad (14.16)$$

However, if each pair of events is not independent, this formula does not work, and the probability of each scenario must be found individually through, for instance, the forecasting methods discussed in [Chapter 8](#).

The third step is to determine the α of each stock. There are two approaches to this, the subjective approach and the event-study approach.

The subjective approach draws on the portfolio manager's personal beliefs. The event-study approach assigns the α based on historical data. To determine α through the event-study approach, we need first to define the event exactly. For example, we may define high unemployment as an unemployment rate of above 5% and low unemployment as an unemployment rate of below 5%. Then we need to find the average abnormal return for each event. During the period from January 2011 to December 2020, 72 months can be categorized as high-unemployment months, and the remaining 48 months can be categorized as low-unemployment months. The abnormal return can be calculated in various ways, but the simplest method is to calculate the excess return of each stock over the market return. Table 14.1 shows the average abnormal returns as well as the standard deviation of selected stocks for low- and high-unemployment months. These average abnormal returns can be used as α 's. When applying the event-study approach, one should be careful not to base the prior on the data that eventually will be used to estimate the stock return model. This issue will be discussed further toward the end of this chapter.

TABLE 14.1

Average Abnormal Returns of Selected Stocks in the High-/Low-Unemployment Scenario

Scenario	AAPL	AMZN	JNJ	MSFT	WMT
Low unemployment	19.49 (21.84)	16.34 (21.62)	1.54 (13.55)	23.02 (12.07)	11.73 (19.19)
High unemployment	13.31 (23.78)	22.62 (25.22)	-1.01 (11.71)	3.96 (18.55)	-8.58 (17.90)

Note: The numbers represent the annualized return of each stock in excess of the market return averaged across the months following a month whose unemployment rate is below 5% ($N = 48$) or above 5% ($N = 72$) between January 2011 and December 2020. The figures in the parentheses are standard errors.

Suppose that we defined S scenarios. Let $P(s)$ be the probability of scenario s and $\alpha_i(s)$ be the α of stock i in scenario s . Then the profile of the probability and the α $\{[\alpha_i(1), P(1)], \dots, [\alpha_i(S), P(S)]\}$ completely determine the distribution of the α . We certainly could stop here as far as the prior is concerned. However, approximating the prior distribution as a normal distribution reduces the computational burden significantly at a later stage of the analysis. The approximate normal distribution is given as follows:

$$\alpha_i \sim N(\mu_i, \sigma_i^2) \quad (14.17)$$

where

$$\mu_i = \sum_{s=1}^S P(s) \alpha_i(s) \quad (14.18)$$

and

$$\sigma_i^2 = \sum_{s=1}^S P(s) [\alpha_i(s) - \mu_i]^2 \quad (14.19)$$

14.7 POSTERIOR COMPUTATION

Once the prior is determined, the posterior can be computed from Bayes' rule. While the formula for the posterior may look complicated, the computation itself is straightforward. This is especially true when the prior is a normal distribution. When the prior is a normal distribution, the posterior is also a normal distribution, and we only have to determine the mean and the variance of the posterior. In this section we focus on the interpretation of the posterior formula.⁴

Suppose that the portfolio manager uses the following factor model to estimate the return of individual stocks:

$$r_i = \alpha_i + \beta_{i1}f_1 + \cdots + \beta_{iK}f_K + \epsilon_i \quad (14.20)$$

where r_i is the return of stock i , $f_1, \dots, f_K$ are factor premiums, $\beta_{i1}, \dots, \beta_{iK}$ are factor exposures of stock i , and ϵ_i is the error component of stock i 's return. α_i is the α of stock i for which the portfolio manager defined the following prior using one of the approaches described in the preceding section:

$$\alpha_i \sim N(\mu_{\alpha_i}, \sigma_{\alpha_i}^2) \quad (14.21)$$

We assume that the portfolio manager did not define the prior for $\beta_{i1}, \dots, \beta_{iK}$. The computation is in fact easier if the prior for $\beta_{i1}, \dots, \beta_{iK}$ is defined as well, but we will focus on the more typical situation in which it is not.⁵

Since the posterior combines the information in the prior and the information in the data [which is summarized in the ordinary least squares (OLS) estimate], for α_i , the posterior mean is the weighted average of the prior mean and the OLS estimate. The weights are the inverse of the respective variances. For the prior mean, the weight is the inverse of the prior variance, and for the OLS estimate, the weight is the inverse of the OLS variance of the estimate (i.e., the squared standard error of the estimate). The inverse of the variance shows how precise the quantity is. If the variance is large, the quantity is less precise, and if the variance is small, the quantity is more precise. For this reason, the inverse of the variance is called the *precision* of the estimate. It is often said that the posterior mean is the *precision-weighted average* of the prior mean and the OLS estimate. The posterior mean of α_i is given as follows:

$$\begin{aligned} \mu_{\alpha_i}^* = & \left(\frac{1}{\sigma_{\alpha_i}^2} + \frac{1}{\sigma_{\epsilon_i}^2} \iota' \mathbf{M} \iota \right)^{-1} \frac{1}{\sigma_{\alpha_i}^2} \mu_{\alpha_i} \\ & + \left(\frac{1}{\sigma_{\alpha_i}^2} + \frac{1}{\sigma_{\epsilon_i}^2} \iota' \mathbf{M} \iota \right)^{-1} \left(\frac{1}{\sigma_{\epsilon_i}^2} \iota' \mathbf{M} \iota \right) \hat{\alpha} \end{aligned} \quad (14.22)$$

where ι is a T -dimensional vector of 1s and T is the number of observations in the estimation, and $\mathbf{M}$ is the residual matrix (defined in Appendix 14A). As explained earlier, the posterior mean is the weighted average of the prior mean (μ_{α_i}) and the OLS estimate ($\hat{\alpha}$). The average is taken using the prior precision ($1/\sigma_{\alpha_i}^2$) and the OLS precision $[(1/\sigma_{\epsilon_i}^2)\iota' \mathbf{M} \iota]$ as the weights. The “denominator” (in the matrix sense), $(1/\sigma_{\alpha_i}^2) + (1/\sigma_{\epsilon_i}^2)\iota' \mathbf{M} \iota$, is simply the sum of the prior precision and the OLS precision. By “dividing” each weight by this quantity, the sum of the weights equals 1.

The posterior variance of α_i is the inverse of the sum of the prior precision and the OLS precision. That is, the posterior precision is the sum

of the prior precision and the OLS precision. The precision shows how much confidence we have in a given quantity. If we have great confidence in our prior (there is high prior precision), then we still will have great confidence after the estimation (high posterior precision). Even if we have little confidence in our prior, if the OLS produces reliable results (there is high OLS precision), then our confidence level will go up (high posterior precision). Thus it makes sense to add the prior precision and the OLS precision to get the posterior precision.

The posterior variance of α_i is given as

$$\left(\sigma_{\alpha_i}^*\right)^2 = \left(\frac{1}{\sigma_{\alpha_i}^2} + \frac{1}{\sigma_{\epsilon_i}^2} \iota' \mathbf{M} \iota \right)^{-1} \quad (14.23)$$

It is the inverse of the sum of the prior precision ($1/\sigma_{\alpha_i}^2$) and the OLS precision $[(1/\sigma_{\epsilon_i}^2)\iota' \mathbf{M} \iota]$.

The formulas for the posterior mean and the variance of $\beta_{i1}, \dots, \beta_{iK}$ (together with other useful quantities) are presented in Appendix 14A.⁶ Here we only note that while the prior is defined only for α_i , the estimates of $\beta_{i1}, \dots, \beta_{iK}$ will be affected by the prior as well. This is so because there is a correlation between the estimate of α_i and the estimates of $\beta_{i1}, \dots, \beta_{iK}$. Since the prior influences the estimate of α_i , it also will indirectly affect the estimates of $\beta_{i1}, \dots, \beta_{iK}$, even though there is no prior for $\beta_{i1}, \dots, \beta_{iK}$.

14.8 THE INFORMATION CRITERION AND BAYESIAN α

The Bayesian approach is an extremely powerful tool for combining two sets of information, typically data information and nondata information. Data information is summarized by the factor model (i.e., the likelihood), and nondata information is summarized by the prior. The two sets of information are optimally combined in the posterior. While the Bayesian approach is quite useful, it is easy to make certain mistakes in applying it, especially (though not only) when the prior is determined from data

information. If the prior is determined from data information and if the data are not properly *excluded* from the model estimation, one will end up double counting certain information. The following are examples of possible *mistakes* that portfolio managers might be tempted to make:

- Screen stocks based on the price-to-earnings ratio, develop the prior for the alpha based on the screen, and use the price-to-earnings ratio again as a factor in the estimation of the factor model.
- Rank stocks based on momentum or some other factor, develop the prior for the alpha based on the ranking, and use momentum again as a factor in the estimation of the factor model.
- Generate the prior based on analyst recommendations, and when estimating the factor model, use the analyst factor again based on the same information.
- Base the prior on the Z-score of certain factors, and use some of these factors again when estimating the factor model.
- Do the scenario analysis based on the past stock performance when inflation was high or low, and use the same inflation data as a factor in the factor model.

The list of possible mistakes is infinite, but the general idea is the same: Information should not be used twice. If a piece of information was used to make the prior, it should not be used in estimating the model. The information criterion we developed in [Chapter 2](#) encapsulates this rule. In [Chapter 2](#) we emphasized that information should not be wasted and that all information should be combined in the most efficient way. Using the same information twice is inefficient. In fact, it violates all the fundamental assumptions of statistical inference. Thus the standard errors will be wrong if the same information is used twice.

14.9 CONCLUSION

Bayesian theory represents a major step forward from classical statistical theory because it allows us to treat any variable as a random variable and assign subjective probabilities, based on qualitative or nondata information, to the variable's potential values. Bayesian theory opens up the possibility of encapsulating such extra information in a Bayesian α that amplifies the

QEPM model of stock returns. In this chapter we first showed how to calculate the Bayesian α 's prior, which summarizes and quantifies extra information that does not originate in the data set with which the factor model was estimated. This extra information can come from stock screens and rankings, buy/sell recommendations, Z-scores, or scenario analysis. We then discussed how to calculate the Bayesian α 's posterior, which summarizes all the available information by combining the prior and the factor model estimation. Having calculated the posterior, we can fully describe the distribution of Bayesian α . As long as all pieces of extra information contained in the α are distinct from the data used to estimate the factor model, adding Bayesian α to the model upholds the information criterion. Bayesian α , with its ability to widen the scope of a purely data-based factor model into diverse kinds of information, is our third source of α mojo.

QUESTIONS

- 14.1. Prostate cancer affects roughly 12.5% of the male population. A prostate-specific antigen (PSA) test is able to detect whether men might have the disease. The test has a type I error of 70% (when the person does not have cancer, there is a 70% chance that the test will conclude that he does) and a type II error of 20% (when the person does have cancer, there is a 20% chance that the test will conclude that he does not). In a recent testing at UCSF Medical Center, a person tested positive for prostate cancer.
- (a) What is the probability that the person has prostate cancer?
 - (b) A company called Grail has a blood cancer screening test called Galleri. Its false-positive rate is 0.5% and its false-negative rate is 0.6%. If 100,000 men took the new test over the old test, how many more people with cancer would be alerted early? How many more people without prostate cancer would avoid the psychological stress and physical inconvenience of a biopsy? (*Note:* The false-positive and false-negative rates were obtained from a study by Grail; however, they are for detecting all cancers and not specifically prostate cancer. The prostate-specific cancer numbers were not available).